The Marked Quaternionic Continuation

Exact reconciliation, positive fibres, and a dynamical completion

Codex verification and continuation of two supplied Pro-mode constructions

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Abstract

We identify two proposed $4+4+4+1$ continuations as the same embedded subspace of $\mathrm{Herm}_3(\mathbb{H})$, with explicit mutually inverse coordinates preserving the original observation matrix. Their common positive cone contains the old complex-Hermitian cone and yields the sphere pair (S^{12}, S^8). We compute the Jordan product with the retained nonunitary marking and prove its fifteen-dimensional closure. In the full positive carrier a split observation to $\mathrm{Herm}_2(\mathbb{C})$ gives the exact bound $\det G \geq |\rho|^2$, with every permitted pair realized. An explicit spectral-floor map transfers this observation to trace-free scalar configurations without identifying them with positive matrices. In the supplied scalar dynamics, however, the omitted complex coordinate is generically sourced: explicit initial data give $\ddot{\rho} = -2\kappa$. We also construct global nonzero invariant sectors of the unrestricted equations: for fixed complex $W \subset u^{\perp}$, retain $b \in W$ and $\Phi W \subset W$, with tangent velocities. Their configuration dimensions are six and eight; no reaction force is needed. A fourth-order escape example shows why low-order tangency tests alone do not prove invariance. An exact periodic moving-kernel family then proves that these fixed-subspace sectors are not exhaustive. Its conserved colour moment identifies the nonzero Gauss residual of the zero-connection gauge lift. We prove that no homogeneous spatial connection in temporal gauge repairs this literal orbit, and derive the exact Gauss-inertia equation for arbitrary temporal connection. A separate two-variable interacting sector solves every gauge and scalar equation with zero connection. We also verify the supplied scalar vacuum transition and its full metric Hessian, with one radial, eight tangent, and five normal directions. Averaging the marked Lorentz determinant over the colour group gives $\mathrm{tr}\,\mathrm{adj}(A)$; on the positive rank-one boundary this is $\|b\|^2$, a variable which ranges continuously along gauge-inequivalent scalar minima. Finally, the retained rank-six period map has complete positive metric fibres of dimensions twelve over H and seventeen over G. An explicit monodromy-compatible metric produces rank loss, whereas bounded relative metric condition number excludes it. We derive the exact constrained dynamics as a separate model, and correct the right-action convention and the anti-Hermitian gauge kinetic sign. This is a finite-dimensional geometric and dynamical audit, not a continuum quantum-field-theory mass-gap theorem. We further prove all-time real-connection and static-frame complex obstructions to the original rotating orbit. A different three-coordinate sector supplies exact current-free null waves and periodic full-gauge solutions with frequency tending to zero and energy tending to the vacuum value at fixed spatial volume. Their amplitude does not tend to zero. The complete eight-branch transverse gauge spectrum includes $\mu_{\mathrm{d}}^2 = (2g^2/3)\|b\|^2$, detected by an explicitly conserved gauge-invariant vector current. Thus this branch is proportional to the colour-averaged Lorentz determinant. Finally, W is geometrically a lattice cometric: Hermitian compatibility gives $h_3 = 2H_3^{-1}$. The unchanged native Euclidean periods give constant $H_3 = 2I_3$ and $G = 2PP^*$ while their source and induced quotient Laplace gaps are asymptotic to $4\pi^2/y^2$ and $8\pi^2/y^2$, respectively. For the original rotating scalar orbit we also exclude a genuinely complex rotating-pair gauge class, including its constant spatial rotations. In the unrestricted rotating-pair equations, Gauss gives the positive inverse $T = 4(g^2\mathcal{I} + 2\mathcal{K})^{-1}[Z_1, Z_2]$, $C = \omega T$, and the scalar equation fixes $\omega^2 = (1/4 + \sum_j \|Z_j q\|^2)/\|Tq\|^2$. We prove a two-sign bijection with the common zero set of four full matrix/vector residuals in all sixteen spatial coordinates

and g^2. Existence or exclusion of that zero set remains open. The unrestricted scalar equations also force the stabilizer coefficients $s_X = -i\,\mathrm{tr}(HX)$ to satisfy $s_C^2 < s_{Z_1}^2 + s_{Z_2}^2$ unless all vanish. Both spatial equations give a positive two-by-two stress matrix with an explicit sum-of-squares discriminant; these exact tests retain all colour coordinates.

For each fixed unrestricted spatial pair, we now prove an explicit bounded coupling interval and a nonnegative residual polynomial of degree exactly twenty-eight whenever its domain polynomial is nonzero. Its admissible positive roots are precisely the allowed original g^2: there are at most fourteen, with both frequency signs reconstructed. Three vector tests decide the identically empty Gauss domain. A nonzero, explicitly defined 24×24 Sylvester determinant polynomial further confines the entire spatial solution projection to a closed, nowhere-dense, measure-zero subset of the sixteen-coordinate carrier. An exact unrestricted witness and rational and finite-field Bezout certificates prove that polynomial is nonzero. The eight-dimensional linear class in which both spatial matrices exchange the matter line with its orthogonal complement lies inside this exceptional set, but a full-equation positivity identity excludes every positive coupling and both frequency signs there. A denominator-free residual identity also gives a strictly positive spatial-error bound on its scalar–Gauss zero set, uniform for bounded spatial norm and coupling bounded away from zero. This scalar–Gauss zero set is now completely parameterized: the classified domain is $\mathcal{D} \times \{1,-1\}^2 \times U(1)^2$, and the literal domain retains the frame phase subject only to $(\zeta, p_+, p_-) \sim (-\zeta, -p_+, -p_-)$. An explicit family attains every original $t = g^2 > 0$, with all scalar and Gauss entries zero. The two spatial residuals cannot vanish simultaneously, and existence on the remaining exceptional set is open.

The literal inverse charts further give the diffeomorphism $\mathcal{S}_{\mathrm{sg}} \cong \{1,-1\}^2 \times (0,\infty) \times \mathbb{R}^2 \times \mathbb{T}^3$ with $t = g^2$ unchanged. Its four connected components are embedded and six-dimensional. At every fixed positive coupling, each of the four components is $\mathbb{R}^2 \times \mathbb{T}^3$, of dimension five (Corollary 1.75). This smooth classification imposes no spatial equation and gives no quantum spectral conclusion.

1 The common marked slice and its dynamics

Two Pro-mode packets dated 4 September 2026 describe a thirteen-dimensional continuation of the marked nine-dimensional Hermitian carrier. One uses the condition $u^*b = 0$ in a complex-plus-triplet decomposition; the other requires the observed quaternionic block PHP^* to be complex. We prove that these are the same embedded space, not merely isomorphic quadratic models. We retain the original nonunitary marking throughout. We then compute the omitted-coordinate force in the packet's scalar mechanics, give a constrained alternative, and correct two action conventions. The results below concern these specified matrices and dynamics; they do not certify every theorem in either supplied packet.

The subsequent two-cross continuation computes the entire scalar–Gauss domain in Section 1.28. Its classified parameters form $\mathcal{D} \times \{1,-1\}^2 \times U(1)^2$; retaining the frame phase ζ gives the literal domain after the sole identification $(\zeta, p_+, p_-) \sim (-\zeta, -p_+, -p_-)$. Corollary 1.70 constructs tuples at every original $t = g^2 > 0$. The two spatial residuals cannot vanish simultaneously on this class by Theorem 1.60; existence on the remaining unrestricted exceptional set is open.

Section 1.29 proves smooth embeddedness and the exact product $\mathcal{S}_{\mathrm{sg}} \cong \{1,-1\}^2 \times (0,\infty) \times \mathbb{R}^2 \times \mathbb{T}^3$, with $t = g^2$ the displayed positive coordinate. The four literal components have dimension six; every fixed positive coupling has four embedded five-dimensional components $\mathbb{R}^2 \times \mathbb{T}^3$. The inverse charts retain all phases and both signs, without a physical gauge quotient or an additional spatial equation.

1.1 Conventions, determinant, and the retained marking

Write $q = a + b\mathbf{j}$, with $a, b \in \mathbb{C}$, $\mathbf{j}^2 = -1$, $\mathbf{j}a = \bar{a}\mathbf{j}$. Thus

$$(a + b\mathbf{j})(c + d\mathbf{j}) = ac - b\bar{d} + (ad + b\bar{c})\mathbf{j}, \qquad \overline{a + b\mathbf{j}} = \bar{a} - b\mathbf{j}.$$

For complex matrices of the appropriate sizes this gives

$$(A + B\mathbf{j})(C + D\mathbf{j}) = AC - B\bar{D} + (AD + B\bar{C})\mathbf{j}, \qquad (A + B\mathbf{j})^* = A^* - B^{\mathsf{T}}\mathbf{j}.$$

In particular a quaternionic Hermitian matrix is uniquely $A + B\mathbf{j}$ with $A = A^*$ and $B = -B^{\mathsf{T}}$. For $b \in \mathbb{C}^3$ define

$$B(b) = \begin{pmatrix} 0 & \bar{b}_3 & -\bar{b}_2 \\ -\bar{b}_3 & 0 & \bar{b}_1 \\ \bar{b}_2 & -\bar{b}_1 & 0 \end{pmatrix}, \qquad \Theta(A, b) = A + B(b)\mathbf{j}.$$

The map $\Theta : \mathrm{Herm}_3(\mathbb{C}) \oplus \mathbb{C}^3 \to \mathrm{Herm}_3(\mathbb{H})$ is consequently a real-linear bijection.

We use the Moore determinant with value one on I_3. Its Hermitian rank-three formula is

$$N\begin{pmatrix} a & q_3 & \bar{q}_2 \\ \bar{q}_3 & d & q_1 \\ q_2 & \bar{q}_1 & f \end{pmatrix} = adf - a|q_1|^2 - d|q_2|^2 - f|q_3|^2 + 2\,\mathrm{Re}(q_1 q_2 q_3). \tag{1}$$

The quaternionic spectral theorem and determinant congruence law used below are recorded in S. Alesker, *Non-commutative linear algebra and plurisubharmonic functions of quaternionic variables*, Theorem 1.1.9, arXiv:math/0104209; see also W. Wang, *The linear algebra in the quaternionic pluripotential theory*, Propositions 2.1–2.3, arXiv:1805.07759. In our signs the faithful complex representation is

$$\mathcal{R}(A + B\mathbf{j}) = \begin{pmatrix} A & B \\ -\bar{B} & \bar{A} \end{pmatrix}.$$

Wang's representation has the opposite signs on both off-diagonal blocks; the exact conversion is conjugation by $\mathrm{diag}(I, -I)$.

Lemma 1.1 (The full cubic)**.** *For every $A = A^* \in \mathrm{Mat}_3(\mathbb{C})$ and $b \in \mathbb{C}^3$,*

$$N(\Theta(A, b)) = \det A - b^* A b. \tag{2}$$

Proof. Write the cyclic off-diagonal entries of A as $A_{23} = z_1$, $A_{31} = z_2$, $A_{12} = z_3$. Then $q_i = z_i + \bar{b}_i\mathbf{j}$ in (1), and $|q_i|^2 = |z_i|^2 + |b_i|^2$. The complex part of their product is

$$z_1 z_2 z_3 - \bar{b}_1 b_2 z_3 - z_1 \bar{b}_2 b_3 - \bar{b}_1 \bar{z}_2 b_3.$$

The three additional cross terms in twice its real part are the negatives of the off-diagonal terms of $b^* A b$. The additional diagonal terms in (1) are $-a|b_1|^2 - d|b_2|^2 - f|b_3|^2$. All remaining terms are precisely the ordinary determinant of A. This proves the identity with all signs. □

Retain the marked matrices

$$\begin{gathered} P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad u = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad C = PP^* = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \\ R = P^* C^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ -1 & 1 \\ -1 & -2 \end{pmatrix}, \quad X = (R\ u), \quad X^{-1} = \begin{pmatrix} P \\ u^*/3 \end{pmatrix}, \quad \det X = 1. \end{gathered} \tag{3}$$

Direct multiplication gives $PR = I_2$, $u^*R = 0$, $RP = I_3 - uu^*/3$. In particular the metric retained by this marking is

$$D = X^*X = \mathrm{diag}(C^{-1}, 3), \qquad C^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \tag{4}$$

It is not the identity matrix.

1.2 The two constructions are the same embedded space

Define, in the original quaternionic matrix space,

$$\mathcal{J}_P = \{H \in \mathrm{Herm}_3(\mathbb{H}) : PHP^* \in \mathrm{Herm}_2(\mathbb{C})\}, \qquad \mathcal{L}_u = \{\Theta(A, b) : u^*b = 0\}.$$

For any (A, b) put

$$X^{-1}AX^{-*} = \begin{pmatrix} G & z \\ z^* & \lambda \end{pmatrix}, \qquad X^*b = \begin{pmatrix} \chi \\ \rho \end{pmatrix}, \qquad \varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \tag{5}$$

Here G is complex Hermitian, $z, \chi \in \mathbb{C}^2$, $\lambda \in \mathbb{R}$, $\rho \in \mathbb{C}$, and $\rho = u^*b$.

Theorem 1.2 (Embedded reconciliation)**.** *The two subspaces satisfy $\mathcal{J}_P = \mathcal{L}_u$. Their complete coordinate relation, with no change of the ambient matrix H, is*

$$H = X\begin{pmatrix} G & z + w\mathbf{j} \\ (z + w\mathbf{j})^* & \lambda \end{pmatrix} X^*, \qquad w = -\varepsilon\bar{\chi}, \qquad \chi = \varepsilon\bar{w}. \tag{6}$$

The inverse in the original complex-plus-triplet coordinates is

$$A = X\begin{pmatrix} G & z \\ z^* & \lambda \end{pmatrix} X^*, \qquad b = X^{-*}\begin{pmatrix} \varepsilon\bar{w} \\ 0 \end{pmatrix}.$$

Thus both thirteen-dimensional constructions have the same positive cone, the same restricted determinant, and the same embedded old complex space, which is $w = 0$, equivalently $\chi = 0$.

Proof. The skew matrices obey the following two identities in the fixed basis:

$$PB(b)P^{\mathsf{T}} = \overline{u^*b}\,\varepsilon, \qquad X^{-1}B(b)X^{-\mathsf{T}} = B(X^*b). \tag{7}$$

For the first identity the upper-right entry is $\bar{b}_3 + \bar{b}_2 + \bar{b}_1$; skew symmetry gives the other entries. For the second, write $B(b)_{ij} = \sum_k \epsilon_{ijk}\bar{b}_k$. For any real invertible T, expansion of the alternating determinant gives

$$\sum_{a,b} T_{ia}T_{jb}\epsilon_{abk} = \det(T)\sum_l \epsilon_{ijl}(T^{-1})_{kl}.$$

Multiplication by $\bar{b}_k$ therefore gives $TB(b)T^{\mathsf{T}} = \det(T)B(T^{-\mathsf{T}}b)$. Set $T = X^{-1}$ and use $\det X = 1$. This proves (7). Its first identity proves the asserted equality of subspaces. At $\rho = 0$ the second identity is the explicit matrix

$$B\begin{pmatrix} \chi \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & -\bar{\chi}_2 \\ 0 & 0 & \bar{\chi}_1 \\ \bar{\chi}_2 & -\bar{\chi}_1 & 0 \end{pmatrix}.$$

Its upper-right column is $-\varepsilon\bar{\chi}$. This is exactly (6); $\varepsilon^2 = -I_2$ gives the inverse. Every claim about the embedded matrix and its positive cone follows from this identity, rather than from matching dimensions or determinant levels. □

Proposition 1.3 (The restricted and unrestricted cubics)**.** *In* (5) *the unrestricted cubic is*

$$N_{15} = \lambda \det G - z^* \operatorname{adj}(G) z - \chi^* G \chi \\ - 2 \operatorname{Re}(\rho \chi^* z) - \lambda |\rho|^2. \tag{8}$$

On the common thirteen-space this becomes

$$N_{13} = \lambda \det G - z^* \operatorname{adj}(G) z - w^* \operatorname{adj}(G) w. \tag{9}$$

Proof. Since $\det X = 1$, the ordinary block determinant gives $\det A = \lambda \det G - z^* \operatorname{adj}(G) z$ first for invertible G and then, as a polynomial identity, for all G. Also

$$b^* A b = (\chi^*, \bar{\rho}) \begin{pmatrix} G & z \\ z^* & \lambda \end{pmatrix} \begin{pmatrix} \chi \\ \rho \end{pmatrix}.$$

Expansion and Lemma 1.1 prove (8). For $G = \left(\begin{smallmatrix} a & c \\ \bar{c} & d \end{smallmatrix}\right)$, direct multiplication gives $\varepsilon^* \operatorname{adj}(G) \varepsilon = G^{\mathsf{T}}$. Substituting $w = -\varepsilon \bar{\chi}$ shows $w^* \operatorname{adj}(G) w = \chi^* G \chi$, proving (9). In the second packet's notation $w = \varepsilon \bar{\psi}_R$, the exact dictionary is consequently $\chi = -\psi_R$. □

1.3 Positive fibres and the embedded sphere pair

Proposition 1.4 (All positive fibres)**.** *The matrix in* (6) *is positive semidefinite exactly when*

$$G \succeq 0, \qquad z, w \in \operatorname{Ran}_{\mathbb{C}} G, \qquad s := \lambda - z^* G^{\dagger} z - w^* G^{\dagger} w \geq 0.$$

Its quaternionic rank is $\operatorname{rank}_{\mathbb{C}} G + \mathbf{1}_{s>0}$*. Here* $G^{\dagger}$ *is the inverse on* $\operatorname{Ran} G$*, extended by zero on* $\ker G$*. In particular the fibre over* $G = 0$ *is exactly* $\{\lambda u u^* : \lambda \geq 0\}$*.*

Proof. Congruence by the invertible real matrix X preserves positivity and rank. Write $\psi = z + w\mathbf{j}$. For any complex Hermitian T, quaternionic multiplication gives

$$\psi^* T \psi = z^* T z + w^* T w;$$

the $\mathbf{j}$ coefficient is $z^* T w - w^{\mathsf{T}} \bar{T} \bar{z} = 0$, while the scalar w term is the conjugate of the real number $w^* T w$. Testing the full Hermitian form on $(v, 0)$ gives $G \succeq 0$. If $v \in \ker G$ and $v^* \psi \neq 0$, testing on (tv, q) with q chosen to make $\operatorname{Re}(v^* \psi q) \neq 0$ makes the form negative for one sufficiently large signed real t. Thus ψ is in the quaternionic range of G, exactly the two complex range conditions above. With $a = G^{\dagger} \psi$, direct multiplication gives

$$\begin{pmatrix} I & 0 \\ -a^* & 1 \end{pmatrix} \begin{pmatrix} G & \psi \\ \psi^* & \lambda \end{pmatrix} \begin{pmatrix} I & -a \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} G & 0 \\ 0 & s \end{pmatrix}.$$

Both triangular factors are invertible, proving positivity and rank in both directions. At $G = 0$ the two spinors vanish, and return through X gives the asserted ray. □

Proposition 1.5 (The old boundary inside the new boundary)**.** *Let* $\mathcal{C} = \mathcal{J}_P \cap \{H \succeq 0\}$*. Its determinant-zero locus is its boundary, and its one-point compactification satisfies*

$$(\mathcal{C}^+, \partial \mathcal{C}^+) \cong (B^{13}, S^{12}), \qquad ((\partial \mathcal{C})^+, (\partial \operatorname{Herm}_3(\mathbb{C})_{\succeq 0})^+) \cong (S^{12}, S^8).$$

The sphere pair is the standard linear pair and $S^{12} \setminus S^8 \cong S^3 \times \mathbb{R}^9$*.*

Proof. The identity belongs to $\mathcal{J}_P$, since $PI_3P^* = C$ is complex. Set $\mathcal{J}_P^0 = \{K \in \mathcal{J}_P : \operatorname{tr} K = 0\}$. If $\lambda_{\min}(K)$ is the least quaternionic Hermitian eigenvalue, the mutually inverse maps are

$$K \longmapsto K - \lambda_{\min}(K) I_3, \qquad H \longmapsto H - \frac{\operatorname{tr} H}{3} I_3.$$

They identify $\mathcal{J}_P^0$ with the positive singular locus. Indeed, the first produces least eigenvalue zero; for singular positive H, the least eigenvalue after the second map is $-\operatorname{tr}(H)/3$. Eigenvalue continuity proves both maps continuous. More generally

$$(K, t) \longmapsto K + (t - \lambda_{\min}(K)) I_3, \quad (K, t) \in \mathcal{J}_P^0 \times [0, \infty)$$

is a homeomorphism onto $\mathcal{C}$, taking $t = 0$ to its boundary. This also proves that no other boundary points occur. Its inverse sends H to $(H - (\operatorname{tr} H/3) I_3, \lambda_{\min}(H))$. The one-point compactification of a closed half-space is a closed ball, with the boundary compactifying to a sphere; for example inverse stereographic projection from a point on the equator gives this pair. On complex matrices the quaternionic eigenvalues are the same ordinary Hermitian eigenvalues. The trace-free pair is therefore the linear pair $(\mathbb{R}^{12}, \mathbb{R}^8)$ under the displayed homeomorphisms. Both compactifications have their infinity point in the smaller sphere, so its complement is $\mathbb{R}^8 \times (\mathbb{R}^4 \setminus \{0\}) \cong \mathbb{R}^8 \times S^3 \times \mathbb{R}$. This proves the full pair assertion, not just the two dimensions. □

1.4 The nonunitary Jordan-product correction

Theorem 1.6 (Completion with the retained metric)**.** *The Jordan subalgebra generated by $\mathcal{J}_P$ under $H \circ K = (HK + KH)/2$ is all of* $\operatorname{Herm}_3(\mathbb{H})$*. In the marked block coordinates the transported product is*

$$M \circ_D N = \frac{1}{2}(MDN + NDM), \qquad D = \operatorname{diag}(C^{-1}, 3), \tag{10}$$

not the ordinary block product.

Proof. For $F_X(M) = XMX^*$, multiplication gives $F_X(M) \circ F_X(N) = F_X((MDN + NDM)/2)$ exactly. Take

$$M = \begin{pmatrix} 0 & \psi \\ \psi^* & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 0 & \phi \\ \phi^* & 0 \end{pmatrix}, \quad \psi = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \phi = \begin{pmatrix} 0 \\ \mathbf{j} \end{pmatrix}.$$

The upper-left block of $M \circ_D N$ is

$$\frac{3}{2} \begin{pmatrix} 0 & -\mathbf{j} \\ \mathbf{j} & 0 \end{pmatrix}.$$

The bottom scalar is zero: its two terms are $\mathbf{j}/3$ and $-\mathbf{j}/3$. The other blocks vanish. Replacing $\mathbf{j}$ by $\mathbf{k} = \mathbf{ij}$ gives the independent second missing direction. The common slice already contains every other coordinate in $\operatorname{Herm}_2(\mathbb{H}) \oplus \mathbb{H}^2 \oplus \mathbb{R}$. Hence the linear span of the slice and these two products is the whole fifteen-dimensional algebra. Any Jordan subalgebra containing the slice contains that span. This proves the closure. The $1/2$ product printed in the quaternionic packet is correct for the ordinary product in the block frame, but the coefficient for the stated embedded slice under the original product is $3/2$. □

1.5 The corrected group-action convention

Put $Z = (z\ w)$. The ordinary direct product has the following left action:

$$(g, k) \cdot (G, Z, \lambda) = (gGg^*, gZk^{-1}, \lambda), \quad g \in SL_2(\mathbb{C}), \quad k \in SU(2). \tag{11}$$

Composition gives $g_1 g_2 Z k_2^{-1} k_1^{-1} = (g_1 g_2) Z (k_1 k_2)^{-1}$, as required. The packet's gZk describes commuting left and right actions, or a left action of $SL_2(\mathbb{C}) \times SU(2)^{\mathrm{op}}$; it is not a left action of the usual direct product with those parameters. For instance $k_1 = \varepsilon$, $k_2 = \mathrm{diag}(i, -i)$ do not commute.

Both conventions give the same transformations after $k \mapsto k^{-1}$. To prove invariance, use

$$\mathrm{adj}(gGg^*) = (g^*)^{-1} \,\mathrm{adj}(G) g^{-1}, \qquad \det g = 1,$$

first on invertible G and then polynomially on all G. The two spinor terms in (9) are $\mathrm{tr}(Z^* \,\mathrm{adj}(G) Z)$; conjugation by k does not change this trace. Positivity under the left factor follows by quaternionic congruence with $\mathrm{diag}(g, 1)$ in the block frame. For the right factor, a unit quaternion $a + b\mathbf{j}$ acts on Z by

$$Z \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}.$$

This follows by multiplying $(z + w\mathbf{j})(a + b\mathbf{j})$ explicitly; it preserves $\psi^* G^\dagger \psi$, a real scalar, and the range conditions in Proposition 1.4. Finally, if an element fixes every G, then g is unitary and commutes with every complex Hermitian matrix; hence $g = \pm I_2$. Fixing every Z then forces $k = g$. The effective group is therefore $(SL_2(\mathbb{C}) \times SU(2))/\{(I, I), (-I, -I)\}$.

1.6 The full carrier still has an exact four-dimensional observation

Requiring the observed quaternionic block to be complex defines the thirteen-slice above. A different operation retains every coordinate of the full fifteen-dimensional carrier and applies a positive observation map. These two operations have different domains, with an exact relation.

Proposition 1.7 (Full positive observation and the hidden-coordinate bound)**.** *Fix the same complex subfield $\mathbb{C} = \mathbb{R} + \mathbb{R}\mathbf{i}$ and real matrix P. For a quaternionic Hermitian 2×2 matrix A put*

$$\pi_{\mathbb{C}}(A) = \tfrac{1}{2}\big(A + (\mathbf{i}I_2)^* A (\mathbf{i}I_2)\big).$$

Then the real-linear map

$$\mathscr{O} : \mathrm{Herm}_3(\mathbb{H}) \longrightarrow \mathrm{Herm}_2(\mathbb{C}), \qquad \mathscr{O}(H) = \pi_{\mathbb{C}}(PHP^*)$$

is positive, onto, and has positive section $G \mapsto RGR^*$. Its real kernel has dimension eleven. On writing $H = \Theta(A_0, b)$, its observed value is $G = PA_0P^*$ and $\rho = u^* b$ obeys

$$H \succeq 0 \quad \Longrightarrow \quad G \succeq 0, \qquad |\rho|^2 \le \det_{\mathbb{C}} G. \tag{12}$$

Every pair (G, ρ) satisfying these inequalities occurs. Thus the possible hidden ρ at a fixed G are exactly the closed complex disc of radius $\sqrt{\det G}$. At a null observed block, $\det G = 0$ forces $\rho = 0$.

Proof. For $q = a + b\mathbf{j}$, $a, b \in \mathbb{C}$, direct multiplication gives $(q - \mathbf{i}q\mathbf{i})/2 = a$. Thus $\pi_{\mathbb{C}}$ extracts the complex matrix part. It is positive as the average of the identity congruence and unitary congruence by $\mathbf{i}I_2$. Compression by P is positive as well. Since $PR = I_2$ and G is complex, $\mathscr{O}(RGR^*) = G$; positivity of this section is again congruence. Rank–nullity gives $15 - 4 = 11$.

The original skew identity (7) gives

$$PHP^* = G + \bar{\rho}\varepsilon\mathbf{j} = \begin{pmatrix} g_{11} & g_{12} + \bar{\rho}\mathbf{j} \\ \bar{g}_{12} - \bar{\rho}\mathbf{j} & g_{22} \end{pmatrix}.$$

Its Moore determinant is exactly $g_{11}g_{22} - |g_{12}|^2 - |\rho|^2 = \det_{\mathbb{C}} G - |\rho|^2$. A Hermitian quaternionic 2×2 matrix with real diagonal a, d and upper entry q is positive semidefinite precisely when $a, d \geq 0$ and $ad \geq |q|^2$: for $a > 0$ its quadratic form is $a|x + a^{-1}qy|^2 + (d - |q|^2/a)|y|^2$; for $a = 0$, positivity forces $q = 0$ by varying x, and then requires $d \geq 0$. Applying this criterion proves (12) and its converse for the displayed compressed block.

To realize every permitted pair in the original coordinates, take any $\lambda \geq 0$ and set

$$A_0 = RGR^* + \lambda uu^*, \qquad b = (\rho/3)u, \qquad H = \Theta(A_0, b).$$

The identities for $X = (R\ u)$ give, with $\chi = R^*b = 0$,

$$X^{-1}HX^{-*} = \begin{pmatrix} G + \bar{\rho}\varepsilon\mathbf{j} & 0 \\ 0 & \lambda \end{pmatrix}.$$

The preceding criterion and congruence prove positivity of H, while its observation and contraction are exactly G and ρ. This proves the claimed complete image and every disc fibre for (G, ρ). It does not discard the further z, w, λ coordinates in a general H. □

In explicit Lorentz coordinates, retain

$$G = \begin{pmatrix} p_0 + p_3 & p_1 - ip_2 \\ p_1 + ip_2 & p_0 - p_3 \end{pmatrix}, \qquad \det G = p_0^2 - p_1^2 - p_2^2 - p_3^2.$$

The bound is therefore an exact inequality between this Lorentz quadratic form and the omitted complex coordinate. It does not supply a positive uniform gap: taking $G = \epsilon I_2$, $\rho = \epsilon/2$, $\lambda = \epsilon$, $\epsilon > 0$, in the realization above gives positive definite H, nonzero ρ, and $\det G = \epsilon^2 \to 0$. An interpretation as physical mass or as a sector with a nonzero threshold requires a state-space, units, dynamics, and sector dictionary. The formula identifies the exact quantity that such a dictionary must retain. It also avoids imposing $\rho = 0$ on unrestricted motion merely to obtain a four-dimensional observation. Positivity of a chosen dynamical trajectory itself must still be proved. In fact a positive semidefinite trace-free Hermitian matrix has all three eigenvalues nonnegative with sum zero, so it is zero. Thus the nonzero scalar configurations below cannot themselves be positive matrices.

There is an exact bridge that retains this domain distinction. Put $\mathcal{J}_{14} = \{Q \in \mathrm{Herm}_3(\mathbb{H}) : \mathrm{tr}\, Q = 0\}$. The restriction $\mathscr{O}|_{\mathcal{J}_{14}}$ is onto all of $\mathrm{Herm}_2(\mathbb{C})$, not only its positive cone: its section is

$$j_0(G) = RGR^* - \frac{\mathrm{tr}(C^{-1}G)}{3} uu^*.$$

Indeed $R^*R = C^{-1}$, $\mathrm{tr}(uu^*) = 3$, and $Pu = 0$, so $\mathrm{tr}\, j_0(G) = 0$ and $\mathscr{O}(j_0(G)) = G$. Its kernel therefore has dimension ten.

For a positive boundary observation of every scalar configuration retain the separate, nonlinear map

$$\mathcal{F}(Q) = Q - \lambda_{\min}(Q) I_3, \qquad \widehat{G}(Q) = \mathscr{O}(\mathcal{F}(Q)) = P\Phi P^* - \lambda_{\min}(Q) C, \quad Q = \Theta(\Phi, b). \tag{13}$$

It maps $\mathcal{J}_{14}$ bijectively and continuously to the positive singular boundary of $\mathrm{Herm}_3(\mathbb{H})_{\succeq 0}$. The inverse is $H \mapsto H - (\mathrm{tr}\, H/3) I_3$: the least eigenvalue of this trace-free matrix is $-\mathrm{tr}\, H/3$. Eigenvalue continuity proves continuity in both directions. This is the full-space version of the earlier sphere-pair construction. The map leaves b, and hence ρ, unchanged, and therefore

$$\widehat{G}(Q) \succeq 0, \qquad \det \widehat{G}(Q) \geq |u^*b|^2.$$

The retained matrix $C = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ is essential in (13). This exact observation of a scalar trajectory is not an assertion that the trajectory preserves a positive linear slice; at repeated least eigenvalues the observation need not even be differentiable. No Hamiltonian intertwining is asserted.

The fixed marked observation is not automatically a gauge-invariant four-momentum. There is an exact covariance rule: under $H' = UHU^*$, $U \in SU(3)$, transform the observer as $P' = PU^*$; then $P'H'P'^* = PHP^*$ and hence its complex observation agrees. Keeping P fixed gives a chosen observer; specifying how that observer and the integral angular lattice participate in a physical model is additional data, not supplied by an equality of dimensions.

1.7 The actual period-to-observation bridge

The common marking also occurs in the rank-three toroidal extension of the native rank-two cusp. To specify an actual map, retain its data

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad Q_{\mathrm{latt}} = B_0^{-1}P = \begin{pmatrix} 0 & -1 & 1 \\ 1 & -1 & 0 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Here $PS = I_2$. Let $C_2(t)$ be the retained bounded holomorphic period correction at $t = 0$. On the logarithmic cover $t = e^{2\pi i s}$ put

$$Z_2 = sB_0 + C_2(t), \quad C_3(t) = SC_2(t)Q_{\mathrm{latt}}, \quad Z_3 = sI_3 + C_3(t), \quad \Phi_{\mathrm{latt}} = P \oplus Q_{\mathrm{latt}}.$$

These are the local corrected periods of the toroidal elliptic bridge; the underlying toric and compact-fibre morphisms are recorded in the workbench's *Higher-rung toroidal elliptic bridge*, equations (TEB.16)–(TEB.35). The new calculation below only needs the displayed matrices and retains the entire correction $C_2(t)$.

Proposition 1.8 (Metric-dependent period observation)**.** *Set $\Omega_3 = (I_3\ Z_3)$ and $\Omega_2 = (I_2\ Z_2)$. For every real symmetric positive-definite 6×6 matrix W_3,*

$$H_3 = \Omega_3 W_3 \Omega_3^*, \quad W_2 = \Phi_{\mathrm{latt}} W_3 \Phi_{\mathrm{latt}}^{\mathsf{T}} \quad \Longrightarrow \quad PH_3P^* = \Omega_2 W_2 \Omega_2^*.$$

Both sides are positive definite. No fixed positive W_3 makes H_3 periodic under $s \mapsto s+1$. Nevertheless every retained positive matrix W_* gives an explicit descending choice

$$M(x) = \begin{pmatrix} I_3 & xI_3 \\ 0 & I_3 \end{pmatrix}, \quad x = \operatorname{Re} s, \quad W_3(s) = M(-x)W_*M(-x)^{\mathsf{T}}.$$

For this choice, with $y = \operatorname{Im} s$, the descended matrix is

$$H_3(t) = (I_3\ iyI_3 + C_3(t))W_*(I_3\ iyI_3 + C_3(t))^*.$$

If $W_* = \left(\begin{smallmatrix} E & F \\ F^{\mathsf{T}} & K \end{smallmatrix}\right)$, then $G(t) = PH_3(t)P^*$ has leading term y^2PKP^*, with $PKP^* \succ 0$, and $\det G(t) = y^4 \det(PKP^*) + O(y^3)$ at the cusp.

Proof. Direct multiplication gives $PZ_3 = Z_2Q_{\mathrm{latt}}$, hence $P\Omega_3 = \Omega_2\Phi_{\mathrm{latt}}$ and the claimed Gram identity. Both P and Q_{latt} have rank two, so Φ_{latt} is surjective and W_2 is positive definite. Each period matrix has an identity block, proving positivity of both Gram matrices.

Write a fixed W_3 in $3+3$ blocks. Since $Z_3(s+n) = Z_3(s) + nI_3$, the coefficient of n^2 in $H_3(s+n)$ is its positive-definite lower diagonal block. Therefore $H_3(s+n)$ cannot be constant in $n \in \mathbb{Z}$.

In particular the period-derived observation cannot be made single-valued by silently retaining a constant lattice metric.

Put $M_3 = M(1)$. The proposed varying metric is positive by congruence and satisfies $W_3(s+1) = M_3^{-1} W_3(s) M_3^{-\mathsf{T}}$. Since $\Omega_3(s+1) = \Omega_3(s) M_3$, its Gram matrix is invariant. Also $\Omega_3(s) M(-x) = (I_3 \; iyI_3 + C_3(t))$, which proves the exact descended formula. The complete expansion, without removing its lower terms, is

$$\begin{aligned} H_3 = y^2 K + iy(F^{\mathsf{T}} - F + KC_3^* - C_3 K) \\ + E + FC_3^* + C_3 F^{\mathsf{T}} + C_3 K C_3^*. \end{aligned}$$

Boundedness of C_3 gives the asserted leading term and determinant estimate. The principal block K is positive definite and P^* is injective, so PKP^* is positive definite. This proves every assertion. □

For the explicitly specified example $W_* = I_6$ the exact expression is $H_3 = y^2 I_3 + iy(C_3^* - C_3) + I_3 + C_3 C_3^*$ and $\det G = 3y^4 + O(y^3)$, retaining $C = PP^*$ and $\det C = 3$. Thus this period-derived observation grows in a timelike, full-rank direction, not toward a null determinant. It is an actual positive matrix-valued map on the punctured cusp, not an inverse recovery of periods or of individual torus points. Its extension through the successor's finite-order overlaps and special fillings, and a dynamical interpretation of the chosen lattice metric, are further constructions.

All positive period-metric fibres and the cost of rank loss

The following calculation keeps the periods in Proposition 1.8 fixed. In particular $C_3(t) = SC_2(t)Q_{\mathrm{latt}}$ still uses the displayed S, not the different right inverse $R = P^* C^{-1}$. Choose a punctured cusp on which the retained correction is bounded, and fix a finite constant $K \geq 0$ such that $\|C_3(t)\| \leq K$ there. Such a constant exists because $\|C_3(t)\| \leq \|S\| \|C_2(t)\| \|Q_{\mathrm{latt}}\|$. All norms below are the Euclidean operator norms. Write

$$t = e^{2\pi i s}, \qquad s = x + iy, \qquad y = -\frac{\log |t|}{2\pi},$$

and restrict to the nonempty smaller cusp $y > \max\{K, 1\}$. This changes no period or coefficient. It identifies a domain where the inverse used below is proved to exist.

Theorem 1.9 (The complete positive metric fibre)**.** *Retain*

$$A_{\mathrm{per}}(s) = xI_3 + \operatorname{Re} C_3(t), \qquad Y_{\mathrm{per}}(t) = yI_3 + \operatorname{Im} C_3(t),$$

and define the real period map and its inverse by

$$\mathcal{N}(s) = \begin{pmatrix} I_3 & A_{\mathrm{per}}(s) \\ 0 & Y_{\mathrm{per}}(t) \end{pmatrix}, \qquad \mathcal{N}(s)^{-1} = \begin{pmatrix} I_3 & -A_{\mathrm{per}}(s) Y_{\mathrm{per}}(t)^{-1} \\ 0 & Y_{\mathrm{per}}(t)^{-1} \end{pmatrix}. \tag{MF.1}$$

For every prescribed Hermitian matrix $H = E + iF \succ 0$, where $E = E^{\mathsf{T}}$ and $F = -F^{\mathsf{T}}$ are real, all real symmetric positive-definite lattice matrices giving the unchanged value $\Omega_3 W_3 \Omega_3^* = H$ are exactly

$$W_3 = \mathcal{N}^{-1} \left[\frac{1}{2} \begin{pmatrix} E & -F \\ F & E \end{pmatrix} + \begin{pmatrix} U & V \\ V & -U \end{pmatrix} \right] \mathcal{N}^{-\mathsf{T}}, \tag{MF.2}$$

where U, V are real symmetric 3×3 matrices and the bracket is positive definite. This is a nonempty open convex twelve-real-dimensional fibre. An explicit description of its entire parameter domain is

$$E_+ = E/2 + U \succ 0, \qquad E/2 - U - (-F/2 + V)^{\mathsf{T}} E_+^{-1} (-F/2 + V) \succ 0. \tag{MF.3}$$

In particular $U = V = 0$ always gives a positive lift.

Every smooth map $H_3(t) \succ 0$ on the smaller punctured cusp has a smooth positive lift (MF.2) with

$$W_3(s+1) = M_3^{-1} W_3(s) M_3^{-\mathsf{T}}. \tag{MF.4}$$

The full set of positive observations $G(t) = PH_3(t)P^*$ obtainable from such metrics is exactly the set of all smooth $\mathrm{Herm}_2(\mathbb{C})_{\succ 0}$-valued maps on that cusp. For fixed $G \succ 0$, all the intermediate positive H are

$$H = X \begin{pmatrix} G & z \\ z^* & z^* G^{-1} z + \tau \end{pmatrix} X^*, \qquad z \in \mathbb{C}^2, \quad \tau > 0. \tag{MF.5}$$

The positive real lattice-metric fibre of G has real dimension seventeen.

Proof. For every real vector v,

$$\|Y_{\mathrm{per}} v\| \geq y\|v\| - \|\operatorname{Im} C_3\, v\| \geq (y - K)\|v\|.$$

The inequality $\|\operatorname{Im} C_3\| \leq \|C_3\|$ follows from $\operatorname{Im} C_3 = (C_3 - \bar{C}_3)/(2i)$ and $\|\bar{C}_3\| = \|C_3\|$. Since $y > K$, Y_{per} has zero kernel and is an invertible square real matrix. Block multiplication now proves both inverse identities in (MF.1). This is the original real period map: if $a, b \in \mathbb{R}^3$, then

$$a + Z_3 b = (a + A_{\mathrm{per}} b) + i Y_{\mathrm{per}} b.$$

Thus the inverse takes $v + iw$ to $(v - A_{\mathrm{per}} Y_{\mathrm{per}}^{-1} w,\ Y_{\mathrm{per}}^{-1} w)$. No correction in C_3 has been removed.

Put $\mathcal{J} = (I_3 \ \ iI_3)$. The exact factorization is $\Omega_3 = \mathcal{J}\mathcal{N}$. Congruence gives a bijection between real positive W_3 and real positive $T = \mathcal{N} W_3 \mathcal{N}^{\mathsf{T}}$. For

$$T = \begin{pmatrix} E_0 & F_0 \\ F_0^{\mathsf{T}} & K_0 \end{pmatrix}$$

one obtains by multiplication

$$\mathcal{J} T \mathcal{J}^* = E_0 + K_0 + i(F_0^{\mathsf{T}} - F_0). \tag{MF.6}$$

Consequently this equals $E + iF$ precisely when $E_0 = E/2 + U$, $K_0 = E/2 - U$, and $F_0 = -F/2 + V$ with U, V symmetric. Indeed, the first equality in (MF.6) fixes $K_0 = E - E_0$, and $F_0^{\mathsf{T}} - F_0 = F$ is equivalent to symmetry of $F_0 + F/2$. These recover U, V uniquely, proving the full fibre formula.

For real v, w, with $q = v + iw$,

$$\begin{pmatrix} v \\ w \end{pmatrix}^{\mathsf{T}} \begin{pmatrix} E & -F \\ F & E \end{pmatrix} \begin{pmatrix} v \\ w \end{pmatrix} = q^* H q.$$

The right side is positive for every nonzero (v, w), so the first bracket term in (MF.2) is positive definite. This proves nonemptiness. The kernel parameters have dimension $6 + 6 = 12$. Positive definiteness is an open convex condition on real symmetric matrices, so its intersection with this affine twelve-dimensional fibre is open and convex and has dimension twelve. Finally, for $B_+ = -F/2 + V$,

$$\begin{pmatrix} E_+ & B_+ \\ B_+^{\mathsf{T}} & E/2 - U \end{pmatrix} = \begin{pmatrix} I & 0 \\ B_+^{\mathsf{T}} E_+^{-1} & I \end{pmatrix} \begin{pmatrix} E_+ & 0 \\ 0 & E/2 - U - B_+^{\mathsf{T}} E_+^{-1} B_+ \end{pmatrix} \begin{pmatrix} I & E_+^{-1} B_+ \\ 0 & I \end{pmatrix}.$$

The two triangular factors are transposes of each other and are invertible; the identity proves the necessity and sufficiency of (MF.3), including every positivity restriction on the kernel parameters.

The original correction is single-valued in t, so

$$\mathcal{N}(s+1) = \mathcal{N}(s) M_3, \qquad \mathcal{N}(s+1)^{-1} = M_3^{-1} \mathcal{N}(s)^{-1}. \tag{MF.7}$$

For any smooth prescribed $H_3(t)$, its real and imaginary parts are smooth and periodic on the cover. The inverse of Y_{per} is smooth, since its entries are its adjugate divided by its nonzero determinant. Formula (MF.2) with $U = V = 0$ is therefore smooth and satisfies (MF.4). More generally, smooth periodic U, V satisfying (MF.3) give all such smooth lifts: conversely (MF.7) and (MF.4) make $\mathcal{N} W_3 \mathcal{N}^{\mathsf{T}}$ periodic, and its blocks recover periodic U, V by (MF.6). No holomorphic condition is being imposed on the real metric or on the prescribed Hermitian map.

If $W_3 \succ 0$ is real, it is also positive on nonzero complex columns: for $p + iq$, its quadratic form is $p^{\mathsf{T}} W_3 p + q^{\mathsf{T}} W_3 q$. Since Ω_3^* is injective by the identity block of Ω_3, one has $H_3 \succ 0$. Since P^* is injective, also $G = P H_3 P^* \succ 0$. This proves necessity of strict positivity; no singular G occurs at an interior point for a positive metric.

For the converse, the marked inverse $X^{-1} = (P;\, u^*/3)$ makes the upper-left block of $X^{-1} H X^{-*}$ equal to PHP^*. With that block fixed at $G \succ 0$, the complete positive block matrices are

$$\begin{pmatrix} G & z \\ z^* & \ell \end{pmatrix}, \qquad \ell - z^* G^{-1} z = \tau > 0.$$

To prove this exact positivity statement, multiply

$$\begin{pmatrix} I_2 & 0 \\ z^* G^{-1} & 1 \end{pmatrix} \begin{pmatrix} G & 0 \\ 0 & \tau \end{pmatrix} \begin{pmatrix} I_2 & G^{-1} z \\ 0 & 1 \end{pmatrix}.$$

The result is the displayed block with $\ell = z^* G^{-1} z + \tau$. The invertible triangular factors prove both directions, and congruence by X gives (MF.5). Taking $z = 0$ and any fixed $\tau = \lambda > 0$ already gives the smooth lift $H_3(t) = R G(t) R^* + \lambda u u^*$. Applying (MF.2) proves every asserted positive observation is realized. The intermediate H fibre has four real coordinates in z and one in τ. Equivalently, the real-linear metric-to-observation map has rank four: the constructions above give a right inverse on all Hermitian matrices before imposing positivity. Its real symmetric domain has dimension $6 \cdot 7/2 = 21$, so its affine kernel has dimension $21 - 4 = 17$. The positive part is nonempty and relatively open, proving the stated seventeen-dimensional fibre without a dimension-only surjectivity argument. □

The freedom above does not replace the retained periods by an identity period matrix. It computes the metric which compensates their full real period map. The correction itself has not lost information in passing from rank two to rank three: the exact inverse on its prescribed image is

$$C_2 = P C_3 S B_0,$$

because $PS = I_2$ and $Q_{\mathrm{latt}} S = B_0^{-1}$. To identify exactly what an unrestricted metric observation cannot recover, take two retained corrections with real period maps $\mathcal{N}$ and $\widetilde{\mathcal{N}}$ on a common sufficiently small cusp. The invertible real transformation

$$T_{\mathrm{per}} = \widetilde{\mathcal{N}}^{-1} \mathcal{N}, \qquad \widetilde{W}_3 = T_{\mathrm{per}} W_3 T_{\mathrm{per}}^{\mathsf{T}} \tag{MF.8}$$

preserves positivity and gives $\widetilde{\Omega}_3 \widetilde{W}_3 \widetilde{\Omega}_3^* = \Omega_3 W_3 \Omega_3^*$ by the two exact factorizations $\Omega_3 = \mathcal{J}\mathcal{N}$. Furthermore $T_{\mathrm{per}}(s+1) = M_3^{-1} T_{\mathrm{per}}(s) M_3$, so (MF.8) preserves the prescribed metric monodromy. This is an explicit correspondence between period–metric pairs with the same H_3 and G. Thus these observations alone cannot recover the period correction when the metric is allowed this freedom. The constants remain in the metric formula; they have not been set to zero.

Proposition 1.10 (Relative-metric bounds obstruct projective rank loss)**.** *For an equivariant positive metric put*

$$W_*(t) = M(x)W_3(s)M(x)^{\mathsf{T}}, \qquad \widetilde{\Omega}_3(t) = (I_3 \;\; iyI_3 + C_3(t)).$$

These are single-valued on the punctured cusp, and $H_3 = \widetilde{\Omega}_3 W_* \widetilde{\Omega}_3^*$*. If fixed constants* $0 < a \le b < \infty$ *obey* $aI_6 \preceq W_*(t) \preceq bI_6$*, then*

$$a\big(1 + (y-K)^2\big)C \preceq G(t) \preceq b\big(1 + (y+K)^2\big)C. \tag{MC.1}$$

In particular the unchanged eigenvalues satisfy

$$\lambda_{\min}(G) \ge a\big(1 + (y-K)^2\big)\lambda_{\min}(C), \qquad \lambda_{\max}(G) \le b\big(1 + (y+K)^2\big)\lambda_{\max}(C). \tag{MC.2}$$

With the original C*,* $\lambda_{\min}(C) = 1$ *and* $\lambda_{\max}(C) = 3$*. More generally, writing* $\kappa_*(t) = \lambda_{\max}(W_*)/\lambda_{\min}(W_*)$*, one always has*

$$\frac{\lambda_{\min}(G)}{\lambda_{\max}(G)} \ge \frac{1 + (y-K)^2}{3\kappa_*(t)(1 + (y+K)^2)}. \tag{MC.3}$$

Thus if the observed eigenvalue ratio tends to zero as $t \to 0$*, the relative metric condition number tends to infinity. Bounded relative condition number excludes this projective rank loss even when the common scale of the metric is not bounded above or below.*

Proof. The identity $M(x+1) = M(x)M_3$, together with (MF.4), proves that W_* is periodic. Multiplication gives $\Omega_3 M(-x) = \widetilde{\Omega}_3$, which proves the stated Gram formula. For every complex vector v,

$$(y-K)\|v\| \le \|(iyI_3 + C_3)^* v\| \le (y+K)\|v\|.$$

Both inequalities are the triangle inequality, its reverse, and $\|C_3^*\| \le K$. Since $\widetilde{\Omega}_3\widetilde{\Omega}_3^* = I_3 + (iyI_3 + C_3)(iyI_3 + C_3)^*$, they give

$$\big(1 + (y-K)^2\big)I_3 \preceq \widetilde{\Omega}_3\widetilde{\Omega}_3^* \preceq \big(1 + (y+K)^2\big)I_3.$$

Congruencing $aI_6 \preceq W_* \preceq bI_6$ by $\widetilde{\Omega}_3$, and then by P, proves (MC.1). The minimum and maximum Rayleigh quotients prove (MC.2). The characteristic polynomial of the original C is $(z-1)(z-3)$, retaining the factors 1 and 3.

At each point take $a(t) = \lambda_{\min}(W_*)$ and $b(t) = \lambda_{\max}(W_*)$ in the same inequalities. Dividing the resulting positive scalar bounds gives (MC.3); this division is a comparison of eigenvalues, not a rescaling or replacement of G or W_*. Its period-dependent factor tends to 1. For fixed a, b, the lower bound tends to $a/(3b) > 0$; for example it is at least $a/(27b)$ once $y \ge 2K$. If the ratio on the left tends to zero, (MC.3) implies

$$\kappa_*(t) \ge \frac{1 + (y-K)^2}{3(1 + (y+K)^2)} \frac{\lambda_{\max}(G)}{\lambda_{\min}(G)} \longrightarrow +\infty.$$

This proves the full asserted obstruction, including metrics whose overall scale varies. □

Proposition 1.11 (A retained-period rank-loss family and its exact cost)**.** *Fix, without rescaling any matrix, four real constants*

$$\alpha_{\mathrm{obs}} > 0, \qquad \beta_{\mathrm{obs}} > 0, \qquad \lambda > 0, \qquad p > 0,$$

and set

$$\delta(t) = \beta_{\mathrm{obs}} y^{-p}, \qquad G(t) = \begin{pmatrix} \alpha_{\mathrm{obs}} & 0 \\ 0 & \delta(t) \end{pmatrix}, \qquad H_3(t) = RG(t)R^* + \lambda uu^*. \tag{RL.1}$$

Take the lift (MF.2) with $U = V = 0$. It is a smooth equivariant positive real lattice metric at every point of the smaller punctured cusp, with the original periods and correction unchanged. Its observation has

$$\det G(t) = \alpha_{\mathrm{obs}}\beta_{\mathrm{obs}} y^{-p} = \alpha_{\mathrm{obs}}\beta_{\mathrm{obs}} \left(-\frac{\log|t|}{2\pi}\right)^{-p} \longrightarrow 0, \qquad G(t) \longrightarrow \operatorname{diag}(\alpha_{\mathrm{obs}}, 0). \tag{RL.2}$$

Put $D_2(t) = \operatorname{Im} C_2(t)$ and $Y = Y_{\mathrm{per}}(t) = yI_3 + \operatorname{Im} C_3(t)$. The relative metric has the exact determinant

$$\det W_*(t) = \frac{\alpha_{\mathrm{obs}}^2 \lambda^2 \beta_{\mathrm{obs}}^2 \, y^{-2p}}{64 \det(Y)^2}, \qquad \det(Y) = y\left[y^2 + y \operatorname{tr}(D_2 B_0^{-1}) + \det D_2\right]. \tag{RL.3}$$

In particular

$$\det W_*(t) \sim \frac{\alpha_{\mathrm{obs}}^2 \lambda^2 \beta_{\mathrm{obs}}^2}{64} y^{-2p-6}. \tag{RL.4}$$

If $L_{\max} = \max\{2\alpha_{\mathrm{obs}}/3,\, 3\lambda\}$, then

$$\lambda_{\min}(W_*) \sim \frac{\beta_{\mathrm{obs}}}{4}\, y^{-p-2}, \qquad \lambda_{\max}(W_*) \longrightarrow \frac{L_{\max}}{2}, \qquad \kappa_*(t) \sim \frac{2L_{\max}}{\beta_{\mathrm{obs}}}\, y^{p+2}. \tag{RL.5}$$

Every constant and the complete correction in the exact formulas (RL.1)–(RL.3) is retained.

Proof. Formula (MF.5) with $z = 0$ and $\tau = \lambda$ gives $H_3 \succ 0$ and $PH_3P^* = G$. Formula (MF.2) proves the metric, smoothness, and monodromy assertions. Equation (RL.2) is the direct determinant of the displayed matrix and uses the original logarithmic-cover factor 2π. The ratio of its least and greatest eigenvalues is eventually $\delta/\alpha_{\mathrm{obs}}$, so this is projective rank loss in the precise sense used in Proposition 1.10.

Write $A_c = \operatorname{Re} C_3$ and

$$\widetilde{\mathcal{N}} = \begin{pmatrix} I_3 & A_c \\ 0 & Y \end{pmatrix}, \qquad \mathcal{N} = \widetilde{\mathcal{N}} M(x).$$

Here H_3 is real symmetric. The relative metric is exactly

$$W_* = \frac{1}{2}\widetilde{\mathcal{N}}^{-1} \begin{pmatrix} H_3 & 0 \\ 0 & H_3 \end{pmatrix} \widetilde{\mathcal{N}}^{-\mathsf{T}} = \frac{1}{2} \begin{pmatrix} H_3 + A_c Y^{-1} H_3 Y^{-\mathsf{T}} A_c^{\mathsf{T}} & -A_c Y^{-1} H_3 Y^{-\mathsf{T}} \\ -Y^{-1} H_3 Y^{-\mathsf{T}} A_c^{\mathsf{T}} & Y^{-1} H_3 Y^{-\mathsf{T}} \end{pmatrix}. \tag{RL.6}$$

Since $H_3 = X \operatorname{diag}(G, \lambda) X^*$ and $\det X = 1$, $\det H_3 = \lambda \alpha_{\mathrm{obs}} \delta$. Taking the real 6×6 determinant in (RL.6) gives $\det W_* = (\det H_3)^2/(2^6 \det Y^2)$, proving the first identity in (RL.3).

The second identity retains the correction rather than replacing Y by yI_3. Indeed $Y = yI_3 + SD_2Q_{\mathrm{latt}}$ and $Q_{\mathrm{latt}}S = B_0^{-1}$. For rectangular U, V, the identity $\det(I + UV) = \det(I + VU)$ follows by eliminating either off-diagonal block of $\left(\begin{smallmatrix} I & U \\ -V & I \end{smallmatrix}\right)$ with triangular determinant-one row operations. Applying it here gives

$$\det Y = y^3 \det\left(I_2 + y^{-1} D_2 B_0^{-1}\right) = y\left[y^2 + y \operatorname{tr}(D_2 B_0^{-1}) + \det D_2\right],$$

because $\det B_0 = 1$. If $D_2 = \left(\begin{smallmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{smallmatrix}\right)$, this is the complete polynomial

$$\det Y = y\left[y^2 + y(d_{12} - d_{21}) + d_{11}d_{22} - d_{12}d_{21}\right].$$

The entries of D_2 remain the prescribed bounded functions of t. Thus $\det Y/y^3 \to 1$, proving (RL.4) with the displayed coefficient.

For the eigenvalue assertions, $H_3 u = 3\lambda u$ because $R^* u = 0$. On the two-dimensional invariant plane $u^\perp$, its trace is

$$\operatorname{tr}(C^{-1}G) = \frac{2}{3}(\alpha_{\text{obs}} + \delta)$$

and the product of the two eigenvalues is $\det H_3/(3\lambda) = \alpha_{\text{obs}}\delta/3$. The invariance of the plane follows also from Hermiticity and the displayed u eigenvector. Its two eigenvalues are therefore exactly

$$h_\pm = \frac{\alpha_{\text{obs}} + \delta \pm \sqrt{\alpha_{\text{obs}}^2 - \alpha_{\text{obs}}\delta + \delta^2}}{3}. \tag{RL.7}$$

They are positive: their sum and product are positive. Rationalizing the numerator of h_- gives

$$\frac{h_-}{\delta} = \frac{\alpha_{\text{obs}}}{\alpha_{\text{obs}} + \delta + \sqrt{\alpha_{\text{obs}}^2 - \alpha_{\text{obs}}\delta + \delta^2}} \longrightarrow \frac{1}{2}.$$

Also $h_+ \to 2\alpha_{\text{obs}}/3$. Consequently $\lambda_{\min}(H_3) \sim \delta/2$ and $\lambda_{\max}(H_3) \to L_{\max}$, with H_3 bounded.

The explicit blocks (RL.6), the bound $\|Y^{-1}\| \le (y-K)^{-1}$, and bounded A_c, H_3 imply

$$\left\| W_* - \tfrac{1}{2}\operatorname{diag}(H_3, 0) \right\| = O(y^{-2}).$$

For real symmetric matrices the maximum eigenvalue changes by at most the operator norm of a perturbation: apply the Rayleigh-quotient maximum to each matrix and subtract. Hence $\lambda_{\max}(W_*) \to L_{\max}/2$.

To control the least eigenvalue without assuming a bounded condition number of H_3, use the exact inverse

$$W_*^{-1} = 2\widetilde{\mathcal{N}}^{\mathsf{T}} \operatorname{diag}(H_3^{-1}, H_3^{-1})\widetilde{\mathcal{N}}.$$

For $y > 1$, the decomposition $\widetilde{\mathcal{N}} = \operatorname{diag}(I_3, yI_3) + \left(\begin{smallmatrix} 0 & A_c \\ 0 & \operatorname{Im} C_3 \end{smallmatrix}\right)$ gives $\|\widetilde{\mathcal{N}}\| \le y + \sqrt{2}K$. Thus

$$\lambda_{\max}(W_*^{-1}) \le \frac{2(y + \sqrt{2}K)^2}{\lambda_{\min}(H_3)}.$$

Its lower-right principal block includes the positive term $2Y^{\mathsf{T}} H_3^{-1} Y$. If v is a unit eigenvector of H_3^{-1} for its greatest eigenvalue, testing that block on $Y^{-1}v/\|Y^{-1}v\|$ gives

$$\lambda_{\max}(W_*^{-1}) \ge \frac{2(y-K)^2}{\lambda_{\min}(H_3)}.$$

All denominators are positive. Taking reciprocals yields

$$\frac{\lambda_{\min}(H_3)}{2(y+\sqrt{2}K)^2} \le \lambda_{\min}(W_*) \le \frac{\lambda_{\min}(H_3)}{2(y-K)^2}. \tag{RL.8}$$

Both bounds are asymptotic to $\delta/(4y^2)$, proving the first assertion of (RL.5). Combining it with the maximum-eigenvalue limit proves the condition-number asymptotic with its full coefficient. □

This is metric-choice freedom, not a selection of a physical momentum or mass. For the same unchanged periods the allowed choice $W_* = I_6$ instead gives exactly

$$H_3 = I_3 + (iyI_3 + C_3)(iyI_3 + C_3)^*, \qquad G = y^2 C + O(y), \qquad \det G = 3y^4 + O(y^3),$$

whereas (RL.1)–(RL.2) give a determinant tending to zero and retain the same C_2, C_3. These two explicit admissible choices, the complete twelve-dimensional metric fibres, and the seventeen-dimensional observed fibres prove that the specified period data and positivity alone do not select an observed trajectory. The eigenvalue ratios above only diagnose rank loss; no field, period, metric, or parameter was replaced by a normalized one. Units, a metric-selection law, dynamics, and an interpretation of the marked observation would be additional specified data. No extension across other finite-order overlaps or special fillings, and no physical or quantum gap conclusion, is asserted here.

Hermitian flat metrics, the original period lattice, and its dual spectrum

Here a flat metric means a translation-invariant metric on each fixed torus fibre. Its coefficients may depend smoothly on the cusp parameter t, but not on the point of that fibre. The word *physical* below distinguishes the original real coordinates of $\mathbb{C}^3$ from integral lattice coordinates; it does not assert a physical field theory. Retain, without changing any basis or period,

$$B_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad Q_{\mathrm{latt}} = B_0^{-1} P, \quad S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{pmatrix},$$

$$C_3 = SC_2Q_{\mathrm{latt}}, \qquad Z_3 = sI_3 + C_3, \qquad Z_2 = sB_0 + C_2, \qquad s = x + iy, \quad t = e^{2\pi i s}, \quad y = -\frac{\log|t|}{2\pi}.$$

Choose finite bounds $K_2, K \geq 0$ on $\|C_2(t)\|, \|C_3(t)\|$, respectively, and restrict to $y > \max\{K_2, K\} + 1$. Norms are the original Euclidean operator norms. Write

$$A = xI_3 + \operatorname{Re} C_3, \quad Y = yI_3 + \operatorname{Im} C_3, \quad \mathcal{N} = \begin{pmatrix} I_3 & A \\ 0 & Y \end{pmatrix}, \quad \mathcal{J} = (I_3 \;\; iI_3), \quad J_0 = \begin{pmatrix} 0 & -I_3 \\ I_3 & 0 \end{pmatrix}.$$

Thus $\Omega_3 = (I_3 \;\; Z_3) = \mathcal{J}\mathcal{N}$ and $\mathcal{N}^{-1} = \left(\begin{smallmatrix} I_3 & -AY^{-1} \\ 0 & Y^{-1} \end{smallmatrix}\right)$. Indeed $\|Yv\| \geq (y - K)\|v\|$ proves invertibility. The same argument, using $\|B_0 v\| = \|v\|$, proves invertibility of $Y_2 = \operatorname{Im} Z_2 = yB_0 + \operatorname{Im} C_2$. For $r = 2, 3$ put

$$\Lambda_r = \Omega_r \mathbb{Z}^{2r}, \qquad T_r = \mathbb{C}^r / \Lambda_r, \qquad \mathcal{N}_2 = \begin{pmatrix} I_2 & \operatorname{Re} Z_2 \\ 0 & Y_2 \end{pmatrix}.$$

The real coordinates of $z \in \mathbb{C}^3$ are $q = (\operatorname{Re} z, \operatorname{Im} z)^{\mathsf{T}} = \mathcal{N}\xi$, where $\xi \in \mathbb{R}^6/\mathbb{Z}^6$. This is an exact change of coordinates, not a replacement of the lattice. For any complex matrix D define its realification by

$$\mathfrak{r}(D) = \begin{pmatrix} \operatorname{Re} D & -\operatorname{Im} D \\ \operatorname{Im} D & \operatorname{Re} D \end{pmatrix}.$$

Block multiplication gives $\mathfrak{r}(D_1 D_2) = \mathfrak{r}(D_1)\mathfrak{r}(D_2)$ and $\mathfrak{r}(D^*) = \mathfrak{r}(D)^{\mathsf{T}}$.

Theorem 1.12 (The precise Hermitian metric in a period fibre)**.** *Let $g_{\mathrm{phys}} \succ 0$ be a real symmetric metric matrix and put*

$$T = g_{\mathrm{phys}}^{-1}, \qquad g_{\mathrm{latt}} = \mathcal{N}^{\mathsf{T}} g_{\mathrm{phys}} \mathcal{N}, \qquad W = g_{\mathrm{latt}}^{-1} = \mathcal{N}^{-1} T \mathcal{N}^{-\mathsf{T}}. \tag{HS.1}$$

Then the earlier period weight W has the exact geometric meaning of a lattice cometric*; the covariant lattice metric is its inverse. The observation is $H = \Omega_3 W \Omega_3^* = \mathcal{J} T \mathcal{J}^*$. For a fixed positive Hermitian $H = E + iF$, its full real cometric fibre is*

$$T = \frac{1}{2}\mathfrak{r}(H) + \mathcal{A}(U,V), \qquad \mathcal{A}(U,V) = \begin{pmatrix} U & V \\ V & -U \end{pmatrix}, \qquad U = U^{\mathsf{T}}, \quad V = V^{\mathsf{T}}, \quad T \succ 0. \tag{HS.2}$$

The real metric is compatible with the given complex structure, meaning $J_0^{\mathsf{T}} g_{\mathrm{phys}} J_0 = g_{\mathrm{phys}}$, if and only if $U = V = 0$. With the explicit Hermitian metric convention

$$g_{\mathrm{phys}}(v,w) = \mathrm{Re}(v^* h_3 w), \qquad v, w \in \mathbb{C}^3,$$

the unique compatible member over H is

$$h_3 = 2H^{-1}, \qquad h_3^{-1} = H/2, \qquad g_{\mathrm{phys}} = 2\mathfrak{r}(H^{-1}), \qquad T = \tfrac{1}{2}\mathfrak{r}(H), \tag{HS.3}$$

$$g_{\mathrm{latt}} = 2\mathcal{N}^{\mathsf{T}} \mathfrak{r}(H^{-1}) \mathcal{N}, \qquad W = \tfrac{1}{2}\mathcal{N}^{-1} \mathfrak{r}(H) \mathcal{N}^{-\mathsf{T}}. \tag{HS.4}$$

For an arbitrary positive real cometric, its observed Hermitian part is precisely the positive J_0-invariant average

$$\frac{T + J_0 T J_0^{\mathsf{T}}}{2} = \frac{1}{2}\mathfrak{r}(H). \tag{HS.5}$$

Proof. Substitution of $q = \mathcal{N}\xi$ into the covariant quadratic form gives the middle formula in (HS.1); multiplication verifies its inverse. The observation is positive: $\mathcal{J}^*$ is injective, and a real positive matrix T is also positive on complex vectors, since for a, b real, $(a+ib)^* T (a+ib) = a^{\mathsf{T}} T a + b^{\mathsf{T}} T b$. For a real symmetric block matrix $T = \left(\begin{smallmatrix} D & B \\ B^{\mathsf{T}} & L \end{smallmatrix}\right)$,

$$\mathcal{J} T \mathcal{J}^* = D + L + i(B^{\mathsf{T}} - B).$$

Thus equality to $E + iF$ is equivalent to $D = E/2 + U$, $L = E/2 - U$, $B = -F/2 + V$ with U, V symmetric. This proves both directions and uniqueness of the parameters in (HS.2).

Because $J_0^{\mathsf{T}} = -J_0$ and $J_0^2 = -I_6$, metric compatibility is equivalent to $g_{\mathrm{phys}} J_0 = J_0 g_{\mathrm{phys}}$, which is equivalent, by multiplying by the inverse on both sides, to $T J_0 = J_0 T$. Direct multiplication shows that $\mathfrak{r}(H)$ commutes with J_0, whereas $\mathcal{A} J_0 = -J_0 \mathcal{A}$. Hence compatibility is equivalent to $J_0 \mathcal{A} = 0$, and the invertibility of J_0 makes this equivalent to $\mathcal{A} = 0$, or $U = V = 0$.

For $z = v + iw$ one has $(v,w)^{\mathsf{T}} \mathfrak{r}(H)(v,w) = z^* H z > 0$ unless $z = 0$. Thus $\mathfrak{r}(H)$ is positive. Realification respects products, so inversion of $T = \mathfrak{r}(H)/2$ gives (HS.3), and (HS.1) gives (HS.4). Finally $J_0 \mathfrak{r}(H) J_0^{\mathsf{T}} = \mathfrak{r}(H)$ and $J_0 \mathcal{A} J_0^{\mathsf{T}} = -\mathcal{A}$, proving (HS.5). This average is an average of cometrics, not an average of covariant metrics. The metric associated to it is the inverse of the average. □

Thus the twelve real fibre coordinates in (HS.2) describe genuine non-Hermitian translation-invariant real metrics, and compatibility removes exactly those coordinates. For example, $H = 2I_3$ and $U = \varepsilon\,\mathrm{diag}(1,0,0)$, $V = 0$ give

$$T_\varepsilon = \mathrm{diag}(1+\varepsilon, 1, 1, 1-\varepsilon, 1, 1), \qquad -1 < \varepsilon < 1.$$

All have the same H; only $\varepsilon = 0$ is Hermitian-compatible. They have different marked cometrics, and the next theorem proves that some marked character eigenvalue therefore changes.

Theorem 1.13 (The lattice cometric is the marked character spectrum)**.** *For any $T \succ 0$ as in (HS.1), define the nonnegative Laplacian by $\Delta = -\operatorname{div}\operatorname{grad}$. Its complete character eigenbasis and eigenvalues on T_3 are*

$$\chi_k([\xi]) = e^{2\pi i k^{\mathsf{T}}\xi}, \qquad k \in \mathbb{Z}^6, \qquad \lambda_k = 4\pi^2 k^{\mathsf{T}} W k = 4\pi^2 p_k^{\mathsf{T}} T p_k, \quad p_k = \mathcal{N}^{-\mathsf{T}} k. \tag{HS.6}$$

Here p_k is in the integer-valued dual of $\mathcal{N}\mathbb{Z}^6$: its pairing with $\mathcal{N}n$ is $k^{\mathsf{T}}n \in \mathbb{Z}$. Conversely every integer-valued dual covector is of this form. The eigenspace multiplicity is exactly the number of integer k with the specified value of (HS.6). The volume, in the original scales, is

$$\operatorname{Vol}(T_3, g_{\mathrm{phys}}) = |\det Y|\sqrt{\det g_{\mathrm{phys}}} = |\det Y| \det h_3 = \frac{8|\det Y|}{\det H} \tag{HS.7}$$

in the compatible case. Without compatibility the first equality still holds.

The marked spectrum recovers every entry of W:

$$W_{aa} = \frac{\lambda_{e_a}}{4\pi^2}, \qquad W_{ab} = \frac{\lambda_{e_a+e_b} - \lambda_{e_a} - \lambda_{e_b}}{8\pi^2} \quad (a \neq b). \tag{HS.8}$$

If $H(t)$ is single-valued and (HS.4) is used, all these metrics descend with the original monodromy

$$\mathcal{N}(s+1) = \mathcal{N}(s)M_3, \quad M_3 = \begin{pmatrix} I_3 & I_3 \\ 0 & I_3 \end{pmatrix}, \quad W(s+1) = M_3^{-1} W(s) M_3^{-\mathsf{T}}. \tag{HS.9}$$

The same character has labels k and $k' = M_3^{\mathsf{T}} k$ in the two markings, and $\lambda'_{k'} = \lambda_k$.

Proof. The covariant metric and its determinant are constant in the fibre coordinate ξ, so the defining divergence formula becomes $\Delta = -\sum_{a,b=1}^{6} W_{ab}\partial_{\xi_a}\partial_{\xi_b}$. Two derivatives of χ_k prove (HS.6). The dual-lattice assertion follows from $p^{\mathsf{T}}\mathcal{N}n \in \mathbb{Z}$ for all integer n: taking the six unit vectors is equivalent to $\mathcal{N}^{\mathsf{T}}p \in \mathbb{Z}^6$.

For completeness, no unlisted eigenfunctions are needed. Integration over $[0,1]^6$ gives zero for $\chi_k\overline{\chi_l}$ when $k \neq l$; the common squared norm in Riemannian measure is the volume. To prove density, the one-dimensional Fejer kernels

$$F_M(\theta) = \frac{1}{M}\left|\sum_{r=0}^{M-1} e^{2\pi i r\theta}\right|^2 = \sum_{|j|<M}\left(1 - \frac{|j|}{M}\right)e^{2\pi i j\theta}$$

are nonnegative and have integral one. For $\operatorname{dist}(\theta,\mathbb{Z}) \geq \eta > 0$ they are bounded by $1/(M\sin^2(\pi\eta))$, using the finite geometric sum. The product of six such kernels also has integral one; its mass outside a coordinate cube of radius η tends to zero by the union bound over its six coordinates. Uniform continuity then proves that convolution of any continuous periodic function with these product kernels converges uniformly to that function. Each convolution is a finite linear combination of characters. Continuous functions are dense in $L^2([0,1]^6)$: step functions on finite unions of boxes are dense by regular approximation in Lebesgue measure, and a box indicator is approximated in L^2 by a continuous function that differs only in a boundary strip of arbitrarily small measure. Multiplication by the constant Riemannian density preserves this argument.

Consequently the characters, divided by the square root of the volume, are a complete orthonormal basis. On Fourier coefficients the closure of Δ is the real diagonal operator with domain $\{(a_k) : \sum_k \lambda_k^2|a_k|^2 < \infty\}$; its adjoint has that same domain, as testing against each coordinate vector forces the displayed weighted square summability. It is therefore self-adjoint. Finite Fourier sums are a core

by truncation of both the ordinary and weighted square sums. For a smooth function, integration by parts identifies the Fourier coefficients of its smooth Laplacian with $\lambda_k a_k$; their square sum is finite. Thus the operator on all smooth functions lies in this diagonal operator and has the same closure as its restriction to finite Fourier sums. Since $\lambda_k \geq 4\pi^2 \lambda_{\min}(W)\|k\|^2$, bounded eigenvalue intervals contain only finitely many integer indices. This proves the completeness, discreteness and multiplicity assertions, including a positive smallest eigenvalue away from $k = 0$.

The parallelepiped $\mathcal{N}[0,1]^6$ has coordinate volume $|\det \mathcal{N}| = |\det Y|$, proving the first equality in (HS.7). For Hermitian h_3, the determinant identity $\det \mathfrak{r}(h_3) = |\det h_3|^2 = (\det h_3)^2$ follows by conjugating the realification over $\mathbb{C}$ to $\operatorname{diag}(h_3, \overline{h_3})$ with the invertible change $(v, w) \mapsto (v+iw, v-iw)$. Positive h_3 has positive determinant. Now $\det(2H^{-1}) = 8/\det H$, proving (HS.7). Expansion of the quadratic form at $e_a + e_b$ proves (HS.8).

The correction $C_3(t)$ is unchanged by $s \mapsto s+1$, giving the first formula in (HS.9). The cometric formula gives the last. Since $q = \mathcal{N}\xi = \mathcal{N}M_3\xi'$, one has $\xi = M_3\xi'$. Hence $k' = M_3^{\mathsf{T}} k$ and substitution in (HS.6) proves the final assertion. All matrix inversions depend smoothly on the entries where their determinants are nonzero, so a smooth $H(t)$ gives the asserted smooth descending family. □

Theorem 1.14 (The original torus quotient is a metric and spectral map)**.** *Let $H_3 \succ 0$, $G = PH_3P^*$, and equip T_3 with $h_3 = 2H_3^{-1}$. Retain the original integral map*

$$L = P \oplus Q_{\mathrm{latt}} : \mathbb{Z}^6 \longrightarrow \mathbb{Z}^4, \qquad P_{\mathbb{R}} = \operatorname{diag}(P, P).$$

Then $f_P : T_3 \to T_2$, $[z] \mapsto [Pz]$, is a surjective holomorphic homomorphism with connected kernel

$$\mathbb{C}u/(\mathbb{Z}u + s\mathbb{Z}u), \qquad u = (1,1,1)^{\mathsf{T}}. \tag{HS.10}$$

There is a unique target translation-invariant metric making this map a Riemannian submersion. Its Hermitian metric, complex cometric, real cometric and lattice cometric are, respectively,

$$h_2 = 2G^{-1}, \qquad h_2^{-1} = G/2 = Ph_3^{-1}P^*, \qquad \mathsf{T}_2 = \tfrac{1}{2}\mathfrak{r}(G) = P_{\mathbb{R}}\mathsf{T}_3 P_{\mathbb{R}}^{\mathsf{T}}, \tag{HS.11}$$

$$W_2 = \mathcal{N}_2^{-1}\mathsf{T}_2\mathcal{N}_2^{-\mathsf{T}} = LW_3L^{\mathsf{T}}. \tag{HS.12}$$

Here $\mathsf{T}_3 = \mathfrak{r}(H_3)/2$ is the real source cometric, distinguished from the torus T_3. The horizontal lift is the exact complex map

$$\eta : \mathbb{C}^2 \longrightarrow \mathbb{C}^3, \qquad \eta = H_3P^*G^{-1}. \tag{HS.13}$$

For every smooth function f on the target,

$$\Delta_3(f \circ f_P) = (\Delta_2 f) \circ f_P. \tag{HS.14}$$

For a target character $\ell \in \mathbb{Z}^4$, its pullback is the source character labelled by $L^{\mathsf{T}}\ell$, and its eigenvalue is unchanged. These are exactly the source characters constant on the fibres. Thus the target spectrum, with each character's multiplicity counted, is a subcollection of the source character spectrum, not an identification with all source modes.

Proof. Direct multiplication gives $PS = I_2$, $B_0Q_{\mathrm{latt}} = P$ and $PZ_3 = Z_2Q_{\mathrm{latt}}$. Thus

$$P\Omega_3 = \Omega_2 L, \qquad P_{\mathbb{R}}\mathcal{N} = \mathcal{N}_2 L. \tag{HS.15}$$

Moreover $\operatorname{diag}(S, SB_0)$ is an integral right inverse of L, since $Q_{\mathrm{latt}}SB_0 = I_2$. Therefore $P\Lambda_3 = \Lambda_2$, proving that the torus homomorphism is well-defined and surjective.

If $Pz \in \Lambda_2$, surjectivity on lattices supplies $\gamma \in \Lambda_3$ with $P\gamma = Pz$, so $z - \gamma \in \ker_{\mathbb{C}} P = \mathbb{C}u$. To find its periods, if $\Omega_3(a,b)$ is in $\mathbb{C}u$, then $\Omega_2(Pa, Q_{\mathrm{latt}}b) = 0$. The invertibility of $\mathcal{N}_2$ forces $Pa = Q_{\mathrm{latt}}b = 0$. Solving the two displayed difference equations gives $a = ru$, $b = vu$ with $r, v \in \mathbb{Z}$. Since $C_3 u = 0$, $\Omega_3(a,b) = (r + sv)u$. This proves (HS.10), including connectedness. Its lattice has real rank two because $y > 0$.

Put $K_3 = h_3^{-1} = H_3/2$, $K_2 = PK_3P^* = G/2$. Surjectivity of P makes P^* injective, so $K_2 \succ 0$. The lift in (HS.13) is $K_3P^*K_2^{-1}$ and satisfies

$$P\eta = I_2, \qquad \eta^* h_3 v = K_2^{-1} P v = 0 \quad (v \in \ker P), \qquad \eta^* h_3 \eta = K_2^{-1} = h_2. \tag{HS.16}$$

Every $w \in \mathbb{C}^3$ has the orthogonal decomposition $w = \eta(Pw) + (w - \eta(Pw))$, whose second term lies in $\ker P$. Hence P restricts to an isometry on the orthogonal complement of its kernel; this is precisely the definition of a Riemannian submersion. It also proves uniqueness of h_2, since each target vector has a unique horizontal lift. Taking realifications proves (HS.11), and multiplying (HS.15) by the inverses on both sides proves (HS.12).

In lattice coordinates $f_P([\xi]) = [L\xi]$. The chain rule gives $\partial_{\xi_a}\partial_{\xi_b} f(L\xi) = \sum_{i,j} L_{ia}L_{jb}(\partial_i\partial_j f)(L\xi)$. Contraction with $-W_{3,ab}$ and (HS.12) proves (HS.14). Also $e^{2\pi i \ell^{\mathsf{T}} L\xi} = \chi_{L^{\mathsf{T}}\ell}(\xi)$; inserting this index in (HS.6) proves equality of the eigenvalues.

The lattice kernel is $\mathbb{Z}u \oplus \mathbb{Z}u$. The characters constant on the connected kernel annihilate its real tangent space, so their indices (m,n) obey $u^{\mathsf{T}}m = u^{\mathsf{T}}n = 0$. Conversely these equalities make the character constant there. Every integral zero-sum triple $m = (m_1, m_2, m_3)$ is $P^{\mathsf{T}}(m_1, -m_3)^{\mathsf{T}}$, and $\mathrm{im}_{\mathbb{Z}}\, Q_{\mathrm{latt}}^{\mathsf{T}} = \mathrm{im}_{\mathbb{Z}}\, P^{\mathsf{T}}$ because B_0 is unimodular. Thus the stated indices are exactly $L^{\mathsf{T}}\mathbb{Z}^4$. Injectivity of L^{T} preserves the number of target characters at each eigenvalue. □

The compatibility restriction still leaves genuine source-metric freedom over a fixed target metric. Indeed, using the original $R = P^*C^{-1}$, $C = PP^*$ and $X = (R \;\; u)$, all positive H_3 with $PH_3P^* = G$ are

$$H_3 = X \begin{pmatrix} G & z \\ z^* & z^*G^{-1}z + \tau \end{pmatrix} X^*, \qquad z \in \mathbb{C}^2, \quad \tau > 0. \tag{HS.17}$$

To prove the assertion, $PX = (I_2 \;\; 0)$ and $X^{-1} = \begin{pmatrix} P \\ u^*/3 \end{pmatrix}$ show that the top-left block of $X^{-1}H_3X^{-*}$ is G. For any remaining blocks z, d, its quadratic form is

$$(v + G^{-1}zw)^* G (v + G^{-1}zw) + (d - z^*G^{-1}z)|w|^2.$$

It is positive exactly when $d = z^*G^{-1}z + \tau$ with $\tau > 0$. This proves (HS.17) in both directions and its five real parameters. The injectivity of inversion makes distinct H_3 distinct compatible source metrics; (HS.11)–(HS.14) prove that the quotient metric and quotient spectrum stay fixed. This specifies exactly what the remaining observation fibre means geometrically.

Proposition 1.15 (Native Euclidean observation and expanding dual spectrum)**.** *Equip the source with its native Euclidean metric in the retained coordinates: $h_3 = I_3$, $g_{\mathrm{phys}} = I_6$. Then*

$$W_3 = \mathcal{N}^{-1}\mathcal{N}^{-\mathsf{T}}, \qquad H_3 = 2I_3, \qquad G = 2C, \qquad C = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}. \tag{HS.18}$$

The induced target metric is $h_2 = C^{-1}$, not I_2. The source eigenvalues, with no correction suppressed, are

$$\lambda_{m,n} = 4\pi^2 \left(\|m\|^2 + \|Y^{-\mathsf{T}}(n - A^{\mathsf{T}}m)\|^2 \right), \qquad m, n \in \mathbb{Z}^3. \tag{HS.19}$$

For $y > K + 1$, its smallest positive eigenvalue is exactly

$$\lambda_1^{(3)} = 4\pi^2 \min_{0\neq n\in\mathbb{Z}^3} \|Y^{-\mathsf{T}} n\|^2, \qquad \frac{4\pi^2}{(y+K)^2} \le \lambda_1^{(3)} \le \frac{4\pi^2}{(y-K)^2}. \tag{HS.20}$$

For the induced target metric, its smallest positive eigenvalue is

$$\lambda_1^{(2)} = 4\pi^2 \min_{\substack{0\neq n\in\mathbb{Z}^3 \\ u^{\mathsf{T}} n=0}} \|Y^{-\mathsf{T}} n\|^2, \qquad \frac{8\pi^2}{(y+K)^2} \le \lambda_1^{(2)} \le \frac{8\pi^2}{(y-K)^2}. \tag{HS.21}$$

In particular the exact leading constants are

$$\lambda_1^{(3)} \sim \frac{4\pi^2}{y^2} = \frac{16\pi^4}{(\log|t|)^2}, \qquad \lambda_1^{(2)} \sim \frac{8\pi^2}{y^2} = \frac{32\pi^4}{(\log|t|)^2}. \tag{HS.22}$$

The source volume is the retained polynomial

$$\mathrm{Vol}(T_3, I) = \det Y = y\left[y^2 + y\,\mathrm{tr}\big((\mathrm{Im}\, C_2)B_0^{-1}\big) + \det(\mathrm{Im}\, C_2)\right] \sim y^3. \tag{HS.23}$$

Proof. In (HS.1) the physical cometric is $T = I_6$, so $H_3 = \mathcal{J}\mathcal{J}^* = 2I_3$. Multiplication by P, P^* and (HS.11) prove all of (HS.18). The exact inverse transpose is

$$\mathcal{N}^{-\mathsf{T}} = \begin{pmatrix} I_3 & 0 \\ -Y^{-\mathsf{T}} A^{\mathsf{T}} & Y^{-\mathsf{T}} \end{pmatrix},$$

which gives (HS.19). The singular values of Y lie between $y - K$ and $y + K$, since $(y - K)\|v\| \le \|Yv\| \le (y + K)\|v\|$. The same singular-value bounds for the inverse transpose give

$$\frac{\|n\|}{y+K} \le \|Y^{-\mathsf{T}} n\| \le \frac{\|n\|}{y-K}. \tag{HS.24}$$

For source modes with $m \neq 0$, (HS.19) is at least $4\pi^2$. The mode $m = 0, n = e_1$ has eigenvalue at most $4\pi^2/(y - K)^2 < 4\pi^2$. Thus a minimizer has $m = 0$. The minimum exists because the inverse-transpose quadratic form is positive definite and only finitely many integer vectors have bounded value. Nonzero integer vectors have norm at least one, and a unit vector exists, giving (HS.20).

Target modes have $m = P^{\mathsf{T}} a$, $n = Q_{\mathrm{latt}}^{\mathsf{T}} b$ for $a, b \in \mathbb{Z}^2$. A nonzero integer zero-sum triple has squared norm at least two: it has at least one positive and at least one negative integer coordinate. The value two is attained by $(1, -1, 0)^{\mathsf{T}}$. Therefore $a \neq 0$ costs at least $8\pi^2$, whereas a mode with $a = 0$ and a shortest nonzero zero-sum n has eigenvalue at most $8\pi^2/(y - K)^2 < 8\pi^2$. The complete index characterization in the preceding theorem now proves (HS.21). Multiplying the bounds by y^2 and taking the limit proves (HS.22), with the original definition of y.

Finally $D_2 = \mathrm{Im}\, C_2$ is real and $Y = yI_3 + SD_2Q_{\mathrm{latt}}$. Since $Q_{\mathrm{latt}}S = B_0^{-1}$, the identity $\det(I + UV) = \det(I + VU)$ gives

$$\det Y = y^3 \det(I_2 + y^{-1} D_2 B_0^{-1}).$$

For completeness that determinant identity follows by taking the determinant of $\left(\begin{smallmatrix} I & U \\ -V & I \end{smallmatrix}\right)$ and eliminating first the upper-right block or first the lower-left block using block triangular matrices with determinant one. Expansion of the two-by-two determinant and $\det B_0 = 1$ give the displayed polynomial in (HS.23). Its sign is positive: $yI_3 + rSD_2Q_{\mathrm{latt}}$ is invertible for $0 \le r \le 1$ by the same bound $y - rK > 0$, so its determinant has the positive sign of y^3. Formula (HS.7) and boundedness of D_2 finish the proof. □

Equations (HS.18)–(HS.23) exhibit the exact relationship rather than a nonidentity alone: the constant observation is twice the native physical cometric and twice its quotient cometric, while the unchanged period map enters the dual lattice in (HS.19). The periods expand, the volume grows, and both displayed positive spectral gaps tend to zero. No period coefficient was replaced and no metric scale was divided out. At fixed periods, H_3 uniquely specifies the Hermitian-compatible source metric; at fixed G the five-parameter family (HS.17) specifies its precise residual freedom. When periods are also allowed to vary, the constant $H_3 = 2I_3$ alone cannot recover their marked cometric; (HS.8) proves that the full marked spectrum does recover it. These are geometric and spectral assertions, not a selection of a preferred dynamical metric or a field-theoretic mass.

Primary reference conventions. The nonnegative Laplacian and integer-valued dual-lattice convention agree with A. Gogolev, *Spectra of flat tori*, Section 2 (27 November 2017), available at `https://andreygogolev.wordpress.com/2017/11/27/spectra-of-flat-tori/`. V. Ivrii, *Partial Differential Equations*, Appendix 4.B, `https://www.math.utoronto.ca/courses/apm346h1/20181/PDE-textbook/Chapter4/S4.B.html`, instead places 2π in the definition of the dual lattice; his dual is exactly $2\pi\mathcal{N}^{-\mathsf{T}}\mathbb{Z}^6$ here. The horizontal-isometry definition and uniqueness of quotient metrics are those in U. Lang, *Lecture Notes on Riemannian Geometry*, 16 June 2020, Definition 4.1 and Proposition 4.4, printed pages 45 and 47, `https://metaphor.ethz.ch/x/2020/fs/401-3532-08L/sc/DG2_16June2020.pdf`. All specialized matrix identities and period/spectral asymptotics used here have been proved above in the retained coordinates.

1.8 The unrestricted and constrained scalar equations

Equation (8) gives the exact normal covector at $\rho = 0$:

$$d_\rho N_{15}(\delta\rho) = -2\,\mathrm{Re}(\delta\rho\,\chi^* z). \tag{14}$$

It need not vanish. For $G = I_2$, $z = \chi = (1,0)^{\mathsf{T}}$, $\lambda = 3$, $\rho = t \in \mathbb{R}$,

$$N_{15}(t) = 1 - 2t - 3t^2.$$

At $t = 0$ this is a positive-definite quaternionic matrix: $w = (0,1)^{\mathsf{T}}$ and the Schur excess is $3 - 1 - 1 = 1$. Thus the nonzero normal covector occurs even in the interior of the positive cone. This alone is a statement about the cubic; to determine motion one must also retain the specified kinetic form.

Theorem 1.16 (Normal force in the supplied scalar model)**.** *Use the triality packet's fourteen-real-dimensional scalar variables and potential*

$$\begin{gathered}\Phi = \Phi^*, \quad \operatorname{tr}\Phi = 0, \quad b \in \mathbb{C}^3, \quad s = \operatorname{tr}\Phi^2 + 2b^*b,\\ V(\Phi, b) = \frac{m^2}{2}s + \frac{\Gamma}{4}s^2 - \kappa(\det\Phi - b^*\Phi b),\\ L = \tfrac{1}{2}\operatorname{tr}(\dot\Phi^2) + \dot b^*\dot b - V, \quad m^2 \geq 0, \quad \Gamma > 0, \quad \kappa \in \mathbb{R}.\end{gathered} \tag{15}$$

This is the explicitly specified spatially constant scalar mechanics with gauge fields omitted, not a reduction claimed for arbitrary gauge fields. For its solutions, with $\rho = u^*b$,

$$\ddot b = -[(m^2 + \Gamma s)I_3 + \kappa\Phi]b, \qquad \ddot\rho = -(m^2 + \Gamma s)\rho - \kappa u^*\Phi b. \tag{16}$$

The phase-space constraint $\rho = \dot\rho = 0$ is not invariant for $\kappa \neq 0$. At initial data

$$\Phi_0 = \operatorname{diag}(1,-1,0), \quad b_0 = (1,-1,0)^{\mathsf{T}}, \quad \dot\Phi_0 = 0, \quad \dot b_0 = 0,$$

one has $\ddot\rho(0) = -2\kappa$.

Proof. For a variation δb,

$$\delta_b V = 2\,\mathrm{Re}\big(\delta b^*[(m^2+\Gamma s)I_3+\kappa\Phi]b\big).$$

The kinetic variation has the same factor two, giving the first equation in (16); multiply by u^* for the second. At the displayed initial data $u^*b_0=0$ but $\Phi_0 b_0=(1,1,0)^{\mathsf T}$, so the acceleration is -2κ. The equations have polynomial right-hand sides and hence a unique local solution for these initial data. Taylor's theorem gives $\rho(t)=-\kappa t^2+o(t^2)$, which leaves the constraint. For comparison with the marked cubic, if $X^{-1}\Phi X^{-*}=\left(\begin{smallmatrix} G & z \\ z^* & \lambda\end{smallmatrix}\right)$, then

$$\mathrm{tr}(C^{-1}G)+3\lambda=0, \qquad u^*\Phi b = 3(z^*\chi+\lambda\rho).$$

Thus at $\rho=0$ the normal acceleration is $-3\kappa z^*\chi$. This factor three is fixed by the marking and the kinetic form; it is not removed by rewriting the cubic. □

Proposition 1.17 (Exact constrained alternative)**.** *Let $\Pi_u=uu^*/3$, $Q_u=I_3-\Pi_u$. Restrict the Lagrangian* (15) *to $b\in u^\perp$ and velocities in $u^\perp$. Its Euler–Lagrange equations are*

$$\begin{aligned}\ddot b &= -(m^2+\Gamma s)b-\kappa Q_u\Phi b,\\ \ddot\Phi &= -(m^2+\Gamma s)\Phi+\kappa\big(\mathrm{adj}\,\Phi-bb^*\big)_0, \qquad T_0=T-\tfrac13(\mathrm{tr}\,T)I_3.\end{aligned} \tag{17}$$

The reaction force added to the unrestricted b equation is exactly $\kappa\Pi_u\Phi b$. The constrained solution is also an unrestricted solution exactly when this force vanishes throughout the solution (or automatically when $\kappa=0$).

Proof. Allowed b variations are projected by the orthogonal matrix Q_u, so projecting (16) gives the first equation. In particular $u^*\ddot b=0$, and initial data tangent to the constraint remain there. For the Φ equation, variations are trace-free Hermitian and

$$\delta_\Phi V=\mathrm{tr}\big([(m^2+\Gamma s)\Phi-\kappa\,\mathrm{adj}\,\Phi+\kappa bb^*]\delta\Phi\big).$$

Projection onto trace-free matrices gives the second equation. The unrestricted Φ equation is unchanged. Subtracting the two b equations gives the stated reaction, proving the last equivalence. □

The unrestricted model retains all six real coordinates of b; the constrained model retains four and supplies the displayed reaction force. Neither is excluded by the failure of the other identification. The old complex sector $b=\dot b=0$ *is* invariant by (16). By contrast, a nonzero fixed complex subspace of $\mathbb{C}^3$ invariant under all possible Hermitian Φ must be the whole space: traceless Hermitian matrices together with I_3 complex-linearly span every matrix unit, and the matrix units applied to a nonzero vector span $\mathbb{C}^3$. Thus a four-real-dimensional complex-linear transverse sector cannot retain all these unrestricted couplings without the explicit constraint or an additional restriction on Φ.

1.9 Preserved interacting sectors of the unrestricted equations

The failure of the whole slice to be invariant does not exclude invariant motions within it. We now construct such motions using the original equations, not by adding the reaction in Proposition 1.17. Write $a(t)=m^2+\Gamma s(t)$, and retain the trace-free Hermitian variable Φ, the standard metric on $b\in\mathbb{C}^3$, and $u^*u=3$.

Lemma 1.18 (Global existence in the scalar mechanics)**.** *Every initial position and velocity in* (15) *has a unique solution for all real time. Its conserved energy is*

$$E=\tfrac12\,\mathrm{tr}\,\dot\Phi^2+\dot b^*\dot b+V(\Phi,b).$$

Proof. The unrestricted equations for both variables are

$$\ddot{\Phi} = -a\Phi + \kappa(\operatorname{adj}\Phi - bb^*)_0, \qquad \ddot{b} = -ab - \kappa\Phi b. \tag{18}$$

They form a polynomial first-order system on positions and velocities, so local existence and uniqueness hold. Put $t_\Phi = \operatorname{tr}\Phi^2$ and $n_b = b^*b$. Direct differentiation of the kinetic energy gives

$$-a\operatorname{tr}(\dot{\Phi}\Phi) + \kappa\operatorname{tr}(\dot{\Phi}\operatorname{adj}\Phi) - \kappa b^*\dot{\Phi}b - 2a\operatorname{Re}(\dot{b}^*b) - 2\kappa\operatorname{Re}(\dot{b}^*\Phi b).$$

The trace projection drops out because $\operatorname{tr}\dot{\Phi} = 0$. Differentiating the displayed potential gives precisely the negative of this expression, including the two terms from differentiating $b^*\Phi b$. Thus E is constant.

For completeness, a bound in the unchanged scalar coordinates suffices to rule out escape. The three real eigenvalues of Φ give $|\det\Phi| \le (t_\Phi/3)^{3/2}$ by the arithmetic–geometric mean inequality applied to their squares. Cauchy–Schwarz and the Frobenius norm give $|b^*\Phi b| \le n_b\sqrt{t_\Phi}$. Since $s = t_\Phi + 2n_b$, these are the explicit inequalities

$$|\det\Phi - b^*\Phi b| \le (t_\Phi/3)^{3/2} + n_b\sqrt{t_\Phi} \le \left(\frac{1}{3\sqrt{3}} + \frac{1}{2}\right) s^{3/2}.$$

This bound is not asserted to be sharp and does not change V. Consequently

$$V \ge \frac{\Gamma}{4}s^2 + \frac{m^2}{2}s - |\kappa|\left(\frac{1}{3\sqrt{3}} + \frac{1}{2}\right) s^{3/2} \longrightarrow +\infty \quad (s \longrightarrow \infty).$$

The energy identity bounds positions; the global lower bound for V then bounds velocities. A polynomial vector field and its derivative are bounded on this compact phase-space set. A finite endpoint therefore has a limiting initial state to which local existence applies, extending the solution. Neither a finite forward nor a finite backward endpoint occurs. □

Theorem 1.19 (Fixed-subspace invariant sectors)**.** *Fix a complex subspace $W \subset u^\perp$ of dimension $r = 1$ or 2, and let E_W be its orthogonal projection in the original metric. Define*

$$\mathcal{T}_W = \{(\Phi, b) : \Phi = \Phi^*,\ \operatorname{tr}\Phi = 0,\ [\Phi, E_W] = 0,\ b \in W\}.$$

With tangent initial velocities, its tangent bundle $T\mathcal{T}_W$ is invariant under the full equations (18) *for all real time. For each fixed W, the configuration dimensions are respectively* 6 *and* 8*, and the phase-space dimensions are* 12 *and* 16*. Every such solution satisfies*

$$\rho = 0, \qquad u^*\Phi b = 0, \qquad \kappa(uu^*/3)\Phi b = 0$$

identically, without any constraint force.

Proof. For Hermitian Φ, preservation of W also preserves $W^\perp$: $\langle\Phi x, w\rangle = \langle x, \Phi w\rangle = 0$ for $x \in W^\perp$, $w \in W$. This is equivalent to $[\Phi, E_W] = 0$. The trace-free adjugate identity is

$$\operatorname{adj}\Phi = \Phi^2 - \tfrac{1}{2}(\operatorname{tr}\Phi^2)I_3. \tag{19}$$

Indeed the characteristic polynomial is $z^3 - \frac{1}{2}(\operatorname{tr}\Phi^2)z - \det\Phi$; Cayley–Hamilton and the adjugate identity prove (19) when Φ is invertible, and polynomial continuity proves it at singular Φ as well. Thus the adjugate commutes with E_W. So does bb^*: it takes W to W and vanishes on $W^\perp$. Scalar trace subtraction preserves this property. Both accelerations in (18) consequently remain in the specified linear spaces. The position derivatives do so when the velocities satisfy $[\dot{\Phi}, E_W] = 0$, $\dot{b} \in W$.

The restricted polynomial vector field has a local solution; its inclusion is a solution of the full system. Uniqueness and Lemma 1.18 give the claimed global invariance.

The Hermitian blocks on W and $W^{\perp}$ have real dimensions r^2 and $(3-r)^2$. The trace imposes one real equation, and b contributes $2r$ real coordinates. Hence $\dim_{\mathbb{R}} \mathcal{T}_W = r^2 + (3-r)^2 - 1 + 2r$. Velocities double this dimension. Finally $b, \Phi b \in W \subset u^{\perp}$ proves each asserted zero contraction. The parameter W is fixed, not an additional moving coordinate: for $r = 1$ the choices form $\mathbb{P}(u^{\perp}) \cong \mathbb{CP}^1$, and for $r = 2$ the only choice is $W = u^{\perp}$. □

Here is an explicit interacting member that also retains a nonzero marked cross-coordinate z. Keep

$$v = (1,-1,0)^{\mathsf{T}}, \quad w_0 = (1,1,-2)^{\mathsf{T}}, \quad E = vv^*/2, \quad F = w_0 w_0^*/6,$$

and take

$$W = \mathbb{C}v, \quad \Phi_0 = E - F + w_0 u^* + u w_0^*, \quad b_0 = v, \quad \dot{\Phi}_0 = 0, \quad \dot{b}_0 = 0. \tag{20}$$

The vectors u, v, w_0 are orthogonal with squared lengths $3, 2, 6$. Thus $\operatorname{tr}\Phi_0 = 0$, $\Phi_0 v = v$, $\operatorname{tr}\Phi_0^2 = 38$, $s_0 = 42$, and

$$b_0^* \Phi_0 b_0 = 2, \qquad \ddot{b}_0 = -(m^2 + 42\Gamma + \kappa)v.$$

For example $\kappa > 0$ gives nonstationary interacting motion. In the original marked coordinates, not a newly chosen basis,

$$\chi_0 = R^* b_0 = (1,0)^{\mathsf{T}}, \qquad z_0 = P\Phi_0 u/3 = (0,3)^{\mathsf{T}}, \qquad \lambda_0 = 0.$$

Hence $z_0 \neq 0$ while $z_0^* \chi_0 = 0$. Preservation of the scalar slice does not require $z = 0$ for every solution. The theorem constructs exact families; it does not assert they exhaust all invariant initial data.

Proposition 1.20 (Tangency through third order is insufficient)**.** *For any unrestricted solution put $c = u^* \Phi b$ and $h = u^* \dot{\Phi} \dot{b}$. The exact identities are*

$$\begin{aligned} \ddot{c} &= -2ac - \frac{\kappa s}{3}\rho + 2h, \\ \rho^{(4)} &= \left(-\ddot{a} + a^2 + \frac{\kappa^2 s}{3} \right) \rho - 2\dot{a}\dot{\rho} + 3a\kappa c - 2\kappa h. \end{aligned} \tag{21}$$

There are initial data with $\rho^{(j)}(0) = 0$ for $0 \le j \le 3$ but $\rho^{(4)}(0) = -12\kappa$.

Proof. Equation (19) gives, with $t_\Phi = \operatorname{tr}\Phi^2$ and $n_b = b^* b$,

$$(\operatorname{adj}\Phi - bb^*)_0 = \Phi^2 - bb^* + \frac{n_b - t_\Phi}{3} I_3, \qquad (\operatorname{adj}\Phi - bb^*)_0 b = \Phi^2 b - \frac{s}{3} b.$$

Differentiate c twice and substitute (18):

$$\ddot{c} = u^*(\ddot{\Phi} b + 2\dot{\Phi}\dot{b} + \Phi \ddot{b}) = -2ac - \frac{\kappa s}{3}\rho + 2h.$$

The two terms $\kappa u^* \Phi^2 b$ cancel exactly. Twice differentiating $\ddot{\rho} = -a\rho - \kappa c$ and substituting this identity proves the second formula. In particular, at $\rho = \dot{\rho} = 0$, vanishing second and third derivatives requires $c = \dot{c} = 0$ when $\kappa \neq 0$; then the fourth derivative is $-2\kappa h$.

For the promised initial data use the same u, v, w_0 as above, but now set

$$\Phi_0 = 0, \quad b_0 = v, \quad \dot{b}_0 = w_0, \quad \dot{\Phi}_0 = (u w_0^* + w_0 u^*)/3.$$

This matrix is trace-free Hermitian, $\dot\Phi_0 v = 0$ and $\dot\Phi_0 w_0 = 2u$. Consequently $\rho_0 = \dot\rho_0 = c_0 = \dot c_0 = 0$, whereas $h_0 = 6$. Formula (21) gives $\rho^{(4)}(0) = -12\kappa$, and Taylor's theorem gives $\rho(t) = -(\kappa/2)t^4 + o(t^4)$. For $\kappa \neq 0$ this solution leaves the slice despite the lower-order tangencies. For $\kappa = 0$, by contrast, the full locus $\rho = \dot\rho = 0$ is invariant because its equation is the homogeneous linear equation $\ddot\rho = -a(t)\rho$ with zero initial data. □

1.10 Periodic motions outside every fixed-subspace family

The families in Theorem 1.19 do not exhaust the motions inside $\rho = 0$. The following exact relative equilibria rotate the instantaneous kernel of Φ while retaining all the original couplings.

Theorem 1.21 (A moving-kernel family)**.** *Choose $\Gamma > 0$, $\kappa \neq 0$, and real x, a satisfying*

$$0 < \frac{x}{\kappa} \le \frac{1}{6\Gamma}, \qquad a > \frac{\kappa x}{3}.$$

Choose $B, d > 0$ with

$$B^2 + d^2 = \frac{3ax}{\kappa} - x^2, \qquad m^2 = a\left(1 - \frac{6\Gamma x}{\kappa}\right). \tag{22}$$

The right-hand sides give a positive sum and $m^2 \ge 0$. Retain u, v, w_0 from (20)*, and put*

$$J_0 = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}, \quad \Pi = uu^*/3, \quad Q_u = I_3 - \Pi, \quad \Omega = \sqrt{a/3}\, J_0,$$

$$O(t) = \Pi + \cos(\sqrt{a}\, t)Q_u + \frac{\sin(\sqrt{a}\, t)}{\sqrt{3}} J_0. \tag{23}$$

Define the original-coordinate initial data and their motion by

$$\begin{aligned} b_0 &= \frac{B}{\sqrt{2}} v, \qquad \Phi_0 = -\frac{x}{6} w_0 w_0^* + \frac{x}{3} uu^* + \frac{d}{\sqrt{18}}(w_0 u^* + u w_0^*), \\ b(t) &= O(t) b_0, \qquad \Phi(t) = O(t)\Phi_0 O(t)^*. \end{aligned} \tag{24}$$

These are global periodic solutions of (18)*. They satisfy*

$$s = \frac{6ax}{\kappa}, \qquad m^2 + \Gamma s = a, \qquad \Phi(t) b(t) = 0, \qquad \rho(t) = 0.$$

No fixed complex $W \subset u^\perp$ contains this trajectory in $T\mathcal{T}_W$.

Proof. Direct multiplication gives $J_0^* = -J_0$, $J_0 u = 0$, $J_0^2 = -3Q_u$, and $\Pi J_0 = J_0 \Pi = 0$. Consequently $O(t)^* O(t) = I_3$, $O(t)u = u$, and $\dot O = \Omega O = O\Omega$. Its determinant is one by continuity from $O(0) = I_3$. This proves that the displayed rotation is in the original Euclidean metric, with its frequency $\sqrt{a}$ unchanged.

For the following verification only, use the explicit orthogonal matrix

$$K = \left(\frac{v}{\sqrt{2}}, \frac{w_0}{\sqrt{6}}, \frac{u}{\sqrt{3}}\right), \quad K^* K = I_3,$$

and retain its inverse K^* and all its factors. It gives

$$K^* \Phi_0 K = H = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -x & d \\ 0 & d & x \end{pmatrix}, \quad K^* b_0 = B e_1, \quad K^* \Omega K = A = \begin{pmatrix} 0 & -\sqrt{a} & 0 \\ \sqrt{a} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus $s = 2(x^2 + d^2 + B^2) = 6ax/\kappa$, as asserted. In these exact coordinates $HBe_1 = 0$ and

$$(\operatorname{adj} H - B^2 e_1 e_1^*)_0 = \operatorname{diag}(-s/3, s/6, s/6).$$

The required accelerations are explicitly

$$[A,[A,H]] = \begin{pmatrix} -2ax & 0 & 0 \\ 0 & 2ax & -ad \\ 0 & -ad & 0 \end{pmatrix} = -aH + \kappa \operatorname{diag}(-s/3, s/6, s/6), \qquad A^2 B e_1 = -aBe_1.$$

Conjugation by K therefore verifies both equations at $t = 0$. The full force commutes with orthogonal conjugation on Φ and the same orthogonal action on b: the adjugate, trace subtraction, norms, and product Φb all transform in this way. Since O commutes with Ω, conjugation by $O(t)$ verifies both equations at every time, not merely to a finite Taylor order. This also proves the stated kernel and normal-coordinate identities.

At times 0 and $\pi/(2\sqrt{a})$, the two values of b are nonzero multiples of v and w_0. They span $u^\perp$ over $\mathbb{C}$. Any fixed W containing the whole trajectory would consequently equal $u^\perp$. But $u^*\Phi_0 w_0 = \sqrt{18}\, d \neq 0$, so Φ_0 does not preserve this plane. This rules out every fixed-W family in the earlier theorem. □

A particularly small exact instance is

$$\Gamma = m^2 = 1, \quad \kappa = 6, \quad x = \tfrac{1}{2}, \quad a = 2, \quad B = d = \frac{1}{2\sqrt{2}},$$
$$b_0 = \frac{1}{4}\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad \Phi_0 = \frac{1}{4}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & -2 \end{pmatrix}. \tag{25}$$

Here $s = 1$ and the period of b is $2\pi/\sqrt{2}$. Its original marked coordinates at zero are $\chi_0 = (1/4, 0)^\mathsf{T}$, $z_0 = (0, 1/4)^\mathsf{T}$, and $\lambda_0 = 1/6$. Although $\det \Phi - b^*\Phi b = 0$ along this solution, its cubic force does not vanish: at zero it is $6(\operatorname{adj}\Phi_0 - b_0 b_0^*)_0 \neq 0$. It is the coupled equations, not removal of the cubic, that sustain the rotating-kernel solution.

Proposition 1.22 (Conserved charge and the precise gauge comparison)**.** *In the scalar mechanics the anti-Hermitian matrix*

$$\mathcal{J} = [\Phi, \dot\Phi] + b\dot b^* - \dot b b^* \tag{26}$$

is constant. Its trace-free part is the $SU(3)$ moment matrix, using the pairing $\operatorname{tr}(\mathcal{J}\xi)$ with $\xi \in \mathfrak{su}(3)$. On (24),

$$\mathcal{J} = -\frac{s}{2}\Omega, \qquad T = \frac{as}{2}, \qquad E = m^2 s + \frac{3\Gamma}{4}s^2.$$

In particular (25) has $\mathcal{J} = -\Omega/2 \neq 0$, $T = 1$, $V = 3/4$, and $E = 7/4$. This scalar solution is not a source-free homogeneous gauge solution with all $\mathcal{A}_\mu = 0$: it fails the $\mathcal{A}_0$ field equation.

Proof. Using (18), polynomial commutation of Φ with its adjugate, and commutation with the scalar trace correction, gives

$$\frac{d}{dt}[\Phi, \dot\Phi] = -\kappa[\Phi, bb^*], \qquad \frac{d}{dt}(b\dot b^* - \dot b b^*) = \kappa[\Phi, bb^*].$$

Their sum vanishes. Under the infinitesimal simultaneous symmetry $\delta\Phi = [\xi, \Phi]$, $\delta b = \xi b$, the kinetic momentum paired with this variation is

$$\operatorname{tr}(\dot\Phi[\xi, \Phi]) + 2\operatorname{Re}(\dot b^* \xi b) = \operatorname{tr}(\mathcal{J}\xi).$$

This proves the pairing, including its coefficient and sign. In the explicit frame K, compute

$$[H,[A,H]] + Be_1(ABe_1)^* - (ABe_1)(Be_1)^* = -(x^2+d^2+B^2)A = -\frac{s}{2}A.$$

It commutes with A, so conjugation by $O(t)$ retains the same constant matrix. Similarly $\frac{1}{2}\operatorname{tr}[A,H]^2 + |ABe_1|^2 = as/2$. The determinant and $b^*\Phi b$ both vanish on the trajectory; substitution of $a = m^2+\Gamma s$ gives the displayed energy.

For the gauge comparison retain the packet's covariant derivatives $D_0\Phi = \dot\Phi + [\mathcal{A}_0,\Phi]$, $D_0 b = \dot b + \mathcal{A}_0 b$. Varying $\mathcal{A}_0$ by $\xi \in \mathfrak{su}(3)$ in the scalar kinetic density gives $\operatorname{tr}(\mathcal{J}\xi)$ at $\mathcal{A}_0 = 0$. With homogeneous fields and every $\mathcal{A}_i = 0$, the electric field vanishes and the gauge kinetic variation contributes zero. Thus its Euler–Lagrange equation requires the trace-free part of $\mathcal{J}$ to vanish. The exhibited matrix is already trace-free and nonzero, so this particular zero-connection lift fails. This is the Gauss equation obtained by variation before imposing temporal gauge; compare D. Tong, *Lectures on Gauge Theory*, Section 2.2.1, equation (2.27), `https://www.damtp.cam.ac.uk/user/tong/gaugetheory/gt.pdf`. Our matter contribution has been derived here in the retained anti-Hermitian convention, not imported from a different convention. □

There is an exact moving-frame map, not an assertion that these theories are unrelated. With $U(t) = O(t)^*$ the gauge transformation rule is

$$\Phi' = U\Phi U^{-1} = \Phi_0, \quad b' = Ub = b_0, \quad \mathcal{A}_0' = U\mathcal{A}_0U^{-1} - \dot U U^{-1} = \Omega.$$

The covariant velocities remain $D_0'\Phi_0 = [\Omega,\Phi_0]$ and $D_0' b_0 = \Omega b_0$. The same nonzero charge and Gauss residual therefore remain; making the ordinary coordinates static does not remove them. A gauge-field lift would have to supply the electric contribution to Gauss' equation and satisfy the remaining coupled equations. Neither its existence nor its nonexistence follows from failure of the zero-connection lift, and no quantum spectral-gap conclusion is inferred from this periodic motion.

1.11 The complete scalar vacuum calculation with its kinetic metric

This subsection validates the homogeneous scalar vacuum theorem in the triality packet. It does not assert a new discovery of that phase diagram, and does not include the packet's subsequent gauge-orbit claims. Set $Q = \Theta(\Phi, b)$ with $\operatorname{tr} Q = 0$, and retain

$$g(Y,Z) = \operatorname{Re}\operatorname{tr}(YZ), \qquad s(Q) = g(Q,Q) = \operatorname{tr}\Phi^2 + 2b^*b, \qquad n(Q) = N(Q) = \det\Phi - b^*\Phi b.$$

The kinetic term is $g(\dot Q,\dot Q)/2$, not the unweighted sum of squares of the independent entries. Explicitly, $g(Y,Y) = \sum_i Y_{ii}^2 + 2\sum_{i<j}|Y_{ij}|^2$ for Hermitian Y.

Lemma 1.23 (Sharp bound and all equality cases)**.** *For every trace-free quaternionic Hermitian Q,*

$$54\,n(Q)^2 \le s(Q)^3.$$

For $Q \neq 0$, equality holds exactly at $Q = x(3P - I_3)$, $x \in \mathbb{R}\setminus\{0\}$, where $P = P^* = P^2$ has quaternionic rank one. There $s = 6x^2$ and $n = 2x^3$.

Proof. Here are the spectral and determinant details in the stated conventions. The real-valued function v^*Qv on the unit sphere of $\mathbb{H}^3$ attains a maximum. Its Lagrange-multiplier equation gives $\operatorname{Re}(w^*(Qv - v\lambda)) = 0$ for every w, with real $\lambda = v^*Qv$. Taking $w = Qv - v\lambda$ proves $Qv = v\lambda$. The quaternionic orthogonal complement of v is invariant because $v^*Qw = \lambda v^*w$. Iteration gives a unitary eigenbasis with real eigenvalues $\lambda_1,\lambda_2,\lambda_3$.

For quaternionic matrices, the real trace is cyclic by $\mathrm{Re}(pq) = \mathrm{Re}(qp)$, entry by entry. Moreover, the original cubic (1) obeys

$$6n(Q) = (\operatorname{tr} Q)^3 - 3\operatorname{tr}(Q)\operatorname{tr}(Q^2) + 2\operatorname{tr}(Q^3). \tag{27}$$

To check it without changing the quaternionic order, in the ordered entries of (1) the cubic trace is the sum of the diagonal cubes, plus $3(d+f)|q_1|^2 + 3(a+f)|q_2|^2 + 3(a+d)|q_3|^2$, plus $6\,\mathrm{Re}(q_1 q_2 q_3)$. Substitution gives each term of six times (1). Unitary conjugation therefore preserves s, n, which are respectively $\sum_i \lambda_i^2$ and $\lambda_1\lambda_2\lambda_3$ in the eigenbasis. Their sum is zero. The exact polynomial identity

$$s^3 - 54n^2 = 2(\lambda_1 - \lambda_2)^2(\lambda_1 - \lambda_3)^2(\lambda_2 - \lambda_3)^2$$

follows by substituting $\lambda_3 = -\lambda_1 - \lambda_2$ and expanding. It proves the bound. For nonzero Q, equality means a repeated eigenvalue, hence spectrum $(2x, -x, -x)$ for some $x \neq 0$. Letting P project onto the simple eigenline gives $Q = x(3P - I_3)$. The converse follows from this same spectrum. □

Theorem 1.24 (All scalar global minima)**.** *For $\Gamma > 0$, $m^2 \geq 0$, and $\kappa \neq 0$, put $m_{\mathrm{c}}^2 = \kappa^2/(27\Gamma)$. The global minima of $V = m^2 s/2 + \Gamma s^2/4 - \kappa n$ are exactly:*

- *$Q = 0$ if $m^2 > m_{\mathrm{c}}^2$;*
- *$Q = 0$ and $Q = \frac{\kappa}{9\Gamma}(3P - I_3)$ for every rank-one quaternionic Hermitian projector P, if $m^2 = m_{\mathrm{c}}^2$;*
- *$Q = x_*(3P - I_3)$ for every such P, if $0 \leq m^2 < m_{\mathrm{c}}^2$, where*

$$x_* = \frac{\kappa + \operatorname{sgn}(\kappa)\sqrt{\kappa^2 - 24\Gamma m^2}}{12\Gamma}.$$

Each nonzero component is diffeomorphic to $\mathbb{HP}^2$. For $\kappa = 0$, the origin is the unique global minimum.

Proof. For $\kappa = 0$, the positive quartic proves the assertion, including $m^2 = 0$. Otherwise at each fixed original norm $r = \sqrt{s}$ the sharp bound is attained with $x = \operatorname{sgn}(\kappa) r/\sqrt{6}$ and $Q = x(3P - I_3)$. For $r > 0$, these are exactly its equality and sign cases. Thus the full minimization is equivalent, with all fibres of the norm map accounted for, to minimizing

$$f(r) = \frac{m^2}{2} r^2 + \frac{\Gamma}{4} r^4 - \frac{|\kappa|}{3\sqrt{6}} r^3, \qquad r \geq 0.$$

Set $r_{\mathrm{c}} = 2|\kappa|/(3\sqrt{6}\Gamma)$. Exact factorization gives

$$f(r) = \frac{\Gamma}{4} r^2 (r - r_{\mathrm{c}})^2 + \frac{m^2 - m_{\mathrm{c}}^2}{2} r^2.$$

This proves the first two cases, including every zero. Below the transition, $f(r_{\mathrm{c}}) < 0$, so a global minimum is nonzero. Put $a_0 = |\kappa|/\sqrt{6}$. The derivative is $f'(r) = r(\Gamma r^2 - a_0 r + m^2)$, with roots

$$r_\pm = \frac{a_0 \pm \sqrt{a_0^2 - 4\Gamma m^2}}{2\Gamma}.$$

Below the transition its discriminant is strictly positive. If $m^2 > 0$, f' has signs $+, -, +$ across these two positive roots, so r_+ is the only positive local minimum. If $m^2 = 0$, $r_- = 0$, and the signs on

$(0, r_+)$ and (r_+, ∞) are $-, +$. Coercivity and the negative value exclude the origin in both cases. Converting r_+ to x proves the third case and the completeness of its matrix equality locus.

For the manifold statement, a right quaternionic line represented by $v \neq 0$ has orthogonal projector $P_v = v(v^*v)^{-1}v^*$. This is unchanged under $v \mapsto vh$, $h \in \mathbb{H}^\times$, and every rank-one Hermitian projector is the unique projection onto its image line. The formula and its inverse line charts are smooth. For a fixed nonzero x, the maps $P \mapsto x(3P - I_3)$ and $Q \mapsto (Q/x + I_3)/3$ are mutually inverse. This proves the claimed $\mathbb{HP}^2$ identification without any gauge quotient. □

Theorem 1.25 (Full metric Hessian)**.** *At every nonzero minimum $Q_* = x(3P - I_3)$, the Hessian relative to g has eigenvalues*

eigenvalue	*real multiplicity*
$x(12\Gamma x - \kappa)$	1
0	8
$3\kappa x$	5

Both nonzero eigenvalues are positive. The entire kernel is the tangent space of the nonzero minimum manifold. At the origin the metric Hessian is $m^2 I$ in all fourteen directions.

Proof. The preceding spectral proof and cubic trace identity show that quaternionic unitary conjugation preserves both g and V. Transport the Hessian isometrically to $Q_* = xD$, $D = \operatorname{diag}(2, -1, -1)$. Every trace-free Hermitian variation has the unique, g-orthogonal decomposition

$$Y = yD + \begin{pmatrix} 0 & v^* \\ v & 0_2 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & B_0 \end{pmatrix}, \qquad y \in \mathbb{R}, \quad v \in \mathbb{H}^2, \quad B_0 = B_0^*, \quad \operatorname{tr} B_0 = 0.$$

Write $B_0 = \left(\begin{smallmatrix} \beta & \omega \\ \bar{\omega} & -\beta \end{smallmatrix}\right)$, $\beta \in \mathbb{R}$, $\omega \in \mathbb{H}$. The respective dimensions are $1, 8, 5$, and the retained norm is

$$g(Y, Y) = 6y^2 + 2v^*v + \|B_0\|^2, \qquad \|B_0\|^2 = 2\beta^2 + 2|\omega|^2.$$

For $h = x + \varepsilon y$, the full cubic, including every mixed direction, is

$$n(Q_* + \varepsilon Y) = 2h^3 + \varepsilon^2 h(v^*v - \|B_0\|^2) + \varepsilon^3 v^* B_0 v. \tag{28}$$

Indeed the diagonal and lower off-diagonal contributions are $2h(h^2 - \varepsilon^2\beta^2) - 2h\varepsilon^2|\omega|^2$. The upper entries contribute $h\varepsilon^2(|v_1|^2 + |v_2|^2) + \varepsilon^3\beta(|v_1|^2 - |v_2|^2)$, and the triangular term is $2\varepsilon^3 \operatorname{Re}(\bar{v}_1 \omega v_2)$. These sum to (28), preserving the quaternionic order. Also

$$s(Q_* + \varepsilon Y) = 6x^2 + 12\varepsilon xy + \varepsilon^2(6y^2 + 2v^*v + \|B_0\|^2).$$

The second derivative of the cubic is therefore $12xy^2 + 2xv^*v - 2x\|B_0\|^2$. Differentiating the potential gives the complete quadratic form before using stationarity:

$$\begin{aligned} D^2 V_{xD}[Y, Y] = {} & 6(m^2 + 18\Gamma x^2 - 2\kappa x)y^2 \\ & + 2(m^2 + 6\Gamma x^2 - \kappa x)v^*v \\ & + (m^2 + 6\Gamma x^2 + 2\kappa x)\|B_0\|^2. \end{aligned}$$

Every nonzero minimum satisfies $m^2 - \kappa x + 6\Gamma x^2 = 0$. Thus this becomes

$$6x(12\Gamma x - \kappa)y^2 + 3\kappa x\|B_0\|^2.$$

There are no mixed quadratic terms; polarization therefore gives no mixed bilinear blocks. Comparison with $g(Y,Y)$ proves every listed eigenvalue and multiplicity, with no rescaling of the kinetic metric.

At coexistence the positive values are $\kappa^2/(27\Gamma)$ and $\kappa^2/(3\Gamma)$. Below it, $12\Gamma x - \kappa = \operatorname{sgn}(\kappa)\sqrt{\kappa^2 - 24\Gamma m^2}$ and x has the same sign as κ, proving positivity. In the line chart

$$P(z) = \frac{1}{1+z^*z}\begin{pmatrix} 1 & z^* \\ z & zz^* \end{pmatrix}, \qquad DP_0[\zeta] = \begin{pmatrix} 0 & \zeta^* \\ \zeta & 0_2 \end{pmatrix},$$

the derivative of $x(3P(z) - I_3)$ spans exactly the eight-dimensional zero block. Isometric conjugation transports this to every P. At the origin, the cubic and quartic contribute zero Hessian, giving $m^2 g$ directly. In particular, at $m^2 = \kappa = 0$ the origin is still the unique minimum, although its Hessian vanishes in all directions. □

These are scalar Hessian values of a specified finite-dimensional model. Their exact multiplicities do not assert a gauge-reduced physical spectrum or an interacting quantum mass gap. The current calculation includes the full tangent and normal blocks, not solely the radial restriction.

1.12 Gauge kinetic sign and precise scope of this audit

In the packet's gauge lift, $\mathcal{A}_\mu \in \mathfrak{su}(3)$ and $D_\mu b = \partial_\mu b + \mathcal{A}_\mu b$ fix anti-Hermitian gauge generators. Retain Minkowski signature $(+---)$, consistent with the positive time-derivative scalar term in that lift. Write $\|T\|^2 = -\operatorname{tr}(T^2)$ for anti-Hermitian T, and $E_i = \mathcal{F}_{0i}$, $B_i = \frac{1}{2}\epsilon_{ijk}\mathcal{F}_{jk}$. Raising the time-space index pair changes its sign and summing both orders of each antisymmetric pair gives

$$\operatorname{tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu}) = 2\sum_i (\|E_i\|^2 - \|B_i\|^2).$$

Consequently the packet's coefficient $-1/(2g_c^2)$ gives negative electric kinetic energy in these conventions. Keeping its anti-Hermitian fields and scalar kinetic terms, the positive-energy gauge density is

$$\mathcal{L}_{\text{gauge}} = +\frac{1}{2g_c^2}\operatorname{tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu}) = \frac{1}{g_c^2}\sum_i (\|E_i\|^2 - \|B_i\|^2).$$

For an immediate check take $\mathcal{F}_{01} = \operatorname{diag}(i, -i, 0)$ and all other independent components zero: the trace contraction is 4, so the two signs give $-2/g_c^2$ and $+2/g_c^2$, respectively. This sign correction changes neither the scalar potential nor its derivative calculated above. It does matter before interpreting the full gauge lift as a positive-energy physical model.

The equal-diagonal thirteen-space in the triality packet remains a different embedded slice: its intersection with $\mathrm{Herm}_3(\mathbb{C})$ has dimension seven, whereas the common Lorentz slice contains all nine complex-Hermitian coordinates. Their precise common ambient space is $\mathrm{Herm}_3(\mathbb{H})$; Theorem 1.2 identifies the *Lorentz* constructions, not the equal-diagonal one. No result here identifies these cone-boundary spheres with the claimed integrable complex structure on S^6, assigns new spinor periods and toroidal fillings, or proves a continuum interacting Yang–Mills spectral gap. The exact next data are the transition maps and lattice extension for the additional spinor, followed by a physical state space and Hamiltonian comparison retaining the couplings above.

1.13 The full homogeneous gauge equations and the literal periodic orbit

Retain signature $(+---)$, $\mathcal{A}_\mu \in \mathfrak{su}(3)$, $\Phi = \Phi^*$, $\operatorname{tr}\Phi = 0$, and $b \in \mathbb{C}^3$. Write $g^2 > 0$ for the gauge coupling in this subsection and keep

$$s = \operatorname{tr}\Phi^2 + 2b^*b, \qquad a = m^2 + \Gamma s, \qquad V = \frac{m^2}{2}s + \frac{\Gamma}{4}s^2 - \kappa(\det\Phi - b^*\Phi b).$$

Thus $m^2 \geq 0$, $\Gamma > 0$, and $\kappa \in \mathbb{R}$ are unchanged. For a matrix Z write $Z_{\mathrm{tf}} = Z - \operatorname{tr}(Z)I_3/3$. The full density is

$$\mathcal{L} = \frac{1}{2g^2}\operatorname{tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu}) + \frac{1}{2}\operatorname{tr}(D_\mu\Phi D^\mu\Phi) + (D_\mu b)^* D^\mu b - V, \tag{29}$$

where $\mathcal{F}_{\mu\nu} = \partial_\mu\mathcal{A}_\nu - \partial_\nu\mathcal{A}_\mu + [\mathcal{A}_\mu, \mathcal{A}_\nu]$, $D_\mu\Phi = \partial_\mu\Phi + [\mathcal{A}_\mu, \Phi]$, and $D_\mu b = \partial_\mu b + \mathcal{A}_\mu b$. We consider spatially homogeneous fields on a finite-volume flat $\mathbb{T}^3$, or locally on $\mathbb{R}^3$. A nonzero constant energy density is not asserted to have finite total energy on $\mathbb{R}^3$.

Proposition 1.26 (Every homogeneous field equation)**.** *Set*

$$E_i = \dot{\mathcal{A}}_i + [\mathcal{A}_0, \mathcal{A}_i], \qquad F_{ij} = [\mathcal{A}_i, \mathcal{A}_j], \qquad X = \dot{\Phi} + [\mathcal{A}_0, \Phi], \qquad v = \dot{b} + \mathcal{A}_0 b,$$

and define the lower-index matter currents

$$J_0 = [\Phi, X] + bv^* - vb^*, \qquad J_i = [\Phi, [\mathcal{A}_i, \Phi]] - bb^*\mathcal{A}_i - \mathcal{A}_i bb^*.$$

The complete homogeneous Euler–Lagrange system is

$$D_0 X - \sum_i [\mathcal{A}_i, [\mathcal{A}_i, \Phi]] + a\Phi - \kappa(\operatorname{adj}\Phi - bb^*)_{\mathrm{tf}} = 0, \tag{30}$$

$$D_0 v - \sum_i \mathcal{A}_i^2 b + (aI_3 + \kappa\Phi)b = 0, \tag{31}$$

$$\frac{2}{g^2}\sum_i [\mathcal{A}_i, E_i] - (J_0)_{\mathrm{tf}} = 0, \tag{32}$$

$$D_0 E_i - \sum_j [\mathcal{A}_j, [\mathcal{A}_j, \mathcal{A}_i]] - \frac{g^2}{2}(J_i)_{\mathrm{tf}} = 0 \quad (i = 1, 2, 3). \tag{33}$$

Here $D_0 X = \dot{X} + [\mathcal{A}_0, X]$, $D_0 v = \dot{v} + \mathcal{A}_0 v$, and similarly for E_i.

Proof. First retain the homogeneous density without deleting $\mathcal{A}_0$:

$$\mathcal{L}_{\mathrm{hom}} = -\frac{1}{g^2}\sum_i \operatorname{tr} E_i^2 + \frac{1}{g^2}\sum_{i<j}\operatorname{tr} F_{ij}^2 + \frac{1}{2}\operatorname{tr} X^2 + v^*v - \frac{1}{2}\sum_i \operatorname{tr}[\mathcal{A}_i, \Phi]^2 - \sum_i |\mathcal{A}_i b|^2 - V.$$

Both signs in the gauge terms follow by raising indices in $(+---)$ and counting the two orders of each antisymmetric pair. For trace-free Hermitian $\delta\Phi$ and arbitrary δb,

$$\delta V = \operatorname{tr}\big[(a\Phi - \kappa(\operatorname{adj}\Phi - bb^*)_{\mathrm{tf}})\delta\Phi\big] + 2\operatorname{Re}\big(\delta b^*(aI_3 + \kappa\Phi)b\big).$$

Covariant integration by parts therefore gives $D_\mu D^\mu\Phi + a\Phi - \kappa(\operatorname{adj}\Phi - bb^*)_{\mathrm{tf}} = 0$ and $D_\mu D^\mu b + (aI_3 + \kappa\Phi)b = 0$. Homogeneity gives the first two displayed equations, including the minus signs in their spatial sums.

For the gauge variation, antisymmetry gives

$$\delta\mathcal{L}_{\text{gauge}} = \frac{2}{g^2}\,\mathrm{tr}((D_\mu\delta\mathcal{A}_\nu)\mathcal{F}^{\mu\nu}).$$

After integration by parts its coefficient is $-2D_\mu\mathcal{F}^{\mu\nu}/g^2$. The scalar coefficient of $\delta\mathcal{A}_\nu$ is

$$J^\nu = [\Phi, D^\nu\Phi] + b(D^\nu b)^* - (D^\nu b)b^*.$$

Indeed cyclicity and anti-Hermiticity respectively give

$$\begin{aligned}\mathrm{tr}(D^\nu\Phi[\delta\mathcal{A}_\nu,\Phi]) &= \mathrm{tr}([\Phi,D^\nu\Phi]\delta\mathcal{A}_\nu),\\ 2\,\mathrm{Re}((D^\nu b)^*\delta\mathcal{A}_\nu b) &= \mathrm{tr}\big((b(D^\nu b)^* - (D^\nu b)b^*)\delta\mathcal{A}_\nu\big).\end{aligned}$$

Variation over $\mathfrak{su}(3)$ consequently yields

$$D_\mu\mathcal{F}^{\mu\nu} = \frac{g^2}{2}(J^\nu)_{\text{tf}}.$$

For $\nu = 0$, $\mathcal{F}^{i0} = E_i$ and $J^0 = J_0$. For $\nu = i$, $\mathcal{F}^{0i} = -E_i$, $\mathcal{F}^{ji} = [\mathcal{A}_j, \mathcal{A}_i]$, and $J^i = -J_i$. These substitutions give exactly (32) and (33). In particular the $\mathcal{A}_0$ equation survives after choosing temporal gauge. This is the same variational principle emphasized by D. Tong, *Lectures on Gauge Theory*, Section 2.2.1, equation (2.27); the matter currents, signs, and coefficient $g^2/2$ above have been derived in the present conventions. □

Theorem 1.27 (No temporal-gauge lift retaining the literal orbit)**.** *Keep exactly the moving-kernel scalar functions in Theorem 1.21, with $B, d > 0$ and $x \neq 0$. There are no homogeneous spatial fields $\mathcal{A}_i(t)$ for which these same scalar functions, together with $\mathcal{A}_0 = 0$, solve the full system* (30)–(33)*.*

Proof. The scalar functions already solve the unrestricted scalar equations. Subtracting them from (30)–(31), with $\mathcal{A}_0 = 0$, gives

$$\sum_i [\mathcal{A}_i, [\mathcal{A}_i, \Phi]] = 0, \qquad \sum_i \mathcal{A}_i^2 b = 0.$$

Taking the trace pairing with Φ and the Hermitian pairing with b gives respectively

$$-\sum_i \mathrm{tr}[\mathcal{A}_i, \Phi]^2 = 0, \qquad -\sum_i |\mathcal{A}_i b|^2 = 0.$$

Each $[\mathcal{A}_i, \Phi]$ is Hermitian, so both are sums of nonnegative squared norms with the displayed common minus sign. Thus $[\mathcal{A}_i, \Phi] = 0$ and $\mathcal{A}_i b = 0$ separately for every i.

To classify this stabilizer use the retained orthogonal matrix $K = (v/\sqrt{2}, w_0/\sqrt{6}, u/\sqrt{3})$ from the moving-family proof, and put $r^2 = x^2 + d^2 > 0$. At time zero the data are

$$H = K^*\Phi_0 K = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -x & d \\ 0 & d & x \end{pmatrix}, \qquad K^* b_0 = Be_1, \qquad A = K^*\Omega K = \begin{pmatrix} 0 & -\sqrt{a} & 0 \\ \sqrt{a} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

An anti-Hermitian traceless matrix killing Be_1 has zero first row and column and a lower $\mathfrak{su}(2)$ block. Commutation with $\left(\begin{smallmatrix} -x & d \\ d & x \end{smallmatrix}\right)$, whose distinct eigenvalues are $\pm r$, forces that block to be its real multiple times i. Therefore the stabilizer Lie algebra is $\mathbb{R}\, i\Phi(t)$. The full stabilizer group is the circle

$$\left\{ P_b + \cos\theta(I_3 - P_b) + i\frac{\sin\theta}{r}\Phi(t) : \theta \in \mathbb{R} \right\}, \qquad P_b = \frac{b(t)b(t)^*}{B^2}.$$

Indeed it fixes b, has eigenvalues $1, e^{i\theta}, e^{-i\theta}$, and exhausts the unitary commutant fixing the zero eigenspace pointwise. Consequently

$$\mathcal{A}_i = ic_i(t)\Phi(t), \qquad c_i(t) \in \mathbb{R}.$$

The original orbit has

$$[\Phi, \dot{\Phi}] = -r^2\Omega, \qquad b\dot{b}^* - \dot{b}b^* = -B^2\Omega, \qquad J_0 = -(r^2 + B^2)\Omega.$$

For verification, $H^2 = r^2\,\mathrm{diag}(0,1,1)$, $HAH = 0$, and hence $[H,[A,H]] = -r^2A$; the vector term is $-B^2A$. Conjugation back by $O(t)K$, with the rotation commuting with Ω, gives the stated original-coordinate identities. Now

$$F_{ij} = 0, \qquad E_i = i\dot{c}_i\Phi + ic_i\dot{\Phi}, \qquad \sum_i [\mathcal{A}_i, E_i] = r^2\left(\sum_i c_i^2\right)\Omega.$$

Thus the full Gauss residual, with precisely the convention in (32), is

$$\boxed{\left(\frac{2r^2}{g^2}\sum_i c_i^2 + r^2 + B^2\right)\Omega.}$$

Its coefficient is strictly positive and $\Omega \neq 0$, proving the obstruction even before the spatial gauge equations are used.

For completeness those equations force all c_i to vanish as well: here $J_i = F_{ij} = 0$, so they require $i(\ddot{c}_i\Phi + 2\dot{c}_i\dot{\Phi} + c_i\ddot{\Phi}) = 0$. At a fixed time conjugate by the retained moving frame. The $(1,1)$ entries of H and $[A,H]$ vanish, while that of $[A,[A,H]]$ is $-2ax \neq 0$. Hence $c_i = 0$. The remaining Gauss residual is then $(r^2 + B^2)\Omega$, still nonzero. □

Proposition 1.28 (Arbitrary temporal connection and exact Gauss inertia)**.** *For arbitrary homogeneous scalar and spatial-gauge positions and velocities, define*

$$J_{\mathrm{sc}} = [\Phi, \dot{\Phi}] + b\dot{b}^* - \dot{b}b^*, \qquad \mathscr{I}_{\Phi,b}Z = \left([\Phi,[\Phi,Z]] + bb^*Z + Zbb^*\right)_{\mathrm{tf}},$$

$$\mathscr{K}_{\mathcal{A}}Z = -\sum_i [\mathcal{A}_i, [\mathcal{A}_i, Z]].$$

Then Gauss' equation is exactly

$$\left(\mathscr{I}_{\Phi,b} + \frac{2}{g^2}\mathscr{K}_{\mathcal{A}}\right)\mathcal{A}_0 = (J_{\mathrm{sc}})_{\mathrm{tf}} - \frac{2}{g^2}\sum_i [\mathcal{A}_i, \dot{\mathcal{A}}_i]. \tag{34}$$

The operator on the left is self-adjoint and positive semidefinite for $\langle Z, W\rangle = -\operatorname{tr}(ZW)$ on $\mathfrak{su}(3)$. Its kernel is exactly

$$\{Z : [Z,\Phi] = 0,\ Zb = 0,\ [Z,\mathcal{A}_i] = 0 \text{ for every } i\}.$$

The right-hand side always annihilates that kernel. Thus Gauss' equation has a pointwise solution for $\mathcal{A}_0$, unique modulo this kernel. This pointwise statement asserts neither a smooth selection across kernel-rank changes nor a solution of the remaining field equations.

Proof. Expansion gives $J_0 = J_{\mathrm{sc}} - \mathscr{I}_{\Phi,b}\mathcal{A}_0$ after trace-free projection, and

$$\sum_i [\mathcal{A}_i, E_i] = \sum_i [\mathcal{A}_i, \dot{\mathcal{A}}_i] + \mathscr{K}_{\mathcal{A}}\mathcal{A}_0.$$

Substitution proves (34). Cyclicity gives

$$\langle Z, \mathscr{I}_{\Phi,b} Z\rangle = \operatorname{tr}[Z,\Phi]^2 + 2|Zb|^2, \qquad \langle Z, \mathscr{K}_{\mathcal{A}} Z\rangle = \sum_i \|[\mathcal{A}_i, Z]\|_{\mathfrak{su}}^2,$$

where $\|T\|_{\mathfrak{su}}^2 = -\operatorname{tr} T^2$. More explicitly, cyclicity gives the bilinear identities

$$\langle Z, \mathscr{I}_{\Phi,b} W\rangle = \operatorname{tr}([Z,\Phi][W,\Phi]) + 2\operatorname{Re}((Zb)^* W b), \qquad \langle Z, \mathscr{K}_{\mathcal{A}} W\rangle = \sum_i \langle [\mathcal{A}_i, Z], [\mathcal{A}_i, W]\rangle.$$

These are symmetric in Z, W and prove self-adjointness; the sums of squares prove the stated kernel. If Z is in the kernel, the matter moment paired with Z vanishes because $[Z,\Phi] = Zb = 0$, and $\operatorname{tr}(Z[\mathcal{A}_i, \dot{\mathcal{A}}_i]) = \operatorname{tr}([Z,\mathcal{A}_i]\dot{\mathcal{A}}_i) = 0$. Thus the right-hand side is orthogonal to the kernel. In a finite dimensional real inner-product space a self-adjoint operator has image equal to the orthogonal complement of its kernel: this follows by diagonalizing its real symmetric matrix. It gives existence and the claimed exact ambiguity. □

For the literal moving orbit, subtracting the original scalar equations while retaining arbitrary $\mathcal{A}_0$ gives the exact additional equations

$$[\dot{\mathcal{A}}_0, \Phi] + 2[\mathcal{A}_0, \dot{\Phi}] + [\mathcal{A}_0, [\mathcal{A}_0, \Phi]] - \sum_i [\mathcal{A}_i, [\mathcal{A}_i, \Phi]] = 0,$$

$$\dot{\mathcal{A}}_0 b + 2\mathcal{A}_0 \dot{b} + \mathcal{A}_0^2 b - \sum_i \mathcal{A}_i^2 b = 0.$$

Their norm consequences are

$$\sum_i \operatorname{tr}[\mathcal{A}_i, \Phi]^2 = \operatorname{tr} X^2 - \operatorname{tr} \dot{\Phi}^2, \qquad \sum_i |\mathcal{A}_i b|^2 = |v|^2 - |\dot{b}|^2.$$

For example the trace pairing of the first equation uses $\operatorname{tr}(\Phi[\dot{\mathcal{A}}_0, \Phi]) = 0$ and $\operatorname{tr}(\Phi[\mathcal{A}_0, \dot{\Phi}]) = -\operatorname{tr}([\mathcal{A}_0, \Phi]\dot{\Phi})$; expansion of X^2 then gives the first identity. The real pairing of the second equation with b gives the second by anti-Hermiticity. These right-hand sides need not be zero; temporal gauge cannot be imposed while silently keeping the same literal scalar functions.

Here is an exact counterexample to extending the temporal-gauge stabilizer deduction to nonzero $\mathcal{A}_0$. For real constants $\alpha, \beta_1, \beta_2, \beta_3$ satisfying $\alpha^2 - \sum_i \beta_i^2 = 1$, keep the same scalar orbit and set

$$\mathcal{A}_0 = (\alpha - 1)\Omega, \qquad \mathcal{A}_i = \beta_i \Omega.$$

All curvatures vanish, while $D_0\Phi = \alpha[\Omega, \Phi]$, $D_0 b = \alpha\Omega b$, $D_i\Phi = \beta_i[\Omega,\Phi]$, and $D_i b = \beta_i \Omega b$. Their covariant wave operators therefore equal the original second derivatives, proving both scalar equations exactly. Taking $\alpha = 5/3$, $\beta_1 = 4/3$, $\beta_2 = \beta_3 = 0$ gives $\mathcal{A}_1 b \neq 0$: the spatial field does not stabilize the scalars. Nevertheless $J_0 = \alpha J_{\mathrm{sc}} \neq 0$, since $|\alpha| \geq 1$, and the zero electric field cannot satisfy Gauss' equation. These are scalar-compatible flat connections, not full gauge solutions.

There is also no lift of the moving orbit with all spatial $\mathcal{A}_i = 0$, even when $\mathcal{A}_0$ is arbitrary. Indeed $J_{\mathrm{sc}} = -\mathscr{I}_{\Phi,b}\Omega$, so (34) gives $\mathscr{I}_{\Phi,b}(\mathcal{A}_0 + \Omega) = 0$. Its kernel description implies $D_0\Phi = D_0 b = 0$. Equation (31) would then require $(aI_3 + \kappa\Phi)b = ab = 0$, impossible because $a > 0$ and $B > 0$. This argument, and Theorem 1.27, do not exclude the general case with both nonzero $\mathcal{A}_0$ and nonzero spatial fields satisfying all coupled equations.

1.14 Constant-matter coordinates and an all-time real-connection obstruction

Retain exactly the parameters and literal scalar functions of Theorem 1.21. In particular $a > 0$, $B, d > 0$, and $x \neq 0$. Keep the same original-coordinate $O(t)$, Ω, and the exact orthogonal map and inverse

$$K = \left(v/\sqrt{2}, w_0/\sqrt{6}, u/\sqrt{3}\right), \qquad K^{-1} = K^*.$$

The squared lengths of v, w_0, u are $2, 6, 3$, their pairwise inner products vanish, and $\det(v, w_0, u) = 6$; hence $K \in SO(3)$. Put

$$H = K^*\Phi_0 K = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -x & d \\ 0 & d & x \end{pmatrix}, \qquad q = K^* b_0 = Be_1, \qquad A = K^*\Omega K = \begin{pmatrix} 0 & -\sqrt{a} & 0 \\ \sqrt{a} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Thus $Hq = 0$. Write $r^2 = x^2 + d^2$ and $\mu = r^2 + B^2$. The retained force, including the cubic term, is

$$F_H = -aH + \kappa(\operatorname{adj} H - qq^*)_{\mathrm{tf}} = \begin{pmatrix} -2ax & 0 & 0 \\ 0 & 2ax & -ad \\ 0 & -ad & 0 \end{pmatrix} = [A, [A, H]], \qquad A^2 q = -aq.$$

Indeed $s = 2(r^2 + B^2) = 6ax/\kappa$, $\operatorname{adj} H = \operatorname{diag}(-r^2, 0, 0)$, and $(\operatorname{adj} H - qq^*)_{\mathrm{tf}} = \operatorname{diag}(-s/3, s/6, s/6)$, which give these identities by direct multiplication.

Proposition 1.29 (An exact constant-matter reformulation)**.** *Set*

$$C = K^*O^*\mathcal{A}_0 OK + A, \qquad Y_i = K^*O^*\mathcal{A}_i OK, \qquad \mathcal{E}_i = \dot{Y}_i + [C, Y_i],$$

$$\mathscr{I}_{H,q} Z = ([H, [H, Z]] + qq^* Z + Zqq^*)_{\mathrm{tf}}, \qquad \mathscr{K}_Y Z = -\sum_i [Y_i, [Y_i, Z]].$$

The prescribed literal scalar functions have a homogeneous gauge lift if and only if the following complete system holds:

$$[\dot{C}, H] + [C, [C, H]] - \sum_i [Y_i, [Y_i, H]] = F_H, \tag{35}$$

$$(\dot{C} + C^2 - \sum_i Y_i^2) q = -aq, \tag{36}$$

$$(\mathscr{I}_{H,q} + 2\mathscr{K}_Y/g^2) C = -\frac{2}{g^2} \sum_i [Y_i, \dot{Y}_i], \tag{37}$$

$$\dot{\mathcal{E}}_i + [C, \mathcal{E}_i] - \sum_j [Y_j, [Y_j, Y_i]] + \frac{g^2}{2} \mathscr{I}_{H,q} Y_i = 0 \quad (i = 1, 2, 3). \tag{38}$$

The inverse connection maps are exactly

$$\mathcal{A}_0 = OKCK^*O^* - \Omega, \qquad \mathcal{A}_i = OKY_iK^*O^*.$$

Neither the literal functions nor their time scale has been changed.

Proof. For a spatially constant $U(t) \in SU(3)$, the transformations are $\Phi' = U\Phi U^*$, $b' = Ub$, $\mathcal{A}_0' = U\mathcal{A}_0 U^* - \dot{U}U^*$, and $\mathcal{A}_i' = U\mathcal{A}_i U^*$. The product rule gives $D_\nu'(Ub) = UD_\nu b$ and $D_\nu'(U\Phi U^*) = U(D_\nu\Phi)U^*$. Their commutators give $\mathcal{F}_{\nu\rho}' = U\mathcal{F}_{\nu\rho}U^*$. Traces, trace projection, and the potential are invariant under this unitary action; adjugate covariance follows from the inverse formula on

invertible matrices and polynomial continuity. Thus all the equations of Proposition 1.26, including Gauss' equation, transform covariantly.

Taking $U = K^*O^*$ gives the constant matter fields H, q and the displayed connection maps, because $-\dot{U}U^* = K^*\Omega K = A$. The scalar covariant derivatives are $[C, H]$ and Cq; their second derivatives are $[\dot{C}, H] + [C, [C, H]]$ and $(\dot{C} + C^2)q$. The trace-free currents are $(J_0)_{\mathrm{tf}} = -\mathscr{I}_{H,q}C$ and $(J_i)_{\mathrm{tf}} = -\mathscr{I}_{H,q}Y_i$. Finally $\sum_i [Y_i, \mathcal{E}_i] = \sum_i [Y_i, \dot{Y}_i] + \mathscr{K}_Y C$. Substitution gives every displayed equation with its stated sign and factor. The explicitly invertible maps prove both directions. □

Theorem 1.30 (No real-connection lift with any spatial time dependence)**.** *The literal moving-kernel scalar functions admit no homogeneous lift with $\mathcal{A}_\nu(t) \in \mathfrak{so}(3)$ for all four indices, on any nonempty open time interval. The temporal connection is arbitrary; the spatial connections may be arbitrary twice differentiable functions, including rotating, multifrequency, and nonperiodic functions. No commutativity or spatial-stationarity assumption is imposed.*

Proof. Since K, O, Ω are real, the exact frame maps preserve real skew-symmetry in both directions. Define

$$L_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad L_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad L(z) = \sum_j z_j L_j \quad (z \in \mathbb{R}^3).$$

Direct multiplication gives

$$L(z)w = z \times w, \quad L(z)L(w) = wz^{\mathsf{T}} - (z \cdot w)I_3, \quad [L(z), L(w)] = L(z \times w), \quad -\operatorname{tr}(L(z)L(w)) = 2z \cdot w.$$

These formulas specify the commutator orientation and every norm factor. Write

$$C = L(c), \quad Y_i = L(y_i), \quad h = \dot{c}, \quad P = \sum_i y_i y_i^{\mathsf{T}}, \quad Q = cc^{\mathsf{T}} - P = (q_{jk})_{j,k=1}^3.$$

Here the scalar entries q_{jk} do not replace the earlier fixed matter vector $q = Be_1$. Since $A = \sqrt{a}L_3$, the residual in (36), divided by B, is

$$R_b = \begin{pmatrix} a - q_{22} - q_{33} \\ h_3 + q_{12} \\ -h_2 + q_{13} \end{pmatrix}.$$

The residual in (35) is a symmetric matrix R_H whose six independent entries are

$$\begin{aligned}
(R_H)_{11} &= 2ax - 4dq_{23} + 2xq_{22} - 2xq_{33}, \\
(R_H)_{12} &= dh_2 + 3dq_{13} + xh_3 - 3xq_{12}, \\
(R_H)_{13} &= -dh_3 + 3dq_{12} + xh_2 + 3xq_{13}, \\
(R_H)_{22} &= -2ax - 2dh_1 + 2dq_{23} + 4xq_{11} + 2xq_{33}, \\
(R_H)_{23} &= ad - 4dq_{11} - dq_{22} - dq_{33} - 2xh_1, \\
(R_H)_{33} &= 2dh_1 + 2dq_{23} - 4xq_{11} - 2xq_{22}.
\end{aligned}$$

These entries follow by multiplying the explicitly given matrices in

$$R_H = [L(h), H] + \sum_{j,k} q_{jk}[L_j, [L_k, H]] - a[L_3, [L_3, H]], \qquad R_b = L(h)e_1 + \sum_{j,k} q_{jk} L_j L_k e_1 + ae_1.$$

Thus no spatial time-dependence ansatz has entered this calculation.

The equation $R_b = 0$ gives $q_{22} + q_{33} = a$, $h_3 = -q_{12}$, $h_2 = q_{13}$. The two equations $(R_H)_{12} = (R_H)_{13} = 0$ then give $dq_{13} - xq_{12} = 0$ and $dq_{12} + xq_{13} = 0$. Since $x^2 + d^2 > 0$, they force $q_{12} = q_{13} = h_2 = h_3 = 0$. Next $(R_H)_{11} = 0$ gives $xq_{22} = dq_{23}$. After these substitutions, $(R_H)_{23} = (R_H)_{33} = 0$ give $xh_1 = -2dq_{11}$ and $dh_1 = 2xq_{11}$. They imply $2(x^2 + d^2)q_{11} = 0$, and hence $q_{11} = h_1 = 0$. Conversely the following values make every residual entry zero:

$$\dot{c} = 0, \qquad Q = \begin{pmatrix} 0 & 0 & 0 \\ 0 & d\tau/x & \tau \\ 0 & \tau & a - d\tau/x \end{pmatrix}, \qquad \tau = q_{23}(t) \in \mathbb{R}. \tag{39}$$

This is the exact necessary-and-sufficient scalar equation within the real algebra. In particular $Qe_1 = 0$ and $\operatorname{tr} Q = a > 0$.

The covariance $P = \sum_i y_i y_i^{\mathsf{T}}$ is positive semidefinite. Suppose $c_1 \neq 0$ at any time. For $v \in e_1^{\perp}$ define $z = v - (c \cdot v)e_1/c_1$. Then $c \cdot z = 0$ and $Qe_1 = 0$, so

$$v^{\mathsf{T}} Q v = z^{\mathsf{T}} Q z = (c \cdot z)^2 - z^{\mathsf{T}} P z = -\sum_i (y_i \cdot z)^2 \leq 0.$$

The lower 2×2 block of Q would therefore be negative semidefinite, contradicting its positive trace a. Equivalently, $P_{11} = c_1^2$ and the Schur complement of P_{11} in $P = cc^{\mathsf{T}} - Q$ is exactly minus that lower block of Q. Thus $c_1 = 0$ at every time. Since $Q_{11} = c_1^2 - \sum_i y_{i1}^2 = 0$, every $y_{i1} = 0$ as well. Differentiation on the open time interval now gives

$$c, y_i, \dot{y}_i \in e_1^{\perp}, \qquad \sum_i y_i \times \dot{y}_i \in \mathbb{R}e_1.$$

The exact inertia on this real algebra is

$$\mathscr{I}_{H,q} L_1 = 4r^2 L_1, \qquad \mathscr{I}_{H,q} L_2 = \mu L_2, \qquad \mathscr{I}_{H,q} L_3 = \mu L_3.$$

To prove these formulas, $H^2 = r^2 \operatorname{diag}(0, 1, 1)$ and $HL_1H = -r^2 L_1$ give $[H, [H, L_1]] = 4r^2 L_1$, while the qq^* terms vanish. For $j = 2, 3$, multiplication gives $HL_jH = 0$, $H^2 L_j + L_j H^2 = r^2 L_j$, and $qq^* L_j + L_j qq^* = B^2 L_j$. Every resulting real-skew matrix is already traceless. This proves the three identities and gives $\mathscr{I}_{H,q} C = \mu C$.

For $Z, W \in \mathfrak{su}(3)$ set $\langle Z, W \rangle = -\operatorname{tr}(ZW)$ and $\|Z\|^2 = \operatorname{tr}(Z^* Z) = -\operatorname{tr} Z^2$. Cyclicity gives $\langle Z, [Y, W] \rangle = -\langle [Y, Z], W \rangle$, so

$$\langle C, \mathscr{K}_Y C \rangle = \sum_i \|[Y_i, C]\|^2 = 2 \sum_i |y_i \times c|^2.$$

Pair the full Gauss equation (37) with C. Its right side vanishes because $c \perp \sum_i y_i \times \dot{y}_i$. The left side, with the exact factor 2 from the trace inner product, is

$$2\mu |c|^2 + \frac{4}{g^2} \sum_i |y_i \times c|^2 = 0.$$

Since $\mu > 0$ and $g^2 > 0$, this forces $c = 0$. Then $Q = -P$ has $\operatorname{tr} Q = -\sum_i |y_i|^2 \leq 0$, contradicting $\operatorname{tr} Q = a > 0$. Every full solution must obey the two scalar equations and Gauss' equation used here. Their contradiction already excludes a full lift, without imposing or deleting any spatial gauge equation. □

Corollary 1.31 (Constant spatial fields with arbitrary complex color)**.** *There is no lift with all Y_i constant in the constant-matter frame, even when $C(t)$ is arbitrary and the fields take values in the full complex algebra $\mathfrak{su}(3)$.*

Proof. Cyclicity and anti-Hermiticity give

$$\langle Z, \mathscr{I}_{H,q} Z\rangle = \operatorname{tr}[Z,H]^2 + 2|Zq|^2, \qquad \langle Z, \mathscr{K}_Y Z\rangle = \sum_i \|[Y_i, Z]\|^2.$$

For completeness, trace projection pairs trivially with traceless Z; cyclicity gives $-\operatorname{tr}(Z[H,[H,Z]]) = \operatorname{tr}[Z,H]^2$, and $-\operatorname{tr}(Z(qq^*Z + Zqq^*)) = -2q^*Z^2q = 2|Zq|^2$. The second formula follows from the same commutator trace identity used in the theorem. The matrix $[Z,H]$ is Hermitian, so all displayed terms are nonnegative.

When $\dot{Y}_i = 0$, Gauss' equation is $(\mathscr{I}_{H,q} + 2\mathscr{K}_Y/g^2)C = 0$. Its pairing with C therefore forces $Cq = 0$, $[C,H] = 0$, and $[C,Y_i] = 0$ for every i at every time. Since q is constant, $Cq = 0$ implies $\dot{C}q = 0$ and $C^2q = 0$. The vector equation becomes $(-\sum_i Y_i^2 + aI_3)q = 0$. Pairing with q gives $\sum_i |Y_iq|^2 + aB^2 = 0$, which contradicts $a > 0$, $B > 0$. □

Corollary 1.32 (Matter-stabilizer gauge saturation)**.** *There is no lift for which a differentiable real $\theta(t)$ makes*

$$C' = SCS^* - i\dot{\theta}H, \qquad Y_i' = SY_iS^*, \qquad S(t) = \exp(i\theta(t)H)$$

all real skew-symmetric. Nor is there a lift when such a transformation makes all Y_i' constant, without a reality restriction.

Proof. H is Hermitian and traceless, so $S \in SU(3)$. It commutes with H, and $Hq = 0$ gives $Sq = q$. Differentiation gives $\dot{S}S^* = i\dot{\theta}H$. Thus these are exactly the gauge transformations proved in Proposition 1.29, retaining the same constant matter fields. They would transform any claimed lift into one excluded by Theorem 1.30 or Corollary 1.31, respectively. □

These are exact exclusions for the stated classes, not an assertion that every $\mathfrak{su}(3)$ connection is real or belongs to their matter-stabilizer gauge saturation. Arbitrary time-dependent complex $\mathfrak{su}(3)$ lifts outside those classes remain open here.

1.15 Constructive homogeneous gauge solutions and their observations

Theorem 1.33 (A real radial family of full gauge solutions)**.** *Let $D = \operatorname{diag}(-2,1,1)$ and retain*

$$\Phi(t) = x(t)D, \qquad b(t) = \beta(t)e_1, \qquad x(t), \beta(t) \in \mathbb{R}, \qquad \mathcal{A}_\mu = 0.$$

This is a full homogeneous gauge–scalar solution exactly when

$$\ddot{x} = -ax - \kappa x^2 + \frac{\kappa}{3}\beta^2,$$
$$\ddot{\beta} = -(a - 2\kappa x)\beta, \qquad a = m^2 + \Gamma(6x^2 + 2\beta^2).$$

For every real initial position and velocity these equations have a global solution. They give finite-energy homogeneous solutions on a finite-volume flat $\mathbb{T}^3$, and local solutions on $\mathbb{R}^3$, without asserting finite total energy on $\mathbb{R}^3$.

Proof. Here $[\Phi, \dot{\Phi}] = 0$ and $b\dot{b}^* - \dot{b}b^* = 0$ because x, β are real. Thus every J_μ vanishes identically and so do all curvatures. The gauge equations hold exactly, not after discarding Gauss' equation. Directly

$$(\mathrm{adj}(xD) - \beta^2 e_1 e_1^*)_{\mathrm{tf}} = \left(-x^2 + \frac{\beta^2}{3}\right) D, \qquad (aI_3 + \kappa x D)\beta e_1 = (a - 2\kappa x)\beta e_1.$$

The two scalar equations are consequently the displayed real system. Its conserved density is

$$\mathcal{E} = 3\dot{x}^2 + \dot{\beta}^2 + V(xD, \beta e_1), \qquad V = \frac{m^2}{2} s + \frac{\Gamma}{4} s^2 - 2\kappa x(\beta^2 - x^2), \quad s = 6x^2 + 2\beta^2.$$

The stated conservation follows by differentiating and substituting the two real equations; the x kinetic coefficient is 3, not 1/2. Moreover $|x| \le \sqrt{s/6}$ and $\beta^2 + x^2 \le s/2$, so $|2x(\beta^2 - x^2)| \le s^{3/2}/\sqrt{6}$ and

$$V \ge \frac{\Gamma}{4} s^2 - \frac{|\kappa|}{\sqrt{6}} s^{3/2}.$$

This is coercive and globally bounded below. Conservation bounds the positions and velocities. The polynomial first-order vector field is locally Lipschitz and bounded with bounded derivative on their compact range; hence no finite-time escape is possible, proving global existence. Multiplying the finite density by the finite volume of $\mathbb{T}^3$ gives finite total energy there. □

Proposition 1.34 (A full-gauge vacuum and its marked Lorentz observation)**.** *At $\Gamma = m^2 = 1$, $\kappa = 6$, let*

$$x_0 = \frac{3 + \sqrt{3}}{12}, \qquad \Phi = x_0 \,\mathrm{diag}(-2, 1, 1), \qquad b = 3x_0 e_1.$$

Together with $\mathcal{A}_\mu = 0$ this is an equilibrium full gauge solution at a global minimum of the scalar potential. For $Q = \Theta(\Phi, b)$ its spectrum is $(4x_0, -2x_0, -2x_0)$. Its spectral-floor matrix $H = Q + 2x_0 I_3$ has rank one and is positive. The complex observation and hidden coordinate are

$$G = 3x_0 \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}, \qquad \rho = 3x_0, \qquad \det G = 9x_0^2 = |\rho|^2.$$

More generally its $SU(3)$ orbit, parameterized by unit vectors $e = ge_1$, has

$$\Phi = x_0(I_3 - 3ee^*), \quad b = 3x_0 e, \quad G = 3x_0 P(I_3 - ee^*)P^*, \quad \det G = 9x_0^2 |u^* e|^2 = |\rho|^2.$$

For fixed P, this one gauge orbit contains both null and timelike observations. The fixed-P determinant is therefore not a gauge-invariant mass observable.

Proof. For a nonzero real equilibrium with $\beta = 3x > 0$, the radial equations reduce to $24\Gamma x^2 - 2\kappa x + m^2 = 0$. The displayed x_0 is the larger root at $(1, 1, 6)$. Its quaternionic matrix is

$$Q = \begin{pmatrix} -2x_0 & 0 & 0 \\ 0 & x_0 & 3x_0 \mathbf{j} \\ 0 & -3x_0 \mathbf{j} & x_0 \end{pmatrix}.$$

The lower block has eigenvalues $x_0 \pm 3x_0$, proving the spectrum. For an independent global-minimum check, every trace-free quaternionic Hermitian matrix satisfies $54N(Q)^2 \le s(Q)^3$: for eigenvalues summing to zero, the exact identity is $s^3 - 54N^2 = 2\prod_{i<j}(\lambda_i - \lambda_j)^2$. Thus, with $r = \sqrt{s} \ge 0$, $V \ge r^2/2 + r^4/4 - 2r^3/\sqrt{6}$, with equality at eigenvalues $(2y, -y, -y)$, $y \ge 0$, $r = \sqrt{6}y$. On this

equality family $V = 3y^2 + 9y^4 - 12y^3$ and $dV/dy = 6y(1 - 6y + 6y^2)$. The larger positive root is $y = (3 + \sqrt{3})/6 = 2x_0$; the derivative is positive, negative, then positive on the three successive intervals, and this minimum has negative value, since at $y = 2/3$ the potential is $-4/9$. It is therefore the global minimum, attained by the displayed Q.

The complex part of H is $3x_0 \operatorname{diag}(0, 1, 1)$, and compression by the retained P gives the displayed G. The $SU(3)$ action on pairs is $(\Phi, b) \mapsto (g\Phi g^*, gb)$; it is exactly quaternionic congruence under Θ, since $B(gb) = gB(b)g^{\mathsf{T}}$ for $g \in SU(3)$. Consequently the spectrum and spectral-floor shift are unchanged. Writing $v = Pe$ and using the determinant lemma in dimension two gives

$$\det(C - vv^*) = \det C\,(1 - v^*C^{-1}v) = 3\left(1 - e^*\left(I_3 - \frac{uu^*}{3}\right)e\right) = |u^*e|^2.$$

This proves the general observation formula. A unit $e \in u^\perp$ gives determinant zero, while $e = e_1$ gives a positive determinant. These vectors are related by $SU(3)$: extend either to a unitary basis and adjust the phase of one of the remaining columns to make determinant one. Thus the examples are genuinely on one gauge orbit.

The exact covariance map retains the observer as part of the data: under $H' = gHg^*$, $P' = Pg^*$ one has $P'H'P'^* = PHP^*$. Complex-component projection therefore has the same value. Simultaneously $u' = gu$, $b' = gb$ gives $u'^*b' = u^*b$. This is covariance of the marked observer, not invariance of a fixed observer under an active transformation of the state. □

Proposition 1.35 (Gauge averaging and distinct minimum orbits)**.** *Let $d\mu$ be Haar measure on $SU(3)$ with its retained positive total mass V_μ, not set to one. For every complex Hermitian A,*

$$\det(PAP^*) = u^* \operatorname{adj}(A)u, \qquad \int_{SU(3)} \det(PgAg^*P^*)\,d\mu(g) = V_\mu \operatorname{tr}\operatorname{adj}(A).$$

For a quaternionic positive rank-one matrix $H = \Theta(A, b)$, $\operatorname{adj}(A) = bb^*$. Hence its averaged observation is $V_\mu\|b\|^2$.

Put $\chi = 2x_0 > 0$ for the preceding minimum and, for $0 \le \theta \le \pi/4$, define

$$v_\theta = (\cos\theta,\ \sin\theta\,\mathbf{j},\ 0)^{\mathsf{T}}, \qquad Q_\theta = \chi(3v_\theta v_\theta^* - I_3).$$

Every Q_θ is a scalar global minimum, and its complex-plus-triplet coordinate satisfies

$$b_\theta = (0, 0, -3\chi\cos\theta\sin\theta)^{\mathsf{T}}, \qquad \|b_\theta\|^2 = \frac{9\chi^2}{4}\sin^2(2\theta).$$

Consequently this minimum family contains pairwise $SU(3)$-inequivalent states with invariant $\|b\|^2$ ranging from 0 to $9\chi^2/4$. The coefficient agrees with the supplied packet's invariant formula; this calculation validates that coefficient rather than claiming it as a newly discovered one.

Proof. The signed 2-by-2 minors of P form precisely $u = (1, 1, 1)^{\mathsf{T}}$. The exterior-square form of the Cauchy–Binet expansion therefore gives $\det(PAP^*) = u^* \operatorname{adj}(A)u$; equivalently this follows by expanding the six entries of PAP^* and collecting the cofactors. The identity $\operatorname{adj}(gAg^*) = g \operatorname{adj}(A)g^*$ holds for invertible A and then for every A by polynomial continuity. The Haar average of these conjugates commutes with every element of $SU(3)$. Commutation with diagonal determinant-one phase matrices kills its off-diagonal entries, and commutation with determinant-one cyclic permutations makes its diagonal entries equal. Its trace fixes the average at $V_\mu \operatorname{tr}(\operatorname{adj} A)I_3/3$. Since $u^*u = 3$, the stated integral follows with all factors retained.

Write $r = b^*b$. The skew convention gives directly $B(b)\overline{B(b)} = bb^* - rI_3$. The complex part of H^2 is therefore $A^2 - bb^* + rI_3$, and $e_2(H) = e_2(A) - r$. For positive rank-one H, its spectral

identity is $H^2 = (\operatorname{tr} H)H$ and $e_2(H) = 0$. Taking complex parts and using $\operatorname{tr} H = \operatorname{tr} A$ gives $A^2 - (\operatorname{tr} A)A + e_2(A)I_3 = bb^*$. The left side is $\operatorname{adj}(A)$, proving the assertion.

Finally $v_\theta^* v_\theta = 1$, so Q_θ has fixed spectrum $(2\chi, -\chi, -\chi)$ and hence the same minimum value. Its $(1,2)$ quaternionic entry is $-3\chi \cos\theta \sin\theta\, \mathbf{j}$, with the other quaternionic off-diagonal coefficients zero. The retained definition of $B(b)$ therefore gives the displayed b_θ with its minus sign. Taking its norm proves the formula. It is strictly increasing on the given interval; since $b \mapsto gb$ preserves its norm, distinct parameters there cannot lie on one $SU(3)$ orbit. The positive spectral-floor matrix is $3\chi v_\theta v_\theta^*$, so the averaging identity applies to the same triplet coordinate. □

The classical $SU(3)$ cohomogeneity-one geometry of $\mathbb{HP}^2$ is discussed by A. Gambioli, *Eight-dimensional $SU(3)$-manifolds of cohomogeneity one*, arXiv:math/0611796, Section 4, the $\mathbb{HP}^2 = M(\mathbb{CP}^2, S^5)$ example, `https://arxiv.org/abs/math/0611796`. The explicit family and its separating invariant above are proved directly; no new classification of this classical orbit space is claimed. The averaging formula supplies the exact relation of that invariant to the marked Lorentz determinant. A continuum of classical potential minima is not, by itself, a calculation of a quantum Hamiltonian spectrum.

1.16 Vacuum kinetics and exact current-free continuations

Retain the real scalar state space

$$\mathcal{E} = \operatorname{Herm}_3(\mathbb{C})_0 \oplus \mathbb{C}^3, \qquad \langle(\phi, c), (\psi, d)\rangle = \operatorname{tr}(\phi\psi) + 2\operatorname{Re}(c^* d).$$

The subscript 0 means trace zero, not a quotient. The scalar kinetic density is exactly one half of this metric:

$$\mathcal{L}_{\mathrm{sc}} = \tfrac{1}{2}\operatorname{tr}(D_\mu \Phi D^\mu \Phi) + (D_\mu b)^* D^\mu b - V, \qquad V = \tfrac{m^2}{2} s + \tfrac{\Gamma}{4} s^2 - \kappa(\det \Phi - b^* \Phi b), \quad s = \operatorname{tr} \Phi^2 + 2b^* b.$$

The spacetime metric is $(+---)$, $m^2 \geq 0$, $\Gamma > 0$; the nonzero vacuum results below use $\kappa > 0$. The gauge action is $(\Phi, b) \mapsto (g\Phi g^*, gb)$, $g \in SU(3)$. Its anti-Hermitian connection is $\mathcal{A}_\mu$, and the retained gauge density is $\operatorname{tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu})/(2g^2)$, with $g^2 > 0$. All constructed connections below are identically zero. We make no claim about connections not considered here.

For $Z_{\mathrm{tf}} = Z - \operatorname{tr}(Z) I_3/3$, variation gives, at $\mathcal{A}_\mu = 0$,

$$\begin{aligned} \Box\Phi &= -a\Phi + \kappa(\operatorname{adj}\Phi - bb^*)_{\mathrm{tf}}, & a &= m^2 + \Gamma s, \\ \Box b &= -(aI_3 + \kappa\Phi)b, & J_\mu &= [\Phi, \partial_\mu \Phi] + b(\partial_\mu b)^* - (\partial_\mu b) b^*. \end{aligned} \tag{40}$$

Indeed

$$dV(\delta\Phi, \delta b) = \operatorname{tr}([a\Phi - \kappa(\operatorname{adj}\Phi - bb^*)_{\mathrm{tf}}]\delta\Phi) + 2\operatorname{Re}[((aI_3 + \kappa\Phi)b)^* \delta b]\,.$$

This proves both scalar equations with their original metric factors. Variation in $\mathcal{A}_\mu$ pairs the matter variation with $\operatorname{tr}(J^\mu \delta\mathcal{A}_\mu)$. Consequently $J_\mu = 0$ and $\mathcal{F}_{\mu\nu} = 0$ satisfy every gauge equation, including Gauss, before any restriction of variations.

We retain the quaternionic convention

$$\Theta(\Phi, b) = \Phi + B(b)\mathbf{j}, \qquad B(b) = \begin{pmatrix} 0 & \bar{b}_3 & -\bar{b}_2 \\ -\bar{b}_3 & 0 & \bar{b}_1 \\ \bar{b}_2 & -\bar{b}_1 & 0 \end{pmatrix}, \qquad \mathbf{j}z = \bar{z}\mathbf{j}.$$

Its real trace metric is exactly the displayed metric on $\mathcal{E}$: the diagonal sum of $B(b)\overline{B(c)}$ has real part $-2\operatorname{Re}(b^*c)$. Its cubic is $N(\Theta(\Phi,b)) = \det\Phi - b^*\Phi b$. To verify the latter identity, write $\Phi_{23} = z_1, \Phi_{31} = z_2, \Phi_{12} = z_3$ and use the Hermitian Moore formula

$$N\begin{pmatrix} a & q_3 & \bar{q}_2 \\ \bar{q}_3 & d & q_1 \\ q_2 & \bar{q}_1 & f \end{pmatrix} = adf - a|q_1|^2 - d|q_2|^2 - f|q_3|^2 + 2\operatorname{Re}(q_1q_2q_3).$$

Here $q_i = z_i + \bar{b}_i\mathbf{j}$. The complex part of the triple product is $z_1z_2z_3 - \bar{b}_1b_2z_3 - z_1\bar{b}_2b_3 - \bar{b}_1\bar{z}_2b_3$. Its extra real terms and the three extra diagonal terms are precisely $-b^*\Phi b$.

Lemma 1.36 (Retained nonzero minimum and its curve)**.** *Suppose $0 \le m^2 \le \kappa^2/(27\Gamma)$, $\kappa > 0$, and put*

$$\chi = \frac{\kappa + \sqrt{\kappa^2 - 24\Gamma m^2}}{12\Gamma} > 0.$$

For a unit $v \in \mathbb{H}^3$, $Q = \chi(3vv^* - I_3)$ is a global minimum of V on the trace-free quaternionic Hermitian state space. The origin is an additional minimum only at $m^2 = \kappa^2/(27\Gamma)$. For $0 \le \theta \le \pi/4$, let

$$v_\theta = (\cos\theta, \sin\theta\,\mathbf{j}, 0)^{\mathsf{T}}, \qquad Q_\theta = \chi(3v_\theta v_\theta^* - I_3).$$

Its pair coordinates and potential value are

$$\begin{aligned} \Phi_\theta &= \chi\,\mathrm{diag}(3\cos^2\theta - 1,\ 3\sin^2\theta - 1,\ -1), \\ b_\theta &= -3\chi\cos\theta\sin\theta\, e_3, \\ V_0 &= 3m^2\chi^2 + 9\Gamma\chi^4 - 2\kappa\chi^3 = \kappa\chi^3 - 9\Gamma\chi^4. \end{aligned} \tag{41}$$

Proof. The quaternionic spectral theorem gives real eigenvalues $\lambda_1 + \lambda_2 + \lambda_3 = 0$ and $s = \sum\lambda_i^2$. On each sphere $s > 0$, extrema of the product have at most two distinct eigenvalues: subtracting two Lagrange equations for the product gives $(\lambda_i - \lambda_j)(-\lambda_k - 2\beta) = 0$. If all three were distinct they would all equal -2β, a contradiction. The two-eigenvalue trace-free possibilities are $(2t, -t, -t)$ and their negatives and permutations, where $t = \sqrt{s/6} > 0$. Thus $N \le 2t^3$, sharply, and

$$V \ge F(t) = 3m^2t^2 + 9\Gamma t^4 - 2\kappa t^3, \qquad F'(t) = 6t(m^2 - \kappa t + 6\Gamma t^2).$$

For $m^2 > 0$ the larger positive root is χ, the only positive local minimum; the smaller root is a local maximum. For $m^2 = 0$, $F'(t) < 0$ on $0 < t < \chi$ and > 0 on $t > \chi$. Using $m^2 = \kappa\chi - 6\Gamma\chi^2$ yields $F(\chi) = \chi^3(\kappa - 9\Gamma\chi)$, negative below the displayed threshold and zero at it. Comparing with $F(0) = 0$ proves the global claim and all equality cases used here. The spectrum of Q_θ is $(2\chi, -\chi, -\chi)$. Its $(1,2)$ entry is $-3\chi\cos\theta\sin\theta\,\mathbf{j}$, giving the stated sign of b_θ. Direct substitution gives the potential value. □

Proposition 1.37 (Horizontal metric and normal acceleration)**.** *Write $q_\theta = (\Phi_\theta, b_\theta)$, $T = \partial_\theta q_\theta$, $N = \partial_\theta^2 q_\theta$. Then*

$$\langle T, T\rangle = 18\chi^2, \qquad \langle T, N\rangle = 0, \qquad \langle N, N\rangle = 72\chi^2.$$

The curve is $SU(3)$-horizontal and its induced kinetic density is $9\chi^2\dot\theta^2$. Every homogeneous unconstrained solution confined to this fixed-χ curve has constant θ. The vector N is normal to the entire fixed-χ minimum manifold. Its radial projection is $-3q_\theta$, and $\|N + 3q_\theta\|^2 = 18\chi^2$; hence there is a nonzero normal component in addition to the overall radial one.

Proof. With $D = \operatorname{diag}(-1, 1, 0)$, differentiation gives

$$T_\Phi = 3\chi \sin(2\theta) D, \quad T_b = -3\chi \cos(2\theta) e_3, \quad N_\Phi = 6\chi \cos(2\theta) D, \quad N_b = 6\chi \sin(2\theta) e_3.$$

Since $\operatorname{tr} D^2 = 2$, the three metric identities follow without changing the coefficient of the triplet metric. All commutators of the diagonal matrices vanish, and $b_\theta T_b^* - T_b b_\theta^* = 0$ because both coefficients are real. For every $\xi \in \mathfrak{su}(3)$,

$$\langle T, ([\xi, \Phi_\theta], \xi b_\theta) \rangle = \operatorname{tr}(([\Phi_\theta, T_\Phi] + b_\theta T_b^* - T_b b_\theta^*)\xi) = 0.$$

Thus this nonzero tangent is horizontal and cannot itself be a nonzero vertical tangent in this positive metric. At a vacuum the potential gradient vanishes, so the homogeneous equation is $T\ddot{\theta} + N\dot{\theta}^2 = 0$. Pairing with T first gives $\ddot{\theta} = 0$, and the nonzero norm of N then gives $\dot{\theta} = 0$. The exact vector that would have to be supplied as an extra force for a constrained constant-speed motion is $N\dot{\theta}^2$. Its force covector is

$$\operatorname{tr}(N_\Phi \dot{\theta}^2\, \delta\Phi) + 2\operatorname{Re}((N_b \dot{\theta}^2)^* \delta b).$$

For the assertion about the whole minimum manifold, $v_\theta^* v_\theta' = 0$, $v_\theta'^* v_\theta' = 1$, and $v_\theta'' = -v_\theta$. Therefore

$$Q_\theta'' = 6\chi(v_\theta' v_\theta'^* - v_\theta v_\theta^*).$$

Every tangent to the rank-one projector manifold has the form $3\chi(z v_\theta^* + v_\theta z^*)$, $v_\theta^* z = 0$. The real trace pairing with both projectors in Q_θ'' is zero, by $v_\theta^* v_\theta' = 0$ and cyclicity of real trace. This proves normality. Finally $\|q_\theta\|^2 = 6\chi^2$ and $\langle N, q_\theta \rangle = -18\chi^2$; subtracting the radial projection gives the asserted remaining norm. Thus the constant-speed curve is an intrinsic geodesic of the minimum manifold but not an unconstrained homogeneous solution. □

Proposition 1.38 (Exact three-real-coordinate full-gauge sector)**.** *Let h, k, β be real fields and set*

$$\Phi = \operatorname{diag}(h + k, h - k, -2h), \qquad b = \beta e_3, \qquad \mathcal{A}_\mu = 0.$$

This ansatz satisfies all full field equations if and only if

$$\begin{aligned} \Box h &= -ah - \kappa h^2 + \tfrac{\kappa}{3}(k^2 + \beta^2), \\ \Box k &= -(a - 2\kappa h)k, \qquad \Box\beta = -(a - 2\kappa h)\beta, \\ a &= m^2 + \Gamma(6h^2 + 2k^2 + 2\beta^2). \end{aligned} \tag{42}$$

In particular it is an unconstrained invariant sector for homogeneous initial data with tangent velocities. Its homogeneous density is

$$\mathcal{L}_{\mathrm{hom}} = 3\dot{h}^2 + \dot{k}^2 + \dot{\beta}^2 - V(h, k, \beta), \quad V = \tfrac{m^2}{2} s + \tfrac{\Gamma}{4} s^2 - 2\kappa h(k^2 + \beta^2 - h^2), \quad s = 6h^2 + 2k^2 + 2\beta^2.$$

Every homogeneous initial condition in this sector has a global solution for $m^2 \geq 0, \Gamma > 0, \kappa \in \mathbb{R}$.

Proof. Direct multiplication gives

$$\det \Phi - b^* \Phi b = 2h(k^2 + \beta^2 - h^2), \qquad \operatorname{adj} \Phi - bb^* = \operatorname{diag}(-2h(h - k), -2h(h + k), h^2 - k^2 - \beta^2).$$

Its trace is $-3h^2 - k^2 - \beta^2$. Removing one third of this trace and substituting in (40) gives exactly (42), including the factor $1/3$ in the h equation. The last component of the triplet force is $-(a - 2\kappa h)\beta$; all other entries remain zero. The currents vanish for each spacetime index: the

matrices and their derivatives are diagonal, while b and its derivative are real multiples of e_3. The zero connection therefore satisfies all gauge equations. This also proves consistency under unrestricted variations, not merely stationary points of a restricted Lagrangian. As a check, that Lagrangian has Euler–Lagrange weights $6, 2, 2$ on h, k, β, respectively.

For homogeneous fields the smooth polynomial vector field is tangent to the displayed real linear sector, so local uniqueness proves invariance. Its conserved energy density is

$$E = 3\dot{h}^2 + \dot{k}^2 + \dot{\beta}^2 + V.$$

If $r^2 = k^2 + \beta^2$, then

$$|2h(r^2 - h^2)| \le 2|h|(h^2 + r^2) \le \frac{s^{3/2}}{\sqrt{6}}, \qquad V \ge \frac{\Gamma}{4}s^2 - \frac{|\kappa|}{\sqrt{6}}s^{3/2}.$$

Here $|h| \le \sqrt{s/6}$ and $h^2 + r^2 \le s/2$. This lower bound is coercive, so a fixed energy bounds all positions and velocities. The polynomial first-order system remains in a compact set on every finite interval and extends globally. This is an ODE existence proof, not a general nonlinear PDE global-existence claim. □

On $r > 0$, introduce $k = r\cos(2\theta)$, $\beta = -r\sin(2\theta)$. The original kinetic density and homogeneous equations become

$$\begin{aligned} \mathcal{L}_{\text{kin}} &= 3\dot{h}^2 + \dot{r}^2 + 4r^2\dot{\theta}^2, \\ \ddot{h} &= -ah - \kappa h^2 + \kappa r^2/3, \qquad \ddot{r} = 4r\dot{\theta}^2 - (a - 2\kappa h)r, \\ (r^2\dot{\theta})^{\cdot} &= 0, \qquad a = m^2 + \Gamma(6h^2 + 2r^2). \end{aligned} \tag{43}$$

These follow by differentiating k, β, resolving the two equations into radial and angular components, and using $k\dot{\beta} - \beta\dot{k} = -2r^2\dot{\theta}$. The Cartesian sector includes $r = 0$; the polar chart does not. Even permitting a varying positive amplitude $Q = \chi(t)(3v_\theta v_\theta^* - I_3)$ gives $r = 3h > 0$. Then the h equation implies $3\ddot{h} = -(a - 2\kappa h)r$, whereas the r equation has the extra term $4r\dot{\theta}^2$. Since $\ddot{r} = 3\ddot{h}$, one obtains $4r\dot{\theta}^2 = 0$. Thus changing this single amplitude does not by itself make angular homogeneous motion consistent.

Theorem 1.39 (Exact gauge-invariant periodic continuation)**.** *Suppose* $0 < m^2 < \kappa^2/(27\Gamma)$, $\Gamma > 0, \kappa > 0$. *Define*

$$d_0 = \sqrt{\kappa^2 - 24\Gamma m^2}, \quad h_0 = \frac{\kappa + d_0}{24\Gamma} = \frac{\chi}{2}, \quad h_0 < h < \frac{\kappa}{6\Gamma}, \quad \Delta(h) = 1 - \frac{6\Gamma h}{\kappa}.$$

Put

$$a(h) = \frac{m^2 + 12\Gamma h^2}{\Delta(h)}, \qquad R^2(h) = \frac{3a(h)h}{\kappa} + 3h^2, \qquad \omega^2(h) = a(h) - 2\kappa h = \frac{m^2 - 2\kappa h + 24\Gamma h^2}{\Delta(h)}.$$

Take the positive roots R, ω. *Then the sector fields*

$$h(t) = h, \quad k(t) = R\cos(\omega t), \quad \beta(t) = R\sin(\omega t), \quad \mathcal{A}_\mu = 0$$

are nonconstant, spatially homogeneous, exact full gauge–scalar solutions. The gauge-invariant observable $\|b(t)\|^2 = R^2\sin^2(\omega t)$ *is nonconstant. Their conserved energy density is exactly*

$$E(h) = R^2\omega^2 + m^2(3h^2 + R^2) + \Gamma(3h^2 + R^2)^2 - 2\kappa h(R^2 - h^2). \tag{44}$$

Let

$$\Delta_0 = 1 - \frac{6\Gamma h_0}{\kappa} = \frac{3\kappa - d_0}{4\kappa} > 0, \qquad V_0 = 12m^2h_0^2 + 144\Gamma h_0^4 - 16\kappa h_0^3.$$

As $h \downarrow h_0$,

$$\begin{aligned} R &\longrightarrow 3h_0, \qquad \omega \longrightarrow 0, \\ \omega^2 &= \frac{2d_0}{\Delta_0}(h - h_0) + O((h - h_0)^2), \\ E(h) - V_0 &= \frac{18h_0^2 d_0}{\Delta_0}(h - h_0) + O((h - h_0)^2) = 9h_0^2\omega^2 + O(\omega^4). \end{aligned} \tag{45}$$

These are finite-volume classical statements. On a flat torus with retained volume $\mathcal{V}_3 > 0$, the total energy and its excess are $\mathcal{V}_3 E(h)$ and $\mathcal{V}_3(E(h) - V_0)$. The amplitude tends to the nonzero value $3h_0$: the family approaches the minimum manifold while traversing a finite range of its invariant $\|b\|^2$, rather than becoming a small oscillation about one fixed vacuum. Its period tends to infinity.

Proof. The denominator is positive on the stated interval; $a > 0$ and $R^2 > 0$. The numerator of ω^2 has roots $(\kappa - d_0)/(24\Gamma)$ and h_0; with its positive leading coefficient it is positive for $h > h_0$. Moreover $d_0 < \kappa$, so $h_0 < \kappa/(12\Gamma) < \kappa/(6\Gamma)$; the interval is nonempty. The identities

$$a = m^2 + \Gamma(6h^2 + 2R^2), \quad -ah - \kappa h^2 + \kappa R^2/3 = 0, \quad \omega^2 = a - 2\kappa h$$

follow by substitution. They are precisely the constant-h and the two oscillatory equations in (42). The previous proposition proves all gauge equations. The triplet norm is preserved by the full $SU(3)$ action, so its nonconstant value rules out a purely gauge motion.

Substitution in the conserved energy gives (44). At $h = h_0$, $m^2 = 2\kappa h_0 - 24\Gamma h_0^2$, so $a = 2\kappa h_0$, $R^2 = 9h_0^2$ and $\omega^2 = 0$. The limiting field at any phase is a global minimum: the 2-by-2 quaternionic block has eigenvalues $h_0 \pm 3h_0$, and the last eigenvalue is $-2h_0$; hence the full spectrum is $(4h_0, -2h_0, -2h_0)$. Writing the frequency numerator as $n(h)$, $n(h_0) = 0$ and $n'(h_0) = 2d_0$. Taylor expansion proves the frequency estimate. The curve $h \mapsto (h, R(h), 0)$ is smooth near h_0, and its endpoint is a critical point of V. Consequently its potential differs from V_0 by $O((h - h_0)^2)$. Its kinetic density is

$$R^2\omega^2 = \left(9h_0^2 + O(h - h_0)\right)\left(\frac{2d_0}{\Delta_0}(h - h_0) + O((h - h_0)^2)\right).$$

This proves both energy expansions. Since the leading coefficient of ω^2 is strictly positive, replacing $h - h_0$ by its smooth inverse as a function of ω^2 gives the last equality. For every $h > h_0$, $E(h) > V_0$ because the kinetic term is positive and the potential is bounded below by V_0. The torus assertion is integration of a constant spatial density. Nonzero homogeneous energy density on all of $\mathbb{R}^3$ is not asserted to have finite total energy. □

Proposition 1.40 (Linearized physical tangent and exact null waves)**.** *At each fixed vacuum q_{θ_0}, the perturbation*

$$\delta\Phi = T_\Phi(\theta_0)\eta, \qquad \delta b = T_b(\theta_0)\eta, \qquad \delta\mathcal{A}_\mu = 0$$

solves the full linearized equations precisely when $\Box\eta = 0$. Its quadratic density is

$$9\chi^2((\partial_t\eta)^2 - |\nabla\eta|^2).$$

For a homogeneous mode on a torus of volume $\mathcal{V}_3$, its momentum and quadratic Hamiltonian are

$$\pi_\eta = 18\chi^2\mathcal{V}_3\dot{\eta}, \qquad H_\eta = \frac{\pi_\eta^2}{36\chi^2\mathcal{V}_3}.$$

There is also an exact nonlinear inhomogeneous continuation at fixed χ: for any smooth real f,

$$\theta(t,x) = f(t-x_1), \qquad q(t,x) = q_{\theta(t,x)}, \qquad \mathcal{A}_\mu = 0$$

solves the full nonlinear field equations wherever these fields are defined. On the rectangular flat torus with sides L_1, L_2, L_3, choose f L_1-periodic. Its excess over the vacuum energy is

$$E_{\text{wave}} - \mathcal{V}_3 V_0 = 18\chi^2 L_2 L_3 \int_0^{L_1} (f'(u))^2\, du, \qquad \mathcal{V}_3 = L_1L_2L_3.$$

Proof. Differentiate $\nabla V(q_\theta) = 0$ in θ. The metric Hessian annihilates T. The linearized currents are

$$([\Phi_{\theta_0}, T_\Phi] + b_{\theta_0}T_b^* - T_b b_{\theta_0}^*)\partial_\mu\eta = 0.$$

Thus the linearized scalar equation is $T\Box\eta = 0$, and $T \neq 0$. The gauge equations have zero current and zero field strength. The kinetic norm $18\chi^2$ proves the quadratic density and its Legendre transform with the stated volume.

For the nonlinear assertion, the chain rule in signature $(+---)$ gives

$$\Box q_\theta = T(\theta)\Box\theta + N(\theta)\partial_\mu\theta\,\partial^\mu\theta.$$

Both coefficients vanish for $\theta = f(t-x_1)$. Every value remains a critical point of V; the trigonometric formulas define it for all real θ, or one can restrict the range of f to the displayed minimum section. The exact currents vanish for all spacetime derivatives as in the sector proposition. This verifies every nonlinear equation. The scalar energy density is the sum of one temporal and one spatial term, each $9\chi^2(f')^2$, plus V_0. Periodicity makes its integral independent of time and gives the formula. For $f(u) = \theta_0 + \varepsilon\cos(2\pi u/L_1)$ with range inside the section, the excess is $9\chi^2\mathcal{V}_3\varepsilon^2(2\pi/L_1)^2$. □

Gauge-invariant coordinate and endpoint scope. The invariant

$$I = \|b_\theta\|^2 = \frac{9\chi^2}{4}\sin^2(2\theta), \qquad \frac{dI}{d\theta} = \frac{9\chi^2}{2}\sin(4\theta)$$

has nonzero derivative on $0 < \theta < \pi/4$. Therefore the displayed linearized mode is visible through the genuine gauge-invariant linear observable $\delta I = (9\chi^2/2)\sin(4\theta_0)\eta$ in the interior. With $M = 9\chi^2/4$, its interior coordinate expression is

$$9\chi^2(\partial\theta)^2 = \frac{9\chi^2}{16I(M-I)}(\partial I)^2.$$

The denominator vanishes at the endpoints because I ceases to be a regular linear coordinate there, not because the norm of T diverges.

For completeness the stabilizers on this section can be checked without importing an orbit classification. In the interior the three diagonal entries of Φ_θ are distinct, so a stabilizer must be diagonal. Fixing the nonzero b_3 and its determinant then gives $\mathrm{diag}(e^{it}, e^{-it}, 1)$, of dimension one. At $\theta = 0$, $b = 0$ and the eigenvalues are $2\chi, -\chi, -\chi$, giving $S(U(1)\times U(2))$, of dimension four. At $\theta = \pi/4$, fixing $b_3 \neq 0$ and the two-dimensional first eigenspace gives $\mathrm{diag}(U, 1)$, $U \in SU(2)$, of

dimension three. The gauge-orbit dimensions are consequently $7, 4, 5$, respectively. The fixed-χ minimum manifold has dimension eight: the unit sphere in $\mathbb{H}^3$ has dimension eleven and the free unit-quaternion action $v \mapsto va$ has dimension three, with vv^* precisely its quotient. Thus the horizontal tangent dimensions within that manifold are $1, 4, 3$, respectively. In particular the interior has exactly one horizontal vacuum direction. At the endpoints the invariant linear function I vanishes to first order; no assertion of a smooth one-dimensional linear quotient there is made. This agrees with the classical cohomogeneity-one description in A. Gambioli, *Eight-dimensional $SU(3)$-manifolds of cohomogeneity one*, arXiv:math/0611796, Section 4, Examples: $\mathbb{HP}^2$ has singular orbit types $\mathbb{CP}^2$ and S^5. The kinetic and current statements above are independently proved here, not imported from that classification.

Fixed observation and classical scope. Retain

$$P = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \qquad u = (1, 1, 1)^{\mathsf{T}}.$$

The positive spectral-floor matrix $H_\theta = Q_\theta + \chi I_3$ has complex part $A_\theta = 3\chi \operatorname{diag}(\cos^2\theta, \sin^2\theta, 0)$. Consequently

$$G_\theta = PA_\theta P^* = 3\chi \begin{pmatrix} 1 & -\sin^2\theta \\ -\sin^2\theta & \sin^2\theta \end{pmatrix}, \qquad \rho = u^* b_\theta = -3\chi \cos\theta \sin\theta, \qquad \det G_\theta = |\rho|^2 = I.$$

These are direct matrix products with the original P, u, χ. They do not assert gauge invariance of a fixed-P observation away from this chosen section: under a general gauge transformation its complex part is $gA_\theta g^*$, so the observation is $PgA_\theta g^* P^*$, not generally $PA_\theta P^*$.

On a rectangular torus the linearized wave equation has classical Fourier frequencies $\omega_{\mathbf{n}}^2 = 4\pi^2 \sum_i n_i^2/L_i^2$; the spatial zero mode has zero linearized frequency. The exact homogeneous periodic family additionally has arbitrarily small positive classical frequency and energy excess at fixed volume. The null waves give exact nonlinear inhomogeneous solutions with arbitrarily small amplitude. None of these statements quantizes the model, identifies a Hilbert-space spectrum, proves a quantum mass gap or its absence, or assigns physical units that the retained parameters have not been supplied. No finite-total-energy assertion on all of $\mathbb{R}^3$ is inferred from a finite-volume solution.

1.17 The gauge-vector spectrum and an invariant vector observable

We now compute masses in the linearized classical field equations of (29), not by assigning a physical meaning to a determinant in isolation. Retain a positive nonzero minimum amplitude χ from Theorem 1.24; thus $\kappa > 0$, $0 \le m^2 \le \kappa^2/(27\Gamma)$ and

$$\chi = \frac{\kappa + \sqrt{\kappa^2 - 24\Gamma m^2}}{12\Gamma}.$$

The nonzero branch is meant at coexistence. Keep $0 \le \theta \le \pi/4$, and introduce the notation $q = \cos(2\theta)$, not a change in the field or its kinetic metric. The same minimum representatives are

$$\Phi_\theta = \chi \operatorname{diag}\left(\frac{1+3q}{2}, \frac{1-3q}{2}, -1\right), \qquad b_\theta = -\frac{3\chi}{2}\sin(2\theta) e_3, \qquad \beta^2 := b_\theta^* b_\theta = \frac{9\chi^2}{4}(1 - q^2).$$

Every constant representative together with $\mathcal{A}_\mu = 0$ solves the full classical field equations.

Theorem 1.41 (All eight gauge-inertia eigenvalues)**.** *Use the positive inner product* $\langle Z, W\rangle = -\operatorname{tr}(ZW)$ *on* $\mathfrak{su}(3)$*. At the displayed minimum, the exact spectrum of* $\mathscr{I}_{\Phi_\theta, b_\theta}$ *is*

generator space	*inertia eigenvalue*	*real multiplicity*
$i\,\mathrm{diag}(1,-1,0)$	0	1
$i\,\mathrm{diag}(1,1,-2)$	$3\chi^2(1-q^2)$	1
$\langle E_{12}-E_{21},\, i(E_{12}+E_{21})\rangle$	$9\chi^2 q^2$	2
$\langle E_{13}-E_{31},\, i(E_{13}+E_{31})\rangle$	$\frac{9}{2}\chi^2(1+q)$	2
$\langle E_{23}-E_{32},\, i(E_{23}+E_{32})\rangle$	$\frac{9}{2}\chi^2(1-q)$	2

(GS.1)

Here E_{ij} *are the actual matrix units. The unscaled generators in the row order with the six off-diagonal generators first have kinetic Gram matrix* $\mathrm{diag}(2,2,2,2,2,2,2,6)$*. The squared masses of transverse solutions of the full linearized field equations are exactly* $g^2/2$ *times the eigenvalues in (GS.1). In particular the diagonal nonzero branch has*

$$\mu_{\mathrm{d}}^2 = \frac{3}{2}g^2\chi^2(1-q^2) = \frac{2g^2}{3}\|b_\theta\|^2. \tag{GS.2}$$

There is one zero eigenvalue for $0 < q < 1$*, four at* $q = 1$*, and three at* $q = 0$*. These are the Lie-algebra dimensions of the stabilizers* $U(1)_{1,-1,0}$*,* $S(U(1)\times U(2))$*, and* $SU(2)$*, respectively.*

Proof. For a diagonal Hermitian $\Phi = \mathrm{diag}(d_1, d_2, d_3)$, the operator $[\Phi,[\Phi, Z]]$ multiplies the ij off-diagonal entry by $(d_i - d_j)^2$. Here $bb^* = \mathrm{diag}(0, 0, \beta^2)$. Consequently $\{bb^*, Z\}$ contributes zero on the 12 root space and β^2 on each of the 13 and 23 root spaces; these entries have zero trace, so no trace-free correction changes them. Retaining the displayed differences gives

$$(d_1-d_2)^2 = 9\chi^2q^2, \quad (d_1-d_3)^2+\beta^2 = \frac{9}{2}\chi^2(1+q), \quad (d_2-d_3)^2+\beta^2 = \frac{9}{2}\chi^2(1-q).$$

For $Z = i\,\mathrm{diag}(z_1, z_2, z_3)$, $\sum z_i = 0$, the diagonal contribution is

$$\mathscr{I}Z = 2i\beta^2 z_3\left(E_{33} - \frac{1}{3}I_3\right).$$

It annihilates $i\,\mathrm{diag}(1,-1,0)$ and multiplies $i\,\mathrm{diag}(1,1,-2)$ by $4\beta^2/3 = 3\chi^2(1-q^2)$. The eight displayed generators are linearly independent and span $\mathfrak{su}(3)$; multiplication and trace give their stated diagonal Gram matrix. This proves the entire generalized Hessian spectrum without replacing that metric by an identity matrix.

For the dynamical assertion, perturb only the connection at first order: $\mathcal{A}_\nu = \varepsilon Z a_\nu$, with both scalar perturbations zero. At the constant scalar background the first-order scalar equations reduce to $[Z, \Phi_\theta]\partial_\nu a^\nu = 0$ and $Z b_\theta\, \partial_\nu a^\nu = 0$. The first-order current is $(J^\nu)_{\mathrm{tf}} = -\varepsilon\mathscr{I}(Z)a^\nu$. For an inertia eigenvector with eigenvalue λ, the full gauge equation is therefore

$$(\Box + g^2\lambda/2)a^\nu - \partial^\nu(\partial_\mu a^\mu) = 0. \tag{GS.3}$$

Choose $a_0 = 0$ and $a_j = e_j\cos(k\cdot x - \omega t)$ with $k\cdot e = 0$. Then $\partial_\mu a^\mu = 0$ and all scalar equations hold. The temporal gauge equation is $\sum_j \partial_j \dot{a}_j = 0$, so Gauss is satisfied as well, not omitted. The remaining equations give $\omega^2 = |k|^2 + g^2\lambda/2$. At nonzero frequency or spatial momentum, the corresponding field strength is nonzero. Thus these are actual transverse solutions, not pure-gauge variations. For a specified rectangular flat spatial torus, $k_j = 2\pi n_j/L_j$, with every length L_j retained. Their positive kinetic sign follows from $\langle Z, Z\rangle/g^2 > 0$ in the original gauge density.

For $0 < q < 1$, every displayed eigenvalue except the first is positive. At $q = 1$ the second and the 23 pair additionally vanish; at $q = 0$ the 12 pair additionally vanishes. A stabilizer satisfies $[Z, \Phi_\theta] = 0$ and $Zb_\theta = 0$ by the proved inertia kernel. For $0 < q < 1$ distinct diagonal entries and nonzero b_θ give $\mathrm{diag}(e^{i\alpha}, e^{-i\alpha}, 1)$. At $q = 1$, $b_\theta = 0$ and the last two entries of Φ_θ coincide, giving the stated $U(2)$ subgroup. At $q = 0$ the first two entries coincide and $b_\theta \neq 0$ fixes e_3, giving the upper-left $SU(2)$. This proves the full group statements, including the endpoints. □

Proposition 1.42 (An exact invariant current and its mass branch)**.** *The full nonlinear gauge–scalar theory has the real, gauge-invariant conserved vector current*

$$j_\nu = -i\left(b^* D_\nu b - (D_\nu b)^* b\right), \qquad \partial_\nu j^\nu = 0. \tag{GS.4}$$

At any displayed vacuum with $\beta^2 > 0$, the transverse perturbation $\mathcal{A}_\nu = \varepsilon i\,\mathrm{diag}(1, 1, -2)a_\nu$, with zero scalar perturbations, has

$$j_\nu = -4\varepsilon\beta^2 a_\nu + O(\varepsilon^2). \tag{GS.5}$$

It therefore detects the branch (GS.2) in the exact linearized system. For the positive rank-one spectral-floor matrix $H_\theta = Q_\theta + \chi I_3 = \Theta(A_\theta, b_\theta)$,

$$\boxed{\mu_{\mathrm{d}}^2 = \frac{2g^2}{3V_\mu}\int_{SU(3)} \det(PgA_\theta g^* P^*)\, d\mu(g).} \tag{GS.6}$$

The Haar volume $V_\mu > 0$ and the original marking P are retained. At $\beta = 0$ the linear coefficient in (GS.5) vanishes; that endpoint does not supply a nonzero linear interpolator through this current.

Proof. Under $b \mapsto Ub$ and $D_\nu b \mapsto U D_\nu b$, each scalar product in (GS.4) is unchanged. Their difference is purely imaginary, so j_ν is real. It is the Noether current for the constant phase action $b \mapsto e^{i\alpha} b$: allowing α to depend on spacetime produces $j^\nu \partial_\nu \alpha$ in the variation of the kinetic term. The potential is phase invariant. Directly, the covariant product rule gives

$$\partial_\nu j^\nu = -i\left(b^* D_\nu D^\nu b - (D_\nu D^\nu b)^* b\right).$$

The exact equation is $D_\nu D^\nu b = -(aI_3 + \kappa\Phi)b$; the matrix in parentheses is Hermitian. The two scalar products are therefore equal, proving conservation in the full interacting theory.

At the specified constant background, $b_\theta^* \mathcal{A}_\nu b_\theta = -2i\varepsilon\beta^2 a_\nu$. Subtracting its conjugate and multiplying by $-i$ gives $-4\varepsilon\beta^2 a_\nu$, with no rescaling of the Cartan generator. Every off-diagonal generator, and $i\,\mathrm{diag}(1, -1, 0)$, has zero 33 entry, so none contributes to this linear current when scalar perturbations are zero. For an arbitrary scalar perturbation δb, its additional linear term is $2\,\mathrm{Im}(b_\theta^* \partial_\nu \delta b)$, a pure gradient at the constant background; the transverse sector used in (GS.3) sets it to zero. Thus the nonzero transverse mode is seen by a gauge-invariant observable. Finally Proposition 1.35 proves the integral in (GS.6) equals $V_\mu \beta^2$. Substitution into (GS.2) proves the exact relation with all coefficients and domains fixed. □

This follows the classical Higgs-phase use of gauge-invariant vector operators discussed by D. Tong, *Lectures on Gauge Theory*, Section 2.7.3, especially the fundamental-scalar example on printed pages 107–108, `https://www.damtp.cam.ac.uk/user/tong/gaugetheory/gt.pdf`. The eight-eigenvalue calculation, current coefficient, and marked determinant identity here are derived for this specified adjoint-plus- fundamental model; they are not transferred from that example. Equation (GS.6) is a relation to a linearized gauge-vector mass in this model, not an identification of all entries of G with spacetime momentum, and not a spectral theorem for pure interacting quantum Yang–Mills.

Corollary 1.43 (Variation of the massive branches along the minimum family)**.** *For $0 < q < 1$, the smallest strictly positive inertia eigenvalue is*

$$\lambda_{\mathrm{pos}}(q) = \begin{cases} 9\chi^2 q^2, & 0 < q \le 1/2, \\ \frac{9}{2}\chi^2(1-q), & 1/2 \le q < 1. \end{cases} \tag{GS.7}$$

It tends to zero at both ends and has unique maximum $9\chi^2/4$ at $q = 1/2$. At that point five eigenvalues coincide at $9\chi^2/4$, two equal $27\chi^2/4$, and one is zero. This positive degeneracy does not enlarge the stabilizer. The retained spectral traces are

$$\operatorname{tr}\mathscr{I} = \chi^2(36 - 15\delta), \qquad \operatorname{tr}(\mathscr{I}^2) = \chi^4(324 - 405\delta + 171\delta^2), \qquad \delta = 1 - q^2. \tag{GS.8}$$

Proof. The 13 eigenvalue is at least the 23 eigenvalue. Subtracting the 12 eigenvalue from the diagonal and 23 eigenvalues gives $3\chi^2(1-4q^2)$ and $\frac{9}{2}\chi^2(1-q-2q^2)$, respectively; both are nonnegative exactly for $0 \le q \le 1/2$ on the stated interval. For $q \ge 1/2$, the diagonal minus the 23 eigenvalue is $\frac{3}{2}\chi^2(1-q)(2q-1) \ge 0$. These comparisons prove (GS.7). The first expression increases strictly and the second decreases strictly, proving the maximum and endpoint limits. Substitution of $q = 1/2$ in (GS.1) gives the multiplicities, while its kernel remains one-dimensional. Adding the eight eigenvalues and their squares gives $\chi^2(21 + 15q^2)$ and $\chi^4(90 + 63q^2 + 171q^4)$, respectively. Replacing the notation q^2 by $1 - \delta$ proves (GS.8). □

There is always a zero transverse branch in this linearized system. The strictly positive branches alone also have no positive lower bound uniform over the minimum family. These statements concern the proved quadratic spectrum and its exact parameters. They do not determine radiative selection of a minimum, a renormalized continuum theory, or the interacting Hamiltonian spectrum.

1.18 The retained system and the scope of the result

The literal constant-matter data are

$$H = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1/2 & 1/(2\sqrt{2}) \\ 0 & 1/(2\sqrt{2}) & 1/2 \end{pmatrix}, \qquad q = \frac{e_1}{2\sqrt{2}}, \qquad A = \sqrt{2}\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The original scalar couplings are $m^2 = \Gamma = 1$, $\kappa = 6$; $s = \operatorname{tr} H^2 + 2q^*q = 1$ and $a = m^2 + \Gamma s = 2$. Write $g^2 > 0$ for the gauge coupling and keep $\omega \ne 0$ real. Every unknown C, Z_1, Z_2 belongs to $\mathfrak{su}(3)$, the real vector space of traceless anti-Hermitian complex 3×3 matrices. For any matrix M, set $M_{\mathrm{tf}} = M - (\operatorname{tr} M)I_3/3$, and define

$$\mathcal{I}X = ([H,[H,X]] + qq^*X + Xqq^*)_{\mathrm{tf}}, \qquad \mathcal{K}X = -[Z_1,[Z_1,X]] - [Z_2,[Z_2,X]],$$

$$F = [A,[A,H]] = -2H + \operatorname{diag}(-2,1,1).$$

The last identity follows from $Hq = 0$, $H^2 = (3/8)\operatorname{diag}(0,1,1)$ and $6(\operatorname{adj} H - qq^*)_{\mathrm{tf}} = \operatorname{diag}(-2,1,1)$. The five equations to be retained are exactly

$$[C,[C,H]] - [Z_1,[Z_1,H]] - [Z_2,[Z_2,H]] = F, \tag{46}$$

$$(C^2 - Z_1^2 - Z_2^2)q = -2q, \tag{47}$$

$$(\mathcal{I} + 2\mathcal{K}/g^2)C = (4\omega/g^2)[Z_1,Z_2], \tag{48}$$

$$-\omega^2 Z_1 - 2\omega[C,Z_2] + [C,[C,Z_1]] + \mathcal{K}Z_1 + (g^2/2)\mathcal{I}Z_1 = 0, \tag{49}$$

$$-\omega^2 Z_2 + 2\omega[C, Z_1] + [C, [C, Z_2]] + \mathcal{K}Z_2 + (g^2/2)\mathcal{I}Z_2 = 0. \tag{50}$$

This section does not solve or exclude all triples in $\mathfrak{su}(3)^3$. It proves necessary identities for every such triple and a no-solution theorem for the precisely stated complex block class below. No remaining general equation is assumed true, and no numerical candidate is a result.

Exact map to the original literal fields

For clarity, keep

$$u = (1,1,1)^{\mathsf{T}}, \quad v_0 = (1,-1,0)^{\mathsf{T}}, \quad w_0 = (1,1,-2)^{\mathsf{T}}, \quad K = \left(v_0/\sqrt{2}, w_0/\sqrt{6}, u/\sqrt{3}\right),$$

$$\Omega = \sqrt{2/3}\begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}, \quad O(t) = e^{t\Omega}, \quad \Phi_0 = \frac{1}{4}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & -2 \end{pmatrix}, \quad b_0 = \frac{1}{4}(1,-1,0)^{\mathsf{T}}.$$

The dot products and determinant give $K \in SO(3)$, and multiplication gives $K^*\Phi_0 K = H$, $K^* b_0 = q$, $K^*\Omega K = A$. The prescribed scalar fields remain $\Phi(t) = O(t)\Phi_0 O(t)^*$, $b(t) = O(t)b_0$; their frequency is $\sqrt{2}$. For a solution of (46)–(50), set

$$Y_1 = \cos(\omega t)Z_1 - \sin(\omega t)Z_2, \quad Y_2 = \sin(\omega t)Z_1 + \cos(\omega t)Z_2, \quad Y_3 = 0,$$

$$\mathcal{A}_0 = OKCK^*O^* - \Omega, \qquad \mathcal{A}_i = OKY_iK^*O^*. \tag{51}$$

The inverse is $C = K^*O^*\mathcal{A}_0 OK + A$ and $Y_i = K^*O^*\mathcal{A}_i OK$. These are gauge transformations by K^*O^*, including the inhomogeneous temporal term.

Here is a direct verification of the complete finite reduction. In the constant-matter frame put $E_i = \dot{Y}_i + [C, Y_i]$. The two scalar equations are $[C,[C,H]] - \sum_i [Y_i,[Y_i,H]] = F$ and $(C^2 - \sum_i Y_i^2)q = -2q$. The projected currents are $J_0 = -\mathcal{I}C$ and $J_i = -\mathcal{I}Y_i$. Gauss is $(2/g^2)\sum_i [Y_i, E_i] + \mathcal{I}C = 0$, and the spatial equations are

$$\dot{E}_i + [C, E_i] - \sum_j [Y_j, [Y_j, Y_i]] + (g^2/2)\mathcal{I}Y_i = 0.$$

These follow by homogeneous substitution in the covariant equations with $D_\mu = \partial_\mu + \mathcal{A}_\mu$, signature $(+---)$ and gauge density $\operatorname{tr}(\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu})/(2g^2)$. Specifically the adjoint derivative is a commutator, and $J_\mu = [\Phi, D_\mu\Phi] + b(D_\mu b)^* - (D_\mu b)b^*$; at constant matter it is minus the displayed inertia. Thus no temporal constraint has been removed. The rotation preserves $\sum_i Y_i^2$ and every quadratic commutator sum, and $\sum_i [Y_i, \dot{Y}_i] = -2\omega[Z_1, Z_2]$. Applying $(\partial_t + \operatorname{ad}_C)^2$ to the rotating pair gives the terms $-\omega^2 Z_i$, the two displayed terms with coefficient 2ω, and $[C,[C,Z_i]]$. The third spatial equation is identically zero. This proves equivalence with all five equations above at every time. The product rule proves covariance under (51), so this equivalence also holds in the original coordinates.

1.19 Universal necessary identities in the unrestricted complex class

Use $\langle X, Y\rangle = -\operatorname{tr}(XY)$ and $\|X\|^2 = -\operatorname{tr} X^2$ on $\mathfrak{su}(3)$. This is a real positive inner product because $X^* = -X$. Define the nonnegative real quantities

$$c = \langle C, \mathcal{I}C\rangle, \quad z = \sum_{j=1}^{2}\langle Z_j, \mathcal{I}Z_j\rangle, \quad k = \sum_{j=1}^{2}\|[C, Z_j]\|^2, \quad d = \|[Z_1, Z_2]\|^2, \quad z_0 = \|Z_1\|^2 + \|Z_2\|^2,$$

and the signed real number $b = \langle C, [Z_1, Z_2]\rangle$.

Proposition 1.44. *Every solution of* (46)–(50) *satisfies*

$$c - z = 2, \qquad (g^2/2)c + k = 2\omega b, \qquad \omega^2 z_0 = k + 2d + g^2 c + (g^2/2)z. \tag{52}$$

In particular c, k, d, z_0 *are strictly positive,* $\omega b > 0$*, and*

$$8kd + 2g^2 kz \geq g^4 c^2. \tag{53}$$

Proof. Cyclicity and anti-Hermiticity give

$$\langle X, \mathcal{I}X\rangle = \operatorname{tr}[X, H]^2 + 2|Xq|^2, \qquad \langle X, \mathcal{K}X\rangle = \sum_j \|[Z_j, X]\|^2.$$

Trace projection pairs trivially with traceless X. The first commutator term follows by cyclically moving H twice, and the vector term is $-2q^*X^2q = 2|Xq|^2$. The second identity follows by applying $\langle X, [Z, Y]\rangle = -\langle [Z, X], Y\rangle$ twice. Because $[X, H]$ is Hermitian, all terms are nonnegative.

Pair (46) with H using ordinary trace and pair (47) with $2q^*$, taking the real part. Their left sides add to $-c + z$. Their right sides add to $\operatorname{tr}(HF) - 4q^*q = -2\operatorname{tr} H^2 - 4q^*q = -2$, proving the first identity. Pair (48) with C to obtain the second. Pair (49) with Z_1 and (50) with Z_2. The gyroscopic terms sum to $4\omega b$, the double C terms sum to $-k$, and the spatial double commutators sum to $2d$. The result is $-\omega^2 z_0 + 4\omega b - k + 2d + (g^2/2)z = 0$. Substitute the second identity to obtain the third.

The first identity gives $c \geq 2$. The second gives $\omega b > 0$. Thus $b \neq 0$, which forces $d > 0$, $z_0 > 0$, and $k > 0$, since $b = \langle [C, Z_1], Z_2\rangle$. Apply Cauchy–Schwarz to the ordered pairs $([C, Z_1], [C, Z_2])$ and $(Z_2, -Z_1)$ to obtain $4b^2 \leq kz_0$. The second identity gives $\omega^2 z_0 \geq (k + g^2 c/2)^2/k$. Subtract this lower bound from the third identity and multiply by $4k$; the result is exactly (53). □

These identities rule out any attempted solution with zero spatial commutator or zero electric coupling to C. They do not rule out a general noncommuting complex triple.

1.20 A new genuinely complex block exclusion

Let $P = qq^*/(q^*q) = \operatorname{diag}(1, 0, 0)$ and put

$$\mathfrak{p} = \{X \in \mathfrak{su}(3) : PXP = (I - P)X(I - P) = 0\}, \qquad \mathfrak{k} = \{X \in \mathfrak{su}(3) : [X, P] = 0\}.$$

Thus $\mathfrak{p}$ consists of arbitrary complex matrices exchanging the q-line with its orthogonal complement; $\mathfrak{k}$ consists of the block-diagonal matrices preserving that line. Neither definition imposes reality of the entries.

Theorem 1.45 (Cross–cross–block exclusion)**.** *There is no solution of all five equations with* $C, Z_1 \in \mathfrak{p}$ *and* $Z_2 \in \mathfrak{k}$*. There is likewise no solution for which a constant real rotation of the ordered pair* (Z_1, Z_2) *puts it in that class while* $C \in \mathfrak{p}$*. The assertion allows all complex phases and every* $g^2 > 0$*,* $\omega \neq 0$*.*

Exact coordinates used only for the proof

Set $x = 1/2$, $d_0 = 1/(2\sqrt{2})$, $r = \sqrt{x^2 + d_0^2} = \sqrt{6}/4$, and

$$S = \frac{1}{\sqrt{2r(r + x)}} \begin{pmatrix} d_0 & -(r + x) \\ r + x & d_0 \end{pmatrix}, \qquad U = \operatorname{diag}(1, S).$$

Multiplication gives $S^{\mathsf{T}}S = I_2$, $\det S = 1$, $U^*HU = \operatorname{diag}(0, r, -r)$, and $U^*q = q$. All connections and A are conjugated by U^*, with inverse conjugation by U. Brackets, trace projection, the force, and the inertia transform by these exact invertible maps; hence all five equations are equivalent. The subspaces $\mathfrak{p}, \mathfrak{k}$ are preserved because U commutes with P. No field, period, kinetic coefficient, or coupling is divided out.

For $v \in \mathbb{C}^2$ define

$$X(v) = \begin{pmatrix} 0 & -v^* \\ v & 0_{2\times 2} \end{pmatrix}.$$

In either of the proof frames below, H has zero first row and column, and its lower block squares to $r^2 I_2$. Since $q^*q = 1/8$,

$$\mathcal{I}X(v) = (r^2 + 1/8)X(v) = \tfrac{1}{2}X(v). \tag{54}$$

Indeed $HX(v)H = 0$, $H^2X(v) + X(v)H^2 = r^2X(v)$, and the two qq^* terms sum to $X(v)/8$; all these matrices are traceless.

Lemma 1.46 (Scalar classification of the block generator)**.** *Under the hypotheses of Theorem 1.45, the two scalar equations put the block generator into exactly one of the following two forms by an explicitly invertible constant unitary map fixing q:*

$$\text{(I)} \quad H = \operatorname{diag}(0, r, -r), \qquad Z_2 = i\operatorname{diag}(0, \sigma, -\sigma), \quad \sigma \in \mathbb{R};$$

$$\text{(II)} \quad H = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & r \\ 0 & r & 0 \end{pmatrix}, \qquad Z_2 = i\operatorname{diag}(e, 0, -e), \quad e \neq 0.$$

In both cases write $C = X(v)$, $Z_1 = X(w)$ and $U_v = vv^* - ww^* = (u_{jk})$. In case (I) scalar compatibility is exactly

$$u_{11} = 1 - \delta > 0, \quad u_{22} = 1 + \delta > 0, \qquad \delta = \frac{1}{2r} = \sqrt{2/3}.$$

In case (II) it is exactly

$$\operatorname{tr} U_v = 2 + e^2, \qquad \operatorname{Re} u_{12} = -\delta \neq 0.$$

Proof. Start in the displayed diagonal-H frame. An arbitrary element of $\mathfrak{k}$ is $Z_2 = i\operatorname{diag}(\eta, B)$, where B is Hermitian, $\operatorname{tr} B = -\eta$, and η is real. Write it without deleting any coefficient as

$$B = -\frac{\eta}{2} I_2 + \begin{pmatrix} z & x_1 - ix_2 \\ x_1 + ix_2 & -z \end{pmatrix}, \qquad x_1, x_2, z \in \mathbb{R}, \qquad D = \operatorname{diag}(r, -r).$$

The lower block of the adjoint scalar equation is

$$-\{U_v, D\} + [B, [B, D]] = I_2 - 2D.$$

Matrix multiplication gives

$$[B, [B, D]] = 4r \begin{pmatrix} x_1^2 + x_2^2 & -z(x_1 - ix_2) \\ -z(x_1 + ix_2) & -(x_1^2 + x_2^2) \end{pmatrix}.$$

Its off-diagonal equation is $z(x_1 - ix_2) = 0$. The two diagonal equations give $u_{11} - u_{22} = -1/r$ and $\operatorname{tr} U_v = 2 + 4(x_1^2 + x_2^2)$. The vector equation gives $\operatorname{tr} U_v = 2 + \eta^2$. Thus $\eta^2 = 4(x_1^2 + x_2^2)$.

If $x_1 = x_2 = 0$, then $\eta = 0$ and $B = \operatorname{diag}(z, -z)$. This is case (I) with $\sigma = z$, and its diagonal scalar equations give the two values stated. They are positive because $\delta^2 = 2/3 < 1$.

Otherwise $z = 0$ and $e := \eta \neq 0$. Let $\gamma = 2(x_1 - ix_2)/\eta$, so $|\gamma| = 1$. Choose either complex square root ζ of $-\bar{\gamma}$ and set

$$W = \frac{\zeta}{\sqrt{2}} \begin{pmatrix} \gamma & \gamma \\ 1 & -1 \end{pmatrix}.$$

Then $|\zeta| = 1$, $W^*W = I_2$, $\det W = 1$, $W^*BW = \operatorname{diag}(0, -e)$, and $W^*DW = r\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. These identities follow by multiplying the two columns, which are the zero and $-e$ eigenvectors of B, with the stated phases. Conjugation by $\operatorname{diag}(1, W)^*$, with inverse its adjoint, therefore gives case (II), fixes q, and preserves both block spaces.

With $\sigma_x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, in this frame the lower scalar equation is $-\{U_v, r\sigma_x\} + e^2 r\sigma_x = I_2 - 2r\sigma_x$. Its diagonal entries give $\operatorname{Re} u_{12} = -1/(2r)$ and its off-diagonal entries give $\operatorname{tr} U_v = 2 + e^2$. The vector equation gives the same trace. Conversely direct substitution sets every scalar residual entry to zero in each of the two cases. □

All gauge equations in the two frames

In either case write

$$Z_2 = i \operatorname{diag}(\eta, d_1, d_2), \quad n_j = \eta - d_j, \quad V_j = |v_j|^2, \quad W_j = |w_j|^2,$$
$$V = V_1 + V_2, \quad W_0 = W_1 + W_2, \quad N = V + W_0,$$
$$p = v^* w = \alpha + ib, \quad b_j = \operatorname{Im}(\bar{v}_j w_j),$$
$$A_j = \frac{g^2}{4} + n_j^2 + W_0 > 0, \quad P_j = 3b + 2\omega n_j, \quad B_j = A_j - (\omega^2 + N).$$

Here b is only the imaginary inner product in this section, not the triplet or the signed pairing in Proposition 1.44. Direct block multiplication gives

$$-[X(w), [X(w), X(v)]] = X\big(W_0 v + w(\bar{p} - 2p)\big), \qquad [X(v), Z_2] = X\big(i \operatorname{diag}(n_1, n_2) v\big). \tag{55}$$

Using (54), the lower first-column equations of Gauss and the first spatial equation are exactly

$$A_j v_j = (\alpha + iP_j) w_j, \qquad B_j w_j = -(\alpha - iP_j) v_j \quad (j = 1, 2). \tag{56}$$

There are no other nonzero blocks in these two equations.

The second spatial equation has only a block-diagonal residual. Write $\mathcal{I} Z_2 = i \operatorname{diag}(\ell_0, \ell_1, \ell_2)$; it is diagonal in both frames. Its lower diagonal entries are

$$-\omega^2 d_j + 2n_j(V_j - W_j) + 4\omega b_j + (g^2/2)\ell_j = 0. \tag{57}$$

Its upper diagonal entry is

$$-\omega^2 \eta - 2\sum_j n_j(V_j - W_j) - 4\omega b + (g^2/2)\ell_0 = 0. \tag{58}$$

Its lower off-diagonal entry is

$$i(n_1 + n_2)(v_1 \bar{v}_2 - w_1 \bar{w}_2) + 2\omega(-v_1 \bar{w}_2 + w_1 \bar{v}_2) = 0. \tag{59}$$

For verification, the lower block of $[C, Z_1]$ is $-vw^* + wv^*$; the lower block contributed by $[C, [C, Z_2]] + \mathcal{K} Z_2$ has entry $i(n_j + n_k)(v_j \bar{v}_k - w_j \bar{w}_k)$. Its upper entry is minus its lower trace. These multiplications give (57)–(59) with every sign and coefficient. They retain the entire second spatial equation.

Lemma 1.47. *Every solution of these gauge equations has* $\alpha = 0$ *and*

$$\omega^2 W_0 = \frac{g^2}{4} N + \sum_j n_j^2 (V_j + W_j), \tag{60}$$

$$V_j - W_j = \frac{V_j + W_j}{\omega^2 + N} \left(\omega^2 + V - W_0 - \frac{g^2}{2} - 2n_j^2 \right). \tag{61}$$

Proof. Conjugate the first equation in (56) and multiply by w_j. Summing gives $p = \sum_j (\alpha - iP_j) W_j / A_j$. Taking real parts gives $\alpha = \alpha \sum_j W_j / A_j$. If $W_0 = 0$ the sum is zero. Otherwise $A_j > W_0$ gives $\sum_j W_j / A_j < \sum_j W_j / W_0 = 1$. Hence $\alpha = 0$ in all cases. The two equations are now $A_j v_j = i P_j w_j$ and $B_j w_j = i P_j v_j$. Their real pairings with v_j and w_j give $A_j V_j = -P_j b_j$ and $B_j W_j = P_j b_j$. Adding them gives $A_j (V_j + W_j) = (\omega^2 + N) W_j$. Summing and inserting A_j proves (60). Subtracting twice this last formula divided by $\omega^2 + N$ from $V_j + W_j$ proves (61). The denominator is positive because $\omega \neq 0$ and $N \geq 0$. □

Proof of Theorem 1.45. Use Lemma 1.46 and retain every gauge equation as just derived.

In case (I), $\eta = 0$, $n_1 = -\sigma$, $n_2 = \sigma$, $\mathcal{I} Z_2 = 0$, and $V_j - W_j = 1 \mp \delta > 0$. If $\sigma = 0$, then $Z_2 = 0$ and Gauss paired with C gives $\langle C, \mathcal{I} C \rangle + (2/g^2) \langle C, \mathcal{K} C \rangle = 0$. Both terms are nonnegative and (54) forces $C = 0$, contradicting the two positive scalar differences. Thus $\sigma \neq 0$. Both v_j are nonzero. Gauss then forces $w_j \neq 0$ and $P_j \neq 0$. Since $\alpha = 0$,

$$w_j = i t_j v_j, \qquad t_j = -A_j / P_j \in \mathbb{R} \setminus \{0\}.$$

Here $A_1 = A_2 = g^2/4 + \sigma^2 + W_0$. Because $n_1 + n_2 = 0$, (59) becomes $2i\omega(t_1 + t_2) v_1 \bar{v}_2 = 0$. Thus $t_1 + t_2 = 0$, which implies $P_1 + P_2 = 0$, or $6b = 0$. But (58) gives

$$b = \frac{\sigma((1 - \delta) - (1 + \delta))}{2\omega} = -\frac{\sigma \delta}{\omega} \neq 0.$$

This contradiction excludes case (I).

In case (II), $\eta = e$, $d_1 = 0$, $d_2 = -e$, $n_1 = e$, $n_2 = 2e$, and scalar compatibility gives $V - W_0 = 2 + e^2$ and $\mathrm{Re}(v_1 \bar{v}_2 - w_1 \bar{w}_2) = -\delta \neq 0$. In particular $V_j + W_j > 0$ for both j. The transformations $(Z_1, Z_2) \mapsto (-Z_1, -Z_2)$ and $(C, \omega) \mapsto (-C, -\omega)$ each preserve all five equations, as substitution verifies. They allow $e > 0$ and $\omega > 0$ without changing the given matter fields or their original frequency.

Direct multiplication in this frame gives

$$\mathcal{I} Z_2 = ie \, \mathrm{diag}(1/6, 2/3, -5/6).$$

The adjoint and trace-projected vector parts, respectively, are

$$ie \, \mathrm{diag}(0, 2r^2, -2r^2), \qquad ie \, \mathrm{diag}(1/6, -1/12, -1/12).$$

Their sum proves the formula with $r^2 = 3/8$. The right side of (60) is positive, so $W_0 > 0$. Since $N = 2W_0 + 2 + e^2 > 2W_0$ and $n_j^2 \geq e^2$, that identity gives

$$\omega^2 W_0 \geq (g^2/4 + e^2) N > (g^2/2 + 2e^2) W_0, \qquad \omega^2 > g^2/2 + 2e^2.$$

Formula (61) consequently implies

$$V_1 - W_1 = \frac{V_1 + W_1}{\omega^2 + N} (\omega^2 + 2 - e^2 - g^2/2) > 0.$$

The $j = 1$ equation in (57) is exactly

$$2e(V_1 - W_1) + 4\omega b_1 + eg^2/3 = 0,$$

so $b_1 < 0$. Gauss gives $b_1 = -P_1 W_1/A_1$. Since $V_1 > 0$, it also forces $W_1 > 0$ and hence $P_1 > 0$. Now $P_2 = P_1 + 2e\omega > 0$; because $V_2 + W_2 > 0$, Gauss forces both $V_2, W_2 > 0$ as well.

Taking imaginary parts in the proof of Lemma 1.47 gives

$$b\left(1 + 3\sum_j \frac{W_j}{A_j}\right) = -2\omega \sum_j \frac{n_j W_j}{A_j} < 0.$$

Therefore $b < 0$ and $0 < P_j = 3b + 2\omega n_j < 2\omega n_j$. Put $\rho_j = A_j/P_j > 0$. Since $A_j > n_j^2$, one has the strict lower bound

$$\rho_1 + \rho_2 > \frac{n_1 + n_2}{2\omega} = \frac{3e}{2\omega}. \tag{62}$$

Gauss also gives $w_j = -i\rho_j v_j$. Substitution into the full off-diagonal spatial equation (59), followed by division by the nonzero number $iv_1\bar{v}_2$, gives

$$3e(1 - \rho_1\rho_2) - 2\omega(\rho_1 + \rho_2) = 0.$$

As $\rho_1\rho_2 > 0$, this proves the contradictory strict upper bound

$$\rho_1 + \rho_2 = \frac{3e}{2\omega}(1 - \rho_1\rho_2) < \frac{3e}{2\omega}.$$

This excludes case (II) with arbitrary complex phases of v, w.

Finally a constant $SO(2)$ rotation of the ordered pair (Z_1, Z_2) preserves their sum of squares, every quadratic commutator sum, and $[Z_1, Z_2]$. It commutes with the spatial rotation generator $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, so the two spatial equations transform by the same invertible rotation. It leaves C, H, q unchanged. Thus a solution that could be put in the excluded class by such a rotation would produce a solution already excluded, proving the final assertion. □

1.21 The precise unresolved part

Theorem 1.45 is a complex restriction, not the real-algebra obstruction restated. In case (II) one has $q^* Z_2 q = ie/8 \neq 0$. A real-skew matrix paired with the real vector q has $q^* Z q = 0$, and any unitary transformation fixing q preserves this pairing. Thus these nonzero block generators cannot be turned into real-skew generators by a matter-stabilizer transformation.

Nevertheless the theorem does not put arbitrary C, Z_1, Z_2 into its block class. In particular it assumes $C \in \mathfrak{p}$ and the stated decomposition of the spatial pair; neither follows here from all five unrestricted equations. The general finite system remains unresolved. The universal identities in Proposition 1.44 continue to apply outside this class. There is no asserted full-complex lift and no asserted full-complex no-lift theorem in this section.

The companion `check_complex_rotation.py` verifies the constant-frame matrix reductions with arbitrary complex block coordinates, the two branch inertia formulas, and the ratio identity by exact arithmetic. It also checks the universal pairing formulas on one exact full-algebra test triple. The universal proofs, branch inequalities, symmetry arguments, and original time-dependent gauge-map equivalence are proved in the text. The checker does not solve the unrestricted finite system.

1.22 Exact unrestricted Gauss and frequency elimination

The word unrestricted in this subsection refers to all complex colour coordinates in the specified constant-C, two-spatial-rotation ansatz, not to all time-dependent homogeneous connections. Retain the literal H, q, F, the positive coupling g^2, and all five equations (46)–(50). No block-class restriction from Theorem 1.45 is imposed here. For arbitrary $Z_1, Z_2 \in \mathfrak{su}(3)$ put

$$\mathscr{B} = [Z_1, Z_2], \qquad L = g^2\mathcal{I} + 2\mathcal{K}.$$

Lemma 1.48 (Positive Gauss inverse on every possible solution)**.** *If $\mathscr{B} \neq 0$, then L is a positive-definite self-adjoint operator on the original real eight-dimensional metric space $(\mathfrak{su}(3), -\operatorname{tr})$. Every solution of the five equations has $\mathscr{B} \neq 0$. Thus its Gauss equation is exactly*

$$C = \omega T, \qquad T = 4L^{-1}\mathscr{B}.$$

Proof. Cyclicity gives the full bilinear identities

$$\langle X, \mathcal{I}Y\rangle = \operatorname{tr}([X,H][Y,H]) + 2\operatorname{Re}((Xq)^* Yq),$$
$$\langle X, \mathcal{K}Y\rangle = \sum_{j=1}^{2} \langle [Z_j, X], [Z_j, Y]\rangle.$$

They are real and symmetric. Their values at $Y = X$ are sums of nonnegative squared norms, since $[X, H]$ is Hermitian. Consequently

$$\ker L = \{X : [X,H] = 0,\ Xq = 0,\ [X,Z_1] = [X,Z_2] = 0\}.$$

For the converse direction in this equality, each displayed vanishing condition makes the corresponding double-commutator or vector term zero directly. For the forward direction, vanishing of their positive weighted sum forces every one of the norms to vanish.

The eigenvalues of H are the three distinct real numbers $0, r, -r$, $r = \sqrt{6}/4$, and q spans its zero eigenline. Conjugate by the exact matrix U already displayed in the block proof. A matrix commuting with this diagonal H has no off-diagonal entries. Anti-Hermiticity makes its diagonal entries imaginary; annihilation of q sets the first entry to zero; trace zero makes the last two opposite. Therefore

$$\{X : [X,H] = 0,\ Xq = 0\} = \mathbb{R}\, iH.$$

If a nonzero element of this line also commuted with both Z_j, both Z_j would commute with H. The same distinct-eigenvalue argument would make them diagonal in its eigenbasis, hence mutually commuting. This contradicts $\mathscr{B} \neq 0$. Thus $\ker L = 0$, and positivity and finite dimension give the asserted inverse.

For any solution, Proposition 1.44 proves $\|[Z_1, Z_2]\|^2 > 0$. Equivalently, if that bracket were zero, Gauss paired with C would give $c = 0$, whereas the scalar contraction gives $c - z = 2$ and $z \geq 0$. Finally multiplying Gauss by g^2 gives $LC = 4\omega\mathscr{B}$, exactly the stated inverse relation. No temporal equation was omitted. □

Theorem 1.49 (Full residual zero set and a two-sign inverse)**.** *On the explicit open domain*

$$\mathcal{D} = \{(Z_1, Z_2, g^2) : Z_j \in \mathfrak{su}(3),\ g^2 > 0,\ [Z_1, Z_2] \neq 0,\ Tq \neq 0\},$$

retain the just-defined T, and set

$$\tau = \frac{1/4 + \|Z_1 q\|^2 + \|Z_2 q\|^2}{\|Tq\|^2} > 0. \tag{63}$$

Define the four full residuals, with their original codomains,

$$\begin{aligned}
R_H &= \tau[T,[T,H]] - \sum_{j=1}^{2}[Z_j,[Z_j,H]] - F \in \mathrm{Herm}_3(\mathbb{C})_0,\\
R_q &= (\tau T^2 - Z_1^2 - Z_2^2)q + 2q \in \mathbb{C}^3,\\
R_1 &= \tau\big(-Z_1 - 2[T,Z_2] + [T,[T,Z_1]]\big) + \mathcal{K}Z_1 + (g^2/2)\mathcal{I}Z_1 \in \mathfrak{su}(3),\\
R_2 &= \tau\big(-Z_2 + 2[T,Z_1] + [T,[T,Z_2]]\big) + \mathcal{K}Z_2 + (g^2/2)\mathcal{I}Z_2 \in \mathfrak{su}(3).
\end{aligned} \tag{64}$$

Let $\mathcal{Z} \subset \mathcal{D}$ be their common zero set. The map

$$(Z_1, Z_2, g^2, \varepsilon) \longmapsto \Big(C = \varepsilon\sqrt{\tau}\,T,\ Z_1, Z_2, \omega = \varepsilon\sqrt{\tau},\ g^2\Big), \qquad \varepsilon \in \{+1,-1\}, \tag{65}$$

is a bijection from $\mathcal{Z} \times \{+1,-1\}$ onto all solutions of the five retained equations. Its inverse forgets C, ω and retains $\varepsilon = \mathrm{sign}(\omega)$.

Proof. For a solution, the preceding lemma gives $C = \omega T$ and a nonzero spatial bracket. Pair its full vector equation with q^*. Anti-Hermiticity of the three matrices gives exactly

$$\omega^2\|Tq\|^2 = 2q^*q + \|Z_1q\|^2 + \|Z_2q\|^2 = \tfrac{1}{4} + \|Z_1q\|^2 + \|Z_2q\|^2 > 0.$$

Thus $Tq \neq 0$ and $\omega^2 = \tau$. The constant $1/4$ is the original $2q^*q$; no scalar or frequency has been scaled to one. This contraction determines the frequency but does not replace the vector equation. Substitution of $C = \omega T$ into all the scalar and spatial equations gives every entry of (64). Hence the original solution determines a point of $\mathcal{Z}$ and its unique sign.

Conversely, start at any point of $\mathcal{Z}$ and choose either sign in (65). By definition $LT = 4\mathscr{B}$, so $LC = 4\omega\mathscr{B}$ and Gauss holds. The substitutions $C = \omega T$, $\omega^2 = \tau$ make each of the other four original residuals identical to its displayed R. They therefore vanish as full matrices or vectors, not only after contraction. The constructions plainly recover all original coordinates and signs, proving the bijection. □

This is an exact seventeen-real-variable formulation (sixteen spatial matrix coordinates and the retained positive g^2), not a proof that its zero set is nonempty or empty. The inverse returns all original fields through (51). In particular the nonzero denominator and bracket domain must be retained in any polynomial elimination. The scalar contraction alone is insufficient: the accompanying exact test point has that contraction zero while its full R_q is nonzero. No such point is reported as a solution.

1.23 Complete scalar-coordinate identities and an unrestricted signature constraint

The following calculation retains all eight real coordinates of each of C, Z_1, Z_2. It is not a cross–cross–block restriction. Let $x = 1/2$, $d_0 = 1/(2\sqrt{2})$, $r = \sqrt{6}/4$, and

$$S = \frac{1}{\sqrt{2r(r+x)}}\begin{pmatrix} d_0 & -(r+x) \\ r+x & d_0 \end{pmatrix}, \qquad U = \mathrm{diag}(1,S), \qquad \delta = \frac{1}{2r} = \sqrt{2/3}.$$

Both columns of the numerator of S have squared length $2r(r+x)$, their inner product is zero, and its determinant is $2r(r+x) > 0$. Multiplication by the lower block of the literal H shows that the columns are eigenvectors for $r, -r$ in that order. Thus

$$U \in SO(3), \qquad U^*HU = H_d = \mathrm{diag}(0, r, -r), \qquad U^*q = q.$$

Conjugate every connection and force by U^*, with inverse conjugation by U. Products, commutators, the adjoint operation, trace projection, and the five equations all transform by these exact inverse maps. In this frame $F_d = \mathrm{diag}(-2, 1-2r, 1+2r)$.

For each original connection X define its eight real coordinates by

$$\widetilde{X} = U^* X U = \begin{pmatrix} ia_X & -\overline{v_{X1}} & -\overline{v_{X2}} \\ v_{X1} & ib_X & z_X \\ v_{X2} & -\overline{z_X} & -i(a_X + b_X) \end{pmatrix}, \qquad a_X, b_X \in \mathbb{R}, \quad v_{X1}, v_{X2}, z_X \in \mathbb{C}, \tag{66}$$

and put $h_X = a_X + 2b_X$. This representation and $X = U\widetilde{X}U^*$ are mutually inverse. In particular no entry or phase has been removed. The stabilizer coefficient has the original-coordinate formula

$$h_X = \frac{\langle iH, X\rangle}{r} = \frac{-i\,\mathrm{tr}(HX)}{r}. \tag{67}$$

Indeed $\mathrm{tr}(H_d\widetilde{X}) = ir(a_X + 2b_X)$.

For a scalar expression $f(X)$ define its signed sum

$$\mathscr{S}(f) = f(C) - f(Z_1) - f(Z_2).$$

This notation retains the temporal plus sign and both spatial minus signs; it is not a positive covariance. Define

$$D_a = \mathscr{S}(a^2), \qquad D_z = \mathscr{S}(|z|^2), \qquad D_j = \mathscr{S}(|v_j|^2) \quad (j = 1, 2),$$

$$m_j = \mathscr{S}(hv_j), \quad m_z = \mathscr{S}(hz), \qquad n_1 = \mathscr{S}(zv_2) + \frac{i}{2}\mathscr{S}(av_1), \quad n_2 = \mathscr{S}(\bar{z}v_1) - \frac{i}{2}\mathscr{S}(av_2).$$

Proposition 1.50 (All scalar equations in unrestricted coordinates)**.** *For arbitrary $C, Z_1, Z_2 \in \mathfrak{su}(3)$, the two scalar equations* (46)–(47) *hold if and only if the following three real and five complex identities hold:*

$$D_a = 4D_z, \qquad D_1 + 2D_z = 1 - \delta, \qquad D_2 + 2D_z = 1 + \delta, \tag{68}$$

$$m_1 = m_2 = m_z = n_1 = n_2 = 0. \tag{69}$$

Proof. For a diagonal Hermitian matrix with entries λ_j,

$$[X, [X, H_d]]_{jk} = \sum_{\ell} X_{j\ell} X_{\ell k} (\lambda_j + \lambda_k - 2\lambda_\ell).$$

Use this identity with $(\lambda_1, \lambda_2, \lambda_3) = (0, r, -r)$ and the full matrix (66). Let Q_H and Q_q denote respectively the left sides minus the right sides of the two scalar equations in this diagonal frame. Their diagonal and matter-line entries are

$$\begin{aligned} (Q_H)_{11} &= 2r(D_1 - D_2) + 2, \\ (Q_H)_{22} &= -2rD_1 - 4rD_z - (1 - 2r), \\ (Q_H)_{33} &= 2rD_2 + 4rD_z - (1 + 2r), \\ (Q_q)_1 &= q_1(2 - D_a - D_1 - D_2), \qquad q_1 = 1/(2\sqrt{2}). \end{aligned}$$

The first of these three matrix entries is minus the sum of the next two, since the full residual is traceless. The latter two vanish exactly when $D_1 + 2D_z = 1 - \delta$ and $D_2 + 2D_z = 1 + \delta$. Their sum gives $D_1 + D_2 = 2 - 4D_z$; the vector's first entry vanishes exactly when $D_a = 4D_z$.

For clarity the individual lower matter-column products are

$$(\widetilde{X}^2)_{21} = i(a_X + b_X)v_{X1} + z_X v_{X2}, \qquad (\widetilde{X}^2)_{31} = -ib_X v_{X2} - \bar{z}_X v_{X1},$$

and the three independent lower/upper matrix products are

$$\begin{aligned}
[\widetilde{X}, [\widetilde{X}, H_d]]_{21} &= r\big(i(a_X - b_X)v_{X1} + 3z_X v_{X2}\big),\\
[\widetilde{X}, [\widetilde{X}, H_d]]_{31} &= r\big(-i(2a_X + b_X)v_{X2} + 3\bar{z}_X v_{X1}\big),\\
[\widetilde{X}, [\widetilde{X}, H_d]]_{23} &= -2irh_X z_X.
\end{aligned}$$

Since $b_X = (h_X - a_X)/2$, the complete remaining residual entries are

$$\begin{aligned}
(Q_q)_2 &= q_1(n_1 + im_1/2), & (Q_H)_{21} &= r(3n_1 - im_1/2),\\
(Q_q)_3 &= q_1(-n_2 - im_2/2), & (Q_H)_{31} &= r(3n_2 - im_2/2),\\
& & (Q_H)_{23} &= -2irm_z.
\end{aligned}$$

The remaining off-diagonal entries of Q_H are the conjugates, since it is Hermitian. Both q_1 and r are nonzero. For either j, the two relevant equations first give $n_j = -im_j/2$ and then $3n_j - im_j/2 = -2im_j = 0$. Thus $m_j = n_j = 0$; the last matrix entry gives $m_z = 0$. Conversely these five complex equalities make every displayed off-diagonal and vector entry vanish. Together with the three real relations already proved, they set the entire scalar residuals to zero. This proves both directions without a rank or phase restriction. □

Theorem 1.51 (Quantitative spacelike stabilizer constraint)**.** *Every scalar-compatible unrestricted triple satisfies the following inequalities. Put*

$$H_s = h_{Z_1}^2 + h_{Z_2}^2, \qquad S_j = \sum_{\ell=1}^{2}(|v_{Z_\ell j}|^2 + 2|z_{Z_\ell}|^2), \qquad c_1 = 1 - \delta > 0, \quad c_2 = 1 + \delta > 0.$$

Then

$$h_C^2(S_j + c_j) \le H_s S_j \quad (j = 1, 2). \tag{70}$$

In particular exactly one of the following possibilities holds:

$$h_C = h_{Z_1} = h_{Z_2} = 0, \qquad \text{or} \qquad h_C^2 < h_{Z_1}^2 + h_{Z_2}^2.$$

Equivalently, the three original-coordinate numbers $\langle iH, C\rangle, \langle iH, Z_1\rangle, \langle iH, Z_2\rangle$ are all zero or have strictly negative squared norm for the retained signature $(+--)$. Every zero of the unrestricted reduced residuals consequently satisfies

$$\tau h_T^2 \le H_s \min\left\{\frac{S_1}{S_1 + 1 - \delta}, \frac{S_2}{S_2 + 1 + \delta}\right\}. \tag{71}$$

All denominators in this formula are strictly positive.

Proof. For each $j = 1, 2$ and each $X \in \{C, Z_1, Z_2\}$ put $V_{Xj} = (v_{Xj}, \sqrt{2}z_X) \in \mathbb{C}^2$. The identities $m_j = m_z = 0$ give

$$h_C V_{Cj} = h_{Z_1} V_{Z_1 j} + h_{Z_2} V_{Z_2 j}.$$

Their squared norms and Cauchy–Schwarz on the ordered pair in $\mathbb{C}^2 \oplus \mathbb{C}^2$ imply

$$h_C^2 \|V_{Cj}\|^2 \le (h_{Z_1}^2 + h_{Z_2}^2)(\|V_{Z_1 j}\|^2 + \|V_{Z_2 j}\|^2) = H_s S_j.$$

The diagonal scalar identities give $\|V_{Cj}\|^2 - S_j = D_j + 2D_z = c_j$, proving (70). Since $0 < \delta < 1$, each c_j is strictly positive. If $H_s = 0$, the inequality forces $h_C = 0$. If $H_s > 0$, division by $S_j + c_j > 0$ gives $h_C^2 \le H_s S_j/(S_j + c_j) < H_s$. This covers $S_j = 0$ as well. The original-coordinate version follows by multiplication by $r^2 > 0$ in (67). Finally $C = \varepsilon\sqrt{\tau}\,T$ gives $h_C^2 = \tau h_T^2$, so both inequalities give the last formula, independently of the frequency sign. □

The quantitative inequalities are attained by scalar-compatible triples; equality in them does not certify the gauge equations. For an exact witness in the original coordinates retain A, H, q and set

$$C = \frac{5}{3}A + \frac{4}{3}iH, \qquad Z_1 = 0, \qquad Z_2 = \frac{4}{3}A + \frac{5}{3}iH.$$

The equality $(5/3)^2 - (4/3)^2 = 1$ makes the signed quadratic expressions equal to those of $(A, 0, iH)$: both cross terms cancel in $C^2 - Z_2^2$ and in $[C,[C,H]] - [Z_2,[Z_2,H]]$. Since $Hq = 0$ and $[iH, H] = 0$, these are exactly the original two scalar equations. The A matrix has only matter-cross entries in the diagonal frame. Their squared moduli are $1 - \delta, 1 + \delta$, either by the displayed U or by applying the scalar diagonal identities to $(A, 0, 0)$. Also $h_A = 0$ and $h_{iH} = 2r$ by (67). Thus

$$h_C = 8r/3, \qquad h_{Z_2} = 10r/3, \qquad S_j = \frac{16}{9}c_j,$$

and both sides of (70) are equal. This witness is not a full gauge solution: $[Z_1, Z_2] = 0$ contradicts the necessary noncommutation proved in Lemma 1.48.

There is a second unrestricted test using both spatial equations, not only their trace pairing. We give all its coefficients and its exact nonnegative discriminant below.

1.24 A two-by-two stress obstruction before evaluating the full residuals

In the coordinates (66), direct multiplication gives the full inertia matrix

$$U^*(\mathcal{I}X)U = \begin{pmatrix} ia_X/6 & -\overline{v_{X1}}/2 & -\overline{v_{X2}}/2 \\ v_{X1}/2 & -ia_X/12 & 3z_X/2 \\ v_{X2}/2 & -3\bar{z}_X/2 & -ia_X/12 \end{pmatrix}. \tag{72}$$

For off-diagonal entries the double commutator multiplies by the square of the eigenvalue difference; these are $r^2 = 3/8$ and $4r^2 = 3/2$. The qq^* terms add $1/8$ to the two matter-cross entries. For the diagonal entries those terms are $i\,\mathrm{diag}(a_X/4, 0, 0)$, whose trace-free part is $i\,\mathrm{diag}(a_X/6, -a_X/12, -a_X/12)$. This proves every entry and also

$$\langle X, \mathcal{I}X\rangle = a_X^2/4 + |v_{X1}|^2 + |v_{X2}|^2 + 3|z_X|^2.$$

The eigenvalues of $\mathcal{I}$ are $0, 1/6, 1/2, 3/2$ with respective real multiplicities $1, 1, 4, 2$. Indeed the h diagonal direction, the a diagonal direction with $b = -a/2$, the four cross-entry directions, and the two lower off-diagonal directions are mutually orthogonal and are multiplied by precisely these scalars in the displayed formula.

For an arbitrary point of $\mathcal{D}$ from Theorem 1.49, define the real symmetric two-by-two matrices and real scalars

$$G_{jk} = \langle Z_j, Z_k\rangle, \qquad D_{jk} = \langle [T, Z_j], [T, Z_k]\rangle, \qquad M_{jk} = \langle Z_j, \mathcal{I}Z_k\rangle,$$

$$d = \|[Z_1, Z_2]\|^2 > 0, \qquad b_T = \langle T, [Z_1, Z_2]\rangle, \qquad N = G + D - 2b_T I_2, \qquad B = dI_2 + (g^2/2)M.$$

All these objects are functions of the unrestricted spatial pair and g^2, with no frequency sign selected. M is positive semidefinite: for $x \in \mathbb{R}^2$, $x^T M x = \langle \sum_j x_j Z_j, \mathcal{I}\sum_j x_j Z_j\rangle \ge 0$. Hence B is positive definite.

Proposition 1.52 (Exact stress matrix and nonnegative discriminant)**.** *Every zero of the four full reduced residuals satisfies*

$$\tau N = B. \tag{73}$$

In particular N must be positive definite. On the open domain $N_{11} > 0$, $\det N > 0$, define

$$\Delta = (B_{11}N_{22} + B_{22}N_{11} - 2B_{12}N_{12})^2 - 4\det B \det N. \tag{74}$$

Then $\Delta \geq 0$, and the stress equation is equivalent to the two scalar conditions

$$\Delta = 0, \qquad \tau = B_{11}/N_{11}. \tag{75}$$

These are necessary tests on the unrestricted reduced zero set, not replacements for R_H, R_q, R_1, R_2.

Proof. For arbitrary C, Z_1, Z_2, ω, g^2, let Q_1, Q_2 be the left sides of the two original spatial equations. Write $\beta = \langle C, [Z_1, Z_2]\rangle$ and $D^C_{jk} = \langle [C, Z_j], [C, Z_k]\rangle$. Cyclicity gives the exact matrix of pairings

$$\langle Z_j, Q_k\rangle = -\omega^2 G_{jk} - D^C_{jk} + (g^2/2)M_{jk} + (d + 2\omega\beta)\delta_{jk}. \tag{76}$$

For the double C commutator the term is $-D^C_{jk}$ by invariance of the trace inner product. The spatial double commutators have diagonal pairing d and off-diagonal pairing zero: expand $\langle Z_j, \mathcal{K}Z_k\rangle = \sum_\ell \langle [Z_\ell, Z_j], [Z_\ell, Z_k]\rangle$. The gyroscopic diagonal pairing in either equation is $2\omega\beta$; the off-diagonal pairings vanish because $\langle X, [C, X]\rangle = 0$. These facts prove every entry of (76), including its symmetry. On a reduced zero, $C = \omega T$, $\omega^2 = \tau$, $D^C = \tau D$, and $\omega\beta = \tau b_T$. Setting the pairing matrix to zero gives $\tau N = B$. Since B is positive definite and $\tau > 0$, N must be positive definite.

Put $n = N_{11} > 0$, $b = N_{12}$, $c = N_{22}$, and $h = nc - b^2 > 0$. Expansion of the following sum of squares is an exact polynomial identity:

$$\begin{aligned} n^2\Delta = {} & \big(B_{11}(nc - 2b^2) + 2bnB_{12} - n^2B_{22}\big)^2 \\ & + 4h(nB_{12} - bB_{11})^2. \end{aligned} \tag{77}$$

Indeed the coefficients of $B_{11}^2, B_{22}^2, B_{12}^2, B_{11}B_{22}, B_{11}B_{12}, B_{22}B_{12}$ on either side are respectively $n^2c^2, n^4, 4n^3c, -2n^3c+4n^2b^2, -4n^2bc, -4n^3b$. Thus $\Delta \geq 0$. Equality holds exactly when $nB_{12} = bB_{11}$ and $B_{11}(nc - 2b^2) + 2bnB_{12} - n^2B_{22} = 0$. Substitution of the first equality in the second gives $n(cB_{11} - nB_{22}) = 0$. Since $n > 0$, the two equalities are exactly $B = (B_{11}/n)N$. Conversely this proportionality makes the sum of squares zero. Finally $B_{11} > 0$, so this proportionality factor is positive, and $\tau N = B$ holds exactly when $\tau = B_{11}/n$. This proves the claimed equivalence and every denominator condition. □

The scalar-coordinate equivalence, the quantitative stabilizer inequality, and the stress discriminant apply to the full complex finite system. They neither force a cross–cross–block ansatz nor establish a zero or emptiness of the unrestricted reduced zero set. Their exact inverse coordinate maps and the reconstruction (65) preserve all original fields, couplings, signs, and both frequency branches.

1.25 An exact bounded finite coupling fibre for every unrestricted spatial pair

Retain the literal matrices H, q, F, all eight real coordinates of each $Z_j \in \mathfrak{su}(3)$, and the original positive coupling g^2. The inner product remains $\langle X, Y\rangle = -\operatorname{tr}(XY)$. The letter t below is a polynomial indeterminate whose evaluated value is the original g^2; no physical constant or amplitude is set to one. This subsection constructs the complete fibre over a fixed ordered pair (Z_1, Z_2) of the four-residual zero set (64). The ordered pair is not constrained to a real subalgebra, a block class, or a prescribed phase relation.

Proposition 1.53 (An explicit upper bound for the original coupling)**.** *For an arbitrary fixed ordered pair with $[Z_1, Z_2] \neq 0$, define*

$$z_0 = \|Z_1\|^2 + \|Z_2\|^2, \qquad z = \sum_{j=1}^{2} \langle Z_j, \mathcal{I} Z_j \rangle, \qquad d = \|[Z_1, Z_2]\|^2,$$

$$c_* = 2 + z, \qquad k_* = 2z_0(z_0 + 4), \qquad t_* = \frac{k_* z + \sqrt{k_*^2 z^2 + 8 k_* d c_*^2}}{c_*^2}.$$

All these numbers are real, with $z \geq 0$ and $z_0, d, c_*, k_*, t_* > 0$. Every full solution over this ordered pair satisfies

$$\|C\|^2 \leq z_0 + 4, \qquad 0 < g^2 \leq t_*, \qquad 0 < \omega^2 \leq \frac{k_* + 2d + t_*(c_* + z/2)}{z_0}. \tag{78}$$

The same bounds apply to both signs of the original frequency.

Proof. We include the signed norm calculation so that the amplitude estimate has no unproved input from a proposed coordinate restriction. In the full coordinates (66), put $h_X = a_X + 2b_X$. Multiplication of the diagonal and off-diagonal entries gives

$$\|X\|^2 = \tfrac{3}{2} a_X^2 + \tfrac{1}{2} h_X^2 + 2(|v_{X1}|^2 + |v_{X2}|^2 + |z_X|^2).$$

The scalar identities (68) give $D_a = 4D_z$ and $D_1 + D_2 = 2 - 4D_z$. Consequently

$$\begin{aligned} \|C\|^2 - z_0 &= \tfrac{3}{2} D_a + \tfrac{1}{2}(h_C^2 - h_{Z_1}^2 - h_{Z_2}^2) + 2(D_1 + D_2 + D_z) \\ &= 4 + \tfrac{1}{2}(h_C^2 - h_{Z_1}^2 - h_{Z_2}^2). \end{aligned}$$

For completeness, the sign of the last correction follows directly from the full scalar cross identities. Write $V_X = (v_{X1}, \sqrt{2} z_X)$, $S_1 = \|V_{Z_1}\|^2 + \|V_{Z_2}\|^2$, and $H_s = h_{Z_1}^2 + h_{Z_2}^2$. Those identities give $h_C V_C = h_{Z_1} V_{Z_1} + h_{Z_2} V_{Z_2}$, while the diagonal identities give $\|V_C\|^2 = S_1 + 1 - \sqrt{2/3}$. Cauchy–Schwarz therefore proves

$$h_C^2 (S_1 + 1 - \sqrt{2/3}) \leq H_s S_1.$$

Since $1 - \sqrt{2/3} > 0$, this implies $h_C^2 \leq H_s$, also when $S_1 = 0$ or $H_s = 0$. This proves $\|C\|^2 \leq z_0 + 4$.

For arbitrary anti-Hermitian matrices X, Y, diagonalize X unitarily, writing its eigenvalues as $i\lambda_a$ with real λ_a. In that same basis,

$$\|[X, Y]\|^2 = \sum_{a,b} (\lambda_a - \lambda_b)^2 |Y_{ab}|^2 \leq 2\Big(\sum_a \lambda_a^2\Big)\Big(\sum_{a,b} |Y_{ab}|^2\Big) = 2\|X\|^2 \|Y\|^2.$$

Here $(\lambda_a - \lambda_b)^2 \leq 2(\lambda_a^2 + \lambda_b^2) \leq 2 \sum_a \lambda_a^2$ proves the inequality term by term. Thus the retained $k = \sum_j \|[C, Z_j]\|^2$ satisfies $k \leq k_*$. The universal identities (52) give $c = c_*$ and

$$(g^2)^2 c_*^2 \leq 8kd + 2g^2 kz \leq 8k_* d + 2g^2 k_* z.$$

The quadratic on moving the right side to the left has positive leading coefficient and roots $(k_* z \pm \sqrt{k_*^2 z^2 + 8 k_* d c_*^2})/c_*^2$. Its negative root is strictly negative, so its nonpositive region at $g^2 > 0$ is exactly $g^2 \leq t_*$. Finally the universal spatial trace identity is

$$\omega^2 z_0 = k + 2d + g^2(c_* + z/2).$$

Substitution of the two upper bounds proves the last assertion. Nonzero bracket implies $z_0, d > 0$; positivity of $\mathcal{I}$ and the definitions prove all remaining strict and weak sign assertions. □

We now eliminate the temporal stabilizer component exactly before clearing any frequency denominator. This preserves the original stabilizer vector $J = iH$, whose squared norm is $3/4$. Set

$$W = \{X \in \mathfrak{su}(3) : \langle J, X\rangle = 0\}, \qquad \Pi X = X - \frac{\langle J, X\rangle}{3/4} J, \qquad \mathscr{B} = [Z_1, Z_2].$$

For the fixed spatial pair define

$$\eta = \langle J, \mathcal{K}J\rangle, \qquad w = \Pi\mathcal{K}J, \qquad \beta = \langle J, \mathscr{B}\rangle,$$

and operators and a vector on the original metric subspace W by

$$\mathcal{I}_7 = \mathcal{I}|_W, \qquad \mathcal{S}_7 = \Pi\mathcal{K}|_W - \frac{w \otimes w}{\eta}, \qquad f_7 = \Pi\mathscr{B} - \frac{\beta}{\eta} w.$$

Here $(w \otimes w)x = w\langle w, x\rangle$; no choice of basis alters any of these maps.

Lemma 1.54 (Full stabilizer elimination and polynomial Gauss numerator)**.** *If $\mathscr{B} \neq 0$, then $\eta > 0$, $\mathcal{S}_7 \succeq 0$, and $\mathcal{I}_7 \succ 0$. Define the polynomial endomorphism and its determinant and adjugate by*

$$\mathscr{M}(t) = t\mathcal{I}_7 + 2\mathcal{S}_7, \qquad D_7(t) = \det_W \mathscr{M}(t), \qquad V_7(t) = 4\,\mathrm{adj}_W(\mathscr{M}(t)) f_7,$$

$$N_7(t) = V_7(t) + \frac{2\beta D_7(t) - \langle w, V_7(t)\rangle}{\eta} J. \tag{79}$$

Then $D_7(t) > 0$ for $t > 0$, D_7 has degree exactly seven and leading coefficient $3/128$, $\deg V_7 \leq 6$, and $\deg N_7 \leq 7$. For every original $g^2 > 0$,

$$T = 4(g^2\mathcal{I} + 2\mathcal{K})^{-1}\mathscr{B} = \frac{N_7(g^2)}{D_7(g^2)}, \qquad Tq = \frac{V_7(g^2)q}{D_7(g^2)}. \tag{80}$$

In particular all temporal coordinates, including the entire stabilizer coefficient, are reconstructed by the displayed formula.

Proof. The positive form of $\mathcal{K}$ gives $\eta = \sum_j \|[Z_j, J]\|^2 \geq 0$. If it were zero, both Z_j would commute with H. The three distinct eigenvalues $0, \sqrt{6}/4, -\sqrt{6}/4$ of H would make both matrices diagonal in its eigenbasis; hence $[Z_1, Z_2] = 0$, a contradiction. Thus $\eta > 0$. The kernel calculation in Lemma 1.48 gives $\ker \mathcal{I} = \mathbb{R}J$. Self-adjointness makes W invariant, and positivity then makes $\mathcal{I}_7$ positive definite.

For $x \in W$ and $s \in \mathbb{R}$, use $\langle J, \mathcal{K}x\rangle = \langle w, x\rangle$ to complete the square:

$$\langle x + sJ, \mathcal{K}(x + sJ)\rangle = \langle x, \mathcal{S}_7 x\rangle + \eta\left(s + \frac{\langle w, x\rangle}{\eta}\right)^2.$$

Taking $s = -\langle w, x\rangle/\eta$ and using positivity of $\mathcal{K}$ proves $\mathcal{S}_7 \succeq 0$. Thus $\mathscr{M}(t)$ is positive definite for every $t > 0$.

Write the full Gauss unknown as $T = x + sJ$ with $x \in W$, with no restriction on s. Pair $(t\mathcal{I} + 2\mathcal{K})T = 4\mathscr{B}$ with J and then apply Π. These operations give respectively

$$s = \frac{2\beta - \langle w, x\rangle}{\eta}, \qquad (t\mathcal{I}_7 + 2\mathcal{S}_7)x = 4f_7.$$

Conversely these two equations recover the J component and the W projection of Gauss; the orthogonal decomposition $\mathfrak{su}(3) = W \oplus \mathbb{R}J$ therefore recovers the entire equation. Inverting the

positive operator on W yields $x = V_7(t)/D_7(t)$, and the formula for s yields (79). Since $Jq = 0$, it also yields the second formula in (80).

Determinants and adjugates may be calculated in any real basis of W; similarity preserves their resulting scalar and endomorphism. The inertia computation (72) gives the seven eigenvalues $1/6, 1/2, 1/2, 1/2, 1/2, 3/2, 3/2$ on W. Thus $\det_W \mathcal{I}_7 = (1/6)(1/2)^4(3/2)^2 = 3/128$. The determinant consequently has degree seven with this leading coefficient; its positivity for $t > 0$ follows from positive definiteness. The adjugate of a seven-by-seven matrix with affine entries has entries of degree at most six, and (79) proves the remaining degree bound. □

Put

$$\alpha = \frac{1}{4} + \|Z_1 q\|^2 + \|Z_2 q\|^2 > 0, \qquad Q_7(t) = \|V_7(t)q\|^2.$$

The real polynomial Q_7 is nonnegative at every real t and has degree at most twelve. Its identically-zero case has the following finite test, which does not require computing a polynomial determinant.

Proposition 1.55 (Three exact vector tests for an empty Gauss domain)**.** *For every fixed non-commuting spatial pair, Q_7 is identically zero if and only if the following three vector equalities all hold:*

$$f_7 q = 0, \qquad (\mathcal{S}_7 f_7)q = 0, \qquad (\mathcal{S}_7^2 f_7)q = 0. \tag{81}$$

When they hold, no positive original coupling over that spatial pair belongs to the original four-residual domain.

Proof. Put $\mathcal{N} = \{X \in W : Xq = 0\}$. In the complete coordinates (66), membership of W sets $h_X = 0$; the first column condition $Xq = 0$ sets $a_X = v_{X1} = v_{X2} = 0$. The only remaining entry is the arbitrary complex z_X. Thus $\dim_{\mathbb{R}} \mathcal{N} = 2$, and the full inertia formula gives $\mathcal{I}_7|_{\mathcal{N}} = (3/2)\,\mathrm{id}_{\mathcal{N}}$. Self-adjointness also makes $\mathcal{N}^{\perp} \cap W$ invariant. Consequently $\mathcal{I}_7^{-1} y \in \mathcal{N}$ if and only if $y \in \mathcal{N}$, and on this subspace the inverse is multiplication by $2/3$.

Suppose first that Q_7 is identically zero. For every $t > 0$ the vector $x(t) = 4(t\mathcal{I}_7 + 2\mathcal{S}_7)^{-1} f_7$ then belongs to $\mathcal{N}$. Set $B_7 = \mathcal{I}_7^{-1}$. Factoring the pencil and using the exact finite inverse identity $(I + A)^{-1} = I - A + A^2 - A^3(I + A)^{-1}$ gives

$$x(t) = \frac{4}{t} B_7 f_7 - \frac{8}{t^2} B_7 \mathcal{S}_7 B_7 f_7 + \frac{16}{t^3} B_7 \mathcal{S}_7 B_7 \mathcal{S}_7 B_7 f_7 + O(t^{-4}).$$

Here the remainder has bounded norm after multiplication by t^4: for $A = 2B_7\mathcal{S}_7/t$ and $t > 4\|B_7\mathcal{S}_7\|$, the inverse has norm at most two, since $\|(I + A)y\| \geq (1 - \|A\|)\|y\| \geq \|y\|/2$. The finite-dimensional subspace $\mathcal{N}$ is closed. Multiplying the expansion by t and taking the limit shows $B_7 f_7 \in \mathcal{N}$, hence $f_7 \in \mathcal{N}$. Subtracting the first term, multiplying by t^2 and taking the limit shows $B_7\mathcal{S}_7 B_7 f_7 \in \mathcal{N}$, hence $\mathcal{S}_7 f_7 \in \mathcal{N}$. Subtracting the next term and multiplying by t^3 similarly gives $\mathcal{S}_7^2 f_7 \in \mathcal{N}$. In the latter two steps every occurrence of B_7 on the already established vectors in $\mathcal{N}$ is exactly multiplication by $2/3$. These are precisely the three displayed tests.

Conversely suppose the three tests hold. If $f_7 = 0$, then $x(t) = 0$ for all $t > 0$ and the conclusion follows. If $f_7 \neq 0$ and $\mathcal{S}_7 f_7$ is dependent on f_7, their one-dimensional span is invariant under $\mathcal{S}_7$. Otherwise those two vectors span all of $\mathcal{N}$, and the third test makes that two-dimensional span invariant as well. In either case let V be the resulting invariant span. It is contained in $\mathcal{N}$, so it is invariant under $\mathcal{I}_7$ and therefore under the positive pencil $t\mathcal{I}_7 + 2\mathcal{S}_7$. Its restriction to V is injective and, in finite dimension, onto V; the full inverse thus carries $f_7 \in V$ back into V. Hence $x(t)q = 0$ for all $t > 0$, and the polynomial Q_7 vanishes on the entire positive half-line. A polynomial with infinitely many roots is identically zero. Formula (80) then proves the asserted exclusion from the original domain for every $g^2 > 0$. □

Define four polynomial residuals with the original matrix and vector codomains:

$$\begin{aligned}
P_H(t) &= \alpha[N_7,[N_7,H]] - Q_7\left(\sum_j [Z_j,[Z_j,H]] + F\right),\\
P_q(t) &= \alpha N_7^2 q + Q_7(2I_3 - Z_1^2 - Z_2^2)q,\\
P_1(t) &= \alpha\left(-D_7^2 Z_1 - 2D_7[N_7,Z_2] + [N_7,[N_7,Z_1]]\right) + Q_7\left(\mathcal{K}Z_1 + \frac{t}{2}\mathcal{I}Z_1\right),\\
P_2(t) &= \alpha\left(-D_7^2 Z_2 + 2D_7[N_7,Z_1] + [N_7,[N_7,Z_2]]\right) + Q_7\left(\mathcal{K}Z_2 + \frac{t}{2}\mathcal{I}Z_2\right).
\end{aligned} \tag{82}$$

All polynomials on the right are evaluated at the same t. Their codomains are respectively $\mathrm{Herm}_3(\mathbb{C})_0$, $\mathbb{C}^3$, $\mathfrak{su}(3)$, and $\mathfrak{su}(3)$ for real t. Define the single real polynomial

$$p_Z(t) = \mathrm{tr}(P_H(t)^2) + P_q(t)^* P_q(t) - \mathrm{tr}(P_1(t)^2) - \mathrm{tr}(P_2(t)^2). \tag{83}$$

Proposition 1.56 (The exact degree and positive leading coefficient)**.** *Retain an arbitrary fixed ordered pair $Z_1, Z_2 \in \mathfrak{su}(3)$ with $\mathscr{B} = [Z_1, Z_2] \neq 0$, and all original definitions above: $J, \eta, \beta, \mathcal{I}_7$, D_7, V_7, N_7, Q_7, α, and P_H, P_q, P_1, P_2, p_Z. Put*

$$\delta_7 = \det_W \mathcal{I}_7 = \frac{3}{128}, \qquad a_J = \frac{2\beta}{\eta}, \qquad A_J X = [J, X],$$

and define an endomorphism of the original real product metric space $\mathfrak{su}(3) \oplus \mathfrak{su}(3)$ by

$$\mathcal{L}_{a_J}(X_1, X_2) = (X_1 + a_J A_J X_2,\ X_2 - a_J A_J X_1).$$

Here a_J is an explicitly defined scalar belonging to the fixed spatial pair; it is not a new coupling or a rescaling of the original fields. The coefficient of t^{28} in the original polynomial p_Z is

$$[t^{28}]p_Z = \alpha^2 \delta_7^4 \left\|\mathcal{L}_{a_J}^2(Z_1, Z_2)\right\|^2. \tag{84}$$

If Q_7 is not identically zero, this coefficient is strictly positive. Consequently $\deg p_Z = 28$ *on every such spatial pair.*

Proof. First write $N_7 = V_7 + s_7 J$. The previously proved bounds are $\deg V_7 \leq 6$, $\deg s_7 \leq 7$, and $\deg Q_7 \leq 12$. Since $[J, H] = 0$ and $Jq = 0$, the exact expressions are

$$[N_7,[N_7,H]] = [V_7,[V_7,H]] + s_7[J,[V_7,H]], \qquad N_7^2 q = V_7^2 q + s_7 J V_7 q.$$

Hence $\deg P_H, \deg P_q \leq 13$. The full spatial formulas give $\deg P_1, \deg P_2 \leq 14$, including the $tQ_7\mathcal{I}/2$ terms. Taking their displayed squared norms yields $\deg p_Z \leq 28$. The established degree statements give

$$D_7(t) = \delta_7 t^7 + \text{terms of degree at most six}, \qquad V_7(t) = \text{terms of degree at most six}.$$

The defining full stabilizer numerator is

$$N_7(t) = V_7(t) + \frac{2\beta D_7(t) - \langle w, V_7(t)\rangle}{\eta} J.$$

Its coefficient of t^7 is therefore exactly $a_J\delta_7 J$, also when $\beta = 0$ and that coefficient is zero. Since $\deg Q_7 \le 12$, its contributions $Q_7(\mathcal{K}Z_j + t\mathcal{I}Z_j/2)$ have degree at most thirteen. It follows directly from the full spatial numerator formulas that

$$[t^{14}]P_1 = \alpha\delta_7^2(-Z_1 - 2a_J A_J Z_2 + a_J^2 A_J^2 Z_1),$$
$$[t^{14}]P_2 = \alpha\delta_7^2(-Z_2 + 2a_J A_J Z_1 + a_J^2 A_J^2 Z_2).$$

Direct composition, with both signs retained, gives

$$\mathcal{L}_{a_J}^2(Z_1, Z_2) = (Z_1 + 2a_J A_J Z_2 - a_J^2 A_J^2 Z_1, Z_2 - 2a_J A_J Z_1 - a_J^2 A_J^2 Z_2).$$

The pair of leading spatial coefficients is thus $-\alpha\delta_7^2\mathcal{L}_{a_J}^2(Z_1, Z_2)$. The scalar numerator degrees satisfy $\deg P_H, \deg P_q \le 13$, so their squared norms contribute no term of degree twenty-eight. The two spatial squared norms therefore give exactly (84).

It remains to prove strict positivity under the stated domain-polynomial hypothesis. Invariance of $\langle X, Y\rangle = -\operatorname{tr}(XY)$ gives

$$\langle X, A_J Y\rangle = -\langle A_J X, Y\rangle.$$

For arbitrary $X = (X_1, X_2)$ and $Y = (Y_1, Y_2)$, this implies

$$\begin{aligned}\langle X, \mathcal{L}_{a_J} Y\rangle &= \langle X_1, Y_1\rangle + \langle X_2, Y_2\rangle + a_J\langle X_1, A_J Y_2\rangle - a_J\langle X_2, A_J Y_1\rangle \\ &= \langle X_1 + a_J A_J X_2, Y_1\rangle + \langle X_2 - a_J A_J X_1, Y_2\rangle = \langle \mathcal{L}_{a_J} X, Y\rangle.\end{aligned}$$

Thus $\mathcal{L}_{a_J}$ is self-adjoint on this positive metric space. If the coefficient in (84) were zero, then $\mathcal{L}_{a_J}^2 Z = 0$ for $Z = (Z_1, Z_2)$. Consequently

$$\|\mathcal{L}_{a_J} Z\|^2 = \langle Z, \mathcal{L}_{a_J}^2 Z\rangle = 0,$$

so $\mathcal{L}_{a_J} Z = 0$. If $a_J = 0$, this gives $Z_1 = Z_2 = 0$, contradicting $\mathscr{B} \neq 0$. Hence $a_J \neq 0$, and the two component equations are

$$a_J[J, Z_1] = Z_2, \qquad a_J[J, Z_2] = -Z_1.$$

The full spatial operator now satisfies the exact identity

$$\begin{aligned}a_J\mathcal{K}J &= -a_J[Z_1, [Z_1, J]] - a_J[Z_2, [Z_2, J]] \\ &= a_J[Z_1, [J, Z_1]] + a_J[Z_2, [J, Z_2]] \\ &= [Z_1, Z_2] + [Z_2, -Z_1] = 2\mathscr{B}.\end{aligned}$$

Apply the original orthogonal projection Π to this equality and use $w = \Pi\mathcal{K}J$ and $a_J/2 = \beta/\eta$. It yields

$$f_7 = \Pi\mathscr{B} - \frac{\beta}{\eta} w = \frac{a_J}{2} w - \frac{a_J}{2} w = 0.$$

The exact definition $V_7(t) = 4\operatorname{adj}_W(\mathscr{M}(t))f_7$ then makes V_7 identically zero, and hence makes $Q_7(t) = \|V_7(t)q\|^2$ identically zero. This contradicts the hypothesis. The coefficient is nonnegative by its squared-norm formula, so it must be strictly positive. Since the global degree bound is twenty-eight, the degree is exactly twenty-eight. □

Theorem 1.57 (Complete finite coupling fibre and its exact inverse)**.** *For every ordered pair $Z_1, Z_2 \in \mathfrak{su}(3)$ the complete fibre of the four-residual zero set is as follows. If $[Z_1, Z_2] = 0$, it is empty. Otherwise form the explicitly defined polynomials above. If Q_7 is identically zero, the fibre is*

empty. If Q_7 is not identically zero, then p_Z is not identically zero, $\deg p_Z = 28$*, and the complete set of admissible couplings is exactly*

$$\{t \in (0, t_*] : Q_7(t) > 0,\ p_Z(t) = 0\}\,. \tag{85}$$

This set has at most fourteen elements. Equivalently, the projection $\mathcal{Z} \longrightarrow \mathfrak{su}(3)^2$ that forgets g^2 has fibres of cardinality at most fourteen, including empty fibres.

For each root in (85) *and each $\varepsilon \in \{+1, -1\}$, the complete original finite-system solution is*

$$g^2 = t, \qquad \omega = \varepsilon \frac{\sqrt{\alpha}\, D_7(t)}{\sqrt{Q_7(t)}}, \qquad C = \varepsilon \frac{\sqrt{\alpha}\, N_7(t)}{\sqrt{Q_7(t)}}, \qquad (Z_1, Z_2) = (Z_1, Z_2). \tag{86}$$

Thus every fibre of the five-equation solution set over the fixed ordered spatial pair has at most twenty-eight elements, counting both frequency signs. No root with $Q_7(t) = 0$ is included.

Proof. A zero spatial bracket is impossible by Lemma 1.48. For a nonzero bracket, (80) gives $\|Tq\|^2 = Q_7(t)/D_7(t)^2$ for every $t > 0$. Thus the exact original domain condition $Tq \neq 0$ is precisely $Q_7(t) > 0$. If Q_7 is identically zero, the domain has no point over this pair. On the remaining domain the unchanged frequency is

$$\tau = \frac{\alpha D_7(t)^2}{Q_7(t)}.$$

Substitution in all four full residuals gives, entry by entry,

$$(R_H, R_q, R_1, R_2) = \frac{1}{Q_7(t)}(P_H(t), P_q(t), P_1(t), P_2(t)). \tag{87}$$

Every denominator here is strictly positive on the asserted domain. For real t, each summand in (83) is a squared norm in its stated real inner-product space. The Hermitian and anti-Hermitian assertions follow directly from the commutators in (82); they do not require a residual to vanish. Therefore $p_Z(t) \geq 0$ for every real t, and its vanishing is equivalent to vanishing of every full polynomial residual. This proves the exact correspondence of roots and full residual zeros, before the bound on the number of roots is established.

To prove the degree bounds, write $N_7 = V_7 + s_7 J$, with $\deg V_7 \leq 6$ and $\deg s_7 \leq 7$. Since $[J, H] = 0$ and $Jq = 0$,

$$[N_7, [N_7, H]] = [V_7, [V_7, H]] + s_7[J, [V_7, H]], \qquad N_7^2 q = V_7^2 q + s_7 J V_7 q.$$

Their degrees are at most thirteen. Hence $\deg P_H, \deg P_q \leq 13$. In the two spatial residuals the terms $D_7^2 Z_j$, $D_7[N_7, Z_k]$, and $[N_7, [N_7, Z_j]]$ have degree at most fourteen; the terms involving $Q_7\mathcal{K}$ and $tQ_7\mathcal{I}$ have degree at most twelve and thirteen. Thus $\deg P_1, \deg P_2 \leq 14$ and $\deg p_Z \leq 28$.

Suppose now that Q_7 is not identically zero but p_Z is identically zero. A nonzero univariate polynomial has only finitely many real roots, so there exists $t > t_*$ with $Q_7(t) \neq 0$. Nonnegativity of Q_7 gives $Q_7(t) > 0$. The established exact residual correspondence then makes this t a full zero; the two-sign inverse of Theorem 1.49 reconstructs a solution of all five original equations. This contradicts Proposition 1.53. Consequently p_Z is a nonzero polynomial. Every real root of a real polynomial that is nonnegative on all of $\mathbb{R}$ has even multiplicity: at a root t_0, factor it as $(t - t_0)^m u(t)$ with $u(t_0) \neq 0$; continuity makes u have a constant nonzero sign near t_0, and an odd m would give opposite signs on the two sides. Thus a nonzero polynomial of degree at most twenty-eight of this type has at most fourteen distinct real roots. Imposing the explicit interval and denominator conditions can only decrease that number.

The stronger equality $\deg p_Z = 28$ follows from the independent leading-coefficient proof in Proposition 1.56.

Finally $D_7(t) > 0$ identifies the sign of the displayed ω with ε and gives $C = \omega T$ exactly. The frequency square is the unchanged τ. Formula (87) proves that all remaining equations vanish, while the Schur inverse proves the full Gauss equation. Conversely every original solution determines its fixed pair, its root $t = g^2$, and its unique sign, and these formulas recover C, ω. This proves the stated bijection and both cardinality bounds. □

The original spacetime reconstruction is retained literally: with the already specified $O(t_{\mathrm{phys}})$, K, and Ω, put

$$Y_1(t_{\mathrm{phys}}) = \cos(\omega t_{\mathrm{phys}})Z_1 - \sin(\omega t_{\mathrm{phys}})Z_2,$$
$$Y_2(t_{\mathrm{phys}}) = \sin(\omega t_{\mathrm{phys}})Z_1 + \cos(\omega t_{\mathrm{phys}})Z_2, \qquad Y_3 = 0.$$
$$\mathcal{A}_0 = OKCK^*O^* - \Omega, \qquad \mathcal{A}_i = OKY_iK^*O^*.$$

The prescribed scalar fields remain $\Phi(t_{\mathrm{phys}}) = O\Phi_0O^*$ and $b(t_{\mathrm{phys}}) = Ob_0$, with their original frequency $\sqrt{2}$. The polynomial indeterminate $t = g^2$ is distinct from the physical time t_{phys}.

This theorem solves the coupling fibre exactly as a bounded finite root set while keeping every unrestricted spatial coordinate. It does not establish that any ordered spatial pair has a nonempty fibre, or that all such fibres are empty. Existence or emptiness of the complete sixteen-coordinate spatial projection remains unresolved by this result.

1.26 A proper algebraic exceptional set for the full spatial projection

We now retain all sixteen spatial coordinates simultaneously. The new statement is about the entire spatial projection, not only a selected block class or one value of the coupling. The five original equations and their prescribed rotating ansatz remain unchanged.

Use the exact fixed matrix U of (66). For each $j = 1, 2$, write $\widetilde{Z}_j = U^*Z_jU$ using the eight real coordinates $(a_j, h_j, \operatorname{Re} v_{j1}, \operatorname{Im} v_{j1}, \operatorname{Re} v_{j2}, \operatorname{Im} v_{j2}, \operatorname{Re} z_j, \operatorname{Im} z_j)$, with $b_j = (h_j - a_j)/2$. This is an invertible linear coordinate map on the original spatial carrier. Denote the resulting sixteen-coordinate vector by ξ. No coordinate is held at zero in the theorem below.

Lemma 1.58 (Polynomial diagonal residuals with retained denominators)**.** *Let $\eta = \langle J, \mathcal{K}J\rangle$ be the original nonnegative polynomial in ξ. On $\eta > 0$ define*

$$F_\xi(t) = \eta^{16}(U^*P_H(t)U)_{11}, \qquad G_\xi(t) = \eta^{16}(U^*P_H(t)U)_{22}.$$

Both extend as polynomials in (ξ, t) with real algebraic coefficients, and their degree in t is at most twelve. On the stated domain the inverse formulas are exactly $(U^*P_HU)_{11} = F_\xi/\eta^{16}$ and $(U^*P_HU)_{22} = G_\xi/\eta^{16}$.

Proof. All real coordinate components of $\mathcal{K}, w, \eta, \beta, \Pi\mathscr{B}$ are polynomials in ξ with fixed real algebraic coefficients; the projection denominator $\langle J, J\rangle = 3/4$ is a fixed nonzero constant. Choose once and for all a fixed basis of W with real algebraic coordinates in the displayed diagonal-frame carrier. The entries of $\mathscr{M} = t\mathcal{I}_7 + 2\mathcal{S}_7$ have denominator dividing η, and those of f_7 have the same property. The determinant expansion shows that η^7D_7 is polynomial. Each adjugate entry is a six-by-six minor, so η^7V_7 is polynomial as well. The full formula (79) therefore makes η^8N_7 polynomial. Complex conjugation here conjugates fixed coefficients and the displayed complex coordinate combinations, while every component of ξ and t is real. Thus $\eta^{14}Q_7$ is polynomial. The full formula for P_H now proves polynomial extension after the displayed multiplication by η^{16}. No value at $\eta = 0$ is treated as an inverse Gauss solution.

To prove the sharper degree bound, work through the exact conjugation by U^*, so $H_d = \operatorname{diag}(0, r, -r)$ and $J_d = iH_d$ are both diagonal. Write $N_7 = V_7 + s_7 J$ as before. Since $[J, H] = 0$,

$$[N_7, [N_7, H]] = [V_7, [V_7, H]] + s_7[J, [V_7, H]].$$

For every matrix Y, $([J_d, Y])_{jj} = 0$ follows by subtracting $(J_d)_{jj}Y_{jj}$ from itself. Consequently the second term contributes zero to each diagonal entry after conjugation. The first term has degree at most twelve since $\deg_t V_7 \le 6$. The other term in P_H is Q_7 times a matrix independent of t, and $\deg_t Q_7 \le 12$. Multiplication by η^{16} does not change the degree in t. This proves every assertion. □

Define a fixed-size polynomial matrix $\mathscr{S}(\xi)$ by the linear map

$$\mathbb{R}[t]_{<12} \oplus \mathbb{R}[t]_{<12} \longrightarrow \mathbb{R}[t]_{<24}, \qquad (A, B) \longmapsto AF_\xi + BG_\xi, \qquad \mathscr{R}(\xi) = \det \mathscr{S}(\xi). \tag{88}$$

Use the increasing-power bases $1, t, \dots$ and put the A summand before the B summand. This specifies the determinant sign and its twenty-four columns, including when either polynomial loses degree. This is the classical Sylvester resultant map; see the Stacks Project, Tag 00UA. The argument below proves the properties needed here directly, without replacing either polynomial by a monic one.

Theorem 1.59 (Generic exclusion in all sixteen spatial coordinates)**.** *The explicitly defined polynomial $\mathscr{R}$ is not identically zero. The spatial projection of every full solution of* (46)–(50) *is contained in*

$$\{\xi \in \mathbb{R}^{16} : \mathscr{R}(\xi) = 0\}.$$

This exceptional set is closed, has empty interior, and has sixteen-dimensional Lebesgue measure zero. Thus the complement is an open dense set of unrestricted spatial pairs, each admitting no positive coupling and no frequency sign solving the five retained equations. This theorem does not decide whether any point of the exceptional set admits a solution.

Proof. Any full solution has $[Z_1, Z_2] \neq 0$, hence $\eta > 0$ by Lemma 1.54. It also has $Q_7(t) > 0$ at the original $t = g^2$, and the exact residual identity (87) makes the entire $P_H(t)$ zero. Therefore $F_\xi(t) = G_\xi(t) = 0$. Evaluation at that real t is a nonzero linear functional on $\mathbb{R}[t]_{<24}$ (it sends 1 to 1), and it annihilates the image of (88). That image cannot be all of the codomain. The determinant is zero. This proof includes every degree-drop case and does not use a converse resultant criterion outside the original denominator domain.

Here is an exact witness that the determinant polynomial is nonzero. Write

$$M(a, h, v_1, v_2, z) = \begin{pmatrix} ia & -\bar{v}_1 & -\bar{v}_2 \\ v_1 & i(h-a)/2 & z \\ v_2 & -\bar{z} & -i(a+h)/2 \end{pmatrix}$$

and set, in the displayed diagonal coordinates,

$$\widetilde{Z}_1 = M(1, 2, 1+i, -1+2i, 2-i), \qquad \widetilde{Z}_2 = M(-2, 1, 2+3i, 1-i, -1+2i).$$

The original pair is $Z_j = U\widetilde{Z}_j U^*$, not a replacement of the prescribed H, q, F. Every one of its sixteen displayed real spatial coordinates is nonzero. Substitution in the original definitions of D_7, V_7, N_7, Q_7, P_H gives two diagonal polynomials of degree twelve in $\mathbb{Q}(\sqrt{6})[t]$. Their coprimality can be verified by the following small certificate, with all coefficient lists in descending powers. Reduce their coefficients by the explicit map $\sqrt{6} \mapsto 39$ to $\mathbb{F}_{101}$, where $39^2 = 6$. Each rational

coefficient denominator is nonzero modulo 101; the exact coefficient file and replay test check them individually. The two reduced polynomials f, g and the Bezout coefficients a, b are

$$\begin{aligned} f &: (39, 85, 79, 93, 67, 20, 38, 98, 26, 56, 8, 25, 28), \\ g &: (1, 87, 96, 7, 76, 88, 75, 93, 53, 73, 64, 41, 16), \\ a &: (4, 60, 31, 59, 74, 31, 34, 18, 59, 97, 95, 96), \\ b &: (46, 85, 47, 5, 14, 36, 38, 21, 35, 48, 34, 53). \end{aligned}$$

Multiplication gives $af + bg = 1$ in $\mathbb{F}_{101}[t]$. Both f and g have degree exactly twelve. If a pair (a_0, b_0) of degrees less than twelve were in the kernel of their Sylvester map, coprimality would imply that f divides b_0 and g divides a_0. The degree bounds force both to be zero. Thus that finite-field map, between spaces of equal dimension twenty-four, has nonzero determinant. Reduction commutes with the determinant polynomial. It follows that the Sylvester determinant of the two original diagonal polynomials is nonzero in $\mathbb{Q}(\sqrt{6})$ as well. This uses the localized coefficient ring with all its denominators inverted, not a homomorphism from the whole field $\mathbb{Q}$ to a finite field.

For explicit verification at the witness, multiplication gives

$$[\widetilde{Z}_1, \widetilde{Z}_2]_{21} = 9 - \tfrac{7}{2}i \neq 0, \qquad \eta = \tfrac{3}{4}(7 + 15) + 3(5 + 5) = \tfrac{93}{2} > 0.$$

Multiplying both diagonal polynomials by the same nonzero scalar η^{16} multiplies each of the twenty-four columns by η^{16}, and hence multiplies their determinant by $\eta^{384} \neq 0$. Consequently $\mathscr{R}$ is nonzero at this exact witness and cannot be the zero polynomial. The finite-field calculation is solely a certificate of this real algebraic statement; it does not change the physical carrier or any coupling.

As a separate certificate, write $j_0 = \langle J, J \rangle = 3/4$. In $W \oplus \mathbb{R}J$, without changing the length of J, the full Gauss endomorphism has block matrix

$$L = \begin{pmatrix} t\mathcal{I}_7 + 2\Pi\mathcal{K}|_W & 2w \\ 2\langle w, \cdot \rangle / j_0 & 2\eta / j_0 \end{pmatrix}.$$

Subtracting $j_0 w/\eta$ times its last row from the upper block rows preserves the determinant and makes the upper-right block zero; its upper-left block becomes $\mathscr{M}(t)$. Thus $\det L = (2\eta/j_0) D_7$ exactly. The independent replay reconstructs both original diagonal polynomials from that matrix modulo 101. It obtains Sylvester determinant 40; after the retained η^{16} multiplication it obtains $40\eta^{384} = 60 \neq 0$ in $\mathbb{F}_{101}$. These residues refer to the exact basis order in (88).

For completeness, a nonzero real polynomial on $\mathbb{R}^n$ has a measure-zero zero set by induction on n. For $n = 1$ it has finitely many roots. For $n > 1$, write it as a polynomial in the last variable whose coefficients are polynomials in the first $n - 1$ variables. Choose one nonzero coefficient. The set where all coefficients vanish is contained in the measure-zero zero set of that coefficient, by induction. At every other base point the last-variable polynomial has only finitely many roots. Applying Fubini on each bounded box shows that the zero set in that box has measure zero; countably many boxes cover $\mathbb{R}^n$. Its complement is open because the polynomial is continuous. Every nonempty open ball has positive measure, so no ball is contained in the zero set, and the complement is dense. This proves the assertions at $n = 16$. □

The exact large-coefficient identity over $\mathbb{Q}(\sqrt{6})$ and the small finite-field identity are retained separately in `PAIR_BEZOUT_CERTIFICATE.json` and `PAIR_MODULAR_CERTIFICATE.json`. The replay `check_pair_bezout.py` constructs the original full pair and every Gauss and scalar numerator, then checks the identities; the integer-only certificate replay checks the displayed finite-field products separately. Neither a finite failed search nor two failed test couplings proves this theorem: the proof uses an explicit nonzero polynomial vanishing on every possible spatial projection, with its nonzeroness certified at the literal witness.

1.27 An eight-dimensional exceptional spatial class with no full solution

1.27.1 Literal system and the explicitly selected class

All vector spaces in this note retain their real structures. In particular, $\mathfrak{su}(3)$ is the eight-real-dimensional space of traceless anti-Hermitian complex 3×3 matrices, with positive inner product $\langle X,Y\rangle = -\operatorname{tr}(XY)$. Retain the original matrices

$$H=\begin{pmatrix}0 & 0 & 0\\ 0 & -1/2 & 1/(2\sqrt{2})\\ 0 & 1/(2\sqrt{2}) & 1/2\end{pmatrix},\quad q=\frac{e_1}{2\sqrt{2}},\quad A=\sqrt{2}\begin{pmatrix}0 & -1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 0\end{pmatrix},$$

and the original scalar constants $m^2=\Gamma=1$, $\kappa=6$. For $Y_{\mathrm{tf}}=Y-\operatorname{tr}(Y)I_3/3$, define

$$\mathcal{I}X=([H,[H,X]]+qq^*X+Xqq^*)_{\mathrm{tf}},\qquad \mathcal{K}X=-[Z_1,[Z_1,X]]-[Z_2,[Z_2,X]],$$

$$F=[A,[A,H]]=-2H+\operatorname{diag}(-2,1,1).$$

The last identity is verified by the two displayed matrix products; also $H^2=(3/8)\operatorname{diag}(0,1,1)$ and $q^*q=1/8$. The indeterminate/value $t=g^2$ is the original positive gauge coupling, not physical time. The original frequency satisfies $\omega\in\mathbb{R}\setminus\{0\}$. The five retained equations are

$$[C,[C,H]]-[Z_1,[Z_1,H]]-[Z_2,[Z_2,H]]=F, \tag{89}$$

$$(C^2-Z_1^2-Z_2^2)q=-2q, \tag{90}$$

$$(t\mathcal{I}+2\mathcal{K})C=4\omega[Z_1,Z_2], \tag{91}$$

$$-\omega^2Z_1-2\omega[C,Z_2]+[C,[C,Z_1]]+\mathcal{K}Z_1+(t/2)\mathcal{I}Z_1=0, \tag{92}$$

$$-\omega^2Z_2+2\omega[C,Z_1]+[C,[C,Z_2]]+\mathcal{K}Z_2+(t/2)\mathcal{I}Z_2=0. \tag{93}$$

Multiplying the original Gauss equation by its retained positive t gives (91); division by that same t is its exact inverse.

Put $P=qq^*/(q^*q)=\operatorname{diag}(1,0,0)$ and define

$$\mathfrak{p}=\{X\in\mathfrak{su}(3):PXP=(I_3-P)X(I_3-P)=0\},\qquad \mathfrak{k}=\{X\in\mathfrak{su}(3):[X,P]=0\}.$$

Thus $\mathfrak{p}$ exchanges the matter line and its orthogonal complement, while $\mathfrak{k}$ preserves them. Every $X\in\mathfrak{p}$ has the unique form

$$X(v)=\begin{pmatrix}0 & -v^*\\ v & 0_{2\times 2}\end{pmatrix},\qquad v\in\mathbb{C}^2,$$

whose inverse takes the lower first column. This is a real linear isomorphism $\mathbb{C}^2\to\mathfrak{p}$, so $\mathfrak{p}\times\mathfrak{p}$ has real dimension eight. Block decomposition gives the orthogonal direct sum $\mathfrak{su}(3)=\mathfrak{k}\oplus\mathfrak{p}$, with both summands of real dimension four.

Theorem 1.60 (Two-cross spatial exclusion)**.** *There is no solution of* (89)–(93) *with both* $Z_1,Z_2\in\mathfrak{p}$. *No hypothesis on* C *is imposed in this statement. Every complex coordinate of each of the two cross vectors is allowed, and both signs of* $\omega\neq 0$ *and every original* $t>0$ *are retained.*

This is an explicitly restricted spatial class, not a claim that arbitrary spatial pairs can be put into this class. The proof below also shows that the entire class lies in the previously defined real algebraic exceptional zero set. Consequently the result excludes an eight-dimensional linear subspace inside that zero set, but does not decide existence in its remaining part.

1.27.2 Gauss and the complete scalar equations force the temporal form

Block multiplication gives $[\mathfrak{p},\mathfrak{p}] \subset \mathfrak{k}$, $[\mathfrak{k},\mathfrak{p}] \subset \mathfrak{p}$. Because H and qq^* are block diagonal, $\mathcal{I}$ preserves both summands. Direct multiplication using H^2 and q^*q gives

$$\mathcal{I}X(v) = \left(\frac{3}{8} + \frac{1}{8}\right) X(v) = \frac{1}{2}X(v). \tag{94}$$

In this computation $HX(v)H = 0$, the two H^2 products sum to $(3/8)X(v)$, the two qq^* products sum to $X(v)/8$, and every term is traceless. If $Z_1, Z_2 \in \mathfrak{p}$, their double commutators therefore make $\mathcal{K}$ preserve both summands as well. Cyclicity gives, for every $Y \in \mathfrak{su}(3)$,

$$\langle Y, \mathcal{K}Y\rangle = \sum_{j=1}^{2} \|[Z_j, Y]\|^2 \geq 0.$$

Project (91) onto $\mathfrak{p}$. Its right side lies in $\mathfrak{k}$, hence

$$\left(\frac{t}{2}\,\mathrm{id}_{\mathfrak{p}} + 2\mathcal{K}|_{\mathfrak{p}}\right) C_{\mathfrak{p}} = 0.$$

Pairing with $C_{\mathfrak{p}}$ gives a sum whose first term is $(t/2)\|C_{\mathfrak{p}}\|^2$ and whose other terms are nonnegative. Since $t > 0$, $C_{\mathfrak{p}} = 0$. Thus Gauss forces $C \in \mathfrak{k}$; this was not an additional restriction on the temporal unknown.

Here are all frame changes, together with their inverses. Put

$$x = \frac{1}{2}, \quad d_0 = \frac{1}{2\sqrt{2}}, \quad r = \frac{\sqrt{6}}{4}, \quad \delta = \frac{1}{2r} = \sqrt{\frac{2}{3}},$$

$$S = \frac{1}{\sqrt{2r(r+x)}} \begin{pmatrix} d_0 & -(r+x) \\ r+x & d_0 \end{pmatrix}, \qquad U = \mathrm{diag}(1, S).$$

The relation $r^2 = x^2 + d_0^2$ shows that the numerator columns have squared length $2r(r+x)$, are orthogonal, and have determinant $2r(r+x) > 0$. Multiplication by the lower block of H gives their eigenvalues $r, -r$ in that order. Hence

$$U \in SO(3), \qquad U^*HU = \mathrm{diag}(0, r, -r), \qquad U^*q = q.$$

For every matrix Y the change is $Y \mapsto U^*YU$, with inverse $Y' \mapsto UY'U^*$. It is applied to H, F, C, Z_1, Z_2 simultaneously. It commutes with products, brackets, adjoints and trace projection, and fixes q. Thus it intertwines $\mathcal{I}, \mathcal{K}$ and each of the five full residuals exactly. It preserves $\mathfrak{p}, \mathfrak{k}$ because it commutes with P. No physical constant is rescaled.

In this first frame write the full temporal block matrix as

$$C_d = \begin{pmatrix} ia & 0 & 0 \\ 0 & i(h-a)/2 & z \\ 0 & -\bar{z} & -i(a+h)/2 \end{pmatrix}, \qquad a, h \in \mathbb{R}, \quad z \in \mathbb{C}.$$

Write the spatial matrices as $X(v), X(w)$ and put $M = vv^* + ww^*$ in this frame only for the following calculation. The vector scalar residual is exactly

$$(C_d^2 - X(v)^2 - X(w)^2)q + 2q = (-a^2 + \mathrm{tr}\, M + 2)q.$$

The complete nonzero block of the adjoint scalar residual, including its upper entry, is

$$\begin{pmatrix} 2-2r(M_{11}-M_{22}) & 0 & 0 \\ 0 & -4r|z|^2+2rM_{11}-1+2r & -2irhz \\ 0 & 2irh\bar{z} & 4r|z|^2-2rM_{22}-1-2r \end{pmatrix}.$$

For example, if the lower Hermitian block $B=-i(C_d)_{\text{lower}}$ and $D=\operatorname{diag}(r,-r)$ are used, its lower block is $-[B,[B,D]]+\{M,D\}-(I_2-2D)$; multiplying these two-by-two matrices gives every entry just displayed. The two scalar equations are therefore equivalent to

$$a^2=2+\operatorname{tr}M, \qquad 2|z|^2-M_{11}=1-\delta, \qquad 2|z|^2-M_{22}=1+\delta, \qquad hz=0. \tag{95}$$

The upper entry is minus the sum of the two lower diagonal entries, so no scalar entry is left unchecked. Adding the middle two equations and comparing with the first gives $a^2=4|z|^2$. Since $M_{22}\geq 0$ and $1+\delta>0$, $z\neq 0$. Consequently

$$h=0, \qquad a\neq 0, \qquad |z|=|a|/2. \tag{96}$$

Set $e=a$, retaining either sign, and $\gamma=-2iz/e$, so $|\gamma|=1$. Choose either square root ζ of $-\bar{\gamma}$ and put

$$W=\frac{\zeta}{\sqrt{2}}\begin{pmatrix}\gamma & \gamma \\ 1 & -1\end{pmatrix}, \qquad V=\operatorname{diag}(1,W).$$

Then $|\zeta|=1$, $W^*W=I_2$ and $\det W=-\zeta^2\gamma=\bar{\gamma}\gamma=1$. The lower Hermitian block of C_d is

$$\frac{e}{2}\begin{pmatrix}-1 & \gamma \\ \bar{\gamma} & -1\end{pmatrix}.$$

Its products with $(\gamma,1)^{\mathsf{T}}$ and $(\gamma,-1)^{\mathsf{T}}$ have eigenvalues $0,-e$, respectively. Those same columns give $W^*\operatorname{diag}(r,-r)W=r\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$. Thus the second exact change $Y_d\mapsto V^*Y_dV$ gives

$$C'=ie\operatorname{diag}(1,0,-1), \quad H'=\begin{pmatrix}0&0&0\\0&0&r\\0&r&0\end{pmatrix}, \quad q'=q, \quad F'=-2H'+\operatorname{diag}(-2,1,1). \tag{97}$$

The inverse for every original matrix is, explicitly,

$$Y=UVY'V^*U^*. \tag{98}$$

For spatial vectors the forward change is the corresponding lower unitary action $v\mapsto W^*v$ and its inverse is $v'\mapsto Wv'$. The unitary parameters are retained; the assertion is not that H can be changed while the other data are left fixed. As with U, this simultaneous conjugation intertwines all five equations. In the remainder of the proof primes are omitted only to ease typography; all matrices are the exact objects in (97) and have inverse (98).

1.27.3 Every gauge entry in the classified frame

Write $Z_1=X(v)$, $Z_2=X(w)$ with arbitrary $v,w\in\mathbb{C}^2$, and define, afresh in the second frame,

$$V_0=v^*v, \quad W_0=w^*w, \quad p_0=v^*w=\alpha_0+ib, \quad M=vv^*+ww^*, \quad N=i(vw^*-wv^*).$$

Both M and N are Hermitian; $M \succeq 0$. Put

$$u = M_{11}, \quad v_0 = M_{22}, \quad m = M_{12}, \quad d_M = \det M = |v_1 w_2 - v_2 w_1|^2 \geq 0.$$

The determinant identity follows by expanding the four products in $M_{11}M_{22} - M_{12}M_{21}$ and collecting the squared modulus. The symbol v_0 here is a real diagonal entry, not the vector v. All eight real components of the two vectors remain independent.

Multiplication gives the full scalar residuals in this frame:

$$Q_H = \begin{pmatrix} 2 - 4r \operatorname{Re} m & 0 & 0 \\ 0 & 2r \operatorname{Re} m - 1 & r(\operatorname{tr} M - e^2 + 2) \\ 0 & r(\operatorname{tr} M - e^2 + 2) & 2r \operatorname{Re} m - 1 \end{pmatrix}, \quad Q_q = (\operatorname{tr} M - e^2 + 2)q.$$

Consequently scalar compatibility is exactly

$$e^2 = 2 + \operatorname{tr} M, \qquad \operatorname{Re} m = \delta > 0. \tag{99}$$

In particular $m \neq 0$. No phase of m, v, w has been set to a chosen value.

Set

$$Q = \operatorname{diag}(1, 2), \qquad E = \operatorname{diag}(2/3, -5/6).$$

Direct multiplication, keeping $r^2 = 3/8$ and $q^* q = 1/8$, gives

$$\mathcal{I}C = ie \operatorname{diag}(1/6, 2/3, -5/6), \qquad \mathcal{I}X(v) = \tfrac{1}{2} X(v). \tag{100}$$

For the first formula the two contributions, respectively from H and from the trace-free q term, are

$$ie \operatorname{diag}(0, 3/4, -3/4), \qquad ie \operatorname{diag}(1/6, -1/12, -1/12).$$

The second formula is (94), or follows from $(H')^2 = (3/8) \operatorname{diag}(0, 1, 1)$ directly. The entire spatial bracket and the entire $\mathcal{K}C$ are

$$[Z_1, Z_2] = \begin{pmatrix} -2ib & 0 \\ 0 & iN \end{pmatrix}, \qquad \mathcal{K}C = \begin{pmatrix} 2ie \operatorname{tr}(QM) & 0 \\ 0 & -ie(QM + MQ) \end{pmatrix}.$$

For example $[C, X(v)] = X(-ieQv)$, and the two double cross products give the displayed anti-commutator. Thus the full Gauss equation is equivalent to the upper entry and lower Hermitian block

$$\frac{et}{6} + 4e \operatorname{tr}(QM) + 8\omega b = 0, \qquad etE - 2e(QM + MQ) - 4\omega N = 0. \tag{101}$$

The upper entry is the negative lower trace, since $\operatorname{tr} N = 2b$. No off-block entries occur. The entries needed below are therefore exact consequences of the full Gauss equation:

$$b = -\frac{e}{2\omega}\left(u + 2v_0 + \frac{t}{24}\right), \qquad N_{12} = -\frac{3e}{2\omega} m, \qquad N_{22} = -\frac{e}{\omega}\left(2v_0 + \frac{5t}{24}\right). \tag{102}$$

All divisions use the original $\omega \neq 0$.

Put

$$D = \left(\frac{t}{4} - \omega^2\right) I_2 - e^2 Q^2.$$

The complete spatial residuals are $X(s_1), X(s_2)$, respectively, where

$$s_1 = (D + W_0 I_2)v + (-\alpha_0 I_2 + i(2\omega e Q - 3bI_2))\, w, \tag{103}$$

$$s_2 = (D + V_0 I_2)w + (-\alpha_0 I_2 - i(2\omega eQ - 3bI_2))\, v. \tag{104}$$

To verify every term, $[C, X(v)] = X(-ieQv)$ gives the two gyroscopic terms and the term $-e^2Q^2$. The inertia gives $tI_2/4$. For the remaining double commutators, block multiplication gives

$$\mathcal{K}X(v) = X\big(W_0 v + (\bar{p}_0 - 2p_0)w\big), \qquad \mathcal{K}X(w) = X\big(V_0 w + (p_0 - 2\bar{p}_0)v\big).$$

Since $\bar{p}_0 - 2p_0 = -\alpha_0 - 3ib$ and $p_0 - 2\bar{p}_0 = -\alpha_0 + 3ib$, these are exactly (103)–(104). Their upper rows are the negative adjoints of their lower first columns, and all other entries are zero. Thus $s_1 = s_2 = 0$ is equivalent to the two full spatial equations.

We now use a consequence of those equations; no converse is needed. Multiplying (103) by v^*, (104) by w^* and adding gives the exact matrix identity

$$s_1 v^* + s_2 w^* = DM + d_M I_2 + (4bI_2 - 2\omega eQ)N. \tag{105}$$

Indeed $\operatorname{tr}(M)M - M^2 = d_M I_2$ follows by multiplying the two-by-two matrix M. Its off-diagonal entries are zero, and its diagonal entries are both $M_{11}M_{22} - M_{12}M_{21}$. The cubic terms on the left of (105) are

$$W_0 vv^* + V_0 ww^* - \alpha_0(vw^* + wv^*) + 3ib(vw^* - wv^*).$$

Those in $\operatorname{tr}(M)M - M^2$ have $-ib$ instead of $3ib$; their difference is $4bN$. The gyroscopic terms sum to $-2\omega eQN$. These expansions prove (105) entry by entry, without a reality or orthogonality restriction on v, w.

Proof of Theorem 1.60. Gauss and the scalar equations have already forced the exact temporal form and (99). In particular $e \neq 0$, $m \neq 0$. For a full solution, (105) is zero. Set

$$\chi = \frac{3e}{2\omega}, \qquad B_2 = 4b - 4\omega e, \quad D_2 = \frac{t}{4} - \omega^2 - 4e^2.$$

Its $(2,1)$ entry is $D_2\bar{m} + B_2 N_{21} = \bar{m}(D_2 - \chi B_2) = 0$ by Gauss. Since $m \neq 0$, it follows that $D_2 = \chi B_2$. Its $(2,2)$ entry is therefore

$$0 = d_M + B_2(\chi v_0 + N_{22}).$$

Using exactly (102) gives

$$\chi v_0 + N_{22} = -\frac{e}{2\omega}\left(v_0 + \frac{5t}{12}\right), \qquad B_2 = -\frac{2e}{\omega}\left(u + 2v_0 + \frac{t}{24} + 2\omega^2\right).$$

Consequently every proposed full solution in this class would satisfy

$$0 = d_M + \frac{e^2}{\omega^2}\left(u + 2v_0 + \frac{t}{24} + 2\omega^2\right)\left(v_0 + \frac{5t}{12}\right). \tag{106}$$

But $u, v_0 \geq 0$, $d_M \geq 0$, $t > 0$, $\omega^2 > 0$ and $e^2 > 0$. Both parentheses are strictly positive. The second term on the right is strictly positive for either sign of e and either sign of ω, contradicting (106). This proves the theorem without selecting a frequency or deleting any cross-vector phase. □

Corollary 1.61 (A further necessary condition on every full solution)**.** *Every solution of the unrestricted five equations satisfies*

$$\sum_{j=1}^{2}\Big(\|PZ_jP\|_{\mathrm{HS}}^2 + \|(I_3 - P)Z_j(I_3 - P)\|_{\mathrm{HS}}^2\Big) > 0.$$

Proof. Each summand is a squared Hilbert–Schmidt norm and is nonnegative. The sum vanishes if and only if both spatial matrices lie in $\mathfrak{p}$, which is excluded by Theorem 1.60. □

The scalar-compatible part of the class is nonempty even with a nonzero spatial bracket. This is not a full solution: in the frame (97), choose the exact values

$$v = \sqrt{\delta}(1,1)^{\mathsf{T}}, \qquad w = (i,0)^{\mathsf{T}}, \qquad e = \sqrt{3+2\delta}.$$

Then $\operatorname{tr} M = 1 + 2\delta$, $M_{12} = \delta$, and $[X(v), X(w)]_{11} = -2i\sqrt{\delta} \neq 0$. Thus both full scalar equations hold by their displayed residuals. To embed this example into the literal frame, use the fixed $W = i \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)/\sqrt{2}$ and (98). The choice of the component i is a specified example, not a normalization of the general class. The theorem proves that no retained $t > 0, \omega \neq 0$ can complete this example to all five equations.

1.27.4 Exact inclusion of the selected class in the exceptional locus

We give the determinant map explicitly to specify the comparison with the full sixteen-coordinate projection, including its degree conventions. In this section the symbols H, q, F again denote their literal original matrices in (89)–(93), identical to the earlier retained data of (46)–(50). For any such matrix write $\widetilde{Y} = U^* Y U$ when using its first, H-diagonal-frame value. The inverse remains $Y = U\widetilde{Y}U^*$. Put

$$J = iH, \quad j_0 = \langle J, J\rangle = \frac{3}{4}, \quad \mathscr{W} = J^{\perp}, \quad \Pi X = X - \frac{\langle J, X\rangle}{j_0} J, \quad \mathscr{B} = [Z_1, Z_2],$$

$$\eta = \langle J, \mathcal{K}J\rangle, \quad w_J = \Pi\mathcal{K}J, \quad \beta = \langle J, \mathscr{B}\rangle, \quad \mathcal{I}_7 = \mathcal{I}|_{\mathscr{W}},$$

$$\mathcal{S}_7 = \Pi\mathcal{K}|_{\mathscr{W}} - \frac{w_J \otimes w_J}{\eta}, \qquad f_7 = \Pi\mathscr{B} - \frac{\beta}{\eta} w_J.$$

Here $(w_J \otimes w_J)X = w_J\langle w_J, X\rangle$ and the original length of J is retained. On $\eta > 0$ define

$$\mathscr{M}_7(t) = t\mathcal{I}_7 + 2\mathcal{S}_7, \qquad D_7 = \det_{\mathscr{W}} \mathscr{M}_7, \qquad V_7 = 4 \operatorname{adj}_{\mathscr{W}}(\mathscr{M}_7) f_7,$$

$$N_7 = V_7 + \frac{2\beta D_7 - \langle w_J, V_7\rangle}{\eta} J, \qquad Q_7 = \|V_7 q\|^2, \quad \alpha = \frac{1}{4} + \|Z_1 q\|^2 + \|Z_2 q\|^2,$$

$$P_H(t) = \alpha[N_7, [N_7, H]] - Q_7 \left(\sum_{j=1}^{2} [Z_j, [Z_j, H]] + F \right).$$

Let $F_\xi = \eta^{16}(\widetilde{P}_H)_{11}$ and $G_\xi = \eta^{16}(\widetilde{P}_H)_{22}$ denote the exact polynomials used for the original sixteen-coordinate exceptional set, with inverse division by η^{16} on $\eta > 0$. The matrix P_H in its definition is computed with the literal original H, q, F; $\widetilde{P}_H = U^* P_H U$ is then its exact conjugate, not a second change of already-conjugated data. They extend polynomially in the sixteen real spatial coordinates ξ. For clarity, their polynomial extension follows from the definitions: $\eta^7 D_7$, $\eta^7 V_7$, $\eta^8 N_7$ and $\eta^{14} Q_7$ are polynomial, by determinant and minor expansions. Hence $\eta^{16} P_H$ is polynomial. All other denominators are fixed nonzero constants. Their two selected diagonal entries are real because P_H is Hermitian for real t. The universal degree bounds are $\deg V_7 \leq 6$ and $N_7 = V_7 + s_7 J$ with $\deg s_7 \leq 7$. Since $[\widetilde{J}, \widetilde{H}] = 0$ and every diagonal entry of $[\widetilde{J}, Y]$ is zero, the diagonal entries of $[\widetilde{N}_7, [\widetilde{N}_7, \widetilde{H}]]$ have degree at most twelve; also $\deg Q_7 \leq 12$. Thus F_ξ, G_ξ have degree at most twelve in t, including every specialization where their degrees drop.

The exceptional polynomial is the determinant $\mathscr{R}(\xi)$ of the fixed map

$$\mathbb{R}[t]_{<12} \oplus \mathbb{R}[t]_{<12} \longrightarrow \mathbb{R}[t]_{<24}, \qquad (A_0, B_0) \longmapsto A_0 F_\xi + B_0 G_\xi, \tag{107}$$

in increasing-power bases, with the first summand before the second. Theorem 1.59 proves this polynomial is nonzero on the ambient sixteen-dimensional carrier and that all full solutions project to its zero set. The following inclusion is proved directly, independently of whether any point of that zero set admits a solution.

Proposition 1.62. *The exact linear embedding $\mathfrak{p} \times \mathfrak{p} \hookrightarrow \mathfrak{su}(3)^2$ has image inside $\{\mathscr{R} = 0\}$. On every nonzero pair in this class the two diagonal Gauss-eliminated polynomials have the common factor $D_4(t)^2$, where*

$$D_4(t) = \det_{\mathfrak{p}} \left(\frac{t}{2} \operatorname{id}_{\mathfrak{p}} + 2\mathcal{K}|_{\mathfrak{p}} \right)$$

has degree exactly four and leading coefficient $1/16$.

Proof. The first-frame $\widetilde{J} = i \operatorname{diag}(0, r, -r)$ satisfies $[\widetilde{J}, X(v)] = X(i \operatorname{diag}(r, -r)v)$. The change by U preserves $\mathfrak{p}$ and the inner product. Thus, for the selected class,

$$\eta = \sum_j \|[Z_j, J]\|^2 = r^2(\|Z_1\|^2 + \|Z_2\|^2).$$

It is positive except at the zero pair. This also shows why this class is compatible with the stabilizer decomposition: $-\operatorname{ad}_J^2$ is exactly $r^2 \operatorname{id}$ on $\mathfrak{p}$.

Put $\mathfrak{k}_0 = \mathfrak{k} \cap J^\perp$. The orthogonal decomposition is $\mathscr{W} = \mathfrak{k}_0 \oplus \mathfrak{p}$, with dimensions three and four. The already-proved block preservation of $\mathcal{I}, \mathcal{K}$, and $J \in \mathfrak{k}$, give $w_J, f_7 \in \mathfrak{k}_0$. Consequently $\mathcal{S}_7$ preserves both summands and equals $\mathcal{K}$ on $\mathfrak{p}$. Thus

$$\mathscr{M}_7 = \begin{pmatrix} \mathscr{M}_3 & 0 \\ 0 & \mathscr{M}_4 \end{pmatrix}, \qquad \mathscr{M}_4 = \frac{t}{2} \operatorname{id}_{\mathfrak{p}} + 2\mathcal{K}|_{\mathfrak{p}}, \quad D_7 = D_3 D_4.$$

Its four-dimensional determinant has degree four and leading coefficient $(1/2)^4 = 1/16$. Moreover $\mathscr{M}_4$ is positive for $t > 0$. For completeness, $\mathcal{I}$ has kernel $\mathbb{R}J$: its quadratic form is $\operatorname{tr}[X, H]^2 + 2\|Xq\|^2$; a zero commutes with the three-distinct-eigenvalue H and annihilates q, so has diagonal entries $0, is, -is$. The same formula proves positivity on $\mathscr{W}$. Completing the square in $\langle X + sJ, \mathcal{K}(X + sJ) \rangle$ gives

$$\langle X, \mathcal{S}_7 X \rangle + \eta \left(s + \frac{\langle w_J, X \rangle}{\eta} \right)^2 \geq 0,$$

so choosing the displayed square to be zero proves $\mathcal{S}_7 \succeq 0$. Thus $\mathscr{M}_7$ and its three-dimensional block are positive for every $t > 0$.

Define $V_3 = 4 \operatorname{adj}(\mathscr{M}_3) f_7 \in \mathfrak{k}_0$. The block inverse for $t > 0$ gives

$$V_7 = D_4 V_3, \qquad N_7 = D_4 N_3, \qquad N_3 = V_3 + \frac{2\beta D_3 - \langle w_J, V_3 \rangle}{\eta} J.$$

These are polynomial identities in t: the two sides agree for every positive t, while a nonzero polynomial has only finitely many roots, as follows by repeated division by its linear root factors. Therefore $Q_7 = D_4^2 Q_3$, where $Q_3 = \|V_3 q\|^2$, and

$$P_H = D_4^2 P_{H,3}, \qquad P_{H,3} = \alpha[N_3, [N_3, H]] - Q_3 \left(\sum_j [Z_j, [Z_j, H]] + F \right). \tag{108}$$

The three-dimensional minors give $\deg V_3 \le 2$, $\deg D_3 \le 3$. Writing $N_3 = V_3 + s_3 J$, one has $[J, H] = 0$. In the diagonal frame, $[J, Y]_{jj} = 0$ for every matrix Y, by subtracting the identical diagonal products. Hence the selected diagonal entries of $[N_3, [N_3, H]] = [V_3, [V_3, H]] + s_3[J, [V_3, H]]$ have degree at most four. Also $\deg Q_3 \le 4$. Thus at each fixed pair with $\eta > 0$,

$$F_\xi = D_4^2 F_3, \qquad G_\xi = D_4^2 G_3, \qquad \deg F_3, \deg G_3 \le 4,$$

where F_3, G_3 retain the scalar multiplier η^{16}. If they are not both zero, $(A_0, B_0) = (G_3, -F_3)$ is a nonzero kernel element of (107), with both degrees less than twelve. If both are zero, $(1, 0)$ is such a kernel element instead. The determinant therefore vanishes for every nonzero pair in the class, including every degree-drop case. At the zero pair it vanishes by continuity of the original polynomial $\mathscr{R}$ along this linear subspace. This proves the entire asserted inclusion without using a converse resultant test or interpreting $\eta = 0$ as an inverse solution. □

1.27.5 An exact certificate retaining nonzero residuals

The exclusion has a quantitative extension. It also gives a polynomial identity before the Gauss or spatial residuals are set to zero. Use the entire second-frame variables above, with every entry of v, w retained. More precisely, return here to the matrices and inverse in (97)–(98), not the literal-frame H used in the preceding determinant comparison. All of $M, N, m, u, v_0, b, s_1, s_2$ in this subsection are computed in that same second frame as in (103)–(104). Define the full spatial residual matrix and the Hermitian Gauss components

$$\mathcal{A} = s_1 v^* + s_2 w^*, \qquad \gamma_0 = et/6 + 4e(u + 2v_0) + 8\omega b, \qquad \mathcal{G} = etE - 2e(QM + MQ) - 4\omega N.$$

Here $\mathcal{A}$ is not the original generator A: its calligraphic symbol denotes exactly this two-by-two residual matrix. The full Gauss residual is $i\,\mathrm{diag}(\gamma_0, \mathcal{G})$, with the second block of size two. Set $B_2 = 4b - 4\omega e$ and

$$\mathcal{P} = d_M + \frac{e^2}{\omega^2}(u + 2v_0 + t/24 + 2\omega^2)(v_0 + 5t/12).$$

This expression is defined for the original nonzero ω; the identity below has no such denominator after multiplication.

Proposition 1.63 (Polynomial residual certificate and strict error bound)**.** *For every $v, w \in \mathbb{C}^2$ and real e, t, ω in the displayed second-frame matrices, the following polynomial identity holds:*

$$\begin{aligned} &4\omega^2(\bar{m}\mathcal{A}_{22} - v_0\mathcal{A}_{21}) + \omega B_2(\bar{m}\mathcal{G}_{22} - v_0\mathcal{G}_{21}) \\ &\qquad + e(v_0 + 5t/12)\bar{m}\gamma_0 \\ &= 4\bar{m}\big[\omega^2 d_M + e^2(u + 2v_0 + t/24 + 2\omega^2)(v_0 + 5t/12)\big]. \end{aligned} \tag{109}$$

No scalar, Gauss or spatial equation is assumed in this identity.

Now restrict to the exactly specified set of original tuples with $Z_1, Z_2 \in \mathfrak{p}$, $t > 0$, $\omega \neq 0$, for which both scalar equations and Gauss hold. Let $R_1, R_2 \in \mathfrak{su}(3)$ be their two full spatial residuals. Then

$$\left(\|R_1\|_{\mathrm{HS}}^2 + \|R_2\|_{\mathrm{HS}}^2\right)^{1/2} \ge \frac{\sqrt{2}\,\mathcal{P}}{\sqrt{\mathrm{tr}\, M\,(1 + v_0^2/|m|^2)}} > \frac{5\sqrt{2}\,e^2 t}{6\sqrt{\mathrm{tr}\, M\,(1 + v_0^2/|m|^2)}}. \tag{110}$$

The full matrix norms are preserved by the two exact unitary frame maps. In particular, on any such set with $\mathrm{tr}\, M \le L$ and $t \ge t_0 > 0$, where $L > 0$, the strictly positive uniform bound is

$$\left(\|R_1\|_{\mathrm{HS}}^2 + \|R_2\|_{\mathrm{HS}}^2\right)^{1/2} > \frac{5\sqrt{2}\,t_0}{3\sqrt{L(1 + L^2/\delta^2)}}, \qquad \delta = 1/(2r). \tag{111}$$

The complete scalar–Gauss solution set is parameterized in Theorems 1.66 and 1.68, and Corollary 1.70 proves it is nonempty at every original $t = g^2 > 0$. No sequence in its displayed bounded portion can make both spatial residuals tend to zero.

Proof. The complete residual formulas above give

$$\mathcal{A}_{21} = D_2\bar{m} + B_2N_{21}, \qquad \mathcal{A}_{22} = D_2v_0 + d_M + B_2N_{22}, \quad D_2 = t/4 - \omega^2 - 4e^2,$$

and, without any equation imposed,

$$\mathcal{G}_{21} = -6e\bar{m} - 4\omega N_{21}, \qquad \mathcal{G}_{22} = -5et/6 - 8ev_0 - 4\omega N_{22}.$$

Subtracting the two indicated spatial entries cancels D_2 exactly:

$$\bar{m}\mathcal{A}_{22} - v_0\mathcal{A}_{21} = \bar{m}d_M + B_2(\bar{m}N_{22} - v_0N_{21}).$$

Multiplication of the indicated Gauss entries gives

$$\bar{m}\mathcal{G}_{22} - v_0\mathcal{G}_{21} = -e(2v_0 + 5t/6)\bar{m} - 4\omega(\bar{m}N_{22} - v_0N_{21}).$$

These cancel the two N entries on the left of (109). The remaining factor is

$$\gamma_0 - 2\omega B_2 = 4e(u + 2v_0 + t/24 + 2\omega^2).$$

It yields exactly its stated right side. Each displayed step is a polynomial identity, including at $m = 0$ or $\omega = 0$; no inverse or solution is assigned at those values.

On the stated scalar–Gauss zero set the preceding proof forces the second-frame form, $e^2 = 2 + \operatorname{tr} M$ and $\operatorname{Re} m = \delta > 0$. Hence $m \neq 0$ and $\operatorname{tr} M > 0$. Setting $\gamma_0 = \mathcal{G} = 0$ in the polynomial identity and dividing by its retained nonzero factors gives

$$\mathcal{A}_{22} - \frac{v_0}{\bar{m}}\mathcal{A}_{21} = \mathcal{P} > 0.$$

Cauchy–Schwarz on these two entries implies

$$\mathcal{P} \leq \sqrt{1 + v_0^2/|m|^2}\,\|\mathcal{A}\|_{\mathrm{HS}}.$$

Let $S = (s_1\ s_2)$ and $V = (v\ w)$ be the two column matrices. Then $\mathcal{A} = SV^*$. Applying Cauchy–Schwarz to each entry and summing proves $\|SV^*\|_{\mathrm{HS}}^2 \leq \|S\|_{\mathrm{HS}}^2\|V\|_{\mathrm{HS}}^2$. The right factors are $\|s_1\|^2 + \|s_2\|^2$ and $\operatorname{tr} M$, respectively. Directly from the two off-diagonal blocks of $X(s)$, $\|X(s)\|_{\mathrm{HS}}^2 = 2\|s\|^2$. Thus $R_j = X(s_j)$ proves the first inequality in (110).

The exact expansion

$$\begin{aligned}&(u + 2v_0 + t/24 + 2\omega^2)(v_0 + 5t/12)\\ &= 5t\omega^2/6 + v_0(u + 2v_0 + t/24 + 2\omega^2) + (5t/12)(u + 2v_0 + t/24)\end{aligned}$$

has nonnegative remainder terms, the last strictly positive for $t > 0$. Together with $d_M \geq 0$ and $e^2 > 0$ it gives $\mathcal{P} > 5e^2t/6$, proving the strict second inequality. Finally $e^2 = 2 + \operatorname{tr} M \geq 2$, $0 \leq v_0 \leq \operatorname{tr} M \leq L$, and $|m| \geq \delta$ give (111). Since unitary conjugation preserves the Hilbert–Schmidt norm, the same inequalities hold for the complete spatial residuals in the original literal matter frame. □

1.27.6 Scope, provenance, and exact replay

Theorem 1.60 and Proposition 1.62 together prove an empty admissible fibre at every point of the explicit eight-dimensional two-cross linear subspace inside the full exceptional set. The result does not prove global emptiness or produce a full solution. It does not assert a result for arbitrary time-dependent gauge connections, and it makes no quantum mass-gap assertion. The earlier cross–cross–block exclusion in Theorem 1.45 instead assumes $C, Z_1 \in \mathfrak{p}$, $Z_2 \in \mathfrak{k}$; here only $Z_1, Z_2 \in \mathfrak{p}$ were assumed and $C \in \mathfrak{k}$ was proved.

The retained literal system is (46)–(50); its complete scalar coordinates are given in (66) and Proposition 1.50. The ambient projection statement and determinant convention are Theorem 1.59 and (88). The complete source proof, exact checker, and replay receipt for this additional exclusion are retained under `supporting_materials/workbench/evidence/two_cross_exceptional_20260905/`. All matrix identities needed for the theorem are derived above; no external classification theorem, numerical root search, or unproved literature identity is used.

The companion `check_two_cross_exceptional.py` checks 26 exact groups: the original force and constants, both frame maps (the second over the entire coordinate ring $\gamma\bar{\gamma} = 1$), every scalar residual before and after temporal classification, both inertia maps, the full bracket and Gauss residual, both complete spatial residuals, the nonnegative determinant identity, the full outer-product identity and the final positivity formula. Every real component of v, w is an independent symbol. It uses no floating point acceptance criterion and writes no files. One serial run passed all 26 groups in 0.416 seconds of measured checker time. The determinant block-factor proof and all positivity/domain arguments are analytic proofs in the preceding sections, not claims inferred from sampled determinant values.

The additional denominator-free residual identity and the full spatial error bound are proved in Proposition 1.63. The companion `check_residual_certificate.py` verifies its polynomial identity and exact positive remainder. The Cauchy–Schwarz estimates and their scalar–Gauss domain are proved in the text, not inferred from tests.

1.28 The complete scalar–Gauss domain at every positive coupling

1.28.1 Literal data, residuals, and coordinate maps

All complex vector spaces below retain their underlying real structures. The adjoint is denoted by $*$, and $[X, Y] = XY - YX$. Keep exactly

$$H_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1/2 & 1/(2\sqrt{2}) \\ 0 & 1/(2\sqrt{2}) & 1/2 \end{pmatrix}, \qquad q = \frac{1}{2\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad A_0 = \sqrt{2} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

The original constants are $m^2 = \Gamma = 1$ and $\kappa = 6$. For $Y_{\mathrm{tf}} = Y - (\operatorname{tr} Y) I_3/3$, define

$$\mathcal{I}_H Y = ([H, [H, Y]] + qq^* Y + Y qq^*)_{\mathrm{tf}}, \qquad \mathcal{K}_{Z_1, Z_2} Y = -\sum_{j=1}^{2} [Z_j, [Z_j, Y]].$$

Direct multiplication gives

$$F_0 = [A_0, [A_0, H_0]] = -2H_0 + \operatorname{diag}(-2, 1, 1), \quad H_0^2 = \frac{3}{8} \operatorname{diag}(0, 1, 1), \quad q^* q = \frac{1}{8}.$$

The scalar–Gauss domain in question consists of

$$(t, \omega, C_0, Z_{1,0}, Z_{2,0}), \quad t = g^2 > 0, \quad \omega \in \mathbb{R} \setminus \{0\}, \quad C_0, Z_{j,0} \in \mathfrak{su}(3),$$

where $P = qq^*/(q^*q) = \mathrm{diag}(1,0,0)$ and

$$PZ_{j,0}P = (I_3 - P)Z_{j,0}(I_3 - P) = 0,$$

satisfying exactly the following three full residual equations:

$$[C_0,[C_0,H_0]] - \sum_j [Z_{j,0},[Z_{j,0},H_0]] - F_0 = 0, \tag{112}$$

$$(C_0^2 - Z_{1,0}^2 - Z_{2,0}^2)q + 2q = 0, \tag{113}$$

$$(t\mathcal{I}_{H_0} + 2\mathcal{K}_{Z_{1,0},Z_{2,0}})C_0 - 4\omega[Z_{1,0},Z_{2,0}] = 0. \tag{114}$$

Equation (114) is the original Gauss equation multiplied by its retained positive t; division by the same t is its inverse. Neither spatial equation is imposed anywhere in the construction.

The exact maps to and from the classified frame are as follows. Retain

$$r = \frac{\sqrt{6}}{4}, \quad \delta = \frac{1}{2r} = \sqrt{\frac{2}{3}}, \quad x = \frac{1}{2}, \quad d_0 = \frac{1}{2\sqrt{2}},$$
$$S = \frac{1}{\sqrt{2r(r+x)}} \begin{pmatrix} d_0 & -(r+x) \\ r+x & d_0 \end{pmatrix}, \quad U = \mathrm{diag}(1,S).$$

The identity $r^2 = x^2 + d_0^2$ gives $S^*S = I_2$, $\det S = 1$; multiplication of H_0 by its two displayed lower columns gives

$$U^*H_0U = \mathrm{diag}(0,r,-r), \qquad U^*q = q.$$

Choose any $\zeta \in \mathbb{C}$ with $|\zeta| = 1$, and set

$$\gamma = -\zeta^{-2}, \qquad W = \frac{\zeta}{\sqrt{2}} \begin{pmatrix} \gamma & \gamma \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -\zeta^{-1} & -\zeta^{-1} \\ \zeta & -\zeta \end{pmatrix}, \qquad V = \mathrm{diag}(1,W), \quad B = UV.$$

Here $|\gamma| = 1$ and $\zeta^2 = -\overline{\gamma}$, retaining exactly the original phase relation. Every unit γ occurs, with its two possible square roots ζ. The displayed entries give

$$W^*W = I_2, \quad \det W = 1, \quad W^* \,\mathrm{diag}(r,-r) W = \begin{pmatrix} 0 & r \\ r & 0 \end{pmatrix}.$$

Thus, simultaneously for all data,

$$Y' = B^*Y_0B, \qquad Y_0 = BY'B^*, \qquad H' = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & r \\ 0 & r & 0 \end{pmatrix}, \quad q' = q, \quad F' = -2H' + \mathrm{diag}(-2,1,1). \tag{115}$$

These are inverse maps, with no constant or unknown rescaled. In fact $B^*(XY)B = (B^*XB)(B^*YB)$, the same equality holds for brackets, trace and adjoints, and $Bq = q$. Therefore $\mathcal{I}$ and $\mathcal{K}$ and every full residual in (112)–(114) are conjugated exactly (the vector residual is multiplied by B^*). This proves that zero residuals in the classified frame give zero residuals in the literal frame, with the displayed inverse recovering every datum. The block-diagonal B commutes with P, preserving the two-cross class.

1.28.2 Every scalar and Gauss entry in the classified frame

For $v, w \in \mathbb{C}^2$ write

$$X(v) = \begin{pmatrix} 0 & -v^* \\ v & 0_{2\times 2} \end{pmatrix}, \qquad Z_1 = X(v), \quad Z_2 = X(w), \quad C = ie\,\mathrm{diag}(1,0,-1), \quad e \in \mathbb{R}\setminus\{0\}.$$

The inverse of $v \mapsto X(v)$ is the lower first column; every $X(v)$ is traceless anti-Hermitian. In this section all unprimed matrices are these classified-frame matrices, with literal inverse (115). Define

$$M = vv^* + ww^* = \begin{pmatrix} u & m \\ \bar{m} & v_0 \end{pmatrix}, \quad N = i(vw^* - wv^*),$$
$$b = \mathrm{Im}(v^*w), \quad Q = \mathrm{diag}(1,2), \quad E = \mathrm{diag}(2/3, -5/6).$$

The subscript on the real scalar v_0 distinguishes it from the vector v.

Here are the complete residuals, rather than only contractions of them:

$$Q_H = \begin{pmatrix} 2-4r\,\mathrm{Re}\,m & 0 & 0 \\ 0 & 2r\,\mathrm{Re}\,m - 1 & r(\mathrm{tr}\,M - e^2 + 2) \\ 0 & r(\mathrm{tr}\,M - e^2 + 2) & 2r\,\mathrm{Re}\,m - 1 \end{pmatrix}, \tag{116}$$

$$Q_q = (\mathrm{tr}\,M - e^2 + 2)q, \tag{117}$$

$$Q_G = i\,\mathrm{diag}(\gamma_0, \mathcal{G}), \tag{118}$$

$$\gamma_0 = et/6 + 4e(u + 2v_0) + 8\omega b, \qquad \mathcal{G} = etE - 2e(QM + MQ) - 4\omega N. \tag{119}$$

In (118) the second diagonal block has size two. To derive every displayed entry, multiplication gives

$$X(v)^2 = \mathrm{diag}(-v^*v, -vv^*), \quad [C, X(v)] = X(-ieQv), \quad [X(v), X(w)] = \mathrm{diag}(-2ib, iN).$$

The first two identities and $H'_{23} = H'_{32} = r$ give (116) and (117); all their omitted entries are zero. The two contributions to $\mathcal{I}_{H'}C$ are

$$ie\,\mathrm{diag}(0, 3/4, -3/4), \qquad ie\,\mathrm{diag}(1/6, -1/12, -1/12),$$

so $\mathcal{I}_{H'}C = ie\,\mathrm{diag}(1/6, 2/3, -5/6)$. Also

$$\mathcal{K}C = \mathrm{diag}\big(2ie\,\mathrm{tr}(QM), -ie(QM + MQ)\big).$$

Substitution of these full blocks gives (118). Finally $\mathrm{tr}\,N = 2b$, since $\mathrm{tr}(vw^*) = w^*v = \overline{v^*w}$. Thus $\mathrm{tr}\,\mathcal{G} = -\gamma_0$; the upper Gauss entry vanishes whenever the full lower block does, and this implication is an identity, not an omitted equation. The two scalar equations are equivalent to

$$e^2 = 2 + u + v_0, \qquad \mathrm{Re}\,m = \delta. \tag{120}$$

1.28.3 The exact realization of the two Hermitian matrices

Lemma 1.64. *For every pair v, w, let $D = (v\ w)$ and $J = \left(\begin{smallmatrix} 0 & i \\ -i & 0 \end{smallmatrix}\right)$. Then $M = DD^*$, $N = DJD^*$ and*

$$M + N = (v - iw)(v - iw)^*, \qquad M - N = (v + iw)(v + iw)^*.$$

If M is positive definite, then

$$\mathrm{tr}(M^{-1}N) = 0, \qquad \det N = -\det M. \tag{121}$$

Proof. Multiplication of the displayed two-by-two blocks gives all three factorizations. If $M > 0$, D is invertible because $z^*Mz = \|D^*z\|^2$ has zero kernel. Consequently $M^{-1}N = D^{-*}JD^*$ is similar to J, whose trace is zero and determinant is -1. Taking the trace and determinant proves (121), with every factor retained. □

Lemma 1.65. *Every classified scalar–Gauss solution has $M > 0$.*

Proof. The scalar equation gives $m = \delta + iy \neq 0$, so $u, v_0 > 0$ and $M \neq 0$. It is already positive semidefinite as DD^*. If M had rank one, the range of $N = DJD^*$ would be contained in its one-dimensional range. A Hermitian matrix with that range and its orthogonal complement in its kernel is a real scalar multiple of M, so $N = \nu M$ for some real ν. Set $k = e/\omega \neq 0$. The full Gauss equation reads

$$N = kT, \qquad T = \frac{t}{4}E - \frac{1}{2}(QM + MQ) = \begin{pmatrix} t/6 - u & -3m/2 \\ -3\bar{m}/2 & -5t/24 - 2v_0 \end{pmatrix}. \tag{122}$$

Its off-diagonal entry gives $\nu = -3k/2$ because $m \neq 0$. The $(1,1)$ entry then gives $t/6 - u = -3u/2$, or $t/6 = -u/2$, which contradicts $t > 0$ and $u > 0$. Thus M has rank two and is positive definite. □

Theorem 1.66 (Complete classified-frame parameterization)**.** *Define the open real parameter set*

$$\mathcal{D} = \{(u, v_0, y) \in \mathbb{R}^3 : u > 0, \quad d = uv_0 - \delta^2 - y^2 > 0, \quad c = 4v_0 - 5u > 0\}.$$

For $(u, v_0, y) \in \mathcal{D}$, retain

$$m = \delta + iy, \qquad M = \begin{pmatrix} u & m \\ \bar{m} & v_0 \end{pmatrix}, \qquad t = \frac{72d}{c}, \tag{123}$$

$$T = \frac{t}{4}E - \frac{1}{2}(QM + MQ), \qquad k = \sigma\sqrt{\frac{d}{-\det T}}, \qquad e = \epsilon\sqrt{2 + u + v_0}, \tag{124}$$

$$\omega = e/k, \qquad N = kT, \qquad \epsilon, \sigma \in \{1, -1\}. \tag{125}$$

The denominator $-\det T$ is strictly positive. The two matrices $A_+ = M + N$ and $A_- = M - N$ are positive semidefinite of rank one, and their $(1,1)$ entries h_+, h_- are strictly positive. Define

$$a_\pm = \begin{pmatrix} \sqrt{h_\pm} \\ (A_\pm)_{21}/\sqrt{h_\pm} \end{pmatrix}, \quad p_\pm \in \mathbb{C}, \quad |p_\pm| = 1, \quad v = \frac{p_+a_+ + p_-a_-}{2}, \quad w = \frac{i(p_+a_+ - p_-a_-)}{2}. \tag{126}$$

These formulas give exactly all classified-frame solutions of (116)–(119). In fact they are a bijection from $\mathcal{D} \times \{1, -1\}^2 \times U(1)^2$ to that solution set with coordinates (t, ω, e, v, w).

Proof. First consider any solution. Lemma 1.65 gives $M > 0$ and $d > 0$. Inverting its two-by-two matrix gives

$$M^{-1} = \frac{1}{d}\begin{pmatrix} v_0 & -m \\ -\bar{m} & u \end{pmatrix}.$$

The trace condition in (121), divided by the retained nonzero k, now gives exactly

$$\begin{aligned} 0 = d\,\mathrm{tr}(M^{-1}T) &= v_0(t/6 - u) + u(-5t/24 - 2v_0) + 3|m|^2 \\ &= \frac{t(4v_0 - 5u)}{24} - 3d. \end{aligned} \tag{127}$$

Hence $c > 0$ and $t = 72d/c$. The determinant condition gives $k^2 \det T = -d$, so $\det T < 0$ and k is precisely one of the two choices in (124). The scalar conditions give precisely the two signs of e stated there. The original $\omega = e/k$ is unchanged, with sign $\epsilon\sigma$.

Conversely, start with parameters in $\mathcal{D}$. Positivity of M follows from the exact factorization

$$M = LL^*, \qquad L = \begin{pmatrix} \sqrt{u} & 0 \\ \bar{m}/\sqrt{u} & \sqrt{d}/\sqrt{u} \end{pmatrix}, \qquad \det L = \sqrt{d} > 0.$$

The specified positive t makes (127) zero. Put $R = L^{-1}TL^{-*}$. This is Hermitian and $\operatorname{tr} R = \operatorname{tr}(M^{-1}T) = 0$, so its full form is

$$R = \begin{pmatrix} a & z \\ \bar{z} & -a \end{pmatrix}, \qquad a \in \mathbb{R}.$$

The nonzero m makes $T_{12} = -3m/2 \neq 0$, hence $R \neq 0$. Thus $\det R = -a^2 - |z|^2 < 0$ and $\det T = d \det R < 0$. This proves that both signs of k are well defined. Let $J_0 = kR$. It is Hermitian, has trace zero and determinant -1, and multiplying the displayed matrix shows $J_0^2 = I_2$. For either sign and every column z_0, one has

$$z_0^*(I_2 \pm J_0)z_0 = \frac{1}{2}\|(I_2 \pm J_0)z_0\|^2 \geq 0.$$

Also $\det(I_2 \pm J_0) = 0$ and $\operatorname{tr}(I_2 \pm J_0) = 2$, so these matrices have rank one. Therefore

$$A_\pm = M \pm N = L(I_2 \pm J_0)L^*$$

are positive semidefinite of rank one, as claimed.

Their first diagonal entries are nonnegative. Suppose $h_s = 0$ for a sign $s \in \{1, -1\}$, where $A_s = M + sN$. Its determinant is zero, so $|(A_s)_{12}|^2 = 0$. But

$$(A_s)_{12} = m(1 - 3sk/2), \quad h_s = u + sk(t/6 - u).$$

Since $m \neq 0$, this forces $sk = 2/3$ and then $h_s = u/3 + t/9 > 0$, a contradiction. Thus $h_\pm > 0$. Using $\det A_\pm = 0$ gives $(A_\pm)_{22} = |(A_\pm)_{21}|^2/h_\pm$; consequently $A_\pm = a_\pm a_\pm^*$ with exactly the vectors in (126).

For those vectors, $v - iw = p_+ a_+$ and $v + iw = p_- a_-$. Their two outer products are A_+, A_- because the phases have unit modulus. Adding and subtracting the identities in Lemma 1.64 proves

$$vv^* + ww^* = M, \qquad i(vw^* - wv^*) = N.$$

Equation (120) holds by construction, so both full scalar residuals vanish. Substitution of $N = kT$ and $\omega k = e$ gives, with all coefficients retained,

$$\mathcal{G} = etE - 2e(QM + MQ) - 4e\left(\frac{t}{4}E - \frac{1}{2}(QM + MQ)\right) = 0.$$

The exact identity $\gamma_0 = -\operatorname{tr}\mathcal{G}$ then checks the remaining upper Gauss entry. Thus every entry of all three original residuals vanishes after applying (115).

For bijectivity, u, v_0, y are recovered from M, ϵ is the sign of e, and σ the sign of e/ω. The two phase factors are recovered without any zero denominator by

$$p_+ = \frac{(v - iw)_1}{\sqrt{h_+}}, \qquad p_- = \frac{(v + iw)_1}{\sqrt{h_-}}.$$

Indeed a column c with $cc^* = a_\pm a_\pm^*$ and $h_\pm > 0$ has $|c_1| = \sqrt{h_\pm}$, and the $(2,1)$ entry forces its second component to be the same unit multiple of $(a_\pm)_2$. These formulas prove both surjectivity and the exact inverse, without setting any spatial phase to a selected value. □

1.28.4 The complete literal scalar–Gauss domain and its phase quotient

The classified-frame bijection of Theorem 1.66 has an exact extension to the entire literal two-cross domain. We give the forced temporal classification, the full parameter map, and the complete fibres of that map. All spaces retain their underlying real structures, $[X,Y]=XY-YX$, and $*$ denotes the adjoint.

The literal domain and the retained constants Retain exactly

$$H_0=\begin{pmatrix}0&0&0\\0&-1/2&1/(2\sqrt{2})\\0&1/(2\sqrt{2})&1/2\end{pmatrix},\qquad q=\frac{1}{2\sqrt{2}}\begin{pmatrix}1\\0\\0\end{pmatrix},\qquad A_0=\sqrt{2}\begin{pmatrix}0&-1&0\\1&0&0\\0&0&0\end{pmatrix},$$

$$m^2=\Gamma=1,\qquad \kappa=6,\qquad F_0=[A_0,[A_0,H_0]]=-2H_0+\operatorname{diag}(-2,1,1).$$

Here $\mathfrak{su}(3)$ consists of the traceless anti-Hermitian complex 3×3 matrices, with positive real inner product $\langle X,Y\rangle=-\operatorname{tr}(XY)$. For $Y_{\mathrm{tf}}=Y-(\operatorname{tr}Y)I_3/3$, define

$$\mathcal{I}_{H_0}Y=([H_0,[H_0,Y]]+qq^*Y+Yqq^*)_{\mathrm{tf}},\qquad \mathcal{K}_{Z_1,Z_2}Y=-\sum_{j=1}^{2}[Z_j,[Z_j,Y]].$$

Multiplication of the retained matrices gives $H_0q=0$, $H_0^2=(3/8)\operatorname{diag}(0,1,1)$ and $q^*q=1/8$. Consequently $\operatorname{tr}H_0^2+2q^*q=1$, so the original scalar coefficient $m^2+\Gamma(\operatorname{tr}H_0^2+2q^*q)$ is 2. Also $\operatorname{adj}H_0=\operatorname{diag}(-3/8,0,0)$ gives $6(\operatorname{adj}H_0-qq^*)_{\mathrm{tf}}=\operatorname{diag}(-2,1,1)$, agreeing with the displayed force.

Put

$$P=\frac{qq^*}{q^*q}=\operatorname{diag}(1,0,0),\qquad \mathfrak{p}=\{X\in\mathfrak{su}(3):PXP=(I_3-P)X(I_3-P)=0\}.$$

Let $\mathcal{S}_{\mathrm{sg}}$ be exactly the set of tuples

$$(t,\omega,C_0,Z_{1,0},Z_{2,0}),\qquad t=g^2>0,\quad \omega\in\mathbb{R}\setminus\{0\},\quad C_0\in\mathfrak{su}(3),\quad Z_{j,0}\in\mathfrak{p},$$

for which the following three full residuals vanish:

$$[C_0,[C_0,H_0]]-\sum_j[Z_{j,0},[Z_{j,0},H_0]]-F_0=0, \tag{128}$$

$$(C_0^2-Z_{1,0}^2-Z_{2,0}^2)q+2q=0, \tag{129}$$

$$(t\mathcal{I}_{H_0}+2\mathcal{K}_{Z_{1,0},Z_{2,0}})C_0-4\omega[Z_{1,0},Z_{2,0}]=0. \tag{130}$$

Their codomains are, respectively, $\mathrm{Herm}_3(\mathbb{C})_0$, $\mathbb{C}^3$, and $\mathfrak{su}(3)$. The original Gauss residual is

$$(\mathcal{I}_{H_0}+2\mathcal{K}_{Z_{1,0},Z_{2,0}}/g^2)C_0-(4\omega/g^2)[Z_{1,0},Z_{2,0}].$$

Multiplication by the same retained positive $t=g^2$ gives (130), and division by t is its inverse. No spatial residual is required to vanish in the definition of $\mathcal{S}_{\mathrm{sg}}$.

Temporal classification and all inverse frame maps For $v \in \mathbb{C}^2$, define

$$X(v) = \begin{pmatrix} 0 & -v^* \\ v & 0_2 \end{pmatrix}, \qquad \mathfrak{k} = \{X \in \mathfrak{su}(3) : [X, P] = 0\}.$$

Every element of $\mathfrak{p}$ has exactly this form, whose inverse is the lower first column. Define the fixed constants and matrix

$$r = \frac{\sqrt{6}}{4}, \qquad \delta = \frac{1}{2r} = \sqrt{\frac{2}{3}}, \qquad x = \frac{1}{2}, \qquad d_0 = \frac{1}{2\sqrt{2}},$$

$$S = \frac{1}{\sqrt{2r(r+x)}} \begin{pmatrix} d_0 & -(r+x) \\ r+x & d_0 \end{pmatrix}, \qquad U = \operatorname{diag}(1, S).$$

The identity $r^2 = x^2 + d_0^2$ shows that the numerator columns are orthogonal, have squared length $2r(r+x)$, and have determinant $2r(r+x) > 0$. Multiplication by the lower block of H_0 gives their respective eigenvalues $r, -r$. Thus

$$U^*U = I_3, \qquad \det U = 1, \qquad U^* H_0 U = \operatorname{diag}(0, r, -r), \qquad Uq = q.$$

Lemma 1.67 (The temporal form is forced by scalars and Gauss)**.** *Every tuple in $\mathcal{S}_{\mathrm{sg}}$ has $C_0 \in \mathfrak{k}$. In the fixed first frame $C_d = U^* C_0 U$, there are unique $e \in \mathbb{R} \setminus \{0\}$ and $z \in \mathbb{C}$ with $|z| = |e|/2$ such that*

$$C_d = \begin{pmatrix} ie & 0 & 0 \\ 0 & -ie/2 & z \\ 0 & -\bar{z} & -ie/2 \end{pmatrix}.$$

Set $\gamma = -2iz/e$. Then $|\gamma| = 1$. For either square root $\zeta^2 = -\bar{\gamma}$, define

$$W(\zeta) = \frac{\zeta}{\sqrt{2}} \begin{pmatrix} \gamma & \gamma \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -\zeta^{-1} & -\zeta^{-1} \\ \zeta & -\zeta \end{pmatrix}, \qquad B(\zeta) = U \operatorname{diag}(1, W(\zeta)).$$

The simultaneous change and its inverse are

$$Y' = B(\zeta)^* Y_0 B(\zeta), \qquad Y_0 = B(\zeta) Y' B(\zeta)^*. \tag{131}$$

They give

$$C' = ie \operatorname{diag}(1, 0, -1), \qquad H' = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & r \\ 0 & r & 0 \end{pmatrix}, \qquad q' = q, \qquad F' = -2H' + \operatorname{diag}(-2, 1, 1).$$

These maps preserve the spatial class and intertwine all three full residuals in (128)–(130)*.*

Proof. Block multiplication gives the orthogonal direct sum $\mathfrak{su}(3) = \mathfrak{k} \oplus \mathfrak{p}$, together with $[\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{k}$ and $[\mathfrak{k}, \mathfrak{p}] \subset \mathfrak{p}$. Since H_0, qq^* are block diagonal, $\mathcal{I}_{H_0}$ preserves both summands. More explicitly, $H_0 X(v) H_0 = 0$; the two H_0^2 products sum to $(3/8)X(v)$, and the two qq^* products sum to $X(v)/8$. Each term is traceless. Thus

$$\mathcal{I}_{H_0}|_{\mathfrak{p}} = \tfrac{1}{2} \operatorname{id}_{\mathfrak{p}}.$$

The two bracket inclusions make $\mathcal{K} = \mathcal{K}_{Z_{1,0}, Z_{2,0}}$ preserve both summands as well. Cyclicity gives, for every $Y \in \mathfrak{su}(3)$,

$$\langle Y, \mathcal{K} Y \rangle = -\sum_j \operatorname{tr}([Z_{j,0}, Y]^2) = \sum_j \|[Z_{j,0}, Y]\|^2 \geq 0.$$

The $\mathfrak{p}$ projection of Gauss is therefore

$$(\tfrac{t}{2}\,\mathrm{id}_{\mathfrak{p}} + 2\mathcal{K}|_{\mathfrak{p}})(C_0)_{\mathfrak{p}} = 0.$$

Pairing with $(C_0)_{\mathfrak{p}}$ gives nonnegative summands, including $(t/2)\|(C_0)_{\mathfrak{p}}\|^2$. Since $t > 0$, $(C_0)_{\mathfrak{p}} = 0$, proving $C_0 \in \mathfrak{k}$.

The arbitrary first-frame temporal block consequently has the form

$$C_d = \begin{pmatrix} ia & 0 & 0 \\ 0 & i(h-a)/2 & z \\ 0 & -\bar{z} & -i(a+h)/2 \end{pmatrix}, \qquad a, h \in \mathbb{R}, \quad z \in \mathbb{C}.$$

Write the spatial matrices in this first frame as $X(\widetilde{v}), X(\widetilde{w})$, and define $\widetilde{M} = \widetilde{v}\widetilde{v}^* + \widetilde{w}\widetilde{w}^*$. The whole vector residual is

$$(-a^2 + \operatorname{tr}\widetilde{M} + 2)q,$$

and the whole adjoint residual is

$$\begin{pmatrix} 2 - 2r(\widetilde{M}_{11} - \widetilde{M}_{22}) & 0 & 0 \\ 0 & -4r|z|^2 + 2r\widetilde{M}_{11} - 1 + 2r & -2irhz \\ 0 & 2irh\bar{z} & 4r|z|^2 - 2r\widetilde{M}_{22} - 1 - 2r \end{pmatrix}.$$

For completeness, if $D = \operatorname{diag}(r, -r)$, the lower spatial contribution is $\widetilde{M}D + D\widetilde{M}$, whose diagonal entries are $2r\widetilde{M}_{11}, -2r\widetilde{M}_{22}$ and whose off-diagonal entries vanish. The temporal double commutator has diagonal entries $-4r|z|^2, 4r|z|^2$ and off-diagonal entries $-2irhz, 2irh\bar{z}$. Subtracting the lower force $I_2 - 2D$ gives the displayed lower block. The upper spatial contribution and force give $2 - 2r(\widetilde{M}_{11} - \widetilde{M}_{22})$. Finally $X(v)^2 = \operatorname{diag}(-v^*v, -vv^*)$ gives the displayed vector residual, with all its other components zero.

Thus both scalar equations are equivalent to

$$a^2 = 2 + \operatorname{tr}\widetilde{M}, \qquad 2|z|^2 - \widetilde{M}_{11} = 1 - \delta, \qquad 2|z|^2 - \widetilde{M}_{22} = 1 + \delta, \qquad hz = 0.$$

The upper adjoint entry is minus the lower trace. Adding the middle equations gives $4|z|^2 = 2 + \operatorname{tr}\widetilde{M} = a^2$. Since $\widetilde{M}_{22} \geq 0$ and $1 + \delta > 0$, $z \neq 0$. Hence $h = 0$ and $a \neq 0$. This proves the claimed temporal block with $e = a$, where its entries recover e, z uniquely.

Now $|\gamma| = 1$ and each root $\zeta^2 = -\bar{\gamma}$ has $|\zeta| = 1$, with $\gamma = -\zeta^{-2}$. The displayed entries give $W^*W = I_2$ and $\det W = -\zeta^2\gamma = 1$. The lower Hermitian block of $-iC_d$ is

$$\frac{e}{2}\begin{pmatrix} -1 & \gamma \\ \bar{\gamma} & -1 \end{pmatrix}.$$

Its products with $(\gamma, 1)^{\mathsf{T}}$ and $(\gamma, -1)^{\mathsf{T}}$ have eigenvalues $0, -e$, respectively. The same columns give $W^*DW = \left(\begin{smallmatrix} 0 & r \\ r & 0 \end{smallmatrix}\right)$. This proves the stated classified matrices.

The block-diagonal unitary $B(\zeta)$ fixes q and commutes with P. Its conjugation therefore preserves $\mathfrak{p}$ and $\mathfrak{k}$. Unitarity gives $B^*(XY)B = (B^*XB)(B^*YB)$; adjoints, commutators and trace projection satisfy the corresponding equalities. It follows that $\mathcal{I}, \mathcal{K}$ and every full matrix residual are intertwined by conjugation. The vector residual is multiplied by B^*. Equation (131) gives the exact inverse on every datum, with no rescaling. □

The full parameter map and the quotient bijection We restate the classified parameters so that the literal map is explicit. Put

$$Q = \operatorname{diag}(1,2), \qquad E = \operatorname{diag}(2/3, -5/6),$$

$$\mathcal{D} = \{(u, v_0, y) \in \mathbb{R}^3 : u > 0,\ d = uv_0 - \delta^2 - y^2 > 0,\ c = 4v_0 - 5u > 0\}.$$

For $(u, v_0, y) \in \mathcal{D}$, independent $\epsilon, \sigma \in \{1, -1\}$, and $p_+, p_- \in U(1)$, define

$$\begin{aligned} &m = \delta + iy, && M = \begin{pmatrix} u & m \\ \bar{m} & v_0 \end{pmatrix}, && t = \frac{72d}{c}, \\ &T = \frac{t}{4}E - \frac{1}{2}(QM + MQ), && k = \sigma\sqrt{\frac{d}{-\det T}}, && e = \epsilon\sqrt{2 + u + v_0}, \\ &\omega = e/k, && N = kT, && A_\pm = M \pm N. \end{aligned} \tag{132}$$

Theorem 1.66 proves that $\det T < 0$, that $A_\pm$ are positive semidefinite of rank one, and that $h_\pm = (A_\pm)_{11} > 0$. Thus the following vectors are defined on the whole displayed domain:

$$a_\pm = \begin{pmatrix} \sqrt{h_\pm} \\ (A_\pm)_{21}/\sqrt{h_\pm} \end{pmatrix}, \qquad v = \frac{p_+ a_+ + p_- a_-}{2}, \qquad w = \frac{i(p_+ a_+ - p_- a_-)}{2}. \tag{133}$$

In particular $vv^* + ww^* = M$ and $i(vw^* - wv^*) = N$. The cited theorem proves that these parameters are in bijection with all solutions of the two full scalar equations and full Gauss in the frame $C' = ie\operatorname{diag}(1, 0, -1)$, $H'_{23} = H'_{32} = r$, $q' = q$, and $F' = -2H' + \operatorname{diag}(-2, 1, 1)$. It also proves the exact inverse phases

$$p_+ = \frac{(v - iw)_1}{\sqrt{h_+}}, \qquad p_- = \frac{(v + iw)_1}{\sqrt{h_-}}. \tag{134}$$

Define the full raw parameter space

$$\mathcal{P}_{\text{lit}} = \mathcal{D} \times \{1, -1\}^2 \times U(1)^2 \times U(1)$$

with coordinates $(u, v_0, y, \epsilon, \sigma, p_+, p_-, \zeta)$. For the last unit phase set $\gamma = -\zeta^{-2}$ and use the matrix $B(\zeta)$ of Lemma 1.67. Define the map $\mathfrak{F} : \mathcal{P}_{\text{lit}} \to \mathcal{S}_{\text{sg}}$ by (132)–(133) and the following entire literal tuple:

$$\begin{aligned} \mathfrak{F}(u, v_0, y, \epsilon, \sigma, p_+, p_-, \zeta) &= (t, \omega, C_0, Z_{1,0}, Z_{2,0}), \\ C_0 &= B(\zeta)(ie\operatorname{diag}(1, 0, -1))B(\zeta)^*, \\ Z_{1,0} &= B(\zeta)X(v)B(\zeta)^*, \\ Z_{2,0} &= B(\zeta)X(w)B(\zeta)^*. \end{aligned} \tag{135}$$

The codomain assertion follows from the classified theorem and the exact residual intertwining in Lemma 1.67.

Theorem 1.68 (Exact quotient parameterization of the entire literal domain)**.** *The map $\mathfrak{F}$ is onto $\mathcal{S}_{\text{sg}}$. The map*

$$\iota(u, v_0, y, \epsilon, \sigma, p_+, p_-, \zeta) = (u, v_0, y, \epsilon, \sigma, -p_+, -p_-, -\zeta) \tag{136}$$

is a free involution of $\mathcal{P}_{\text{lit}}$. Each fibre of $\mathfrak{F}$ is exactly one two-element ι-orbit. Consequently (135) *induces a bijection of sets*

$$\overline{\mathfrak{F}} : \mathcal{P}_{\text{lit}}/\langle\iota\rangle \ \longrightarrow\ \mathcal{S}_{\text{sg}}. \tag{137}$$

The inverse of this bijection is recovered from the literal matrices by the formulas in the proof, with both square-root representatives retained.

Proof. We first construct a preimage of every literal tuple, recording all inverse coordinates. Given $(t, \omega, C_0, Z_{1,0}, Z_{2,0}) \in \mathcal{S}_{\mathrm{sg}}$, compute

$$C_d = U^* C_0 U, \qquad e = -i(C_d)_{11}, \qquad z = (C_d)_{23}, \qquad \gamma = -2iz/e.$$

Lemma 1.67 proves $e \in \mathbb{R} \setminus \{0\}$ and $|\gamma| = 1$. Choose either square root $\zeta^2 = -\bar{\gamma}$. Define

$$Z'_j = B(\zeta)^* Z_{j,0} B(\zeta), \qquad v = \begin{pmatrix} (Z'_1)_{21} \\ (Z'_1)_{31} \end{pmatrix}, \qquad w = \begin{pmatrix} (Z'_2)_{21} \\ (Z'_2)_{31} \end{pmatrix}.$$

The lemma proves that these are classified solutions with the original t, ω, e. Compute

$$M = vv^* + ww^*, \qquad N = i(vw^* - wv^*), \qquad u = M_{11}, \quad v_0 = M_{22}, \quad y = \operatorname{Im} M_{12},$$

$$\epsilon = \operatorname{sign} e, \qquad \sigma = \operatorname{sign}(e/\omega).$$

Theorem 1.66 proves $(u, v_0, y) \in \mathcal{D}$, $M_{12} = \delta + iy$, and all equations in (132) with this unchanged t, ω, e. It proves $h_\pm = (M \pm N)_{11} > 0$, so (134) recovers unique unit p_+, p_-. The vector formulas recover the displayed v, w and the inverse frame map then recovers all three literal matrices. Thus (135) sends the recovered raw parameter to the original tuple. This proves surjectivity.

Next we prove that (136) preserves each fibre. It preserves the parameter domain, and applying it twice is the identity. It has no fixed point because $\zeta = -\zeta$ would imply $\zeta = 0$, contrary to $|\zeta| = 1$. Replacing ζ by $-\zeta$ preserves $\gamma = -\zeta^{-2}$ and gives $W(-\zeta) = -W(\zeta)$. Put $D_0 = \operatorname{diag}(1, -I_2)$. Then

$$\begin{aligned} B(-\zeta) &= B(\zeta) D_0, \\ D_0 (ie \operatorname{diag}(1, 0, -1)) D_0 &= ie \operatorname{diag}(1, 0, -1), \\ D_0 X(v) D_0 &= X(-v). \end{aligned}$$

Replacing (p_+, p_-) by $(-p_+, -p_-)$ replaces (v, w) by $(-v, -w)$ in (133), while leaving M, N, t, e, k, ω fixed. The preceding matrix equalities therefore give $\mathfrak{F} \circ \iota = \mathfrak{F}$ on the entire raw parameter domain. This proves that $\overline{\mathfrak{F}}$ is well defined.

To determine every fibre, suppose two raw parameters have the same literal image. The common temporal matrix C_d recovers the same e, z, γ by the inverse formulas above. Their frame roots therefore satisfy either $\zeta_2 = \zeta_1$ or $\zeta_2 = -\zeta_1$, since they square to the same nonzero complex number. If $\zeta_2 = \zeta_1$, the classified temporal and spatial matrices obtained by that common inverse frame map coincide. The classified bijection of Theorem 1.66 then forces equality of all other raw parameters.

If $\zeta_2 = -\zeta_1$, the identity $B(-\zeta) = B(\zeta) D_0$ implies

$$Z'_{j,2} = D_0 Z'_{j,1} D_0, \qquad v_2 = -v_1, \qquad w_2 = -w_1.$$

Hence $M_2 = M_1$ and $N_2 = N_1$, so the inverse coordinates u, v_0, y are equal. The original e, ω are equal, which also gives equal ϵ, σ. Consequently $h_\pm$ agree, and (134) gives

$$p_{+,2} = -p_{+,1}, \qquad p_{-,2} = -p_{-,1}.$$

Thus the two raw parameters differ by exactly ι. No further case is possible. Each fibre is therefore precisely one two-element orbit. This proves injectivity of $\overline{\mathfrak{F}}$ as well as its already-proved surjectivity.

Finally the inverse formulas starting from a literal tuple give the same orbit for either root choice, by the second fibre case. Starting from a raw parameter, applying $\mathfrak{F}$ and then those inverse formulas gives its orbit, by the two fibre cases. Starting from a literal tuple, applying the inverse formulas and then $\overline{\mathfrak{F}}$ gives that literal tuple by the first paragraph. These prove both compositions are identities on their stated domains, completing the quotient bijection. □

The construction retains $\operatorname{sign} e = \epsilon$ and $\operatorname{sign}\omega = \epsilon\sigma$. The involution changes neither sign. The two independent vector phases and the frame phase remain coordinates, subject only to the proved simultaneous-sign identification.

For explicit scope, the additional full spatial residuals would be

$$R_1 = -\omega^2 Z_{1,0} - 2\omega[C_0, Z_{2,0}] + [C_0,[C_0, Z_{1,0}]] + \mathcal{K}_{Z_{1,0},Z_{2,0}} Z_{1,0} + (t/2)\mathcal{I}_{H_0} Z_{1,0},$$
$$R_2 = -\omega^2 Z_{2,0} + 2\omega[C_0, Z_{1,0}] + [C_0,[C_0, Z_{2,0}]] + \mathcal{K}_{Z_{1,0},Z_{2,0}} Z_{2,0} + (t/2)\mathcal{I}_{H_0} Z_{2,0}.$$

Neither $R_1 = 0$ nor $R_2 = 0$ is part of $\mathcal{S}_{\mathrm{sg}}$ or used in either proof above. The bijection therefore classifies exactly the two scalar equations and Gauss; it does not assert a solution of these two additional spatial equations.

1.28.5 A short exact family with every equation checked

Proposition 1.69. *Choose arbitrary $p_+, p_- \in U(1)$ and independent $\epsilon, \sigma \in \{1, -1\}$. Put*

$$M = \delta \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad t = 24\delta, \quad e = \epsilon\sqrt{2 + 3\delta}, \quad k = \frac{2\sigma}{3\sqrt{13}}, \quad \omega = \frac{3\sigma\sqrt{13}}{2} e,$$

$$N = \frac{\sigma\delta}{\sqrt{13}} \begin{pmatrix} 2 & -1 \\ -1 & -6 \end{pmatrix}, \qquad a_\pm = \sqrt{\delta} \begin{pmatrix} \sqrt{1 \pm 2\sigma/\sqrt{13}} \\ (1 \mp \sigma/\sqrt{13})/\sqrt{1 \pm 2\sigma/\sqrt{13}} \end{pmatrix}.$$

Define v, w by (126)*. For any retained unit frame phase ζ, these give original tuples by*

$$C_0 = B(ie\,\mathrm{diag}(1, 0, -1))B^*, \qquad Z_{1,0} = BX(v)B^*, \qquad Z_{2,0} = BX(w)B^*.$$

Every such tuple satisfies (112)–(114) *at the positive original coupling $g^2 = 24\delta$.*

Proof. The first diagonal factors are positive because $2/\sqrt{13} < 1$. For a sign s, the exact identity

$$(1 - s\sigma/\sqrt{13})^2 = (1 + 2s\sigma/\sqrt{13})(2 - 6s\sigma/\sqrt{13})$$

follows by multiplying out and using $\sigma^2 = 1$. Hence $a_\pm a_\pm^* = M \pm N$. The construction therefore has exactly the displayed M, N for every pair of phases. In particular

$$u = \delta, \quad v_0 = 2\delta, \quad m = \delta, \quad \operatorname{tr} M = 3\delta, \quad d = \delta^2, \quad b = \tfrac{1}{2}\operatorname{tr} N = -\frac{2\sigma\delta}{\sqrt{13}}.$$

The scalar identities are $e^2 = 2 + 3\delta$ and $2r\delta = 1$; substitution in (116) and (117) makes every entry zero. For the full Gauss block, direct multiplication gives

$$QM + MQ = \delta \begin{pmatrix} 2 & 3 \\ 3 & 8 \end{pmatrix}, \qquad tE - 2(QM + MQ) = \delta \begin{pmatrix} 12 & -6 \\ -6 & -36 \end{pmatrix}.$$

The same matrix is $(4\omega/e)N$, since $\omega/e = 3\sigma\sqrt{13}/2$. Thus $\mathcal{G} = 0$ entry by entry, and the upper entry is explicitly

$$\gamma_0 = 4e\delta + 20e\delta + 8\left(\frac{3\sigma\sqrt{13}}{2} e\right)\left(-\frac{2\sigma\delta}{\sqrt{13}}\right) = 0.$$

Finally (115) proves all three literal equations exactly. As an additional check on the construction, its determinant data are

$$T = \frac{3\delta}{2} \begin{pmatrix} 2 & -1 \\ -1 & -6 \end{pmatrix}, \quad \det T = -\frac{117}{4}\delta^2, \quad k^2 = \frac{4}{117},$$

agreeing with the general parameterization. Both signs of the original frequency are present because $\operatorname{sign}\omega = \epsilon\sigma$. □

Corollary 1.70 (Every positive original coupling occurs)**.** *For every specified original $\tau = g^2 > 0$, the literal two-cross scalar–Gauss domain contains tuples at $t = \tau$, with either original frequency sign and both temporal signs. It contains continuous phase families at each such coupling.*

Proof. For $\rho > 1/\sqrt{2}$ choose

$$M_\rho = \delta \begin{pmatrix} \rho & 1 \\ 1 & 2\rho \end{pmatrix}.$$

Its parameters obey $u = \delta\rho > 0$, $d = \delta^2(2\rho^2 - 1) > 0$, $c = 3\delta\rho > 0$ and $m = \delta$. Thus Theorem 1.66 gives the complete solution family at

$$t = \frac{72\delta^2(2\rho^2 - 1)}{3\delta\rho} = \frac{24\delta(2\rho^2 - 1)}{\rho}.$$

For a given $\tau > 0$, set

$$\rho = \frac{\tau + \sqrt{\tau^2 + 4608\delta^2}}{96\delta}.$$

Because $\sqrt{4608}\delta = 48\sqrt{2}\delta$, this is strictly greater than $1/\sqrt{2}$. Squaring the displayed defining equality gives $48\delta\rho^2 - \tau\rho - 24\delta = 0$, which is precisely $t = \tau$. No limiting passage or change of coupling is used. The independent choices of $\epsilon, \sigma, p_+, p_-$ supply the stated signs and phases; the inverse in Theorem 1.66 ensures different phase pairs yield different classified tuples. □

1.28.6 Exact scope and reproducibility

The results concern the simultaneous zero set of both scalar equations and Gauss. The construction never imposes either spatial equation and does not claim a full rotating-ansatz solution. It proves that the previously specified scalar–Gauss domain of the two-cross quantitative residual certificate is nonempty, in fact at every original positive coupling. This is an algebraic statement about the retained classical residuals and makes no quantum assertion.

The companion `check_scalar_gauss_domain.py` is a finite exact symbolic replay. It checks the full scalar and Gauss matrix formulas with eight independent real spatial coordinates, the determinant/trace identities, the explicit factors and both signs, and the original unitary frame maps. The rank, positivity, parameter domains, completeness and inverse-map proofs are given in this document; they are not inferred from sampled values or from a numerical residual tolerance.

The exact full literal parameter map and its inverse are proved in Theorem 1.68; its only identification changes the signs of the frame phase and both vector phases together. Combining Corollary 1.70 with Proposition 1.63 makes the stated spatial-error bound non-vacuous at every positive original coupling. This does not alter the full-solution exclusion in Theorem 1.60.

The complete source versions, the independently checked literal-domain addendum and the exact symbolic replay are retained under `supporting_materials/workbench/evidence/scalar_gauss_domain_20260905/`. The retained equations are those derived from the parent model in (46)–(50); this proof neither changes that model nor replaces its two spatial equations by the Gauss equation.

1.29 Smooth literal inverse charts and all positive-coupling fibres

The exact algebraic bijections of Theorems 1.66 and 1.68 admit a smooth refinement. We prove the parameter domain's convexity, give the phase quotient its explicit smooth coordinates, and construct smooth inverse maps from the literal matrix entries. We then give a global product description in which the first continuous coordinate is the unchanged original coupling $t = g^2$. The topology of every fixed-coupling fibre will follow from explicit inverse formulas.

1.29.1 Retained objects and the exact parameter domain

Keep the literal matrices and constants

$$H_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1/2 & 1/(2\sqrt{2}) \\ 0 & 1/(2\sqrt{2}) & 1/2 \end{pmatrix}, \quad q = \frac{1}{2\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad A_0 = \sqrt{2} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$m^2 = \Gamma = 1, \qquad \kappa = 6, \qquad F_0 = [A_0, [A_0, H_0]] = -2H_0 + \operatorname{diag}(-2, 1, 1),$$

and set

$$r = \frac{\sqrt{6}}{4}, \quad \delta = \frac{1}{2r} = \sqrt{\frac{2}{3}}, \quad x = \frac{1}{2}, \quad d_0 = \frac{1}{2\sqrt{2}}, \quad U = \operatorname{diag}\left(1, \frac{1}{\sqrt{2r(r+x)}} \begin{pmatrix} d_0 & -(r+x) \\ r+x & d_0 \end{pmatrix}\right).$$

Thus $U^*U = I_3$, $\det U = 1$, $Uq = q$, and $U^*H_0U = \operatorname{diag}(0, r, -r)$, as proved in Lemma 1.67. Write

$$P = \operatorname{diag}(1, 0, 0), \qquad X(v) = \begin{pmatrix} 0 & -v^* \\ v & 0_2 \end{pmatrix}, \qquad \mathfrak{p} = X(\mathbb{C}^2) \subset \mathfrak{su}(3).$$

The original real ambient manifold for this selected spatial class is

$$\mathcal{E} = (0, \infty) \times (\mathbb{R} \setminus \{0\}) \times \mathfrak{su}(3) \times \mathfrak{p} \times \mathfrak{p}.$$

It has real dimension $1 + 1 + 8 + 4 + 4 = 18$. For $Y_{\mathrm{tf}} = Y - (\operatorname{tr} Y)I_3/3$, put

$$\mathcal{I}_{H_0} Y = ([H_0, [H_0, Y]] + qq^*Y + Yqq^*)_{\mathrm{tf}}, \qquad \mathcal{K}_{Z_1, Z_2} Y = -\sum_{j=1}^{2} [Z_j, [Z_j, Y]].$$

The subset $\mathcal{S}_{\mathrm{sg}} \subset \mathcal{E}$ is exactly the simultaneous zero set

$$\begin{aligned} [C_0, [C_0, H_0]] - \sum_j [Z_{j,0}, [Z_{j,0}, H_0]] - F_0 &= 0, \\ (C_0^2 - Z_{1,0}^2 - Z_{2,0}^2)q + 2q &= 0, \\ (t\mathcal{I}_{H_0} + 2\mathcal{K}_{Z_{1,0}, Z_{2,0}})C_0 - 4\omega[Z_{1,0}, Z_{2,0}] &= 0. \end{aligned}$$

Neither additional spatial residual is included in this definition. The topology of $\mathcal{S}_{\mathrm{sg}}$ initially means its subspace topology in $\mathcal{E}$; its embedded smooth structure will be proved.

For real coordinates (u, v_0, y) define

$$h = \delta^2 + y^2, \qquad m = \delta + iy, \qquad M = \begin{pmatrix} u & m \\ \bar{m} & v_0 \end{pmatrix}, \qquad d = uv_0 - h, \qquad c = 4v_0 - 5u,$$

$$\mathcal{D} = \{(u, v_0, y) \in \mathbb{R}^3 : u > 0,\ d > 0,\ c > 0\}.$$

Here h is a scalar, distinct from the two factor entries $h_\pm$ used below.

Lemma 1.71 (Convexity of the entire continuous parameter domain)**.** *The set $\mathcal{D}$ is a nonempty open convex subset of $\mathbb{R}^3$. In particular it is connected and has real dimension three.*

Proof. The inequalities are strict inequalities for continuous functions, so $\mathcal{D}$ is open. For $u > 0$ and $(a, b) \in \mathbb{C}^2$, the exact quadratic-form identity is

$$\begin{pmatrix} \bar{a} & \bar{b} \end{pmatrix} M \begin{pmatrix} a \\ b \end{pmatrix} = u \left| a + \frac{m}{u} b \right|^2 + \frac{d}{u} |b|^2.$$

It proves $M > 0$ when $u > 0, d > 0$. Conversely, $M > 0$ gives $u > 0$ by the first coordinate vector, and evaluation at $(a, b) = (-m/u, 1)$ gives $d/u > 0$. Thus the conditions $u > 0, d > 0$ are exactly positive definiteness of the Hermitian matrix in the affine slice $\operatorname{Re} M_{12} = \delta$.

If M_0, M_1 are positive definite and $0 \le \lambda \le 1$, then for every nonzero $z \in \mathbb{C}^2$,

$$z^*((1-\lambda)M_0 + \lambda M_1)z = (1-\lambda)z^* M_0 z + \lambda z^* M_1 z > 0.$$

The affine slice is preserved by this combination, and the additional linear inequality $4v_0 - 5u > 0$ is preserved as well. The coordinate map $(u, v_0, y) \mapsto M$ is affine and injective, so these facts prove convexity in the original real coordinates. Finally $(u, v_0, y) = (\delta, 2\delta, 0)$ has $u = \delta > 0$, $d = \delta^2 > 0$, and $c = 3\delta > 0$. It supplies a point and, by openness, a three-dimensional neighbourhood. Straight segments inside a convex set join any two points, proving connectedness. □

1.29.2 The phase quotient with every original phase retained

Write $\mathbb{T} = U(1) = \{z \in \mathbb{C} : |z| = 1\}$. On the original three phases, in the order (ζ, p_+, p_-), define

$$\Psi : \mathbb{T}^3 \longrightarrow \mathbb{T}^3, \qquad \Psi(\zeta, p_+, p_-) = \left(\zeta^2, \frac{p_+}{\zeta}, \frac{p_-}{\zeta}\right) = (\eta, \alpha, \beta). \tag{138}$$

Lemma 1.72 (Exact smooth phase quotient)**.** *The map Ψ is a surjective smooth group homomorphism with kernel $\{(1,1,1), (-1,-1,-1)\}$. It induces a diffeomorphism*

$$\mathbb{T}^3 / \{(1,1,1), (-1,-1,-1)\} \;\longrightarrow\; \mathbb{T}^3.$$

The quotient smooth charts have explicit local inverses retaining both original square-root representatives.

Proof. Products, inverses on $\mathbb{T}$, and the square map are smooth; the displayed formulas commute with coordinatewise multiplication. An element is in the kernel precisely when $\zeta^2 = 1$ and $p_+ = p_- = \zeta$, giving exactly the two listed elements. Given any $(\eta, \alpha, \beta) \in \mathbb{T}^3$, choose either root $\zeta^2 = \eta$ and set $p_+ = \zeta\alpha$, $p_- = \zeta\beta$. This gives all preimages, so Ψ is onto and each fibre is exactly one simultaneous-sign pair.

Here are the smooth local charts, rather than only a set quotient. Fix $\eta_0 \in \mathbb{T}$ and one root $\zeta_0^2 = \eta_0$. On the open arc $\eta = \eta_0 e^{i\theta}$ with $-\pi < \theta < \pi$, the two lifts are exactly

$$\zeta_\pm = \pm\zeta_0 e^{i\theta/2}, \qquad p_{+,\pm} = \zeta_\pm \alpha, \qquad p_{-,\pm} = \zeta_\pm \beta.$$

Each formula is smooth. The two root arcs are disjoint and open in the preimage, and each maps diffeomorphically to the displayed arc times $\mathbb{T}^2$, using (138) as its inverse. Their simultaneous-sign interchange identifies exactly these two charts. The quotient topology agrees with these charts: the quotient projection is open because the inverse image of the image of any open set is that set united with its simultaneous-sign translate. On overlapping arcs the quotient coordinates are the same (η, α, β), so their transitions are the ordinary smooth torus transitions. These charts prove the asserted diffeomorphism. They do not select a global square root or discard either representative. □

1.29.3 Global formulas for literal matrices and their smooth inverses

We record all the algebraic parameters used by the smooth map. For $(u, v_0, y) \in \mathcal{D}$, set

$$Q = \operatorname{diag}(1, 2), \qquad E = \operatorname{diag}(2/3, -5/6), \qquad t = \frac{72d}{c}, \qquad T = \frac{t}{4}E - \frac{1}{2}(QM + MQ).$$

For independent signs $\epsilon, \sigma \in \{1, -1\}$ put

$$e = \epsilon\sqrt{2 + u + v_0}, \qquad k = \sigma\sqrt{\frac{d}{-\det T}}, \qquad \omega = e/k, \qquad N = kT, \qquad A_\pm = M \pm N.$$

Theorem 1.66 proves on this entire domain that $-\det T > 0$, that $A_\pm$ are positive semidefinite of rank one, and that $h_\pm = (A_\pm)_{11} > 0$. Consequently

$$a_\pm = \begin{pmatrix} \sqrt{h_\pm} \\ (A_\pm)_{21}/\sqrt{h_\pm} \end{pmatrix}$$

are defined everywhere in $\mathcal{D}$ for both signs.

For $(\eta, \alpha, \beta) \in \mathbb{T}^3$ put

$$v^\flat = \frac{\alpha a_+ + \beta a_-}{2}, \qquad w^\flat = \frac{i(\alpha a_+ - \beta a_-)}{2}.$$

$$L(\eta) = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 & -1 \\ \eta & -\eta \end{pmatrix}, \qquad B^\flat(\eta) = U \operatorname{diag}(1, L(\eta)).$$

The superscript $\flat$ names these exact intermediate coordinates; it does not denote a metric dual or a change of magnitude. Multiplication gives

$$L(\eta)^* L(\eta) = I_2, \qquad \det L(\eta) = \eta, \qquad \det B^\flat(\eta) = \eta.$$

In particular these auxiliary matrices are unitary with their displayed determinant; they are not asserted to have determinant one. The precise relation to the original determinant-one frame is, for either root $\zeta^2 = \eta$,

$$\gamma = -\zeta^{-2}, \qquad W(\zeta) = \frac{1}{\sqrt{2}}\begin{pmatrix} -\zeta^{-1} & -\zeta^{-1} \\ \zeta & -\zeta \end{pmatrix}, \qquad L(\eta) = \zeta W(\zeta).$$

If $p_+ = \zeta\alpha$, $p_- = \zeta\beta$, the original vectors satisfy $v = \zeta v^\flat$ and $w = \zeta w^\flat$. Thus every phase and every determinant factor is retained.

Let

$$\mathcal{X} = \{1, -1\}^2 \times \mathcal{D} \times \mathbb{T}^3$$

have coordinates $(\epsilon, \sigma, u, v_0, y, \eta, \alpha, \beta)$. Define $\mathcal{F}^\flat : \mathcal{X} \to \mathcal{E}$ by

$$\begin{aligned} \mathcal{F}^\flat(\epsilon, \sigma, u, v_0, y, \eta, \alpha, \beta) &= (t, \omega, C_0, Z_{1,0}, Z_{2,0}), \\ C_0 &= B^\flat(\eta)\big(ie \operatorname{diag}(1, 0, -1)\big)B^\flat(\eta)^*, \\ Z_{1,0} &= B^\flat(\eta)X(v^\flat)B^\flat(\eta)^*, \\ Z_{2,0} &= B^\flat(\eta)X(w^\flat)B^\flat(\eta)^*. \end{aligned} \tag{139}$$

Theorem 1.73 (Four smooth connected components in literal coordinates)**.** *The image of $\mathcal{F}^\flat$ is exactly $\mathcal{S}_{\mathrm{sg}}$, and $\mathcal{F}^\flat$ is a diffeomorphism onto an embedded six-real-dimensional submanifold of $\mathcal{E}$. The submanifold has exactly four connected components, indexed by*

$$\epsilon = \operatorname{sign} e, \qquad \sigma = \operatorname{sign}(e/\omega).$$

Each component is diffeomorphic to $\mathcal{D} \times \mathbb{T}^3$. The smooth structure and inverse charts refer to the literal matrix subset; no quotient of literal configurations by a physical gauge action is being taken.

Proof. First verify exactly which literal matrices the displayed formulas give. Let $D_\zeta = \operatorname{diag}(1, \zeta I_2)$ and let $B(\zeta) = U \operatorname{diag}(1, W(\zeta))$ be the original frame. Then

$$B^\flat(\eta) = B(\zeta) D_\zeta,$$
$$D_\zeta (ie \operatorname{diag}(1,0,-1)) D_\zeta^* = ie \operatorname{diag}(1,0,-1).$$

$$D_\zeta X(v^\flat) D_\zeta^* = X(\zeta v^\flat) = X(v), \qquad D_\zeta X(w^\flat) D_\zeta^* = X(w).$$

Thus (139) gives exactly the same literal tuple as the original map for each of its two phase lifts. Lemma 1.72 and the already-proved literal quotient bijection, Theorem 1.68, therefore show that its image is $\mathcal{S}_{\mathrm{sg}}$ and that $\mathcal{F}^\flat$ is bijective.

Here is its inverse directly in literal matrix entries. Given a tuple in $\mathcal{S}_{\mathrm{sg}}$, compute

$$C_d = U^* C_0 U, \qquad e = -i(C_d)_{11}, \qquad z = (C_d)_{23}, \qquad \eta = -\frac{2i\bar{z}}{e}. \tag{140}$$

The temporal classification proves that e is real and nonzero, $|z| = |e|/2$, and hence $|\eta| = 1$. The sign is checked by multiplying the lower temporal block in (139):

$$(C_d)_{\mathrm{lower}} = -\frac{ie}{2} \begin{pmatrix} 1 & \bar{\eta} \\ \eta & 1 \end{pmatrix}.$$

In particular $z = -ie\bar{\eta}/2$, so $-2i\bar{z}/e = \eta$ exactly.

Define the two matrices and their entire lower first columns by

$$Z_j^\flat = B^\flat(\eta)^* Z_{j,0} B^\flat(\eta), \qquad v^\flat = \begin{pmatrix} (Z_1^\flat)_{21} \\ (Z_1^\flat)_{31} \end{pmatrix}, \qquad w^\flat = \begin{pmatrix} (Z_2^\flat)_{21} \\ (Z_2^\flat)_{31} \end{pmatrix}.$$

Then recover

$$M = v^\flat (v^\flat)^* + w^\flat (w^\flat)^*, \qquad N = i(v^\flat (w^\flat)^* - w^\flat (v^\flat)^*),$$
$$u = M_{11}, \quad v_0 = M_{22}, \quad y = \operatorname{Im} M_{12}, \qquad \epsilon = \operatorname{sign} e, \quad \sigma = \operatorname{sign}(e/\omega),$$
$$h_\pm = (M \pm N)_{11}, \qquad \alpha = \frac{(v^\flat - iw^\flat)_1}{\sqrt{h_+}}, \qquad \beta = \frac{(v^\flat + iw^\flat)_1}{\sqrt{h_-}}. \tag{141}$$

Indeed multiplication by the common unit ζ preserves both M and N. The original phase inverse therefore gives $\alpha = p_+/\zeta$, $\beta = p_-/\zeta$ in these formulas, with $h_\pm > 0$, $|\alpha| = |\beta| = 1$. The algebraic inverse proves that all recovered coordinates lie in their exact domains. The displayed formulas recover the original parameter from (139), and the forward map recovers the original literal tuple from them.

All forward functions are real smooth on each sign sector: $c > 0$, $d > 0$, $-\det T > 0$, $2+u+v_0 > 0$, and $h_\pm > 0$ are precisely the denominators and square-root arguments appearing above, while $k \neq 0$. Matrix products and conjugates are real smooth. In the inverse, $e \neq 0$ and $h_\pm > 0$ ensure

smoothness of every non-discrete formula. The signs are locally constant. The inverse formulas in particular give a continuous inverse for the subspace topology on $\mathcal{S}_{\mathrm{sg}}$.

We check embeddedness, rather than just transferring an abstract smooth structure through a bijection. At any one image point choose open torus arcs around η, α, β, with their real angle coordinates. The inverse formulas extend as real smooth functions to an open neighbourhood in $\mathcal{E}$ as follows. Keep $e \neq 0$ and the computed $h_{\pm} > 0$ by taking a sufficiently small neighbourhood. The same matrix formulas define the complex quantities η, α, β there, even when the tuple is off the zero set. For a unit base point η_0, on the complex neighbourhood $\mathrm{Re}(\eta/\eta_0) > 0$, its local angle is the smooth real function

$$\arctan\left(\frac{\mathrm{Im}(\eta/\eta_0)}{\mathrm{Re}(\eta/\eta_0)}\right).$$

Use the corresponding expressions about α_0, β_0. On the zero set these are exactly its three local torus coordinates. Together with u, v_0, y, they give a real smooth six-coordinate map from an ambient neighbourhood whose composition with $\mathcal{F}^{\flat}$ on the parameter chart is the identity. The derivative of that composition is the identity, so the derivative of $\mathcal{F}^{\flat}$ has rank six at every point.

For clarity, the real inverse function theorem [1, §1, pp. 4–5] now applies with its full-rank hypothesis verified: choose six ambient linear coordinate projections whose composite with this rank-six parametrization has invertible derivative. Locally the image is a smooth graph over those six coordinates. The continuous inverse already proved prevents other parameter patches from contributing additional image points arbitrarily near this point outside its chart: the inverse image of the chart is open in the subspace topology. Thus the graph is a neighbourhood of the point in the whole subset $\mathcal{S}_{\mathrm{sg}}$. This proves it is an embedded six-dimensional submanifold and that the displayed inverse is smooth for its embedded structure.

It remains to count components exactly. By Lemma 1.71, $\mathcal{D}$ is path connected. Each circle is path connected: for chosen real arguments of two unit endpoints, interpolating those arguments gives an explicit path on the circle. Products of these paths prove $\mathbb{T}^3$ is path connected. Hence each of the four sign sectors in $\mathcal{X}$ is nonempty and connected. On the literal domain the two sign functions in (141) are continuous with values in $\{1, -1\}$ because e, ω never vanish. They cannot change on a connected subset, so distinct sectors cannot belong to the same component. This proves there are exactly four connected components.

Finally the only quotient used to construct the coordinates was the two-element kernel of the parameter map. Its two representatives give identical literal matrices, as the first displayed equalities proved. Distinct literal tuples remain distinct under the bijection. No further gauge equivalence or spatial-pair equivalence has been imposed. □

1.29.4 The original coupling is a surjective submersion

On $\mathcal{D}$ the original coupling function is

$$t(u, v_0, y) = \frac{72(uv_0 - \delta^2 - y^2)}{4v_0 - 5u} = \frac{72d}{c}.$$

Theorem 1.74 (Exact differential and global coupling coordinates)**.** *The function $t : \mathcal{D} \to (0, \infty)$ is a smooth surjective submersion. Its exact partial derivatives are*

$$\begin{aligned} \frac{\partial t}{\partial u} &= \frac{72(4v_0^2 - 5h)}{c^2}, & \frac{\partial t}{\partial v_0} &= \frac{72(4h - 5u^2)}{c^2}, \\ \frac{\partial t}{\partial y} &= -\frac{144y}{c}, & h &= \delta^2 + y^2. \end{aligned} \tag{142}$$

In fact $\partial t/\partial u > 0$ throughout $\mathcal{D}$.

Define the fixed open set

$$\mathcal{B} = \{(v_0, y) \in \mathbb{R}^2 : v_0 > \tfrac{\sqrt{5}}{2}\sqrt{\delta^2 + y^2}\}.$$

The map

$$\Theta : \mathcal{D} \longrightarrow (0,\infty) \times \mathcal{B}, \qquad \Theta(u, v_0, y) = (t(u, v_0, y), v_0, y) \tag{143}$$

is a diffeomorphism, with exact inverse

$$u = \frac{4\tau v_0 + 72(\delta^2 + y^2)}{72v_0 + 5\tau}, \qquad (\tau, v_0, y) \in (0,\infty) \times \mathcal{B}. \tag{144}$$

Moreover $\mathcal{B}$ is diffeomorphic to $\mathbb{R}^2$ by the explicit inverse pair

$$\begin{aligned} (y, s) &\longmapsto \left(\tfrac{\sqrt{5}}{2}\sqrt{\delta^2 + y^2} + e^s,\ y\right), \\ (v_0, y) &\longmapsto \left(y, \log\left(v_0 - \tfrac{\sqrt{5}}{2}\sqrt{\delta^2 + y^2}\right)\right). \end{aligned} \tag{145}$$

Here $\tau = t = g^2$ is the unchanged positive original coupling.

Proof. Since $c > 0$, the quotient defining t is smooth. The derivatives $d_u = v_0$, $d_{v_0} = u$, $d_y = -2y$, $c_u = -5$, $c_{v_0} = 4$, $c_y = 0$ give

$$\frac{\partial t}{\partial u} = \frac{72(v_0 c + 5d)}{c^2} = \frac{72(4v_0^2 - 5h)}{c^2}, \qquad \frac{\partial t}{\partial v_0} = \frac{72(uc - 4d)}{c^2} = \frac{72(4h - 5u^2)}{c^2},$$

and the last derivative in (142) is $72(-2y)/c$. On $\mathcal{D}$, $v_0 = (d + h)/u > 0$, $c > 0$, and $d > 0$. Hence $v_0 c + 5d > 0$, proving $\partial t/\partial u > 0$. The differential therefore has rank one everywhere, proving the submersion assertion independently of any fibre-topology argument. In particular its three partial derivatives cannot simultaneously vanish. The proposed algebraic simultaneous-zero test also agrees: the first two zero equations would give $4v_0^2 = 5h$ and $4h = 5u^2$, so positivity of u, v_0 would imply $v_0 = 5u/4$, and then both $c = 0$ and $d = 0$, excluded from $\mathcal{D}$.

We now prove the exact global inverse and surjectivity. For a point of $\mathcal{D}$, the identity

$$4v_0^2 - 5h = v_0 c + 5d > 0$$

and $v_0 > 0$ imply $(v_0, y) \in \mathcal{B}$. Writing $\tau = 72d/c > 0$, multiply by the retained c to obtain

$$\tau(4v_0 - 5u) = 72(uv_0 - h), \qquad (72v_0 + 5\tau)u = 4\tau v_0 + 72h.$$

Its denominator $72v_0 + 5\tau$ is strictly positive, giving (144).

Conversely fix any $\tau > 0$ and $(v_0, y) \in \mathcal{B}$, and use (144). Here $h > 0$, $v_0 > 0$, and the numerator and denominator of u are positive. Exact multiplication, with no sign or factor removed, gives

$$\begin{aligned} c = 4v_0 - 5u &= \frac{72(4v_0^2 - 5h)}{72v_0 + 5\tau} > 0, \\ d = uv_0 - h &= \frac{\tau(4v_0^2 - 5h)}{72v_0 + 5\tau} > 0. \end{aligned} \tag{146}$$

Thus the recovered point is in $\mathcal{D}$, and $72d/c = \tau$ exactly. These calculations prove both compositions of (143) and (144) are identities. Both maps are smooth on their stated open domains.

The set $\mathcal{B}$ is nonempty; for example $(v_0, y) = (\sqrt{5}\delta/2 + 1, 0)$ belongs to it. Thus every $\tau > 0$ has a preimage, proving surjectivity. Finally $\delta > 0$ makes $\sqrt{\delta^2 + y^2}$ smooth for all real y, and the defining inequality of $\mathcal{B}$ makes the logarithm argument in (145) strictly positive. Exponential and logarithm are inverse smooth functions on their stated real domains. Substitution into (145) proves both compositions are identities, establishing the diffeomorphism with $\mathbb{R}^2$. □

1.29.5 Exact scope and topology of the positive-coupling level sets

Corollary 1.75 (All positive-coupling levels and their literal inverse charts)**.** *In the literal embedded manifold $\mathcal{S}_{\mathrm{sg}}$, the original coupling projection*

$$\pi : \mathcal{S}_{\mathrm{sg}} \longrightarrow (0,\infty), \qquad (t,\omega,C_0,Z_{1,0},Z_{2,0}) \longmapsto t$$

is a smooth surjective submersion. There is an explicit diffeomorphism

$$\mathcal{S}_{\mathrm{sg}} \;\cong\; \{1,-1\}^2 \times (0,\infty) \times \mathbb{R}^2 \times \mathbb{T}^3 \tag{147}$$

under which π is projection onto the displayed unchanged $(0,\infty)$ coordinate. For every specified original $\tau > 0$, the complete fixed-coupling level $\pi^{-1}(\tau)$ is an embedded five-real-dimensional submanifold, with exactly four connected components. Each component is diffeomorphic to $\mathbb{R}^2 \times \mathbb{T}^3$. These statements concern precisely the scalar–Gauss domain with both original signs and all three phase coordinates retained.

Proof. Compose the literal diffeomorphism of Theorem 1.73 with the diffeomorphisms of Theorem 1.74. More explicitly, given

$$(\epsilon,\sigma,\tau,y,s,\eta,\alpha,\beta) \in \{1,-1\}^2 \times (0,\infty) \times \mathbb{R}^2 \times \mathbb{T}^3,$$

first put

$$v_0 = \frac{\sqrt{5}}{2}\sqrt{\delta^2+y^2} + e^s, \qquad u = \frac{4\tau v_0 + 72(\delta^2+y^2)}{72v_0 + 5\tau}.$$

Then apply (139) with these u, v_0, y and the retained signs and phases. Equation (146) proves that its coupling is exactly τ and every denominator is positive. The inverse begins with the literal matrix formulas (140) and (141), keeps $\tau = t$ from the original tuple, and recovers

$$s = \log\left(v_0 - \frac{\sqrt{5}}{2}\sqrt{\delta^2+y^2}\right).$$

The preceding theorems prove that these maps are smooth, mutually inverse, and defined on every point of their stated domains. This proves (147).

In these exact product coordinates π is the projection onto τ. Its derivative is onto the one-dimensional tangent space of $(0,\infty)$, and every $\tau > 0$ occurs. The fixed-coordinate slice is a smooth embedded codimension-one submanifold. Transport through the diffeomorphism makes $\pi^{-1}(\tau)$ an embedded five-dimensional submanifold of $\mathcal{S}_{\mathrm{sg}}$, and hence of the original ambient manifold $\mathcal{E}$. Its inclusion into the fixed-coupling ambient hyperplane has the same smooth structure, since that coordinate is identically τ.

The formula identifies the entire fibre, without a missing parameter portion, with $\{1,-1\}^2 \times \mathbb{R}^2 \times \mathbb{T}^3$. The real plane is path connected by line segments; the three circles are path connected by the explicit angle paths used earlier. Their product is therefore connected. The two continuous nonvanishing sign invariants prevent any connection between distinct sign sectors. This proves both the exact count of four components and their stated diffeomorphism type. The fibre topology follows from these explicit inverse maps, not merely from the fact that π is a submersion. □

These manifolds are subsets of the space of ordered literal tuples. The only phase identification was equality of two parameter representatives producing the same tuple. No spatial equation has been imposed, and no physical gauge quotient, frequency-sign identification, or exchange of $Z_{1,0}$ and $Z_{2,0}$ has been made. The construction and all level-set statements use the open domain $t > 0$; they assign no boundary fibre at $t = 0$.

1.30 Sources, independent checks, and reproducibility

The evidence is under `supporting_materials/workbench/evidence/`, in subdirectory `quaternionic_4441_audit_20260904/`. The supplied files are preserved unchanged in its `source/` directory. The source claims used here are the complex-plus-triplet and Lorentz-slice sections of *Triality 4-4-4-1 Dynamic Completion*, and the coordinates, symmetry, positive fibres, and Jordan-completion sections of *Quaternionic 4441 Continuation*, both dated 4 September 2026. Their source TeX and original verification results remain distinguishable from this audit and from the independent checker `verify_exact.py` in that evidence directory. It implements quaternionic matrix multiplication independently, expands the full fifteen-real-variable determinant identity, checks the marked skew transport and charge conjugation, and verifies the positive-cone counterexample, retained Jordan coefficient, group composition, actual scalar acceleration, and electric sign. These checks supplement the proofs above. The separate `verify_invariant_sectors.py` expands the trace-free adjugate and the full complex tangency identity, and checks the retained nonzero-sector coordinates and fourth-order escape coefficient. The invariant-family theorem is proved by tangency and uniqueness, not inferred from finite tests. These checks do not substitute for checking other packet claims such as the gauge-orbit part of the vacuum theorem or the finite-lattice quantization theorem. The separate moving-sector and vacuum verifiers test the exact periodic equations, conserved charge, fourteen-coordinate Hessian cubic, and metric eigenvalues. The scalar global-minimum theorem is proved above, independently of those gauge and quantum claims. The portable `verify_gauge_lift.py` checks all fourteen potential derivatives symbolically and all forty-six real homogeneous variational components at a retained exact two-jet, as well as the orbit, radial, and observation identities. This finite variational test supplements, and does not replace, the full variational proof above. The portable `exact_period_metric_checker.py` verifies the unchanged periods, inverse real period map, full affine fibres, monodromy, rank-loss constants, and a nonreal rational positive-definite example. The smoothness and operator inequalities are proved above rather than inferred from finite tests. The further portable checks `exact_hermitian_metric_spectrum_checker.py`, `verify_vacuum_kinetics.py`, `verify_mass_spectrum.py`, and `verify_arbitrary_gauge_lift.py` test the retained metric/cometric factors, integral quotient and spectral identities, all current-free continuation equations, gauge-generator Gram matrix and eigenvalues, and the complete real-connection residual elimination. Their exact assertion counts and replay logs are in `CLASSICAL_SPECTRUM_EXACT_REPLAYS.json`. Full proofs of existence, spectral completeness, limits and exclusions are given above; finite symbolic tests do not replace those arguments. The further portable `check_complex_rotation.py` verifies the constant-frame blocks, retained original-coordinate map, weighted Gauss inverse, and both sign reconstructions at an exact unrestricted test pair. `check_complex_constraints.py` expands the full unrestricted scalar entries, inertia, spatial stress and polynomial discriminant. Neither checker certifies a zero of the remaining system. The strict inequalities, equivalences, domains and exclusion proofs are the analytic arguments above.

The 5 September finite-fibre and projection evidence is in `supporting_materials/workbench/evidence/finite_coupling_20260905/`. The two original symbolic checkers retain the full eight-dimensional Gauss equation and all residual codomains. The distinct `check_projection_independent.py` uses only standard-library rational-pair and finite-field arithmetic, independently reconstructs the full eight-dimensional Gauss matrix, and reproduces the nonzero Sylvester determinant. Its 101 evaluations verify degree-twelve polynomial identities, not a sampled real coupling exclusion. Complete definitions and proofs are in Theorems 1.57 and 1.59; no existence inside the exceptional algebraic set, or general complex time-dependent lift, is inferred. The original scripts and certificates are retained in the evidence directory's `checkers/` subdirectories.

The separately dated scalar–Gauss continuation and independent review are retained in `supporting_materials/workbench/evidence/scalar_gauss_domain_20260905/`. The maintained proof is `scalar_gauss_domain_shared.tex`; it includes the full temporal-classification and literal-phase-quotient addendum. The portable `check_scalar_gauss_domain.py` replay passes all 25 exact groups, including every scalar and Gauss entry, both sign choices, the continuous phases, inverse frames and the prescribed-coupling formula. Positivity, rank exclusion, completeness and the two-element fibres are proved analytically in Theorems 1.66 and 1.68. This continuation supersedes the earlier absence of a scalar–Gauss nonemptiness assertion; it does not alter the historical proof snapshot: the two spatial residuals still cannot vanish simultaneously. The unrestricted exceptional-set existence problem remains open, and no quantum spectral conclusion follows from these classical residuals.

The subsequent shared source `scalar_gauss_smooth_shared.tex` proves the smooth phase quotient, embedded literal inverse charts, and the global trivialization over the unchanged positive coupling axis (Theorems 1.73 and 1.74, Corollary 1.75). These analytic proofs establish the component and fibre topology beyond the earlier set bijections; the earlier exact replay receipt retains its original scope.

References

[1] J. W. Milnor, *Topology from the Differentiable Viewpoint*, University Press of Virginia, 1965, §1, pp. 1, 4–5. `https://www.maths.ed.ac.uk/~aar/papers/milnortop.pdf`.